

Orchestrating the Single-cell Transcriptome: From Cells to Genes

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Single-cell RNA-sequencing (scRNA-seq) is a technique that measures the transcriptional profile of the samples at single-cell resolution. Such a breakthrough technique brings the genomic study to an entirely new level, providing potential to understand some of the fundamental problems in biology, including gene expression, cell differentiation, cancer progression, and many more. With the help from statisticians, bioinformaticians and computer scientists, pipelines have been developed to process and analyze the huge amount of information from scRNA-seq experiments. However, challenges remain as customization is still needed depending on specific input dataset.

In this study, we applied scRNA-seq to mouse embryos at early organogenesis to understand the cell type dynamics and gene regulations during development, using a variety of data analysis tools with experimental validation. In Chapter 1, we introduced our data analysis pipeline for analyzing large-scale high-heterogeneity dataset obtained from mouse whole embryos at E11.5, and presented a complete catalog of cell types using an integrated annotation approach. In Chapter 2, we extended such a pipeline to time-series datasets from mouse embryos at different stages, by successful removal of batch effects and integration of multiple datasets we provided insights into how cell types emerge and disappear during embryogenesis. In Chapter 3, we focused on the gene expression pattern revealed by scRNA-seq datasets, and presented a novel approach to

identify housekeeping genes from these datasets. In Chapter 4, we have investigated the scRNA-seq datasets from a completely different perspective, treating genes as samples and clustered genes into groups with similar biological functions named 'metagene', and successfully using metagene as a novel approach to understand genes with unknown functions.

Our results showed the possibility of using scRNA-seq as a powerful tool to understand mouse embryonic development from different perspectives, and provided valuable data that could be used for more detailed studies on specific cell types or genes.

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Chapter 1: Cell type atlas of E11.5 mouse embryo based on scRNA-seq

Introduction

Until recently, scientists studying embryonic development at the cellular level had a big problem: they could not get to the single-cell resolution, as the results of bulk RNA sequencing (RNA-seq) could only be interpreted as the average expression level across all cells. During the past decade, the development of high-throughput sequencing technology has allowed for the study of the entire transcriptome in great detail. Specifically, single-cell RNA-sequencing (scRNA-seq) allows for the single-cell resolution of a transcriptome profile. A wide range of methods has been developed for scRNA-seq. Two broad sets of methods include 'plate-based' methods[1,2] and 'drop-based' methods[3,4]. Both methods have been commercialized, by Fluidigm C1 and 10x Genomics[5], respectively. Each approach has advantages and disadvantages in terms of library quality, cellular throughput, and costs.

Tremendous efforts have been made to refine scRNA-seq techniques. However, various challenges remain before achieving the ambitious goal of fully understanding cellular complexity, such as the high level of false negatives[6], amplification bias[7], and difficulties manipulating large data sets. More variables exist during data analysis, such as dimension reduction algorithms, clustering methods, differential gene expression (DGE) detection, and cellular identity annotation, and the choices made between two methods may significantly affect the results. To date, computer scientists have developed several pipelines for scRNA-seq analyses to address the difficulties mentioned above, such as the R package Seurat from the Satija lab[8] and CellRanger from 10x Genomics (<https://support.10xgenomics.com/single-cell-gene-expression/>). However, considering

the massive complexity involved in scRNA-seq analysis, users should be aware of the limitations associated with these convenient pipelines.

A critical task during scRNA-seq analysis is the annotation of cell-type identities based on the gene expression profile. A typical method for annotating cell types is the application of DGE assays to identify differentially expressed genes (DEGs) in individual cell clusters. This analysis has been extensively used in bulk RNA-seq studies to identify changes in expression levels between groups. Examples of widely used bulk RNA-seq DGE tools include DESeq[8] and edgeR[9]. As scRNA-seq technology continues to develop, new challenges have emerged, such as increased signal-noise ratios and dropout events[10,11]. Specific features have been added to available tools to address these problems, and new tools have been developed using multiple approaches, such as MAST[12], Monocle[13], and D3E[14]. Gene set enrichment analysis (GSEA) can be used to analyze gene functions based on a set of well-studied marker genes. This approach works well in most published literature due to the small scale and late developmental stages examined. As we expand to larger scales and multiple-stage samples, the precise annotation of cell-type identities is likely to remain challenging.

In this study, we built a data analysis pipeline to perform large-scale, scRNA-seq on whole mouse embryos using an integrated approach to the investigation of cell type identification, to shed light on significant developmental events.

Result and discussion

1.1 Whole mouse embryo scRNA-sequencing pipeline overview and rationale

To study cell-type heterogeneity in mouse development, we chose mouse embryos at embryonic day E11.5 (TS19) as our target. Embryos were dissected into four samples, including head, trunk, tail, and whole embryo. The 10x Genomics Chromium Single Cell 3' Assay was applied to all samples, and primary analysis was performed using 10x Genomics CellRanger software.

Here, we present a general overview of our scRNA-seq secondary data analysis pipeline, which was specifically tailored for a large-scale, whole-embryo dataset based on our understanding of a variety of existing tool kits (**Figure 1.1**). The digital expression matrix [unique molecular identifier (UMI) count matrix] from 10x Genomics CellRanger was used as the input for this pipeline. The critical steps taken to develop our pipeline and the related considerations are described as follows.

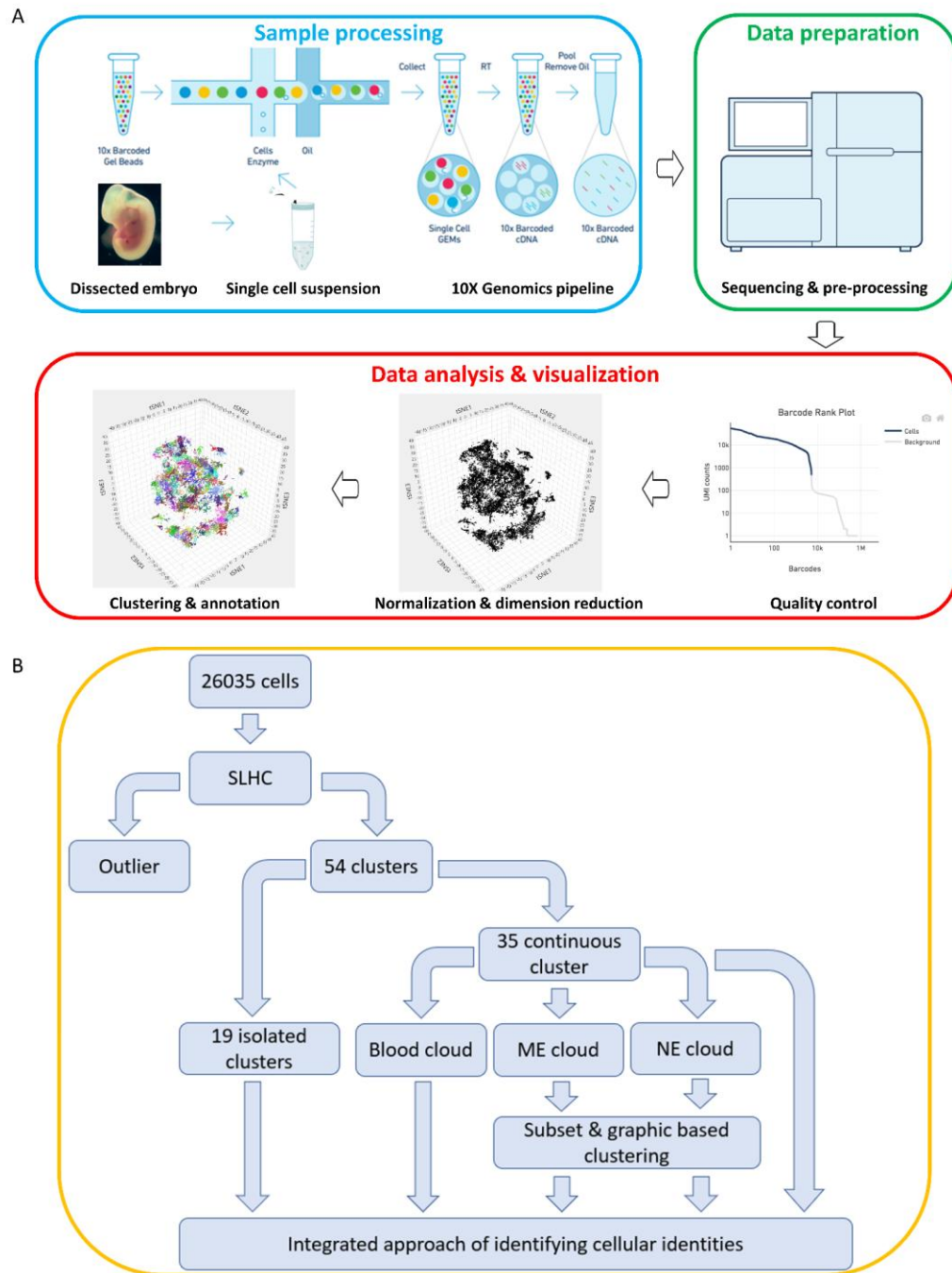


Figure 1.1: Schematic view of scRNA-seq experiment and data analysis pipeline. (A) A schematic figure showing the workflow from processing embryos to library preparation, sequencing, and data analysis. Figures adapted from 10x Genomics: <https://www.10xgenomics.com/products/single-cell-gene-expression>. (B) Schematic

view of clustering of cells in this study. SLHC: single-linkage hierarchical clustering; ME: mesenchyme/mesoderm; NE: Neuroectoderm.

1.1.1 Model-based quality control

The general purpose of quality control (QC) is to eliminate low-quality libraries (in this context, 'cells') before performing any additional secondary downstream analyses. Due to technical limitations associated with the scRNA-seq technique, QC is considered a more challenging task than when performed during canonical batch RNA-seq. Cells featuring technical limitations, including but not limited to low sequencing depth, cell damage, doublets, and inefficient reverse transcription/PCR amplification[15,16], must be removed before proceeding to downstream analyses.

The quality of the cells cannot be determined a priori. Only the results from downstream analyses should be used to judge the performance of QC. Thus, QC matrices must be adjusted back and forth based on downstream results[16]. From our experience, several signals indicate low-quality cells in the pipeline:

1. Low-quality cells form clusters with no specific biological meanings. A set of highly related DEGs are difficult to extract from such cell clusters, as they arise from different cell types.
2. The normalization scaling factor is too small, and the DEG analysis is difficult to perform. Most low-quality cells have a small library size, and the inclusion of these cells in the analysis will bias normalization and make the differences between normal cells less detectable.

3. The data structures of the lower dimensions are distorted. For example, when using principal component analysis (PCA) to reduce dataset dimensionality, the differences between good and poor cells will be captured in the first few principal components (PCs), which are useless for understanding cell-type heterogeneity.

Previous studies have provided various strategies for performing QC on scRNA-seq datasets. Using housekeeping genes is a widely accepted approach during bulk RNA-seq and has been applied in some early scRNA-seq studies[17–19]. However, the assumption that housekeeping genes are constantly expressed in all samples is not necessarily true in scRNA-seq. Housekeeping gene expression has been reported to vary between individual cells[20], and such an approach has rarely been used in more recent studies. Instead, some standard QC matrices are now used to detect low-quality cells. Because spike-ins are not available for 10x Genomics assays and because the use of spike-ins remains debatable, we will not discuss spike-in-related approaches. Most secondary analysis toolkits focus on the following three matrices[15,16,21,22].

1. Library size, defined as the total number of counts per cell. Cells with small library sizes come from various sources, such as broken cells, inefficient cDNA capture, and poor amplification, which are considered low-quality. Cells with abnormally high counts may come from doublets.
2. The number of genes expressed in each cell. Cells with low numbers of detected genes may indicate poor library preparation, whereas higher numbers of genes are considered to represent doublets.

3. The proportion of mitochondrial genes per cell. A high proportion of these genes may indicate low-quality cells whose cytoplasmic mRNA has been lost while mitochondrial RNA has been conserved.

These matrices are obviously correlated; thus, many studies use only one or two matrices for QC. Although the use of fixed cutoffs for these matrices typically works well, occasionally, a seemingly reasonable cutoff can result in the loss of useful information. For example, by plotting the number of genes and library sizes in our TS19 dataset, we identified a population with a significantly low number of detected genes (**Figure S1.1A**), which might be removed in other studies. However, we identified this population as erythrocytes undergoing differentiation, a unique population with a small set of expressed genes and a similar library size. Such unique features would be missed if the matrices had not been fully inspected or if default cutoff values had been applied.

In addition to these standard practices for scRNA-seq QC, we believe that model-based QC might work even better for specific cases. In our dataset, in addition to the primary QC provided by the 10x Genomics pipeline CellRanger, we found that the distribution of UMI fits well with a 3-normal mixture model (**Figure S1.1B**). We believe that the lower normal distribution is derived from a poor library, whereas the upper normal distribution could represent doublets from sample preparation. We removed all cells below the lower 95% tail of the lower distribution and above the upper 95% tail of the upper distribution. We also removed all non-expressed genes and pooled all samples, which resulted in a final UMI matrix with a dimension of 26,305 cells by 20,347 genes. By applying such model-based QC, we have observed improved results.

Statisticians and computer scientists have also been addressing QC issues. Toolkits based on more sophisticated statistical and mathematical analyses have been developed to address QC-related issues, such as detecting doublets[23–25], solving dropout events by imputation[26–28], and using other combined matrices to detect cell outliers. Because these approaches are not widely used, and our initial tests on these toolkits produced only marginal differences, we did not incorporate these toolkits.

1.1.2 Normalization

The general purpose of normalization is to minimize the technical differences between libraries, such as cDNA capture efficiency, reverse transcription efficiency, and PCR amplification efficiency, to compare gene expression profiles between cells without bias.

Normalization methods have been extensively addressed for bulk RNA-seq analyses using well-developed toolkits, such as DESeq2 and EdgeR[9,29]. Although some of these tools have previously been extended to scRNA-seq studies, some fundamental differences between bulk RNA-seq and scRNA-seq must be recognized. For example, one of the most important drawbacks of scRNA-seq is known as dropout events, meaning that several missing values (or 'dropout zeros') can be observed in the digital expression matrix, which does not reflect biological conditions[7,10,26,30]. Another issue is batch effects. Although some toolkits treat batch correction as a separate concept because it considers technical biases in addition to biological differences[15], which involve different computational methods, the overall goal of identifying both types of differences is to make all cells comparable in downstream analyses.

The most widely used and often sufficient normalization method is global scaling normalization, which originated from bulk RNA-seq and relies on a scaling factor to normalize library size[8]. By dividing counts by a scaling factor, this method addresses the differences associated with the sequencing depths of each library and makes gene expression levels comparable between cells. In some more complex experimental designs, non-linear normalization may be preferred, which may involve more sophisticated modeling and is believed to show better performance when strong batch effects exist[31].

Similar to QC, whether a normalization process is sufficient can only be determined by the results of downstream analysis and is highly dependent on the nature of the input dataset. No method can be uniformly applied to all scRNA-seq datasets. In our case, choosing complex model-based approaches did not improve the downstream results; thus, we used the simplest global scaling normalization method, based on the median count per cell, to obtain the scaling factor. In our experimental design, we added a 'whole embryo' sample as a control to allow for the inspection of batch effects. The alignment of all samples appears as expected in the lower-dimensional space (**Figure 1.2**), which convinced us that few batch effects existed; therefore, we did not apply batch correction to this specific dataset. Although we did observe dropout events in our dataset, none of the imputation tools we attempted to use improved our results. Most of the tools tested would over-impute the data, possibly due to high heterogeneity. Because the application of imputation to scRNA-seq remains debatable, we decided not to use it. To summarize, starting from our UMI matrix and after QC, we used median normalization, followed by a

log transformation of $\log(1+UMI)$. When using a more complex experimental design, a different normalization strategy might be considered, in addition to revisiting the application of batch corrections.

1.1.3 Dimensionality reduction and visualization

The general purpose of dimensionality reduction is to reduce noise, ease the computational workload, and provide better visualization of the data. A typical scRNA-seq dataset, based on mouse or human samples, can contain over 25,000 genes, many of which have mostly zero counts and are described as dropout events. In this context, each gene represents one dimension of the dataset. Although downstream analyses, such as clustering, aims to compare gene expression profiles between cells, such a large number of dimensions would be considered a computational burden. Moreover, not all dimensions will be useful, knowing that the dataset is zero-inflated due to dropout events.

Usually, an additional step can be helpful before dimensionality reduction, known as feature selection. The basic idea is to keep only the ‘variable genes’ that contain useful information in the dataset and to remove genes that are considered noise or background. Different algorithms have been developed for selecting such genes[32], and the most straightforward approach is to compute the variance-to-mean ratio for each gene, after binning them based on expression, and then selecting for genes with a high ranking[27]. The number of variable genes that should be kept depends on the complexity of the dataset, although a high number is usually recommended. When varying a number in a certain range (on the high end), the low dimensional structure has been reported to

remain the same. In our dataset, we tested a range from 1,000 to all genes, and the low-dimensional structures tended to stabilize when over 3,000 genes were chosen (**Figure S1.2**). Given that the complexity of the dataset is by far the highest we have seen, and we do not have concerns about computational efficiency, we kept all detected genes in our dataset to prevent any loss of information and allowed the dimensionality reduction step to do the job of denoising the data.

Once the genes were selected, we used a matrix with X genes and Y cells, referred to as an X -dimensional dataset. As mentioned above, not all dimensions are useful. Biologically speaking, many genes are highly correlated because they control the same biological processes. In this case, using X dimensions would be a waste of computational power and introduces the potential for noise. Dimensionality reduction would collapse such uninformative dimensions. In contrast, visualizing data with high dimensionality can be challenging; thus, summarizing and describing the dataset in two or three dimensions is important. The most widely used approach is a two-step dimensionality reduction. The first step uses a small number of dimensions, Z , to describe the entire dataset while retaining most of the underlying structures, resulting in a matrix with Z dimensions and Y cells. The second step is to further collapse Z dimensions into two or three dimensions so that a scatter plot can be used to present the dataset.

Multiple methods exist for addressing the problem of dimension reduction. During the first step, the simplest but most powerful approach is PCA, which has a long history and is widely used in other fields[33]. PCA is a linear dimensionality reduction method that aims to capture axes (PCs) that explain the largest variations. Other approaches exist, such as

diffusion maps[34], which are non-linear algorithms designed to identify continuous trajectories rather than discrete subpopulations. It could be advantageous in studies focusing on the temporal ordering of cells, such as cell differentiation[35,36]. Another approach is known as non-negative matrix factorization (NMF)[37], which is similar to PCA but relies less heavily on feature selection. NMF is more statistically sophisticated but also more difficult to interpret and, thus, less widely used in general dimensionality reduction.

In this study, we have tested both PCA and diffusion mapping, and we preferred PCA, which resulted in a more discrete representation because the goal was to identify different cell types. When performing PCA, the first PC chosen should capture the greatest variance across all cells, and the second PC should be orthogonal to the first PC and explain the greatest remaining variance. This process is repeated until the number of PCs reaches the total dimensions of the dataset. Because most of the variance exists in the top PCs, the remaining PCs are generally considered noise and discarded. In the scRNA-seq dataset, the biological structures showed more variation because they arise from the correlated behaviors of many genes; thus, these structures appear in earlier PCs, whereas technical and biological noise shows less variation, as they affect each gene independently and accumulate in later PCs[38]. When applying PCA to scRNA-seq, the goal is to identify the top PCs (in our example, the number will be Z) that sufficiently explain the variance of the dataset. Usually, the number Z depends on the dataset's complexity. Most toolkits default to approximately 10 to 50 PCs and should be increased if more features were selected in the previous step or not enough variation is captured.

Although this process can represent a somewhat arbitrary decision, similar to feature selection, a simple, data-driven heuristic known as the ‘elbow plot’[16,27,34] exists in our dataset, as shown in **Figure S1.1C**. Because the biological signals explain much more variance than noise, we expect to observe a sharp drop in the percentage of variance explained when identifying PCs associated with noise. This drop is especially obvious in datasets for more specific tissues[39] but less obvious in our dataset due to the large heterogeneity. After testing a range of PCs, between 10 and 200, the plot structure stabilized after 80 PCs (**Figure S1.2**); thus, we chose 100 as the number of PCs to keep for downstream analyses.

For the visualization of the second step of dimensionality reduction, a non-linear method is often used. The standard practice for this step is the t-distributed stochastic neighbor embedding (t-SNE) method[40]. This method attempts to identify low-dimensional representations (usually 2 or 3 for visualization purposes) of high-dimensional data (in this example, a dataset with Z dimensions of PCs), which preserve the structures between data points. A few things should be considered when applying t-SNE. First, t-SNE is based on non-linear transformations; thus, the distances between data points (or clusters, as discussed in the next section) do not accurately reflect their relationships. Second, t-SNE requires some parameters as inputs, the most important one being perplexity. The perplexity determines the granularity of the presentation and could result in a very misleading presentation if not properly chosen[41]. Unlike PCA, t-SNE is not reproducible because it uses random initialization, and a seed is necessary if the results must be repeated. Third, t-SNE focuses more on local structures, which means that the data points

tend to repel each other; thus, the connections between clusters might be ignored. A popular alternative to t-SNE exists, known as uniform manifold approximation and projection (UMAP)[42]. UMAP aims for the same goal as t-SNE but uses a different mathematical approach. UMAP differs from t-SNE in a couple of aspects. In addition to better computational efficiency, UMAP attempts to emphasize global structures over local structures and might be a better choice when the goal of the study is to explore the relationships or track the trajectory of cell clusters, which will be discussed further in the next chapter.

In this study, we performed both t-SNE and UMAP analyses on our dataset, using the top 100 PCs identified by PCA (**Figure S1.2-S1.3**). For t-SNE, we tested perplexity across a wide range, and we believe that the default settings provided us with very informative structures (**Figure S1.4**). We also attempted to use both 2 and 3 dimensions for downstream analyses (**Figure S1.5**), and we concluded that a t-SNE plot with 3 dimensions serves as a better starting point for downstream analyses, as our eyes were able to identify more structures when rotating the plot in the 3-dimensional (3D) view. However, the 2-dimensional (2D) figures are advantageous in terms of presentation and publication. To identify a suitable balance between 2D and 3D, we processed our data in 3D but displayed a collapsed 2D view in all figures. We opted against the use of UMAP because the globally-oriented structures did not show advantages for the identification of different cell types.

1.1.4 Clustering analysis

The general purpose of clustering analysis is to group cells into clusters based on similarities in their expression profiles. After the clusters have been assigned, we can perform downstream analyses to annotate each cluster with accurate biological information. As with many of the steps described above, whether a clustering analysis is successful cannot be determined until the results are analyzed during later processes; therefore, testing a variety of methods using various parameters is recommended. Clusters are defined by purely mathematical terms, unlike the definition of cell types, which is based on biology. We attempted to divide cells into clusters through clustering, such that each cluster represented a single cell type. However, the resolution and definition of cell types can vary from study to study. The clustering analysis parameters should be tuned to fit the purpose of the study, and the sub-clustering of specific clusters is often used to increase the resolution of specific cells.

As an unsupervised learning process, a classic approach that was used during early scRNA-seq analyses was k-means clustering, which remains widely used in recent studies. This algorithm attempts to divide all data points into k-clusters by assigning each data point to its closest centroid, thus, minimizing the total intracluster distance. A large advantage of k-means clustering is its simplicity, as this algorithm is typically applied on simple, Euclidean distance matrices from lower dimensionality, such as the PCA matrix. Whether clustering should be applied to PC space or to lower-dimensional space, such as t-SNE or UMAP, remains debatable; although, both have been applied in previous publications[43]. The disadvantage of this simple clustering algorithm is obvious. First,

this algorithm requires a number, k , as an input parameter, which can be difficult to predict. When k is lower than the actual cell type number, even strong separation observed visually may not be separate. In addition, this algorithm favors spherical clusters with equal radii[38], which can lead to separations that are not only unintuitive but also misleading when complex structures exist in the input data. As shown in **Figure S1.6**, the k -means clustering applied to our dataset in t-SNE space had difficulty identifying visually obvious structures.

Another classical approach for clustering is known as hierarchical clustering. Unlike k -means clustering, hierarchical clustering is a bottom-up approach, which means that each data point is considered its own cluster at the bottom level, and then similar points are pooled together to form larger clusters. Another important feature of hierarchical clustering is the production of a dendrogram, which is often considered to serve as a good summary of the dataset and can be used in various downstream analyses. Based on these features, a predicted cluster number k is not required, and hierarchical clustering can be easily adjusted by ‘cutting’ the dendrogram tree at different heights. Various types of hierarchical clustering exist, differing in how they agglomerate smaller clusters into larger ones, such as ‘ward,’ ‘single linkage,’ and ‘average.’ However, despite these advantages, this method is less computationally efficient, and analyses can take a long time if the input dataset is large.

After years of clustering approach development, the most promising clustering approach that is currently available is likely graph-based clustering. The popular graphics-based clustering approach originated from the Louvain algorithm and represents a combination

between the K-nearest neighbor approach (KNN) and the extraction of community structure[44]. Each cell in the dataset is considered to be a node and is represented by KNN. Then, the communities are extracted based on modularity optimization. This approach has a relatively efficient computation time, and a resolution parameter can be used to adjust the scale of clustering results. However, sometimes these results can be misleading because the resolution depends on the cell density[38].

In a specific study, multiple clustering rounds might be used to optimize the resolution of specific groups, even the application of different clustering approaches. The sub-clustering of a set of clusters derived from primary clustering is a widely used method to investigate further differences within a cell type or among cells with similar identities. In this study, we compared different clustering approaches in both PCA and t-SNE space (**Figure S1.6**). By analyzing the cluster annotation, we opted to use single-linkage hierarchical clustering (SLHC) in t-SNE space. We tested a range of numbers for the resulting clusters, including 8, 16, 32, 64, 128, 256, 512, and 1024 clusters, to obtain the optimal resolution necessary to separate different cell types. The advantage of using SLHC is that the identified clusters align very well with virtual observation, which does a better job than k-means clustering. In addition, the sizes of the clusters do not tend to equalize, unlike during ward hierarchical clustering. Another feature of this approach is that it serves as a denoising step, as cells that are far apart from the primary clusters form their own smaller clusters; thus, a cluster size threshold was applied to remove this noise before later steps. Most of the identified clusters were very small. In our finalized version of clustering, we decided to divide up to 512 clusters, most of which were deemed to be

noisy (<10 cells). We only kept 22 clusters for downstream analyses. Compared to the whole population, these dropped cells represented only a small proportion (2,483 out of 26,305 cells or 9.4%) and were considered more transcriptionally different from the major cells. To avoid dropping small, unique cell clusters, we tuned the threshold number to maintain the smallest meaningful cell cluster that we were able to identify during the preliminary analysis. Before we finalized our clustering approach, we performed some preliminary analyses using simple ward hierarchical clustering. We found that the three, large, continuous clouds represented three large populations of related cell types, which were maintained as three large clusters in our finalized clustering approach. In summary, after we performed clustering analysis, we obtained a total of 512 clusters, some of which were considered to be noisy clusters. Finally, we identified 3 large clusters, which were represented by 3 continuous clouds, and 19 isolated clusters, which will be annotated in later steps (**Figure 1.1B**).

1.1.5 Differential Gene Expression (DGE)

The general purpose of DGE analysis is to identify significant genes that are differentially expressed in each cluster, sometimes described as identifying gene signatures or marker genes associated with each cluster. These gene sets are essentially the driving force that defines the cluster separations. In a perfect scenario, each cell type would be associated with its own cluster, and the marker genes for each cell type would be differentially expressed. However, extracting true DEG signatures can be challenging due to the limitations of scRNA-seq, such as dropout events, low sequencing depth, and large gene panels. Another notable issue is that DGE analysis occurs after ‘feature selection,’ which

limits the gene panel to a relatively small size; therefore, genes that have previously been discarded will not be identified.

The ideas underlying DGE were addressed, in-depth, during the early development of bulk gene expression assays. A variety of statistical solutions have been developed to answer the question ‘Which genes are up/down?’[45–48], especially microarray-based analyses. Mature toolkits with built-in DGE functions are now available to perform these bulk expression analyses, such as the R packages ‘DESeq2’ and ‘EdgeR’[9,29]. Although the methodologies that are commonly applied to bulk analyses can provide satisfactory results for some scRNA-seq studies[49–52], fundamental differences exist between these two analyses. ScRNA-seq data suffer from different types of technical noise, such as dropout events, as described above, and these artifacts should be taken into consideration when performing modeling. Because of the level of dropout events, scRNA-seq-data features a completely different level of heterogeneity compared with bulk analyses, which results in a much more complicated distribution. Additionally, during bulk analyses, comparisons are typically performed between a few samples or conditions, whereas scRNA-seq features a much larger sample size, and the comparisons are generally formed between clusters. Taking all of these differences into consideration, new models and approaches have been developed for the analyses of scRNA-seq datasets[12,14,30,53–56]. However, some existing approaches for bulk analysis have also been adjusted to analyze scRNA-seq datasets, such as ‘DESeq2’, ‘EdgeR,’ and ‘Limma’[9,29,49]. Although comprehensive comparisons have been performed among various tools[57,58], the optimal approach remains under debate because the selection

of appropriate DGE approaches depends highly on the nature of the dataset being analyzed.

Given the heterogeneity of the dataset in this study, we did not use the typical bulk RNA-seq-derived mean expression and fold change approach. During the preliminary analysis, we found that the DEGs identified by such approaches included a group of low expressing genes, which were only detected in a few cells. Ideally, these genes would be dropped due to the p-value threshold; however, due to the nature of zero-inflated datasets, canonical approaches are unable to identify such genes. Eventually, we selected Monocle as our DGE tool, which was ranked highly by a recent review article[58]. For each identified cluster and subcluster, we performed monocle DGE analysis by setting the fold change threshold to 2 and the p-value threshold to 0.05. The top 400 genes were considered DEGs and were used for downstream analyses. We tested the performance, ranging from 50 to 800 genes, and 400 genes was determined to be the optimal number of genes to balance between weak signals and strong noises.

1.1.6 Pipeline summary and discussion

In summary, we developed a pipeline to process a large-scale, whole-embryo scRNA-seq dataset, following primary data processing from the 10x platform CellRanger.

Each of the four samples (head, trunk, heart, and whole embryo) was fitted to a three-normal distribution, and presumptive doublets and bad libraries from the top and bottom populations were discarded. All samples were pooled together and normalized using median log transformation. PCA was applied to a normalized expression matrix, and the

number of PCs used was set to 100 by plotting the variance explained by each PC. The top 100 PCs were used to perform t-SNE to obtain a final matrix of 26,305 cells by 3 t-SNEs. The cells were clustered by SLHC using the identified t-SNE coordinates. The number of clusters was set to 512, and noisy clusters containing fewer than 20 cells were discarded, yielding 54 final clusters. The resulting clusters contained three clouds, pooled from 35 continuous clusters, and 19 isolated clusters, which will be explained in the next section. **Figure 1.1B** shows a schematic view of the dataset clustering and analysis process. We used the R package Monocle for each cluster to perform DGE analysis by comparing the target cluster with all other clusters. Genes with adjusted p-values < 0.05 and a fold change > 2 were considered DEGs, and the top 400 DEGs were used for downstream analysis (if fewer than 400 DEGs were identified, all DEGs were used). Thanks to the hard work from bioinformaticians and computer scientists, various analysis tools for scRNA-seq datasets have been developed during the past few years. With these established pipelines, data analysis has been significantly simplified for biologists. However, certain parameterization is still necessary based on specific dataset. In this study, we have presented a large-scale scRNA-seq including mouse whole embryo, and we demonstrated how such parameterization was done for such highly heterogeneous dataset including determination of number of PCs, perplexity and number of dimensions used in dimension reduction, selecting proper dimension reduction algorithm, and clustering strategy (**Figure S1.1-S1.6**). Next, we will present an approach to obtain useful biological information from the processed dataset using our pipeline.

1.2 An integrated approach to defining cellular identity

Once the DEGs for each cell cluster were identified, the most challenging data analysis step was the interpretation of the identified genes and the assignment of cell identities to each cluster.

The cellular identity of each cluster is typically defined by studying the gene signature associated with the identified DGE pattern. In most studies, the marker genes associated with various cell types are used to annotate the identities of each cluster, which is a very useful and intuitive approach that has some limitations. First, our understanding of cell type markers is very limited. Thus far, the best cell type database we have identified (<https://discovery.lifemapsc.com/>) features only 7,435 documented genes (861 of which are considered marker genes) for 672 cell types, which is fewer than one-third of the genes identified in the transcriptome. Second, a single gene can be identified as a marker for multiple cell types or even be associated with different cell types by different reference databases and literature. Third, the markers will change as the embryo develops, and the genes play different roles at different time points; however, most databases do not include annotations of developmental time.

Another widely used approach is Gene Set Enrichment Analysis (GSEA), which attempts to match not only cell type markers but also all DEGs with existing databases, including some widely used databases, such as Gene Ontology (GO)[59], Reactome[60], NCBI Biosystems[61], and the Kyoto Encyclopedia of Genes and Genomes (KEGG)[62]. Note that we are not specifically referring to the canonical GSEA from the Broad Institute[63,64],

which is a computational method widely used in RNA-seq. Instead, the GSEA mentioned in this thesis refers to a more broad idea that uses bioinformatics to find enriched biological signatures from input gene lists. During this approach, any existing knowledge regarding a gene will be considered. This approach also suffers from problems that arise from under-annotated databases. For example, in the GO database, of the 22k documented genes, over 70% of the terms are based on non-experimental data. Another challenge is the noise, especially for genes that are widely expressed in multiple cell types[65]. To more precisely take advantage of GSEA, manual inspection is often used. However, the effectiveness of manual inspection is dependent on each individual's understanding of specific cell type identities and consumes a large amount of time, even with the help of various toolkits.

A third approach compares the target dataset's expression profile with published datasets that have been annotated with cellular identities. Thus far, a few studies have published broad analyses of mouse embryonic cell types[66,67]. However, as more analysis tools and databases continue to be developed, past literature based on outdated tools and reference databases might not be sufficiently precise. In addition, even when comparing dataset annotations for similar cell sources, the results do not always perfectly align, as the platforms used and analysis tools chosen are rarely the same.

All of the methods mentioned above can be used as references, but we do not believe that a single method is likely to provide us with a precise annotation of the cell clusters from such a large-scale dataset. To address this challenge, we have employed an integrated approach using multiple combined approaches, which is described as follows.

1.2.1 Embryo dissection information helps validate cell type identities

During embryo preparation, we dissected embryos into three parts: head, trunk, and heart (**Figure 1.2A**). When we performed sequencing, we prepared four samples, including these three dissected regions and whole embryos, for two primary reasons.

First, the use of various embryo sections can allow for the validation of the quality of all secondary data processing steps through dimensionality reduction. Ideally, in a reduced dimensional space (for example, t-SNE), the three separated segments should only overlap with each other across very few cells, as they should contain very few shared cell types, whereas the whole embryo sample would be expected to cover the same space as all three dissected segments. After optimizing the pipeline, we observed the expected distribution pattern for all four samples (**Figure 1.2, B-C**), except for a few heart cell clusters that were underrepresented in whole embryo samples. We realized that this phenomenon might be related to the dissection and digestion step because cardiomyocytes in the heart sample require additional time to be digested into single-cell suspension, whereas the whole embryo preparation was submitted to a shorter digestion time to compensate for other cell types. As a result, a few clusters from the heart were represented by a low number of cells in the whole embryo preparation.

Second, this sample information facilitates the validation of preliminary cell cluster identity annotation. As mentioned above, a cell cluster is rarely present in all three samples from different body sections. However, if such a cluster is identified, we would assume that it represents a cell type that is shared by all dissection parts, such as immune cells or blood

lineage cells. In addition, the sample information can also remove noisy signals from cell clusters. Because the data processing involves the entire transcriptome, and some genes are multi-functional, the signals from a gene set could be very noisy, regardless of what bioinformatics tools are applied. When the origin of the cells is known, some impossible cell type signals can be discarded during manual inspections.

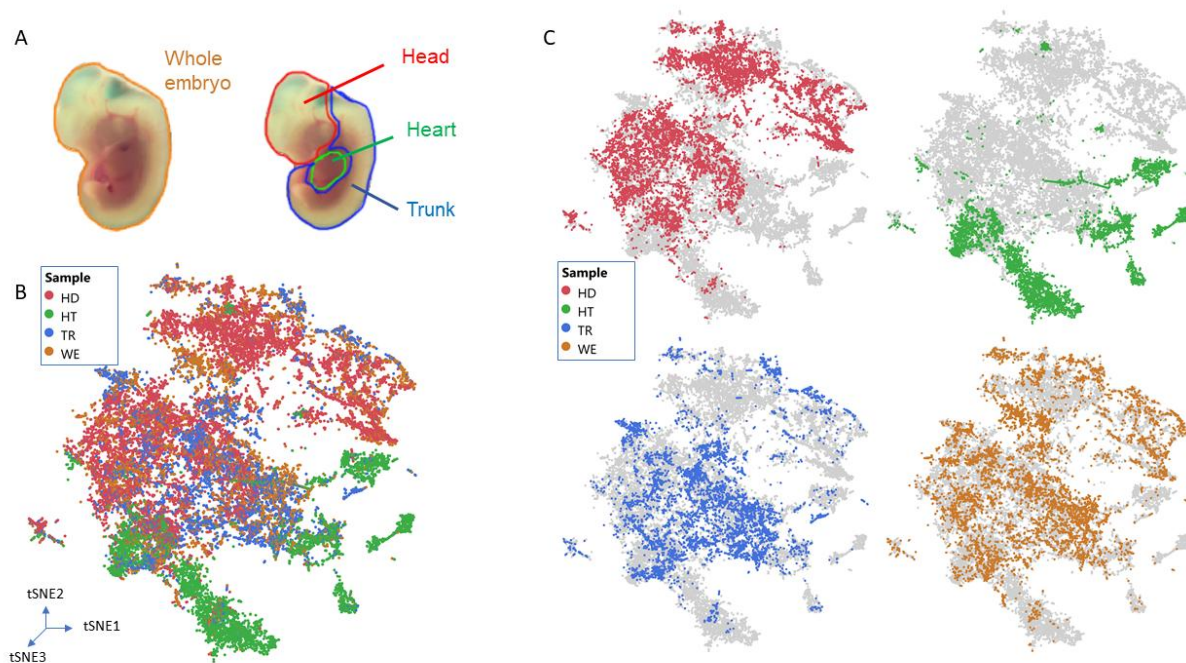


Figure 1.2: Embryo dissection information helps validate cell type identities

The dissected embryo provides regional information for annotation, and the whole embryo serves as a control for evaluating pipeline performance and data quality.

(A) Samples used in this study. Undissected whole embryo (orange), head (red), heart (green), and trunk (blue) labeled in different colors. **(B)** t-SNE projection colored by cells from different samples (head-HD, heart-HT, trunk-TR, whole embryo-WE). To simplify presentation, only t-SNE1 and t-SNE2 showed. **(C)** t-SNE projection highlighted by each sample showing WE has a good representation in most regions.

1.2.2 Integrated gene set enrichment assay (GSEA) gathers cell cluster information from multiple aspects

To obtain details regarding the biological inferences of each cluster, we tested various tools designed for GSEA. We've also considered the canonical GSEA from the Broad Institute, but it was initially designed for bulk RNAseq, and needs specific manipulations to suit for scRNA-seq datasets, therefore not discussed in this study. Eventually we decided to collaborate with LifeMap®, using their online tool GeneAnalytics (<https://ga.genecards.org/>) for our enrichment analysis. This online tool incorporated a collection of over 120 databases, enabling not only GO term analysis, but also tissue and cell expression profiles, human diseases, human and mouse phenotypes, and cellular pathways.

The 'Tissue & cell' analysis tool was based on LifeMap Discovery (<https://discovery.lifemapsc.com/>), a database that uses a combined model, based on scientific literature and high-throughput experiments. This model includes the following sources: peer-reviewed scientific literature; high-throughput experiments, including microarray and RNA-seq; lists of DEGs from large-scale datasets; and *in situ* hybridization (ISH) and immunostaining (IS) datasets. Entities within this database have been divided into three sections: 'Organ/Tissues,' 'Anatomical compartments,' and 'Cells,' with resolutions ranging from low to high. Based on the embryo preparation method we performed and our annotation goal, we largely referred to the Organ/Tissues and Cells databases. All gene annotations were based on information reported in the scientific literature or on bioinformatics calculations based on expression data and were divided

into the following categories: specific genes (genes that are specific to or expressed in only a few organs/tissues), enriched genes (genes that are expressed in many entities of the same organ/tissue), selective genes (established cell-specific markers or genes that have been suggested to be characteristic of a cell type), and expressed gene (genes known to be expressed in the cell but not defined as selective cell markers). Depending on the category, different scores were given to each matched gene. Three other labels have been used to adjust the matching score: abundant genes (genes that are expressed in a large number of different organs/tissues), housekeeping genes (genes that appear on a list of housekeeping genes that has been established by integrating information reported by several studies), and low-confidence-level genes (genes that originate from the analysis of large-scale datasets but do not have strong supporting evidence). An overall matching score for each entity was calculated based on the matching score of all matched genes. Then, the matching score was assessed by statistical models and categorized as high-, medium-, or low-quality matches, based on a normalized log-transformed score.

The 'human disease' analysis tool was based on MalaCard (<https://www.malacards.org/>), the most comprehensive human disease database that is currently available, with over 60 integrated data sources. This database categorizes diseases into global and anatomical diseases, the latter of which could be useful when inferring cellular identity. The query gene set will be matched against genes associated with diseases. A matching score will be determined by both the normalized number of matched genes and the quality and type of gene-disease relationships. The gene-disease relationships are divided into

3 types: genetic associations, DEGs, and 'GeneCards-inferred.' Because DEGs can be either up- or downregulated, and 'GeneCards-inferred' does not necessarily imply causality between the gene and the disease, we primarily focused on genetic associations.

The 'Gene Ontology (GO)' analysis tool is based on the GO project[68], which documents all genes and gene products. Three large domains are covered: cellular component (GO:CC), molecular functions (GO:MF), and biological process (GO:BP). The GO:BP domain is primarily focused on understanding the identities of cell clusters with given gene sets and has been the primary target of various GSEA tools. In GeneAnalytics, a binomial distribution is used to compute a matching score for each entity, which is finally represented in a log-transformed p-value format. The quality of matching is divided into high, medium, and low categories based on the scores. In our study, we primarily focused on the top 20 high-quality matches.

The 'Phenotypes' analysis tool is based on either the Human Phenotype Ontology (HPO) database or the Mouse Genome Informatics (MGI) phenotype database, with a scoring system similar to GO. We primarily focused on mouse phenotypes and only used human phenotypes as references.

In this study, we automated GSEA analysis by sending the top 400 DEGs from each cluster to GeneAnalytics (**Figure 1.3A**). All results were tabulated, and we gathered all top entities from 'Tissue & cell,' 'Gene Ontology (GO),' and 'MGI phenotypes' for detailed analyses, whereas the other results were used as references during the manual curation step. Our preliminary analysis found that the 'GO:BP' category from GO and 'MGI

phenotypes' provide the most valuable information for cell identity annotation. By clustering the matching scores of all entities from 'GO:BP' and 'MGI phenotypes' in every cell cluster, we were able to identify unique patterns for each cell cluster that we identified (**Figure 1.3, B-C**), which indicated our successful capture of distinct signatures from each cluster. The 'Cell type' category from 'Tissue & cell' also provided useful information, which was directly related to our goal, although with much noisier signals (**Figure 1.3D**) due to the limitations associated with the use of cell-type markers and the multifunctionality of genes.

To visualize the GSEA results of each cell cluster, we transformed the matching scores of the top entities into word clouds and phrase clouds. The font size of a word or phrase cloud is proportional to the matching score. For each cell cluster, we generated both word and phrase clouds for 'Cell type,' 'GO:BP,' and 'MGI phenotypes' (**Figure 1.3E**).

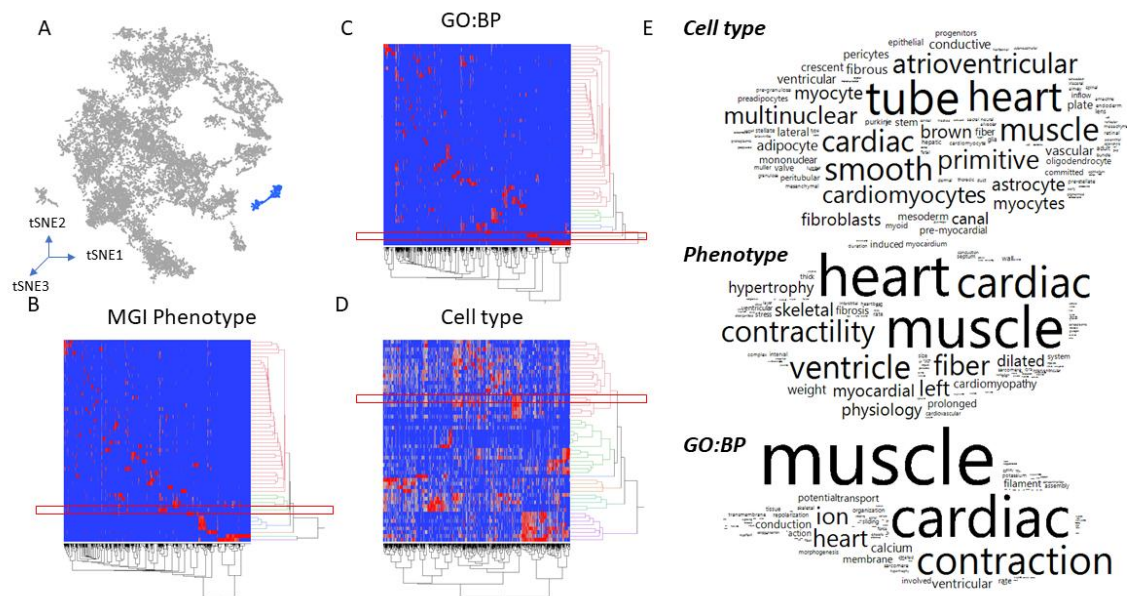


Figure 1.3: GSEA with various databases gathers cell cluster information from multiple aspects

GSEA for an individual cluster helps to identify unique biological signatures from different categories, and could be quantified and visualized by word clouds.

(A) A sample cluster highlighted in t-SNE space in blue. **(B-D)** The hierarchical clustering of GSEA enrichment scores returned from various databases, the x-axis shows different matched terms, the y-axis shows cell clusters. Red: high enrichment scores; Blue: low enrichment scores. Boxed rows show the location of the sample cluster. **(E)** The word clouds showing GSEA results of the sample cluster, font size proportional to the enrichment scores.

1.2.3 Projection of marker genes/gene lists highlights specific cell types and broad cell categories

A common practice in cell cluster annotation is the use of marker genes in almost all single-cell studies. The most straightforward approach is to compute the expression level of certain cell type markers in each cluster and to then determine whether any clusters exist with significantly high marker expression, thus, relating it to the cell type. Although this approach has been shown to be powerful and is widely used, we cannot ignore its disadvantages. First, most literature-derived markers came from bulk gene expression studies, which are fundamentally different from single-cell studies. Second, several marker genes lack developmental information. Some genes play roles within very restricted time frames during embryo development, and these genes might mark specific cell types during one stage of development but be associated with different cell types during later stages. Third, even within a single cell type, the cells in different states could be divided into multiple clusters[16]. Based on these reasons and others, such as the use of different expression assays and platforms, marker genes identified in one study often

do not align well with those identified in another study. We observed that the ‘Tissue & cell’ analysis from GSEA did not outperform the MGI phenotypes and GO:BP in terms of identifying unique patterns for each cell cluster (**Figure 1.3B**), which is essentially a collected cell type and tissue marker gene database. Together, all of these points indicate that cell type markers are great resources that can help define cellular identities, with certain limitations.

In this study, we used a combination of automated and manual marker gene inspection techniques. Furthermore, we employed marker lists containing multiple markers to address potential bias due to single markers (**Figure 1.4**). Specifically, for each cell cluster, we aligned the DEGs with the cell type marker database using GSEA from GeneAnalytics, which provided us with a basic sense of which cell types were related to each cluster. In addition, we used the cell type marker gene list for each cell type in this database and projected the expression levels of these genes onto our cell plot, which identified the clusters that were related to each cell type. Because our embryo was still at an early stage of embryonic development, we included marker gene lists with broad cell type definitions, such as those for the three germ layers during their early stages. This helped us to define the identities of large continuous cell clusters that contained fewer structures. To validate cell clusters with strong signals associated with more differentiated cell types, we manually selected markers based on MGI literature searches and *in situ* images of the developmental stages associated with the identified genes. We then aligned a projection of these genes with our t-SNE plot to confirm our primary annotations of the cell clusters.

In summary, we employed marker genes at different resolutions, ranging from single markers to broad marker gene lists, depending on the primary GSEA results and the topology of cell clusters in t-SNE space. We projected the expression levels of marker DEGs associated with a specific cluster onto t-SNE space to confirm the identity of the cell cluster, in addition to projecting the expression level of a marker list for a specific cell type onto the t-SNE space, to determine which cluster(s) was most likely to align with this cell type. For cell clusters with strong signals, we also manually selected markers and *in situ* images to confirm their identities. For cell clusters with weak or broad signals, we pooled the clusters together to perform sub-clustering analysis.

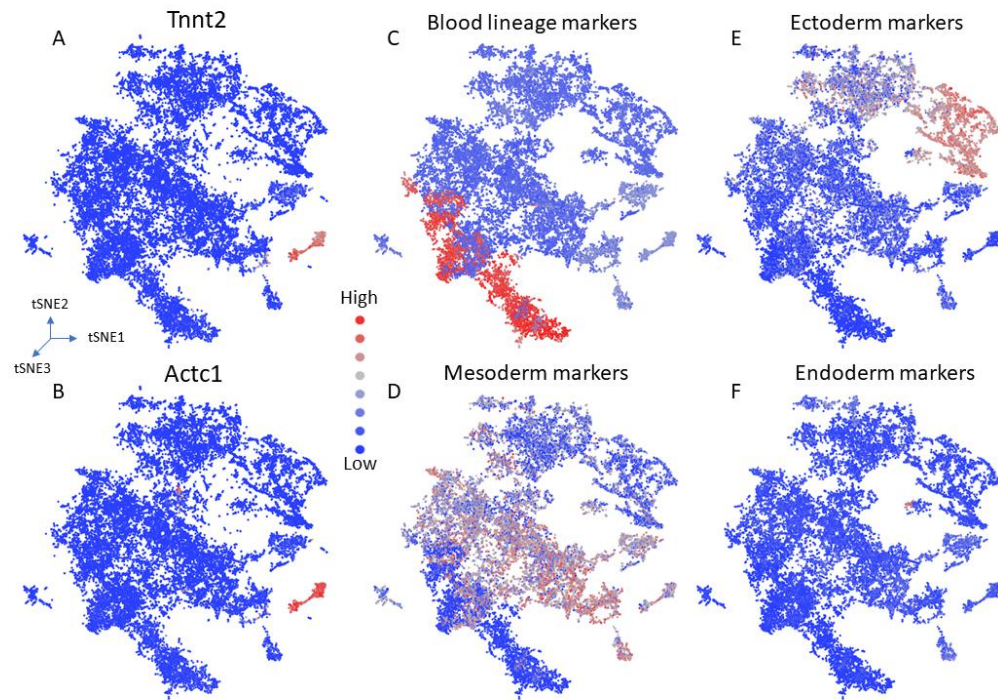


Figure 1.4: Projection of marker genes/gene lists highlights specific cell types and broad cell categories

By projecting certain marker genes or gene lists onto the cell plot, we could find potential cell clusters with such cell types.

(A-B) Projections of two cardiomyocytes markers *Tnnt2* and *Actc1* highlight one cell cluster in t-SNE space. **(C-F)** Projections of normalized total expression levels of four broad cell marker lists highlight different regions in t-SNE space.

1.2.4 Cell lineage map scoring provides a developmental view of cell cluster identity

Using the approaches described above, we were able to obtain relatively precise annotations for most of the cell clusters. However, we would like to further study the identities of these cell types from a developmental point of view. For the example shown in **Figure 1.4A-B**, if we observed a cluster with high expression levels of *Actc1*, *Tnnt2*, and various myosin genes, such as those that mark cardiomyocytes, we can obtain a basic idea of the cellular identity of this cluster. However, many other related cell types could be potential candidates for cellular identities, such as cardiomyocyte progenitors, primitive heart tube cells, or pre-myocardial cells. Considering other related cell types provides at least two advantages. First, this step can align our knowledge of cell lineage progression with real transcriptome data, which can help to build a precise database of cell lineages. Second, for cell clusters with weak signals, the collection of multiple, related cell clusters would likely strengthen these signals.

To achieve this goal, we took advantage of the best cell lineage tree we could find, which was provided by LifeMap discovery. As briefly mentioned in the previous section, this platform provides the ‘Tissue & cell’ analysis tool for GeneAnalytics. In addition to this

valuable expression database, this platform also provides an ontology tree structure that can shed light on the hierarchical order of embryonic development. The database organizes the developing embryo's complexity by using a 'Tri-leveled structure,' in which the top-level, 'The Organ/Tissue Level,' provides a low-resolution view of embryo development, as listed in the simplified tree structure shown in **Figure 1.5A**. The intermediate level, 'The Anatomical Compartment Level,' focuses on specific temporospatial regions within an organ/tissue. The bottom level, 'The Cell Level,' provides detailed cell type differentiation paths toward specific lineages. Because our study aims to characterize cellular identities, we primarily focused on the highest resolution level, which was 'The Cell Level.' The only graphic representation provided by LifeMap Discovery is associated with 'The Organ/Tissue Level'; therefore, we attempted to expand that representation to the cell type resolution level. We took the relational database from LifeMap and reconstructed that database into a hierarchical tree structure (**Figure 1.5B**), in which the root node at the top represents a zygote, and the terminal leaves represent more differentiated cells. As in the database, the same cell type could originate from different anatomical compartments; thus, we pooled all nodes with identical names into a single node, connected to the most undifferentiated ancestor node. Through these manipulations, we generated a hierarchical tree, consisting of 672 nodes, with each node representing a unique cell type and connected to the other nodes by cellular differentiation paths. The vertical axis of the lineage tree was ordered by the 'level' label from the LifeMap database, which was numbered to represent the developmental timeline. Thus, the nodes at the top represented cell types that appeared earlier during development, whereas the nodes at the bottom were representative of more differentiated cell types.

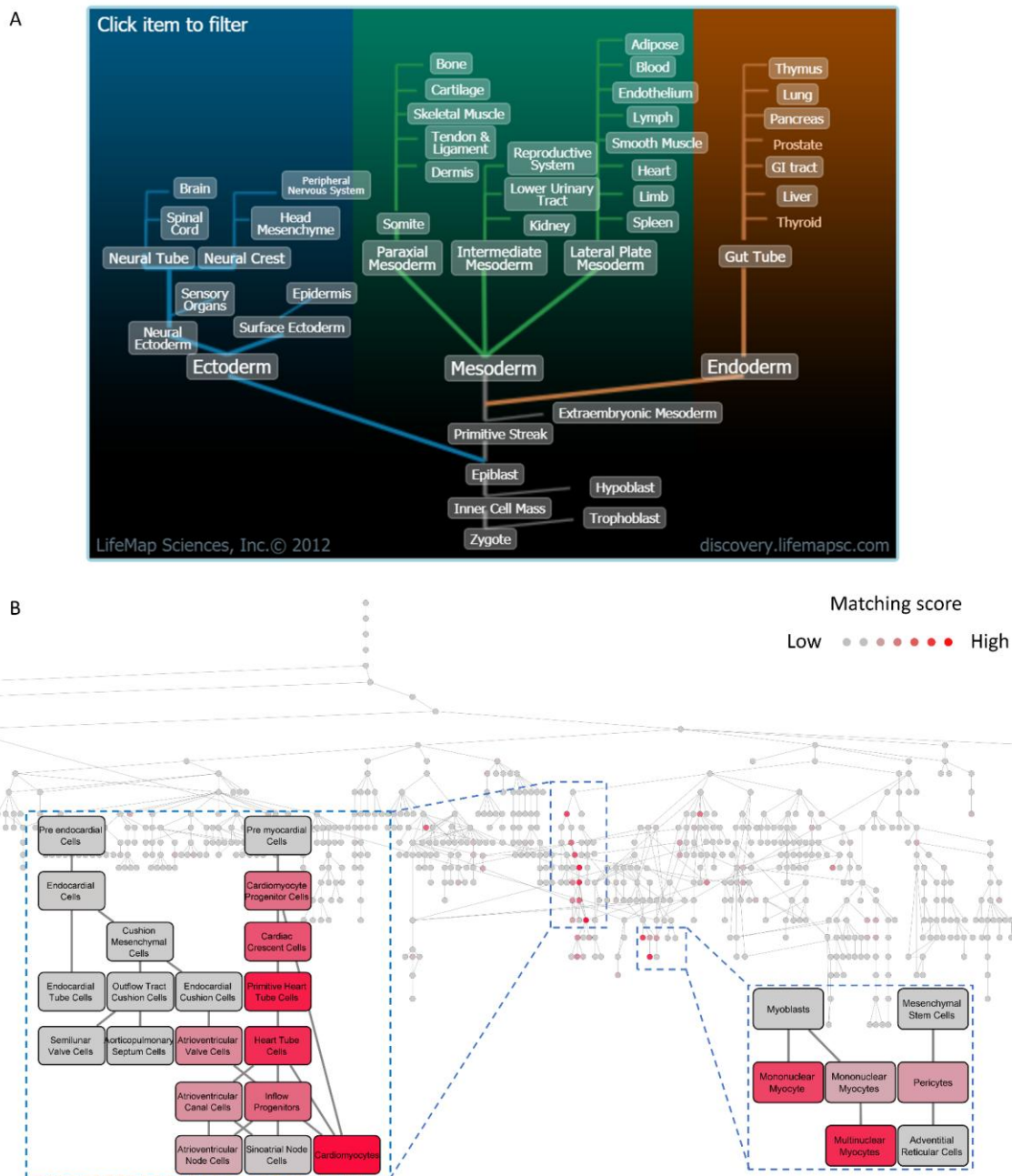


Figure 1.5: Cell lineage map scoring provides a developmental view of cell cluster identity

Mapping the cell cluster to the lineage tree highlights specific regions with cell lineage information.

(A) The lineage tree provided by Life Map (adapted from <https://discovery.lifemaps.com/in-vivo-development/cellular>). (B) The hierarchical lineage tree was constructed based on the Life Map lineage tree. Nodes indicate cell types, edges indicate a differentiation path, less differentiated cell types at the top, more differentiated cell types at the bottom. The coloring of nodes is based on the matching scores of a sample cluster (shown in Fig 1.3A). Dashed boxes show the zoom-in view of two highly-scored regions as cardiomyocytes-related cell types (left) and myocytes-related cell types (right).

To embed our cell clusters into this cell lineage tree, for each cluster, we computed a matching score for all nodes on the tree using the gene expression data for each node. In the LifeMap Discovery database, each cell type has a gene expression profile, as described above. We took advantage of their gene labeling system, and the matching score was calculated by comparing the DEGs in each cell cluster with the gene annotation of each cell type node. We picked three categories, 'selective marker,' 'specific gene,' and 'expressed gene,' giving each match a score of 3, 2, and 1. For each cell cluster, we are able to score the nodes on the lineage tree by calculating matching scores based on the expression profile (**Figure 1.5B**). Specifically, the expression matrix was first binarized, with 0 as not detected and 1 as detected. We scored every cell in our dataset for each documented cell type that was included in this lineage tree by summing up the matching scores of all genes related to each cell type. The scores were then normalized by z-score and averaged within each cell cluster to color the lineage tree. In total, we targeted 7,435 documented genes during this analysis, of which 861 were considered either a 'specific gene' or a 'selective marker.' We compared the results obtained between the inclusion of all documented genes in the analysis (7,435 genes in total) and restricting the analysis to 'specific gene' and 'selective marker' genes (861 genes in total). We found

that although some information is lost when we drop the less valuable ‘expressed genes,’ the visualization becomes much more focused, identifying fewer nodes with better signals, which ultimately narrowed down our candidates. We used both analyses when making our annotations; however, the version that included fewer genes showed a better signal-noise ratio and proved to be more useful.

This approach not only provided a straightforward view regarding which cell types might be related to the target cell cluster but also provided information associated with the cellular differentiation path. Because the lineage tree was constructed based on a collection of various assays at different resolutions, our data could also serve as a good reference resource for refining the structure of the lineage tree.

In this study, the scored lineage tree for each cell cluster helped to validate the cellular identities for more differentiated cell types while also providing clues regarding the identities of less differentiated cells. For example, the lineage tree scoring presentation of the cardiomyocyte cluster, C18, is displayed in **Figure 1.5B**, which highlights two regions of the lineage tree. The cardiomyocyte lineage was identified, which shows that the gene expression pattern of this cluster is similar to some other related cell types. The second region, identified as cell types in skeletal muscle, indicated that the expression pattern of this cluster also shared similarities with myocytes, as expected. In summary, by referring to the cell type scored lineage tree, we are able to compare the expression profile of an unknown cluster with published expression data, contributing to precise cell cluster annotation.

1.2.5 Cellular identity annotation: summary and discussion

A huge challenge for most scRNA-seq based studies is to determine the identity of cell clusters. For simpler datasets with a limited number of cell types, using a list of cell type markers might be sufficient. However, with the increasing heterogeneity and sample size in more recent scRNA-seq studies, such a simple approach is less reliable and likely to introduce bias. We demonstrate that in our dataset with great heterogeneity and less differentiated cell types, it is more of a problem, especially in regions representing undifferentiated cells that lack clear structures. To solve this problem, we've developed an integrated approach that not only rely on known marker genes, but also take advantage of modern bioinformatics databases including cell type, phenotype, gene ontology, and cell lineage tree (**Figure 1.3-1.5**). We've also demonstrated how dissection of embryos during experimental design could help in cell cluster identity annotation (**Figure 1.2**). By comparing the different databases used, we found that the cell type database which is basically a database of cell types and their markers, is actually more noisier compared to phenotype and gene ontology databases (**Figure 1.3B-D**), indicating that the use of markers is less powerful when dealing with high heterogeneity datasets. Other than that, our approach also provided information about cell clusters in developmental context using the lineage map (**Figure 1.5B**).

By combining all of the approaches described above, we are able to divide all clusters into two types based on the preliminary GSEA results. One type included the differentiated cell clusters, featured as isolated 'islands' in the t-SNE plot, indicating that they have distinct transcriptional profiles compared with other cells. The other type was

characterized by inclusion in the continuous ‘clouds’ on the t-SNE plot, indicating that the cell clusters in each cloud shared transcriptional profile similarities. Among the three identified clouds, one showed very strong erythrocyte signals, whereas the other two clouds showed less differentiated cell signals. By comparing the cell plot patterns with the projection of marker gene lists associated with less differentiated cell types and erythrocytes, we were able to identify the three clouds as belonging to ectoderm cells, mesoderm/mesenchyme cells, and blood lineage cells, whereas the other isolated islands have strong and unique signals.

To further validate the more differentiated cell types, we projected marker genes at different resolutions, ranging from manually picked single genes to gene lists from databases, in addition to referring to *in situ* and immunostaining figures from the MGI database.

To further explore the less differentiated clusters and the erythrocyte cluster, we first pooled all associated clusters into three large clouds, referred to as the ‘ectoderm cloud,’ ‘mesoderm/mesenchyme cloud,’ and ‘blood cloud.’ Then, we sub-clustered each cloud and analyzed each sub-cluster using the same approach.

Simultaneously, we generated cell lineage trees with scored nodes for each cell cluster, which provided more precise annotation information along the developmental timeline for those with strong signals and offered potential guesses for those with weak signals.

By combining all of these analyses, including the percentage of each sample in every cell cluster, three embryologists were able to manually annotate all 19 differentiated clusters,

independently. They then discussed their annotations to reach an agreement and provided insights for the less differentiated clouds, which will be described in the following section.

1.3 A catalog of cellular identities for mouse embryonic cells at early organogenesis

The mouse has long been used as a reliable model organism for the study of mammalian embryo development. The clear genetic background and relatively short development time of mice have provided several advantages to researchers. The biological similarity between mice and humans has also allowed the use of mouse models to shed light on the development and progression of human diseases. In this study, we focused on mouse embryos at Theiler stage 19 (TS19), which is equivalent to embryonic day E11.5[69]. During embryonic days E9.5 to E13.5, the embryo is believed to undergo the most dramatic morphological changes during development, including a 100-fold expansion in total cells and the formation of almost all major organ systems[66]. This period of early organogenesis has been extensively studied; however, very little literature has provided a detailed global view of these developmental processes. Recently, a paper describing the use of scRNA-seq to address these events during organogenesis was published[66]. However, due to the large degree of heterogeneity observed in such datasets, addressing the dynamics of cellular identities in a single study is impossible. We hope that by using a different data analysis pipeline in this study, we were able to obtain a more precise and comprehensive view from a different perspective, to complement the findings of this previous study.

In this study, embryos at E11.5 were harvested and dissected into four samples: head, heart, trunk, and whole embryo (**Figure 1.2A**). All four samples were processed and sequenced using 10x Genomics, single-cell RNA-seq. The libraries were prepared following the guidance of the official user manual. The pre-processing of scRNA-seq data was performed using CellRanger, following the recommended settings, including reads mapping and primary quality control. By applying our data analysis pipeline, as described above, we generated a t-SNE plot containing 26,305 cells (**Figure 1.2B**), associated with 20,347 genes. By applying the SLHC analysis and the integrated annotation approach described above, we were able to primarily cluster all cells into 19 differentiated cell clusters and 3 large clouds, with 2,483 cells defined as outliers being discarded prior to downstream analyses. In this section, we describe the comprehensive catalog of all the differentiated cell clusters, including supporting evidence, and the sub-clustering analysis of the clouds. Additionally, we highlight some interesting observed developmental events, providing new perspectives on cell type dynamics during embryo development.

1.3.1 scRNA-seq identifies 20 unique cell types from E11.5 mouse embryos

Among the cells included in the 3D t-SNE representation of our dataset, we identified 19 clusters and one large cloud with strong signals, which indicated differentiated cell types. Due to significant differences in the cluster size, the large cloud will be discussed in a different section. In the t-SNE space, these cells were all isolated from the rest of the dataset, indicating that these clusters have unique transcriptional profiles.

Among the 19 identified clusters, 5 of the clusters originated from the neuroectoderm, including C1 oligodendrocytes and Schwann precursor cells, C2 epithelial cells/dental placodes, C3 epithelial cells/tic placodes, C8 sensory neuron cells, and C14 ganglionic eminence cells. We also observed that these 5 clusters were spatially located near our less well-differentiated neuroectoderm cloud (**Figure 1.6A**), suggesting that these clusters share transcriptional similarities. C1 was annotated as oligodendrocytes and Schwann precursor cells, with the high expression levels of both oligodendrocyte markers, such as *Cnp1*, *Mbp*, *Plp1*, and *Zfp488*[70–73], and Schwann precursor cell markers, including *Cdh19*, *ErbB3*, *Ngfr*, and *Sox10*[74] (**Figure S1.7**, C1). These two cell types myelinate cells in the central and peripheral nervous systems, and our data indicated that these functionally similar cells also share similar transcriptional profiles during this developmental stage. C2 and C3 were both epithelial cells that are located spatially next to each other. These two clusters both returned strong epidermis signals in the GO:BP and MGI phenotypes analyses. We believe that these two clusters are epithelial cells that are specifically found in craniofacial structures, with C2 more related to teeth based on the expression of the dental placode cell marker *Edar*[75], whereas C3 may be more related to the inner ear, expressing high levels of the inner ear markers *Oto1* and *Oc90*[76,77] (**Figure S1.7**, C2/C3). C8 and C14 both showed strong signals associated with nervous system development, which were also observed in the neuroectoderm cloud. We believe that these two clusters represented neurons at later stages of differentiation that have committed to specific fates, whereas the neuroectoderm cloud represented neurons/glia cells at an earlier stage that were still actively proliferating, the details of which will be discussed in the sub-cluster analysis. C8 was annotated as sensory neurons,

with marker genes *Ppp1r1c*, *Syt13*, *Shox2*, and *Pcbp3* identified as DEGs[78,79] (**Figure S1.7**, C8). C14 was annotated as ganglion eminence progenitor cells, which are groups of proliferating cells at the ventral telencephalon in early embryos[80]. Early studies have shown that the ganglionic eminences follow a temporal order, with the medial ganglionic eminence (MGE) arising first, followed by the lateral ganglionic eminence (LGE)[80,81]. Marker genes for ganglionic eminence cells were identified in C14, including both LGE and MGE markers, such as *Dlx1*, *Dlx2*, and *Nkx2-1*[80,82] (**Figure S1.7**, C14). The elongated structure of C14 might indicate a separation between LGE and MGE. We noted that the expression patterns of the LGE-specific markers *Gsx2* and *Is/1* and the MGE-specific markers *Lhx6* and *Lhx8* were spatially different, but the differences were not significant enough to support the hypothesis of separation.

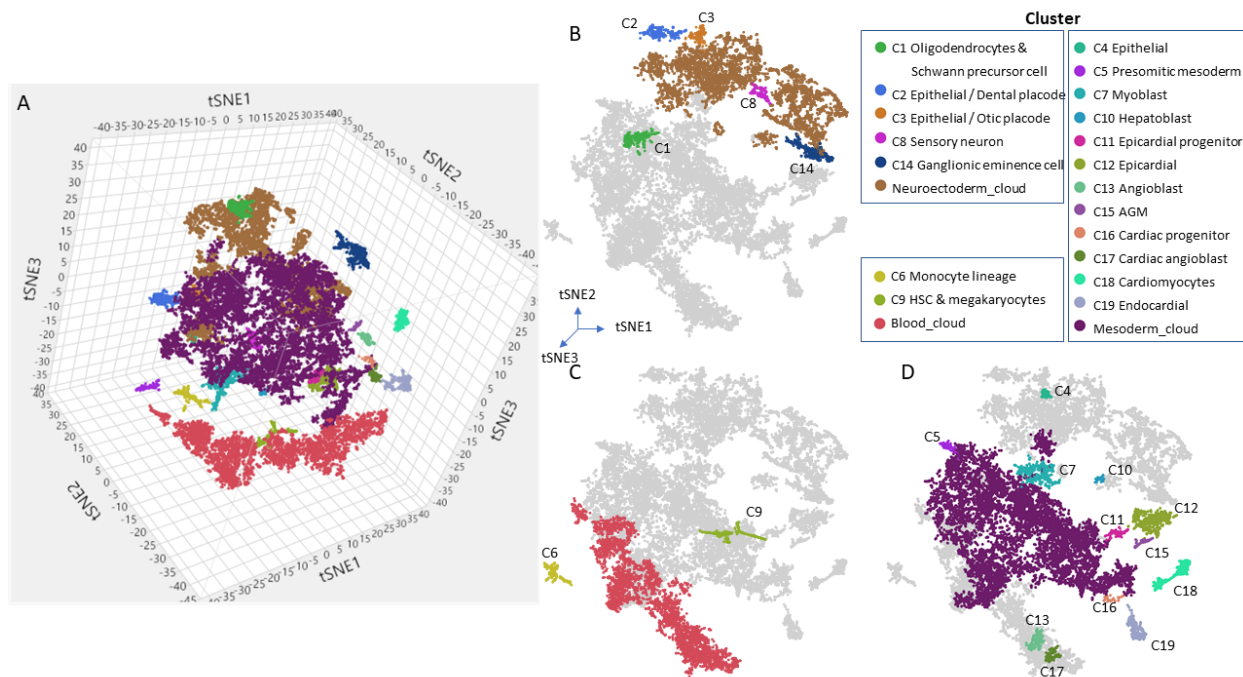


Figure 1.6: scRNA-seq identifies 20 unique cell types from E11.5 mouse embryo

Annotation of clusters identifies 20 isolated cell clusters representing unique cell types, with two large clouds of undifferentiated cell types.

(A) 3D t-SNE projection of 20 unique cell types and two clouds. **(B)** 2D t-SNE projection of 5 neuroectoderm derived cell clusters, and less differentiated cells in 'neuroectoderm cloud'. **(C)** 2D t-SNE projection of two blood-related cell clusters and erythrocytes in 'blood cloud'. **(D)** 2D t-SNE projection of 12 mesoderm/mesenchyme derived cell clusters, and less differentiated cells in 'mesoderm/mesenchyme cloud'. AGM: Aorta-gonad-mesonephros cell.

11 of the clusters originated from the mesoderm, most of which were associated with heart structures. Mesodermal-derived clusters included C4 epithelial cells, C5 paraxial mesoderm cells, C7 myoblasts, C11 epicardial progenitor cells, C12 epicardial cells, C13 angioblasts, C15 aorta-gonad-mesonephros cells, C16 cardiac progenitor cells, C17 cardiac angioblasts, C18 cardiomyocytes, and C19 endocardium cells. Two of the clusters originated from blood lineages, which should, strictly speaking, be included in the mesoderm category, including C6 monocyte precursors and C9 hematopoietic stem cells (HSCs) and megakaryocytes.

Seven of these mesodermal-derived clusters were heavily related to cell types found in the heart, which is unsurprising because the cardiovascular system is believed to be the first organ system to form and function[83]. As early as embryonic day E8.0, the developing heart can be recognized as the primary heart tube[84–86]. At embryonic day E10.5, a heart chamber consisting of a left and right atria and ventricles is morphologically obvious, with a variety of tissue/cell types originating from different precursors[87] due to the complexity of the heart, such as the epicardium from the proepicardial organ[88], the endocardium from a variety of debatable origins, including anterolateral plate mesodermal

precursors[87], cardiovascular progenitors[84,89], and vascular endothelial cells[89], and myocardium from first and second heart field precursors[90]. In addition to these major tissues and cell types, fibroblasts, angioblasts, and blood lineages are also expected to be found in the developing heart. C18 contained very clear signals indicating cardiomyocytes, featuring GO:BP terms such as 'Cardiac muscle contraction,' 'Sarcomere organization,' and 'Heart development,' with high expression levels of most cardiomyocyte markers, including *Tnni3*, *Actc1*, *Myh6*, and many other myosin genes[91–93] (**Figure S1.7**, C18). Surprisingly, most of the cells within this cluster were derived from heart samples (97.3%), with fewer than expected cells deriving from whole embryo samples. We believe that this bias was the result of the sample preparation step, as heart samples are more rigid and require longer digestion times, whereas the whole embryo samples were digested for shorter periods of time to compensate for other, less rigid tissues.

C17 and C19 both presented endocardial cell signals but with fundamental differences. Although they shared similar GO:BP terms, the matching scores varied for different terms. C17 was much more strongly related to vasculogenesis, as represented by GO:BP terms including 'Angiogenesis' and 'Vasculogenesis,' whereas C19 more strongly matched the GO:BP terms including 'ECM organization' and 'Cell adhesion.' Because endocardium cells during this stage are likely undergoing epithelial to mesenchymal transitions (EMT), through this migration, we expected to observe heterogeneity among cell types[88]. By comparing the gene expression levels of both clusters, we believe that they all originated from endocardial cells but that C19 cells were undergoing EMT and were, thus, annotated

as endocardial-derived cushion cells, with high expression levels of *Has2* and *Sox9*[94]. In contrast, C17 cells were annotated as endocardial tube cells that do not undergo EMT transition, with the high expression levels of *Tek* and *Tie1*[83] (**Figure S1.7**, C17/C19). As both cell lines expressed the endocardium marker *Pecam1*[87,95], a differentiation branch was observed here. C17 also shared several terms with C13, which will be discussed later.

C15 showed signals for cell types from both male and female gonads, according to the GO:BP terms 'Male gonad development' and 'Female gonad development,' and the gene expression pattern matched the patterns observed for multiple gonadal cells, including Leydig cells, Sertoli cells, and granulosa cells, which are not differentiated at this stage[96,97]. We validated this cluster as consisting of gonadal somatic cells using the marker genes *Wt1*, *Lhx9*, *Nr5a1*, and *Dmrt1*[96–99] (**Figure S1.7**, C15). As reported in the previous literature, the gonadal fate is generally not determined until approximately embryonic day E11.5-E12[97], which is supported by our data that C15 contained both future testis and ovary cell types.

C11, C12, and C16 were difficult to characterize at first glance, as we noticed that all three clusters featured strong GO:BP matches with the terms 'Cell adhesion' or 'ECM organization.' During mouse cardiogenesis, the formation of the epicardium from freely floating proepicardial cells occurs between embryonic days E9.5 and E11.5, during which time cell adhesion molecules play significant roles[93,100,101]. At the developmental stage we were examining in this study, the epicardial cells are also undergoing EMT to form epicardium-derived cells, which will further differentiate into various cell lineages.

Thus, we hypothesized that C11, C12, and C16 were cell types related to this process. We also noticed that C16 featured stronger signals for ‘Angiogenesis,’ and the high expression level of the vascular smooth muscle markers *Acta2*, *Cald1*, *Cnn1*, *Talgn*, and *Vc*[93,102–104] (**Figure S1.7**, C16), which suggested that C16 might represent epicardium-derived vascular smooth muscle cells. In contrast, C11 and C12 covered a broader range of GO:BP terms, with high levels of the epicardium markers *Tbx18*, *Aldh1a2*, *Tcf21*, and *Wt1*[105–107] (**Figure S1.7**, C11/C12). As C11 and C12 were spatially connected, they could represent two clusters of epicardial cells undergoing differentiation. Considering the sizes of these clusters, the developmental event we observed here shows a population of less differentiated epicardium cells differentiating from C11 to C12 and a small sub-population of more differentiated epicardium-derived vascular smooth muscle cells in C16. This pattern might indicate that the smooth muscle cell is the first cell to be differentiated in the epicardium, whereas other cells, such as cardiac fibroblasts, are differentiated later. Based on the distinct transcriptional profiles for endocardium- and myocardium-derived cells, our data supported the possibility that myocardial and endocardial progenitors might have distinct developmental origins.

The 7 clusters mentioned above were derived from the lateral plate mesoderm, whereas the remaining 4 clusters were derived from the intermediate mesoderm and paraxial mesoderm. C4 showed a strong match with epithelial cells, similar to the epicardial clusters described above, but with much lower cardiac signals. From the top matches, we observed cell types from the early kidney, including ureteric bud cells and ureteric bud tip cells. Based on published studies examining mouse kidney development[108–110], the

nephric ducts emerge as early as embryonic day E8.5 and then undergo epithelialization to form the metanephric mesenchyme, at approximately embryonic day E10. By embryonic day E10.5, the ureteric bud forms from the nephric duct, which also induces the metanephric mesenchyme to undergo the mesenchymal-epithelial transition (MET). By embryonic day E11.5, the ureteric bud begins to branch into a T-like structure, which is considered to be the beginning of kidney formation. Although the gene expression pattern of this cluster was not clear or specific, we did observe the expression of the ureteric bud tip cell markers *Gfra1*, *Wnt7b*, and *Itga3*[111,112] (**Figure S1.7**, C4) in this cluster; thus, we annotated this cluster as ureteric bud tip cells or, more broadly speaking, early kidney cells.

C13 showed very strong matches with the GO:BP terms 'Angiogenesis' and 'Vasculogenesis'; however, compared with C16, C13 was less specific for heart samples (17.79% heart). From a developmental perspective, angiogenesis and vasculogenesis represent two distinct mechanisms involved in embryonic vascular pattern formation, in which vasculogenesis refers to the formation of blood vessels de novo through the aggregation of angioblasts, primarily at the heart, whereas 'angiogenesis' refers to the formation of new vessels from pre-existing vessels in all organs[113,114]. Combined with the featured expression of the angioblast markers *Cd34*, *Cdh5*, *Flt1*, *Kdr*, and *Tek* (**Figure S1.7**, C13/C17), we confirmed that this cluster represented angioblasts derived from mesoderm progenitor cells[113,115]. C13 also shared several terms with C17 endocardial tube cells, and many of the markers mentioned were expressed in both clusters, including *Cd34*, *Cdh5*, *Flt1*, *Kdr*, *Tie1*, *Tek*, and *Pecam1*. After comparing the expression levels of

these markers and other GSEA evidence, we determined that C17 was more specific to the endocardium, whereas C13 was more generalized to angioblasts.

C5 showed a significant match with pattern formation-related GO:BP terms, including ‘Anterior/posterior Pattern Specification’ and ‘Somitogenesis.’ From the MGI phenotypes, we also observed a large number of rib- and vertebrae-related terms. We also noticed that no cells from the heart or head samples were included in this cluster, further indicating that this cell cluster was specific to the trunk. The DEGs had a strong match for presomitic mesoderm cells (PSM), which play significant roles in somitogenesis and pattern formation. Although the embryo at this stage features a number of well-developed somites, the PSM cells are maintained in an immature, mesenchymal state[115]. Based on the featured expression of PSM markers, including *Cyp26a1*, *Fgf4*, *Fgf8*, *Hes7*, *Lfng*, *Nkd1*, *Notch1*, *Nrarp*, *Tbx6*, and *Chst7*[116–121] (**Figure S1.7**, C5), we confirmed that this cluster represented PSM cells, particularly posterior PSM cells at the tail bud.

The last cluster, C7, was the easiest to identify, with a significant GO:BP match to ‘Muscle contraction’ and ‘Muscle filament sliding.’ Confirmed by the expression pattern of the myoblast markers *Myf5*, *Myf6*, *Myod1*, and *Myog*[122–126] (**Figure S1.7**, C7), we annotated this cluster as myocytes.

Interestingly, only one cluster demonstrated strong endodermal signals, which was C10 hepatoblasts. Moreover, in the less-differentiated clouds, we did not observe any strong endodermal signals. The endoderm has been documented to emerge from the anterior primitive streak at embryonic day E7.5 in the mouse[127]; however, in this study and in

other published single-cell studies, no notable populations of endodermal cells has been identified other than hepatoblasts[66]. This finding indicates that the liver bud is perhaps the most well-differentiated endodermal organ by embryonic day E11.5, whereas the other cells are mixed together with undifferentiated mesodermal cells. The mouse liver bud has been reported to form at approximately embryonic day E9.5, followed by the expansion of hepatoblasts, which differentiate into hepatocytes at approximately embryonic day E13.5[127,128]. To further investigate the identity of this cell cluster, we examined well-documented marker genes for hepatic lineages. The strong expression of hepatoblast markers *Afp* and *Alb*[127] was observed, but the level of their descendant cholangiocyte marker *Hnf1b* was negligible. The expression of other descendant hepatocyte markers, *Prox1* and *Hnf4a*, was also identified in this cluster, though at low levels, further validating these cells as undifferentiated hepatoblasts that may potentially be in the process of differentiating into hepatocytes. In addition, we examined *Pdx1*, *Onecut1*, and *Ptf1a*, which are markers for pancreatic progenitors, a cell type that is differentiated from the same ancestor as hepatoblasts[129,130]. All of these markers were expressed in a scattered manner and at low levels in mesenchymal cells, with no specific expression patterns, which indicated either that this cell type has not yet differentiated or that it is associated with a very small population. Mouse pancreatic development has been reported to start at embryonic day E9.0, followed by development into a single organ by embryonic day E12.5[130]; however, during the early stage of development, very little ultrastructural differentiation can be observed at the pancreatic buds until approximately embryonic day E12[129]. One study has been published using scRNA-seq to examine pancreatic development, although the earliest stage they

investigated was embryonic day E12.5[18], which also addresses the dynamic of marker expression along the timeline. Our data showed that pancreatic progenitor cells do not demonstrate a unique expression pattern compared with other mesenchymal cells at this stage, unlike hepatoblasts.

The remaining two clusters were associated with blood lineages, which should also be considered mesodermal-derived cell types. According to several immune-associated GO:BP terms from the GSEA analysis, including 'Inflammatory response,' 'Innate immune response,' and 'Immune system process,' C6 was annotated as monocyte precursor cells, confirmed by the DEGs *Cd14*, *Cd33*, *Itgam*, *Ccr2*, and *Cx3cr1*[131–133]. The DEGs in C9 showed strong matches with HSCs, whereas the GO:BP terms also identified platelet-related processes, such as 'Platelet activation,' 'Platelet degranulation,' and 'Blood coagulation.' Considering the branched structure of this cluster and the mixed clues provided by multiple cell types, we believed this to represent a cluster of differentiating cells, likely to be populated by HSCs and megakaryocytes. The details of these two blood lineage clusters will be discussed later, together with the blood cloud.

1.3.2 Cell type dynamics in the differentiating mesoderm/mesenchyme cloud

Next, we examined the mesoderm/mesenchyme cloud, the structure of which represented a continuous cloud, indicating that the transcriptional profiles within the cloud were not as distinct as the 19 clusters we described above. By highlighting samples from the different dissection segments, we observed that the cells from the trunk and the head were separated, whereas the cells from the whole embryo samples covered both areas,

indicating that this split is due to the transcriptional differences between the head and the trunk, rather than from bias associated with the data processing procedures. The pattern remained, even after we subset the cloud and reran our pipeline (**Figure 1.7A**). To understand the cell type dynamics within this cloud, we subset the cloud from the entire dataset and reperformed the analysis with subclustering. During this second-round analysis, we modified our approach to suit the goal, with less heterogeneity. We used a graph-based clustering approach from the R package, Seurat. Because we were more interested in the connections between sub-clusters, we used UMAP instead of t-SNE for visualization.

The mesoderm cloud consisted of 10,381 cells from all 4 samples, with 21 SLHC clusters, before pooling the cloud. After subsetting and subclustering analysis, we identified 15 sub-clusters, optimizing the resolution to 0.9 (**Figure 1.7B**). As a reference, we also inspected the 21 SLHC results and incorporated significant signals into our cellular identity annotation (far right arrow in the schematic in **Figure 1.1B**).

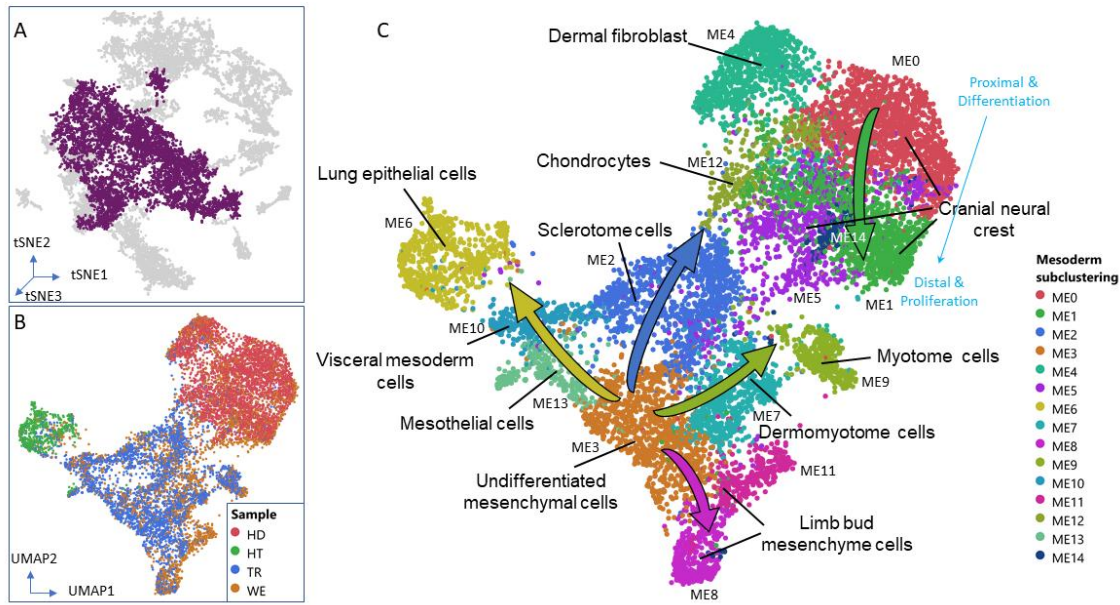


Figure 1.7: Cell type dynamics revealed by sub-clustering of the mesoderm/mesenchyme cloud

Sub-clustering enables higher resolution views of the mesenchyme cloud, and uncovers various cell types.

(A) Mesoderm/mesenchyme cloud highlighted in 2D t-SNE projection of the entire dataset. **(B)** UMAP projection of the mesoderm/mesenchyme cloud cells, after subset and reran the dimension reduction in UMAP. Colored by sample origin of the cells. **(C)** Sub-clustering of the mesoderm/mesenchyme cloud subset, colored by sub-clusters. Labels and arrows indicate cell type annotation and differentiation trajectories.

Of the 15 sub-clusters, six primarily originated from head samples, which we referred to as head mesenchyme cells, including ME0, ME1, ME4, ME5, ME12, and ME14. Eight of the sub-clusters primarily originated from trunk samples, which we referred to as trunk mesenchyme cells, including ME2, ME3, ME7, ME8, ME9, ME10, ME11, and ME13. One of the sub-clusters featured a mixed sample origin, derived from heart and trunk cells, ME6.

ME6 showed the most unique pattern among these sub-clusters and was annotated as early lung mesenchyme cells. Although considered to be an endodermal-derived organ, the lung remains in a very early stage of development, making the transcriptional profile less unique; thus, lung cells were not identified as one of the 19 isolated clusters. Before embryonic day E12.5, the evagination of a group of epithelial cells gives rise to the trachea and the lung buds[134,135]. At embryonic day E11.5, the branching process has not yet been initiated; thus, some early markers, such as *Sox2* and *Sox9*, do not show specific expression patterns. However, the GSEA results revealed some DEGs responsible for lung development, indicated by the high scores of the GO:BP terms ‘Lung development’ and ‘Lung Alveolus development.’ Though the early markers for the lung mesenchyme have not been well-documented, we identified many genes whose loss of function results in significant lung phenotypes, including *Foxf1*, *Tcf21*, and *Tbx3* (**Figure S1.8**, ME6), and some hox genes, such as *Hoxa5*, *Hoxb2*, *Hoxb4*, and *Hoxb5*[134,136,137](**Figure S1.9**). In addition, the adjacent cluster, ME10, presented some endodermal signatures, characterized by the specific expression of *Nkx2-3* (**Figure S1.8**, ME10), which is a marker for the gut mesoderm[138], as well as endothelial compartments in the spleen[139,140]. The GSEA analysis of ME10 DEGs returned signals from multiple visceral organs, including ‘Lung morphogenesis,’ ‘Pancreas morphogenesis,’ ‘Gland morphogenesis,’ and ‘Spleen development.’ Although many of the matched genes were not considered to be specific organ markers, we were not surprised to observe the absence of those markers at such an early stage of development. Another adjacent cluster, ME13, showed the differential expression of *Upk3b* and *Alcam* (**Figure S1.8**, ME13), which serve as specific markers for the mesothelial cells that cover the visceral

organs[141,142]. This finding was also confirmed by the high score for the GO:BO term 'Epithelial cell differentiation.' Therefore, we believe that this cluster contains mesenchymal cells that present mesothelial cell potential but remain undifferentiated. The clusters mentioned above express a mixture of signals from both the mesoderm and the endoderm. As described in the previous section, the only endodermal signal identified among the 19 isolated clusters derived from hepatocytes in C10. Combined with the sub-clusters from the mesoderm/mesenchyme cloud, the observed mixture of mesodermal and endodermal signatures indicated the transcriptional similarity of these less-differentiated cells. No additional endodermal signals were observed, other than in the clusters that have been discussed, which agrees with other single-cell studies performed at a comparative stage[66]. Although the endoderm can form as early as the egg cylinder stage[143], our data indicated that at embryonic day E11.5, the cell population derived from the definitive endoderm was small. Epidermal cells may not be transcriptionally different from other undifferentiated mesenchymal cells. Bipotent mesoderm cells are able to give rise to both mesodermal and definitive endodermal cells[144]. In this study, the 3 clusters discussed here might represent a developmental trajectory from less-differentiated mesenchymal cells with mesothelial potential (ME13), more differentiated lung epithelial cells (ME6), and visceral mesoderm cells (ME10), which eventually give rise to multiple organs, including the lung, pancreas, spleen, and liver[145].

The rest of the trunk-derived sub-clusters all showed clear relationships with somitogenesis. The first somite arises at approximately embryonic day E8, and at embryonic day E11.5, over 40 somites can be identified in the developing embryo[69,146].

In the previous section, we described an isolated cluster, C5, with a strong posterior PSM signal, which arises from the segmentation of the front paraxial mesoderm. In this sub-clustering analysis, ME3 is likely the most undifferentiated cell type, still undergoing proliferation to differentiation. The GSEA of ME3 returns many morphogenesis GO:BP terms. Specifically, we identified multiple members from the mesodermal commitment pathway, including *Gata6*, *Tbx3*, *Tbx5*, *Bmp4*, *Ptn*, and *Pitx1*[147–150](**Figure S1.8**, ME3). The neighboring clusters of ME3 present cell types that are more differentiated, including ME7 dermomyotome cells, ME9 myotome cells undergoing myogenesis, ME2 sclerotome cells, and ME8/ME11 limb bud mesenchyme cells. Somitogenesis is a process that has been well-characterized, and we can easily match the clusters to somite-derived cell types using well-studied markers. In ME2, we found *Pax1* differentially expressed as a sclerotome cell marker[151](**Figure S1.8**, ME2), together with the hox family genes *Hoxc6*, *Hoxc8*, and *Hoxc9*[152]. In addition, *Sox9* and *Col2a1* were highly expressed in this cluster, which indicated that these cells were differentiating into chondrocytes[153]. The adjacent clusters ME12 and ME4, which we will describe later, further validated the identity of ME12. ME7 and ME9 were associated with cells differentiating from the dermomyotome to the myotome, based on the transition from the marker genes *Dmrt2*, *En1*, and *Pax3*[154–156] to *Zic1*, *Zic3*, and *Zic4*[157](**Figure S1.8**, ME7/ME9). ME8 and ME11 (two clusters at the far bottom end) showed clear signals characteristic of the limb bud mesenchyme, with the featured expression of multiple Hox genes (**Figure S1.9**) and other genes involved in limb morphogenesis, such as *Shox2* and *Tbx5*[158,159]. The hox gene expression pattern of the limb bud mesenchyme will be discussed in detail later.

When examining the head sample-derived clusters, fewer structures were revealed by the UMAP plot. Most of these cells could be identified as cranial neural crest-derived at different states. ME0 showed strong GSEA results associated with cranial morphogenesis, which we believe could be cranial neural crest cells. Interestingly, ME1 returned very strong signals for cell proliferation, indicated by the GO:BP terms 'Cell division' and 'Cell cycle,' whereas prior to subsetting the mesenchyme cloud, ME1 showed similar results as ME0. This finding likely indicates that ME0 and ME1 are both cranial neural crest cells, but ME0 might be undergoing differentiation, whereas ME1 is undergoing proliferation. By comparing the gene expression patterns, we observed that both clusters expressed cranial neural crest development-related markers, including *Msx1*, *A/x3*, and *A/x4* in ME0[160,161] (**Figure S1.8**, ME0), and *Dlx1*, *Dlx2*, *Dlx5*, and *Dlx6* in ME1[162,163] (**Figure S1.8**, ME1). We were able to observe a nested expression pattern of *Dlx* family genes in our UMAP plot, with *Dlx5* and *Dlx6* expressed in the tip of ME1, whereas *Dlx1* and *Dlx2* were expressed in a more broad region, which aligns with the literature[163]. The expression of cell-cycle related genes, including *Ccnb1*, *Cenpf*, *Cdk1*, *Cdc20*, and *Cdca8*, supports that ME1 is undergoing rapid cell proliferation. The same group of cell cycle genes is also relatively high in the limb bud mesenchyme cluster, ME3, another region that undergoes rapid cell proliferation. ME5 and ME14 are spatially connected to the two clusters above but do not show clear signals, likely due to a more restricted background. Taking all this information into account, we are able to identify a differentiation trajectory from ME0 to ME1, ME5, and ME14, going from the more proximal differentiating cranial neural crest cells to the more distal proliferating cranial neural crest cells. As described above, the trunk sample-derived ME3 cluster was annotated as

sclerotome cells with strong chondrocytes signals. The two adjacent clusters, ME4 and ME12, which were derived from head samples, also showed similar signals, indicating chondrocytes and dermal fibroblasts. With the high expression level of various collagen genes, including *Col1a1*, *Col1a2*, *Col2a1*, *Col3a1*, *Col4a1*, and *Col4a2*[164–166], together with the other chondrocyte markers *Atp1a2*, *Epha7*, and *Lamc3*[66] (**Figure S1.8**, ME4/ME12), we confirmed that these clusters are neural crest-derived chondrocytes and dermal fibroblasts. These chondrocyte-like epithelium signals appeared in 3 clusters, derived from either head or trunk samples, which indicated that these cells come from different origins, with slightly different transcriptional profiles.

In summary, our subclustering approach to the mesoderm/mesenchyme cloud revealed the dynamics of cell differentiation for the developing trunk and head mesenchyme. Starting from undifferentiated mesenchymal cells, at least four developmental trajectories could be identified in our data, including the limb bud mesenchyme, dermomyotome, sclerotome, and visceral mesoderm (**Figure 1.7C**).

1.3.3 Cell type dynamics in the differentiating neuroectoderm cloud

The subclustering of the neuroectoderm cloud was the most challenging step of this analysis. After subsetting, the cloud, which contained 5,396 cells, was split into two parts, the structure of which was similar to that before subsetting (**Figure 1.8A**). Using the same strategy as was applied to the mesoderm/mesenchyme cloud, we sub-clustered this cloud into 9 sub-clusters by optimizing the resolution parameter to 0.3 (**Figure 1.8B**). Then, we analyzed each cluster independently. Most of the cells in this cloud were derived from

head samples, but we observed one cluster that was overpopulated by whole embryo samples, which was rare during this study and might indicate an artifact associated with the pooling of different samples together, although we did not observe such events in other clusters. This cluster, NE3, was annotated as unknown due to the potential artifact; however, according to the signals from our integrated approach, they appear to be neural crest cells. The nerve system is typically considered the most complex organ in the body, and how this complexity is generated during embryonic development remains a mystery. The first neurons of the central nervous system are born between embryonic days E9 and E10, which marks the initiation of neurogenesis[167]. The radial glial cells arise from the neuroepithelial progenitors, which are considered to serve as the primary progenitor cells for most neurons and other cell types, including astrocytes and oligodendrocytes[168,169]. Embryonic day E14 has been reported to be the peak of neurogenesis, whereas at the stage examined during our study, the onset of neurogenesis was expected to have begun only the day prior, and most of the related cell types are likely still undergoing expansion. Therefore, we did not expect to observe a wide variety of neurons except for radial glial cells. In addition, the development of the cortex is still considered to be at the very early stage, without the layered organization[167]. However, we observed several axes with the potential fates of different neurons, which might serve as early clues of cell lineage progression. Because the signals for specific neurons are either mixed or weak, based on the GSEA analysis, we applied gene expression projections for various neuronal markers, which were obtained from various sources, to gain insights into each cell cluster, including the Abcam neuron marker

list[170], the Allen brain atlas[171], the Linnarsson lab mouse brain atlas[172], and the literature.

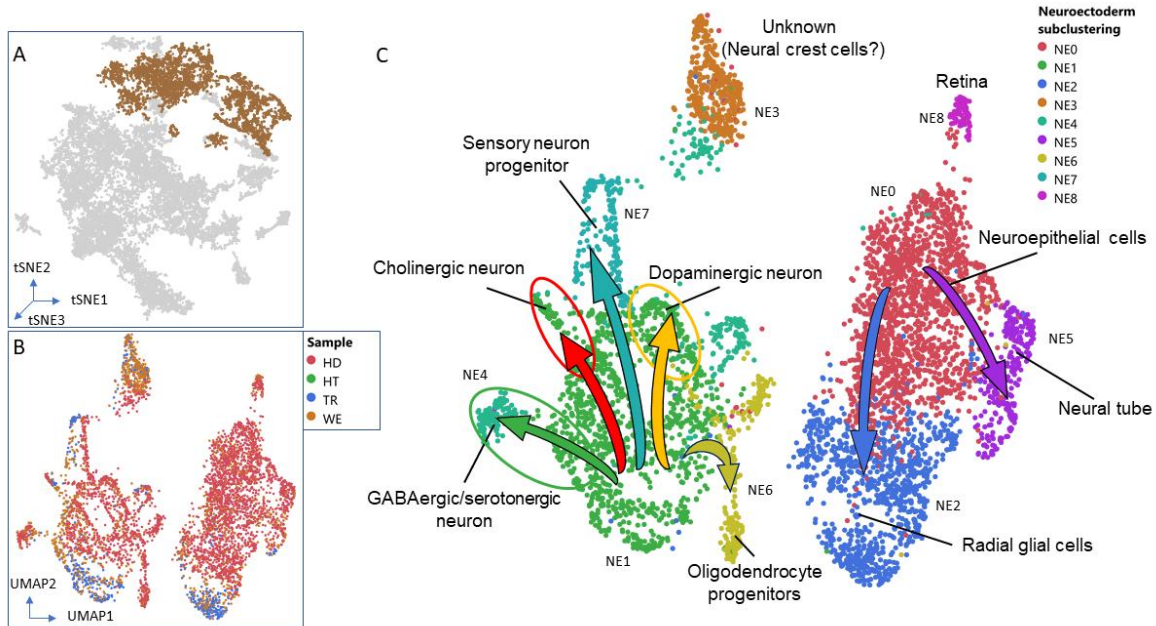


Figure 1.8: Cell type dynamics revealed by sub-clustering of the neuroectoderm cloud

Sub-clustering enables higher resolution view of neuroectoderm cloud, and uncovers both differentiated and undifferentiated neuron cell types

(A) Neuroectoderm cloud highlighted in 2D t-SNE projection of the entire dataset. **(B)** UMAP projection of the neuroectoderm cloud cells, after subset and reran the dimension reduction in UMAP. Colored by sample origin of the cells. **(C)** Sub-clustering of the neuroectoderm cloud subset, colored by sub-clusters. Labels and arrows indicate cell type annotation and differentiation trajectories.

The subclustering analysis identified at least two large populations, one containing fewer structures (NE0, NE2, and NE5) and the other containing more complex structures (NE1, NE4, NE6, and NE7). We observed very strong cell cycle signals in NE0, NE2, and NE5, which indicated that these cells are actively undergoing proliferation. Both before and

after the radial glial cells transit into a neurogenic state, at approximately embryonic day E12, proliferation is a key feature. Symmetric proliferation would expand the progenitor pool, whereas asymmetric proliferation gives rise to cells that are able to differentiate[173,174]. With only broad neural stem cell signals and strong cell cycle gene expression, we annotated NE0 as neuroepithelial cells undergoing proliferation. The adjacent cluster, NE2, also showed active proliferation but with some radial glial markers as DEGs, including *Fabp7*, *Pax3*, *Fzd10*, and *Hes5*[175–177] (**Figure S1.10**, NE2). NE5 showed mixed signals from various GO:BP terms and cell types; however, we found several DEGs in this cluster that were highly related to neural precursors in the neural tube but not to differentiated neural cells, such as *Id1* and *Id3*[178–180] (**Figure S1.10**, NE5). The *Rspo* family genes, *Rspo1*, *Rspo2*, and *Rspo3* were also identified as DEGs in this cluster and have been reported to be expressed specifically in the dorsal neural tube on embryonic day E10.5[181]. Because we already identified other cell clusters in the mesoderm/mesenchyme cloud associated with specific signals for migrating neural crest cells, we believe that NE5 represents neural crest cells at the beginning of migration. The other population, however, showed specific signals associated with more mature neurons. The structure of this population is also more complex compared with the one discussed above; however, of the several clustering approaches that we tested, none was able to perfectly dissect the population into subclusters with unique DEGs. To better understand the nature of these cells, we compared the marker gene projection patterns with the UMAP plot. The majority of this population comes from NE1, which showed strong signals for the GO:BP term ‘Nerve system development.’ The entire cluster

showed a strong expression of *Tubb3*, which marks neural progenitors that are differentiating into mature neurons[182]. By examining the expression of various mature neurons, we believe that the complex structure of this population is derived from neurons differentiating into different subpopulations. The top right part of NE1 shows the high expression of dopaminergic neuron markers, including *Th*, *Nr4a2*, and *Foxa2*[170,183–185] (**Figure S1.10**, NE1), which might indicate the future fate of this small sub-population. The top left branch of NE1 shows the concentrated expression of *Slit2* and *Slit3*, which marks cholinergic neurons[177]. The left part of NE1 and the adjacent part of NE4 show differentially expressed gamma-aminobutyric acid (GABA)ergic neurons and serotonergic neuron markers, such as *Pax2*, *Slc32a1*, *Gabbr1*, *Gad2*, and *Fev*[170,186–190] (**Figure S1.10**, NE4). NE6, at the right side of NE1, has a unique structure but less clear signals and shows the specific expression of *Tbr1* and *Eomes* (**Figure S1.10**, NE6). The upregulation of *Eomes* has been associated with the downregulation of *Pax6* when radial glial cells transition into intermediate progenitor cells, and the subsequent transition from intermediate progenitor cells to postmitotic neurons is associated with the downregulation of *Eomes* and the upregulation of *Tbr1*[191]. Although both *Eomes* and *Tbr1* are expressed in NE6, *Eomes* highlights the top section, and *Tbr1* highlights the bottom tip, which could be explained as a differentiation trajectory. Thus, NE6 might represent the bridge between the two large populations of this subset, marking a key transition during neurogenesis. NE7, above NE1, specifically expresses *Atoh1*, *Barh1*, and *Barh2*[192,193] (**Figure S1.10**, NE7), which marks sensory neuron progenitors and general migrating neurons. The bottom part of NE1 showed the high expression levels of less-differentiated

neural progenitor markers, such as *Gadd45g*, *Cdkn1c*, and *Btg2*[66,194,195] (**Figure S1.10**, NE1), which could be the root of all neurons described above.

Other than the two large populations and the unidentified NE3, NE8 was another isolated cluster identified during this sub-clustering analysis, which indicated that these cells have relatively unique expression profiles and, thus, might represent a more differentiated cell type. The GSEA results showed high score matches to retina progenitor cells, confirmed by the expression of the retina progenitor markers *Rax*, *Six6*, and *Pax6*[196] (**Figure S1.10**, NE8). The mature retina is a tissue with great cell type heterogeneity, including at least 6 neuronal and 1 glial cell types[197]. The retinal neurogenesis timeline matches the embryonic wave of neurogenesis, marked by the production of the retinal ganglion, amacrine cells, cone photoreceptor, and other related cell types, at approximately embryonic day E14[198]. Whether specific progenitors give rise to certain cell types or whether the progenitors are simply multipotent remains under debate. Because we only observed one cell cluster that appeared to be clearly related to the retina, and the gene expression patterns are specific, our finding would support the opinion that retina progenitor cells are multi-potent, based on lineage tracing experiments[199]. We also observe high scores for the GO:BP related to translation, featuring the high expression of ribosomal proteins, which might be a unique feature for progenitor cells that are preparing for differentiation into multiple subpopulations.

In summary, the neuroectoderm-derived cells remain at the early stages of differentiation. We identified two major populations, one associated with neuroepithelial cells, derived radial glial cells, and neural crest cells demonstrated signs of active cell proliferation. The

other population revealed a potential axis for different types of neurons, including GABAergic/serotonergic neurons, cholinergic neurons, sensory neurons, dopaminergic neurons, and oligodendrocytes. We also identified a population of undifferentiated retina progenitor cells that were undergoing active translation. Our data showed that even at such an early stage of neurogenesis, the progenitor cells already showed heterogeneity, with specific lineage potentials. Combining our data with data from later stage mouse brains, these cells are likely to undergo rapid differentiation and gain a much higher degree of heterogeneity in only a few days[200].

1.3.4 scRNA-seq reveals the blood lineage developmental trajectory

Hematopoiesis has been a popular topic for single-cell biology due to the complexity of this process. During the 150-year history of hematopoiesis study, multiple models have been generated using various approaches. In the bifurcation model, the common lymphoid progenitors (CLPs) are believed to give rise to B-cells, T-cells, and natural killer cells, whereas common myeloid progenitors (CMPs) give rise to erythrocytes, megakaryocytes, and all white blood cells[201,202]. In the myeloid-based model, HSCs give rise to MyE (myeloid progenitor with erythroid potential) and MyBT progenitors (myeloid progenitor with B-cell and T-cell potential). MyBT progenitors then further differentiate into MyB (myeloid progenitor with B-cell potential) and MyT (myeloid progenitor with T-cell potential) progenitors; however, megakaryocytes are not addressed in this model[202,203]. In the lymphoid-primed multipotent progenitor model, HSCs give rise to megakaryocyte and erythrocyte progenitors (MEPs) and lymphoid-primed multipotent progenitors (LMPPs), and LMPPs further differentiate into granulocyte and

macrophage progenitors (GMPs) and CMPs[202,204]. The complete landscape of hematopoiesis remains under debate, and, during the past few decades, new evidence has emerged to support multiple models and has spurred the constant refinement of the differentiation tree for HSCs, without reaching complete consensus[205]. The heterogeneity of cell types during hematopoiesis has been well-appreciated; thus, single-cell technology could represent a revolutionary tool for the systematic characterization of the cellular identities during hematopoiesis. Tremendous efforts have been made using scRNA-seq to address the heterogeneity observed during hematopoiesis during the past few years; however, fully understanding the dynamics of cellular identity during this process remains challenging[206–208].

In this study, we found that one of the three large clouds showed strong and clear signals for erythrocytes, which made this cloud an exception among the less-differentiated cell types. Even before pooling the SLHC clusters into this cloud, each of the clusters showed comparable erythroid signals, including the GO:BP terms ‘Heme biosynthetic process’, ‘Oxygen Transport,’ and ‘Porphyrin-containing Compound Biosynthetic.’ We then confirmed the cellular identities by projecting the erythrocytes markers hemoglobin Hba-a1 and Hbb-bh1, which both highlighted the entire cloud. The expansion of this relatively differentiated cell type over such a large cloud, rather than a compact cluster, was a surprising finding. To address this question, we subset the cloud and performed further analyses. One unique feature of this cloud in our dataset was the extremely low number of genes detected (**Figure S1.1A**). This more restricted gene expression pattern might have been more sensitive to noise and, thus, did not collapse into one compact cluster.

In addition, at the stage of our sample, embryonic erythrocytes can be found in the circulation system and are undergoing differentiation into mature erythrocytes, with cells decreasing in diameter and preparing to undergo enucleation[209–211]. The crescent-like structure of this cloud might indicate the ongoing transition of differentiating erythrocytes along its axis, from top to bottom. To test this hypothesis, we projected several hemoglobin genes onto this structure (**Figure 1.9B**). As we expected, the expression level of *Hba-a1* shows a gradient along the crescent axis with a higher level at the bottom end, whereas embryonic β -globin *Hbb-bh1* shows the opposite pattern. The gradient of these gene expression levels matches the pattern of reported gene switching during primitive erythropoiesis[210,212] and supports our hypothesis.

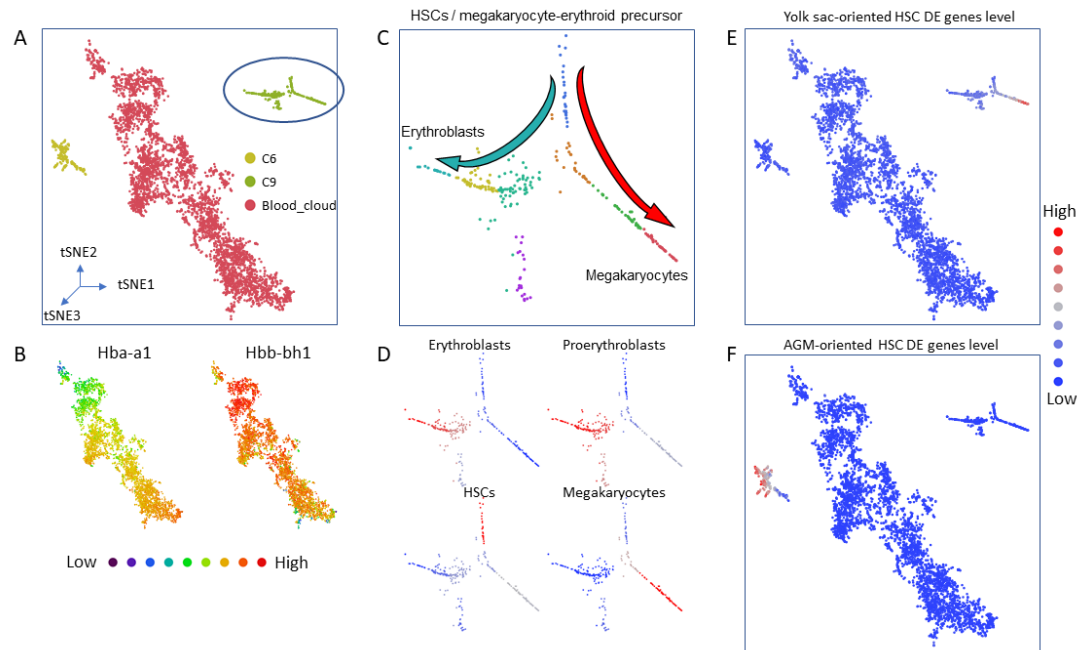


Figure 1.9: scRNA-seq reveals the blood lineage developmental trajectory

A large crescent-like cluster represents maturation of erythrocytes, and sub-clustering of the ‘fork’ reveals different lineage paths of HSC.

(A) 2D t-SNE projection of the three blood-related clusters/cloud subset from the entire dataset. **(B)** Expression level of two erythrocytes markers projected to the 'blood cloud' suggests a differential trajectory along the crescent. **(C)** 2D t-SNE projection of a zoom-in view of C6 (circled in panel A). Colored by sub-clusters. Arrows indicate two possible differential paths. **(D)** Expression level of normalized sum of blood-related cell type markers on C6. Different cell types highlight different axes. **(E-F)** Expression level of normalized sum of DEGs oriented from yolk sac and AGM highlight different clusters.

We also found clusters associated with other blood-related signals, which might be potential cell types that have differentiated from HSCs, including monocytes (**Figure 1.9A**, C6) and megakaryocytes (**Figure 1.9A**, C9). Although the first monocytes have been reported to initially arises in the fetal liver at approximately embryonic day E8.5[213], evidence suggests that myeloid cells examined at approximately embryonic day E10 remain poorly differentiated and that immature monocytes continue to be detected at embryonic day E12[214,215]. Our study, performed on samples from embryonic day E11.5, represents an interesting time point at which the two waves of hematopoiesis are both occurring, including primitive hematopoiesis, which occurs in the yolk sac, and definitive hematopoiesis, which occurs in the aorta-gonad-mesonephros (AGM) region[216,217]. The HSCs from both regions will colonize the fetal liver, which becomes the major hematopoietic site at approximately embryonic day E11.5[216]. Of the two clusters that we previously identified as blood lineages, C6 showed strong signals for monocytes, as described above, with some monocyte markers serving as DEGs. Markers for other parallel cell types were also high in this cluster, such as *Cd28* for T-cells[218], *Cebpa* for granulocytes[219], and *Ptprc* for HSCs[220] (data not shown). When considering these variations, although C6 has been annotated as monocytes, we believe

that this cell cluster may not be fully committed, continuing to express weak HSC and other related cell type signals.

C9 showed strong signals for HSCs and megakaryocytes, with a remarkable three-axis structure. However, the megakaryocytes markers *Itga2b* and *Itgb3* among the DEGs only highlight one axis of the cluster (data not shown), indicating the complex structure that is likely due to a mixture of multiple cell types. To further study the underlying meaning of this structure, we divided this cluster into sub-clusters (**Figure 1.9C**). By analyzing the sub-clusters using GSEA, the three axes showed unique signals for HSCs, megakaryocytes, and erythrocytes. To validate this finding at the gene level, we used the Lifemap cell type database to score each sub-cluster based on the gene expression profile data for several potential cell types, including erythroblasts, proerythroblasts, HSCs, megakaryocyte-erythroid precursor cells, and megakaryocytes (**Table 1.1**). A higher matching score indicates better alignment with known cell-type expression data. We obtained the same results as our GSEA analysis, indicating that the top branch might contain HSCs with a megakaryocyte-erythroid fate, the left branch contains cells differentiating into erythroblasts, and the right branch contains cells with megakaryocyte fates (**Figure 1.9D**). These results represent another example of differentiation along cell lineage, showing that HSCs differentiate into two different cell types.

Cluster	Erythroblasts	Proerythroblasts	HSCs	Megakaryocyte- Erythroid Precursor Cells	Megakaryocytes
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C9-1	1	2.64	6.81	0.5	4.35
C9-2	1.4	2.64	6.95	0.47	4.35
C9-3	1.4	0.35	11.35	3.32	0.81
C9-4	1.99	1.74	5.52	0.97	2.43
C9-5	5.03	4.71	5.82	1.41	0
C9-6	3.41	3.68	4.28	1.41	0.81
C9-7	6.04	5.24	4.31	1.65	0
C9-8	7.24	5.59	2.75	0	0

Table 1.1: Matching scores for the selected blood lineage cell types of the C9 sub-clusters

To better understand the HSC signals in our data set, we examined DEGs associated with HSCs that originated from the yolk sac and those that originated from the AGM[221]. We used the DEGs from the yolk sac and the AGM that were identified from this microarray data, calculated the sum UMI of related genes from the two origins, and projected them on our t-SNE plot (**Figure 1.9**, E-F). The projection of genes related to AGM-originated HSCs largely overlapped with our monocyte progenitors cluster, which agrees with the literature[221,222]. In contrast, the projection of genes related to yolk sac-

originated HSCs highlighted the megakaryocyte axis of C9, described above (**Figure 1.9D**), which also agrees with the literature[223].

1.3.5 Summary: a cell type catalog for the mouse embryo at early organogenesis

By applying our integrated approach of annotating cellular differentiation to our E11.5 mouse embryo dataset, we identified 20 isolated clusters, representing cell types that were differentiated and transcriptionally unique. Of all the organs that are eventually formed in adult mice, the heart is the most differentiated organ at this stage of development, as indicated by the multiple heart-related cell types identified in these 20 clusters, including epicardial cells, endocardial cells, cardiac angioblasts, cardiomyocytes, and some progenitor cells. Another relatively differentiated organ is the liver, with a cluster of hepatoblasts identified. However, pancreatic progenitor cells, which are thought to share the same progenitors as hepatoblasts, did not show any unique transcriptional profiles. In addition, marker genes for other organs, including the lung, kidney, and pancreas, did not show specific expression patterns; thus, related cell types remained mixed within the mesoderm/mesenchyme cloud. The blood lineage also showed a clear separation from other cell types. The blood cloud was annotated as erythrocytes undergoing differentiation, whereas the cluster C9, which presented a fork-like structure, showed a perfect example of precursor cells differentiating into two different daughter cell types.

By applying sub-clustering analysis to two clouds, we were able to identify differences between cells within the cloud. In the mesoderm/mesenchyme cloud, we observed the

clear separation between the head mesenchyme and the trunk mesenchyme, indicating that these two populations with different potentials begin to show differences long before they are fully differentiated. We also identified axes showing the path of cells from differentiation to proliferation, which aligns well with the spatial proximal-distal axis in craniofacial structures. An undifferentiated mesenchymal cell cluster was identified in the trunk mesenchyme, with its neighboring clusters most likely representing its descendants, including sclerotome cells, dermomyotome cells, limb bud mesenchyme cells, and mesothelial cells.

The same differentiation-proliferation axis could be observed in the neuroectoderm cloud sub-clustering analysis. The clear division between radial glial cells and neuron/neuron progenitor cells reflected a snapshot of transcriptional differences during early neurogenesis. The neuron clusters on the left showed finger-like structures. Based on the analysis of marker genes associated with more differentiated neuron types, we considered these structures to be indicative of neuron differentiation paths, with the neural progenitor cells as their root.

1.3.6 scRNA-seq reveals Hox gene nested expression patterns in the mesoderm

The Hox gene family is a well-conserved gene family that plays a significant role in embryonic development, especially the anterior-posterior pattern formation. The Hox genes have been extensively studied, and the idea of a 'Hox code' was first introduced almost 30 years ago, referring to the combination of Hox genes that specifies the identity of vertebral segments[224]. A key feature of Hox genes is 'colinearity,' which refers to the

correspondence between the organization of genes in the chromosome and the order of their expression along the body axis[225,226]. Interestingly, in this study, we observed significantly different expression patterns between Hox genes in the mesoderm/mesenchyme cloud, which indicates that our cell plot might contain useful spatial information regarding embryonic pattern formation and could potentially reflect the colinearity of Hox genes. When we examined the expression regions of these Hox genes and compared them with our cluster annotation, they appeared to follow the anterior-posterior axis. Specifically, the bottom end of the subset has a significantly higher level of posterior Hox gene expression, whereas moving along the axis to sub-cluster ME6 is associated with a decrease in posterior Hox gene expression and an increase in anterior Hox gene expression (**Figure S1.9**). In ME6, we identified *Hoxa4*, *Hoxb4*, and *Hoxb5* as DEGs, which are involved in lung development[227]. In ME8 and ME11, the extreme posterior *Hoxd* genes, including *Hoxd12* and *Hoxd13*, are DEGs, which we believe marks the limb bud (**Figure S1.9**). In addition, within the clusters that we annotated as limb bud mesenchyme cells, the proximal-distal axis might also be reflected. The posterior Hox genes (*Hox9-13*) are also responsible for limb development, especially two clusters of Hox genes, *Hoxa* and *Hoxd*[228,229]. In the mouse, because the limb bud emerges at early embryonic day E9, these Hox genes have been observed to be expressed in a nested pattern, with *Hox9/10* expressed proximally, *Hox11* expressed medially, and *Hox13* expressed distally[229]. In this study, we also observed this nested expression pattern for posterior Hox genes (**Figure 1.10**). Although less obvious for *Hoxb* and *Hoxc*, the more posterior Hox genes are clearly expressed at the distal part of cluster ME8. Additionally, we examined the expression of *Shh*, which is known to be expressed in the

zone of polarizing activity (ZPA), a small group of mesenchyme cells at the very posterior part of the limb bud[230], which highlights the distal margin of ME8.

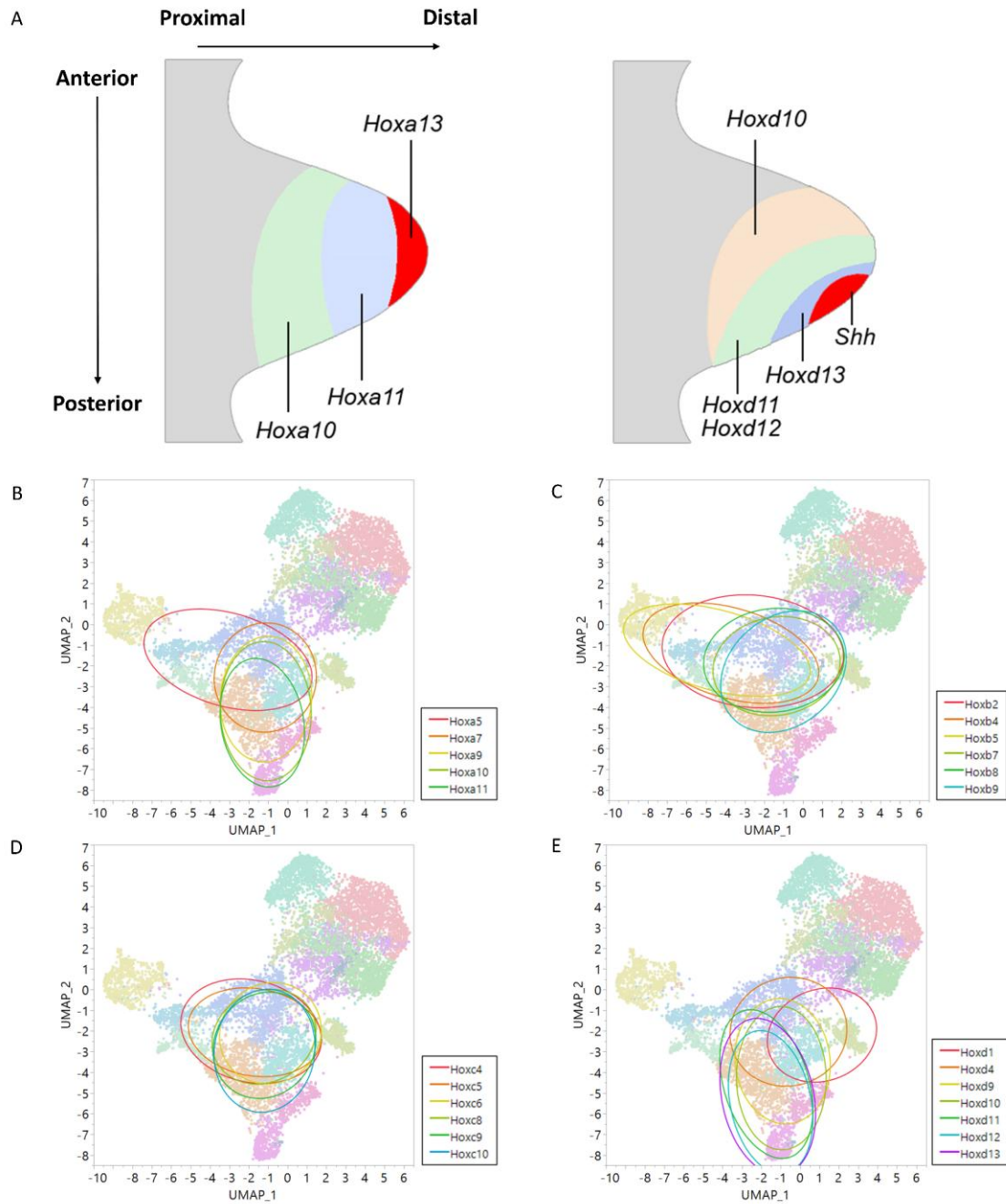


Figure 1.10: Nested Hox gene expression observed in t-SNE space

The Hox family genes show nested gene expression in the mesoderm cloud, indicating the cell plot contains spatial information.

(A) Schematic representation of nested Hox gene expression in the limb bud. **(B-E)** Expression patterns of various hox genes in the mesenchyme cloud subset.

In summary, the Hox gene family contains significant transcription factors for which pattern formation contains valuable spatial information and showed specific expression patterns in our cell plot of the mesoderm/mesenchyme cloud. The anterior-posterior axis expression in lateral plate mesoderm cells and the nested proximal-distal axis expression in the limb bud mesenchyme aligns well with our sub-cluster annotation. This expression pattern not only serves as an additional reference for our understanding of the sub-cluster but also confirms that the spatial information we observe in a reduced dimensional cell plot might reflect the real spatial information of a developing embryo.

1.3.7 Cardiomyocyte cluster and sub-cluster analyses show region-specific cardiomyocytes with different proliferation rates

As the first organ to form in vertebrates, heart development initiates at a very early stage. In this study, we identified multiple clusters related to the heart, most of which were included among the 19 well-isolated clusters, such as C11 epicardial progenitor cells, C12 epicardial cells, C16 cardiac progenitor cells, C17 cardiac angioblasts, C18 cardiomyocytes, and C19 endocardial cells. Of all these clusters, we noticed that C18 cardiomyocytes not only featured the strongest signals from GSEA but also displayed a unique dumbbell structure in t-SNE space. To examine the significance of this unique

structure and to determine whether this structure is associated with developmental events, we subset this cluster and analyzed it one step further.

To investigate which genes drive the dumbbell structure, first, we subsetting the C18 cluster and performed PCA analysis on the cluster (**Figure 1.11**, A-B). By coloring the top few PCs onto our t-SNE plot, we found that PC1 separates the two ends of the dumbbell. By examining the loading of PC1, we were able to capture the differences between the two ends. By performing GSEA on the top 100 genes for positive and negative PC1 loading, we found that the top negative PC1 genes were primarily signals for cardiomyocytes, including several marker genes, such as *Myh6*, *Myl4*, *Myl7*, *Tnni3*, and *Tnnt2*. In contrast, the top positive PC1 genes were more related to cell proliferation and protein translation. Although some cardiomyocytes markers were included, such as *Myh7*, *Myl2*, and *Myl3*, the cardiomyocyte signature was much weaker. Instead, the majority of these top-loaded genes in positive PC1 were associated with protein translation and cell cycle regulation, including approximately 20 ribosomal protein genes and approximately 20 cell cycle marker genes. Based on these results, one end of C18 might contain cells that are committed to cardiomyocyte fates but are still at an early stage of differentiation and, thus, are undergoing rapid proliferation. The other end of C18 might contain more mature cardiomyocytes, at later differentiation stages. From a different perspective, the segmentation of the developing heart into different chambers might result in different clusters at the single-cell level. Early during heart development, the linear heart tube is polarized along the A-P axis[231]. However, the four chambers that emerge after looping are not simply symmetric; instead, some genes have chamber-specific expression

patterns, and the developmental timelines of the ventricular and atrial chambers are different[232,233]. The atrial and ventricular cardiomyocytes also have different functions, morphologies, and gene expression patterns[234]. For example, studies have shown that the atrial cardiomyocytes have lower myofibril densities and lower resting tensions compared with ventricular cardiomyocytes[235–237]. One end of C18 might represent cardiomyocytes in ventricular chambers, whereas the other end represents atrial chambers.

To further validate this hypothesis, we split C18 into 2 sub-clusters and performed DGE analysis to identify DEGs in each sub-cluster (**Figure 1.11C**). The results matched well with our PCA analysis on the C18 subset. The sub-cluster C18-1, corresponding to the negative end of PC1, showed stronger signals for cardiomyocytes, whereas sub-cluster C18-2, corresponding to the positive end of PC1, showed mixed signals for cardiomyocytes and for a group of DEGs related to cell proliferation, cell cycle, and protein translation. The DGE analysis identified two groups of cardiomyocyte markers that were explicitly expressed in only one of the two sub-clusters (**Figure 1.11D**). Specifically, the six myosin genes we identified represent paired isoforms that have previously been reported serve as either atrial or ventricular markers, including myosin regulatory light chain gene *MyI7* (atrial) and *MyI2* (ventricular)[238,239], myosin essential light chain gene *MyI4* (atrial) and *MyI3* (ventricular)[240], and myosin heavy chain *Myh6* (atrial) and *Myh7* (ventricular)[241,242]. Based on these findings, C18-1 appeared to represent cardiomyocytes in the atria, whereas C18-2 represented those in ventricles. The strong signals for cell proliferation in C18-2 also agree with the report that cardiomyocytes in the

ventricle show a higher proliferation rate compared with those in the atria[239], and the cardiomyocyte proliferation centers identified during cardiogenesis were also discovered in the ventricular segment[239].

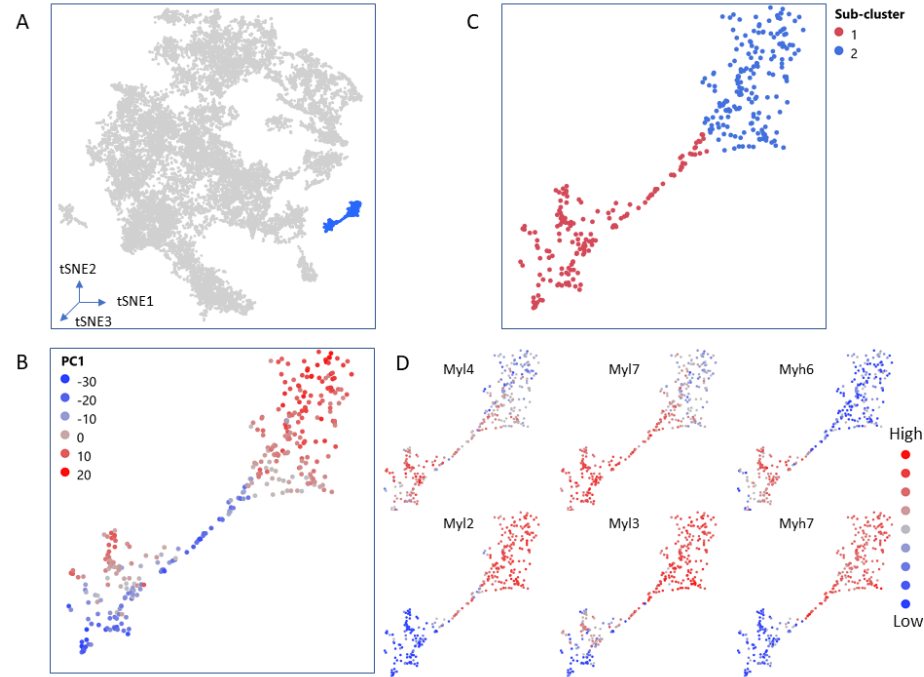


Figure 1.11: Cardiomyocyte cluster and sub-clustering analysis show region-specific cardiomyocytes with different proliferation rates

A unique structure is observed in the cardiomyocyte cluster, by sub-clustering analysis, independent PCA, and projection of marker genes, the two ends of the structure represent cells from different heart chambers

(A) The C18 cardiomyocytes highlighted from 2D t-SNE of the entire dataset. **(B)** A zoom-in view of the C18, colored by PC1. PCA performed on the C18 subset. **(C)** C18 sub-clustered into two clusters using t-SNE coordinates, colored by sub-clusters. **(D)** Expression level of several cardiomyocytes markers highlight different sub-clusters.

In summary, in this study, we identified a cluster of cardiomyocytes with unique structures in t-SNE space. By subsetting this cluster and applying our integrated annotation

approach, combined with a PCA loading analysis, we confirmed that the two sub-clusters from this dumbbell-like structure represented cardiomyocytes from the atria and ventricle. This result demonstrates that scRNA-seq is not only able to identify the same cell types with spatial differences but is also able to reveal important cell state information, such as cell cycle and proliferation rate. This type of analysis could provide valuable information for studying embryonic development and cell differentiation.

1.3.8 Cell type atlas of E11.5 mouse embryo: summary and discussion

By using the approach described in 1.2 combined with sub-clustering at a higher resolution, we were able to precisely annotate the cell clusters identified from this E11.5 mouse whole embryo scRNA-seq dataset, and provide a catalog of existing cell types at this specific time point of mouse embryonic development (**Figure 1.6-1.9**). We found that the lower dimensional projection from scRNA-seq could precisely represent the status of cell differentiation, and the more differentiated cell types can be separated from the undifferentiated ones. With higher resolution subclustering analysis, we were able to find more details within such undifferentiated regions, showing transitions between various related cell types in both neuroectoderm and mesoderm. The spatial information embedded in the tSNE projection was also supported by the example of nested Hox gene expression (**Figure 1.10**). We have also analyzed a few elongated structures we've observed in this dataset, turned out to be perfect examples of showing cellular differentiation paths and ability of scRNA-seq to reveal differences between similar cell types (**Figure 1.9-1.10**).

In summary, our study provided a precise approach to analyze and annotate large-scale scRNA-seq datasets with great heterogeneity and undifferentiated cell types that are difficult to separate. We've demonstrated how to obtain the rich information embedded in lower dimensional space from such a dataset, and provided examples how that would help us better understand the process of mouse embryonic development. The approaches used in this study could be extended to other large-scale datasets to gain more biological insights.

Materials and methods

Animals

Six-week-old CD1 mice were purchased from Charles River Laboratories and were housed and bred in our vivarium. Animal procedures were approved by the Institutional Animal Care and Use Committee at the University of Connecticut and conformed to the animal care guidelines established by the National Institutes of Health and the Association for Assessment and Accreditation of Laboratory Animal Care.

Embryo harvesting and single-cell preparation

CD1 mice were mated, females were sacrificed on embryonic day E11.5, and the uterine horns were collected. Embryos were dissected from the decidua, and extraembryonic membranes were removed under a dissection microscope in ice-cold Dulbecco's modified Eagle's medium (Life Technologies). Embryos were assessed, and the developmental stage was assigned using anatomical features, as described by the Theiler staging criteria[69,243]. Theiler stage 19 embryos (E11.5) were rinsed with phosphate-buffered

saline (PBS) and dissociated into single-cell suspensions using AccuMax (StemPro) at room temperature with gentle rocking and trituration using a P1000 filtered micropipette. Single-cell suspensions were filtered through a pre-wetted, 40- μ m nylon mesh cell strainer (Corning) and resuspended in 0.4% bovine serum albumin in PBS. Cell concentrations and viability were assessed using 0.4% Trypan Blue dye to label dead cells on a Countess II automated cell counter (Life Technologies), and single-cell suspensions were confirmed by visual inspection of cells by microscopy and by examination of Countess images for clumps of cells and debris. The sample characteristics are summarized in Table S1.

Droplet-based sequencing using 10x Genomics Single-cell 3' Assay and Chromium controller

Samples were loaded into individual channels of a 10x Genomics Chromium Single-cell 3' Assay microfluidic chip and processed according to the manufacturer's protocol. After loading cells and lysis/reverse-transcription reagents, using the Chromium controller, emulsion quality was assessed by light microscopy. Reverse-transcription and the breaking of emulsions were performed according to the manufacturer's protocol. Samples were indexed, and libraries were prepared for sequencing using the Nextera XT kit (Illumina), according to the manufacturer's protocol. Reverse-transcription and subsequent library quality were assessed using a 2100 BioAnalyzer and high-sensitivity DNA assay (Agilent Technologies). Sequencing was performed on an Illumina NextSeq 500 (high-output mode, 75 cycle reagent kit, read 1:56 - transcript, Index 1:4 – cell barcode sequence, Index 2:8 – sample index, Read 2:10 – unique molecular identifier).

Sequencing data were demultiplexed and processed using CellRanger software (version 1.0, 10x Genomics), using the recommended settings to generate a raw UMI expression matrix.

Model-based quality control and normalization

In addition to the primary quality control mechanism provided by 10X Genomics CellRanger, using a fixed UMI cutoff, we analyzed histograms of the numbers of UMIs detected per cell from each sample. From all models tested, we selected the 3-normal mixtures model to exclude cells with low library quality. We believe that the lower normal distribution includes potential bad libraries, as indicated by low UMIs, whereas cells in the upper normal distribution include potential doublets from sample preparation and, thus, skew subsequent analyses. We removed all cells below the lower 95% tail of the lower distribution and all cells above the upper 95% tail of the upper distribution. Therefore, only single-cell libraries of similar quality were used in our downstream analyses. We removed all non-expressed genes and pooled all samples by concatenating data tables from individual samples, which finally resulted in a UMI matrix with a dimension of 26,305 cells by 20,347 genes. To normalize UMI counts, we used median normalization by scaling the sum of UMIs detected in each cell toward the grand median number of UMIs detected for all cells, followed by a log transformation of $\log(1+UMI)$.

Dimension reduction, clustering, and sub-clustering

PCA was applied to extract the first 100 PCs using the basic R function 'prcomp.' To visualize the data, we further used the R package 'Rtsne'[244] to perform t-SNE analysis

on the first 100 PCs, extracting 3 t-SNEs. To cluster the cells, we used SLHC analysis to extract 512 clusters from the t-SNE space. We removed all clusters with less than 50 cells (2,483 cells in total were removed) and finally yielded 54 clusters. The cutoff threshold was chosen based on the smallest meaningful cluster from preliminary GSEA results with all cells included. Of all 54 clusters, 19 were labeled as isolated clusters, based on their relatively large spatial distance from neighboring clusters and the unique features identified by the preliminary GSEA results. The remaining 35 clusters were divided into three categories and were pooled into three clouds (named the mesoderm/mesenchyme cloud, neuroectoderm cloud, and blood cloud), based on the expression of ectodermal, mesodermal, and blood lineage markers, as well as the primary GSEA results. The sub-clustering analysis was applied to the mesoderm/mesenchyme cloud, neuroectoderm cloud, and the C9 isolated cluster. The UMI expression matrix of targeted cells was subsetted from the raw UMI matrix. With reduced data sizes and heterogeneity, the R package 'Seurat' was used to process the subset using standard log normalization with recommended settings. To better illustrate the relationship between sub-clusters, the subset was reduced to 30 PCs, and a 2D UMAP was applied instead of 3D t-SNE. The sub-clustering was performed using the 'FindClusters' function from the Seurat package, with manually optimized resolution parameters for each subset.

Differential Gene Expression (DGE) analysis and Gene Set Enrichment Assay (GSEA)

We used the R package 'Monocle' to extract DEGs from the initial clustering[13]. Each cluster was compared to the whole population, and genes with q-value >0.05 and fold

change <2 were discarded. The top 400 genes that passed the threshold were used in GSEA. If less than 400 genes were extracted, then all extracted genes were used.

We used GeneAnalytics (<http://geneanalytics.genecards.org>) for GSEA. For each cluster, scores of related matches for LM:Cell Types, GO:BP, and LM:Phenotypes were extracted.

To correct for over or under-represented terms in the GO database for GSEA, we ran 1,000 permutations of random gene lists through GeneAnalytics to generate “background” datasets for GO:BP, LM:Phenotype, and LM:Cell Types. For GO:BP, the scores for each term in the background dataset were averaged and subtracted from the corresponding enrichment score in the foreground dataset for each cluster to generate a “background-subtracted” score for that term. Because all entities in the LM:Cell Types categories were represented in the background dataset, we were able to calculate a z-score for each significant match by first subtracting the average background dataset score for a given entity from its score in the foreground dataset (as for GO:BP) and then dividing by the standard deviation of the background dataset scores.

Word clouds were generated from the matched terms for each category using JMP with entities sized proportionally to their scores. To obtain a meaningful interpretation, certain stop words with minimal biological information were dropped from the word cloud, including the following: 'use', 'feel', 'like', 'inner', 'cell', 'cells', 'mass', 'regulation', 'positive', 'negative', 'response', 'activation', 'process', 'development', 'abnormal', 'morphology', 'number', 'absent', 'decreased', 'increased', 'level', 'short', and 'long'. No stop word was applied to the phrase cloud.

Assignment of functional identity to cell clusters

Cell type annotations for all clusters were made based on an integrated approach. First, we referred to the origin of the cells based on the embryo dissection information, narrowing down the possible cell types for each cluster. Second, we analyzed high score matches from the GSEA analysis for different categories, including Cell Types, GO:BP, and Phenotypes, from GeneAnalytics. Third, we used marker genes and gene lists from multiple databases, including the LifeMap database, R&D database, and manual literature searches, and analyzed their expression patterns in t-SNE space. Finally, we confirmed our annotation by mapping the expression profile to a lineage map derived from the LifeMap database. We simplified the lineage map by removing related anatomical compartments and tissue/organ information, which finally resulted in 672 cell types. CytoScape was used to generate a hierarchical tree view of a trimmed lineage map. Three categories of genes were documented in the database and contributed to the scoring of our cell clusters, including 'selective marker,' 'specific gene,' and 'expressed gene,' giving each match a score of 3, 2, and 1. To map the cluster onto the lineage tree, the UMI expression matrix was binarized as either 0 and 1, and each cell was scored by summing the matching score for each cell type on the lineage tree, which resulted in a cell by cell type score table. The scores were then normalized into z-scores and averaged within each cell cluster and projected onto the lineage map, with cell type nodes colored by matching scores. A colored projection was made for each cell cluster, and the cell types with high scores could be visualized by color and used as references for cell-type annotation.

Sub-clustering of clouds and cell-type annotation

We sub-clustered clouds using different clustering algorithms, depending on the structure. For the hematopoietic cell cluster C9, we used Ward Hierarchical clustering to obtain 8 clusters. For the mesenchyme/mesoderm cloud and neuroectoderm cloud, we subsetting the cells and clustered using the graph-based clustering approach from R package Seurat. With clustering resolution manually adjusted, we finally obtained 14 sub-clusters from the mesenchyme/mesoderm cloud and 9 sub-clusters from the neuroectoderm cloud.

Chapter 2: Mouse cell type dynamics at early organogenesis from scRNA-seq data

Introduction

Since the cell was discovered as the basic unit of life, scientists have been trying to characterize cell types based on different properties. From the view of developmental biology, building a cell lineage tree remains a long-standing goal of understanding how a fertilized egg develops into a mature adult. Decades ago, the fate of all cells from *Caenorhabditis elegans* was determined by time-lapse microscopy[245]. However, certain limitations restrict the study of more complex organisms. Various approaches have been developed to overcome these limitations, such as dye injection, cell transplantation, and the use of fluorescent proteins, which all have their pros and cons[246].

During the fast development of the mouse embryo, the most dramatic changes take place during gastrulation and organogenesis. Once the germ layers are established in gastrulation, a series of events occur in only a few days to expand the embryo: early heart starts to develop at E6.5 and heart tube form at E8.5, neural plate forms at E7.5, and neuropore closure take place at E9. Limb buds begin to form at E9.5, lens indentation and progression initiate at E10.5[247]. Most of these events have been extensively studied but still lack a global view of how cell types progress during early organogenesis. With the development of next-generation sequencing (NGS) technology, especially single-cell gene expression analysis, we are able to study individual cells at the transcriptome level. Single-cell technology gives us a robust approach to understanding developmental processes globally, with higher resolutions and a larger scale.

One challenge to face is that transcriptional profiling only provides a snapshot from the development timeline. Attempts have been made to gather data from multiple time points, thus inferring the dynamic of lineage progression, followed by various computational biology approaches to create a lineage map by linking these stages[66,248–250]. However, even with the sophisticated tools developed by statisticians and computer scientists, it is still challenging to integrate different experiments and investigate the real heterogeneity. In 2018, a group of scientists from Harvard University published a series of three papers in *Science*, addressing this challenge in zebrafish and frogs using Drop-seq[248–250]. The idea of reconstructing developmental trajectories was raised in these publications, and an algorithm named "URD" was developed. Based on diffusion maps, URD orders cells in pseudotime to discover underlying branching trees and gives visualization. Specifically, a k-nearest-neighbor (KNN) graph is generated between transcriptomes, and transition probabilities are assigned to edges and then used as weights in a random walk along each edge. Using user-defined starting points and end points, the cells can be assigned a pseudotime reflecting developmental progress. The author successfully reconstructed a developmental tree, which shed light on zebrafish embryogenesis and related gene cascades. Using a similar KNN based approach, the same developmental tree was constructed in *Xenopus*. However, these approaches do not resolve the problem of integrating different datasets. Instead, a walk-around was made by treating each dataset as its own, then stitching them together based on transcriptome similarity. A similar study was published a year later using the mouse as the model, with a new version of the R package "Monocle3"[66]. Slightly different from URD, Monocle3 used top differentially expressed genes (DEGs) to construct the

pseudotime trajectories by first clustering the cells using the Louvain algorithm, then merging adjacent clusters and identifying trajectories using a KNN based approach. A few improvements have been made in this study, including shorter computation time, excluding small branches, and allowing for convergence or loops. Again this study did not use a particular approach to address the integration problem; instead, the authors processed all samples together to minimize the batch effects. In the same year, Satija lab published their update on the R package "Seurat", in which a data integration tool was implanted[27]. The basic idea is to take advantage of a previously published approach in scRNA-seq batch effect correction named mutual nearest neighbors (MNNs)[251], to identify a subset of cells that represent shared biological states across two datasets, then treat these cell pairs as "anchors." The difference in the anchor pairs' expression profiles is considered a batch vector that indicates the batch effect between two datasets. Then by averaging and subtracting such batch vectors computed from multiple anchor pairs, a corrected expression matrix would be made to facilitate integration between two datasets.

As the data analysis of scRNA-seq data continues to develop, various computational methods have been developed to order cells along pseudotemporal axes using different strategies[252]. Notably, many of these approaches were not designed for time-series datasets[253]. By going through published studies employing these trajectory inference tools, we observe huge benefits from time-series datasets over the 'snapshot' dataset from one specific time point inferred by the pseudotemporal trajectories. We believe the most elegant and simple way to understand the lineage progression still relies on the precise integration of time-series scRNA-seq datasets. With attempts from various

pioneers described above, we believe that we can now solve the alignment issue using these recently developed statistical tools, thus determining the temporal biological differences between our samples without being biased by technical variations. In this study, we show the challenges in aligning samples from E9.0 to E11.5 mouse embryos and how they could be solved using certain tools. By applying computational tools and biological interpretations, we revealed the cell type dynamics during mouse early embryogenesis and provided valuable transcriptional data on various differentiating cell types.

Result and discussion

2.1 Integrating multiple scRNA-seq datasets to understand mouse development

2.1.1 Canonical scRNA-seq pipeline has significant limitations on pooled datasets

To study cell lineage progression, it is necessary to obtain datasets from different experiments covering a series of stages. A widely used first step to process these datasets is by pooling or integrating various experiments. However, without further modification, directly combining data tables will cause significant bias in downstream analyses. Depending on the experimental design, this might be acceptable; however, if comparisons of different experiments will be made in reduced dimensional space, such bias should be removed if possible. To demonstrate the problem, we pooled the expression matrices obtained from two 10x scRNA-seq runs on mouse embryos from the same stage with different chemistry versions. Briefly speaking, these are two biological replicates sequenced by slightly different approaches. Then we process the pooled dataset with a standard scRNA-seq pipeline using PCA and t-SNE from the R package

"Seurat" without further modifications. In that case, two sets show completely non-overlapping (**Figure 2.1, A-B**). It is unlikely that these two sets revealed biological differences as they were simply biological replicates. Instead, by plotting the number of genes detected per cell and the number of transcripts captured per cell, it is evident that later version chemistry showed much better sequencing sensitivity with more genes detected (**Figure 2.1C**), which biased the dimension reduction algorithms. The t-SNE projection of the cells would indicate that there are almost no shared cell types between two biological replicates, which is not real.

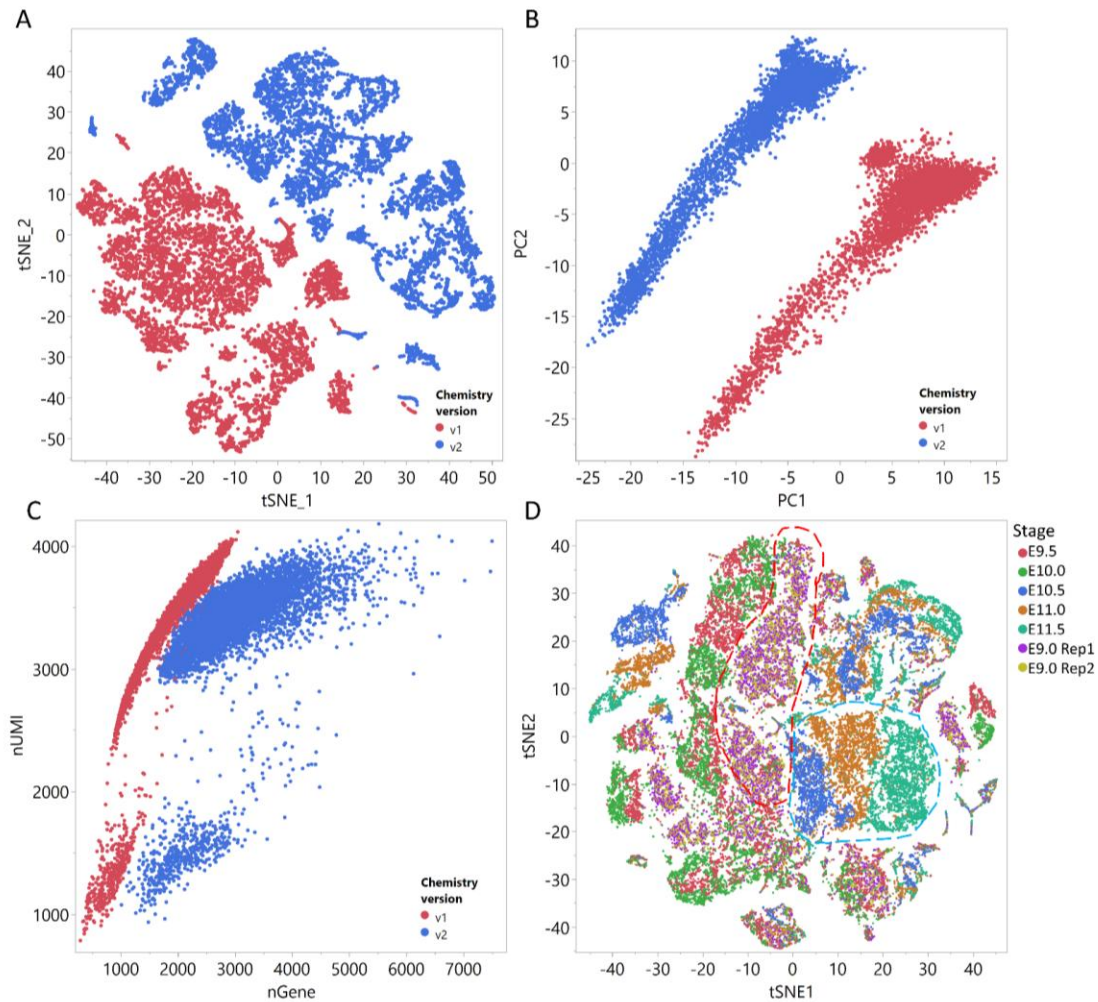


Figure 2.1: Batch effects in canonical scRNA-seq pipeline

Without specific data manipulation, time-series dataset or even biological replicates show huge batch effects in reduced dimensions.

(A) t-SNE projection of two scRNA-seq datasets of E11.5 mouse embryo using different versions of chemistry. **(B)** PC1 and PC2 reveal two distinct populations. **(C)** Two sets of E11.5 scRNAseq show differences in sensitivity. **(D)** Pooling scRNA-seq datasets revealed misalignment from different mouse stages. Red dotted line: Two biological replicates from E9.0 showed excellent alignment. Cyan dotted line: Misalignment among three different stages.

The same problem remains if we use datasets gathered from the same technique on the same day but at different stages of embryos. With similar sequencing sensitivity and sample size, we still observed multiple gaps between some of the stages (**Figure 2.1D**), which contradicts our common understanding of cell-type heterogeneity. The only exception is two biological replicates in the same run, which minimized all possible variations and showed perfect alignment (**Figure 2.1D**, red dotted line). But in other clusters, cells from neighboring stages tend to spread out with almost no overlapping (**Figure 2.1D**, cyan dotted lines), indicating that either the cellular identities were undergoing rapid changes or a misalignment issue. Considering the biological background and estimating the technical variations, it is unlikely that the misalignment revealed true biological differences.

In conclusion, the canonical scRNA-seq pipeline could be biased by both biological aspects and related technical variance when multiple datasets are included. One big challenge of analyzing time-series data is minimizing the technical variations and fairly comparing samples from different sources without batch effects.

2.1.2 The use of Canonical correlation analysis (CCA) improves the alignment of different stages

The algorithms and toolkits for correcting batch effects in scRNA-seq datasets have been developing fast during the past few years. As reviewed by Tran et al., the performance of these tools varies by specific scenarios[254].

To solve the misalignment problem caused by batch effects, we applied Canonical correlation analysis (CCA) from the R package Seurat to normalize and integrate datasets from different runs after testing a variety of tools[255]. With CCA, the biological replicates from two chemistry versions showed good alignment in t-SNE space (**Figure 2.2A**). Further differential gene expression (DGE), gene set enrichment assay (GSEA), and marker gene projections confirmed that the overlapping cells from different sets were actually the same cell type. Other than the regular log-normalization applied in many studies, a more sophisticated scRNA-seq tailored normalization approach designed for UMI-based scRNA-seq datasets was proposed, known as 'SCTransform'[255]. With only a marginal improvement in reduced dimensional space after testing this SCTransform normalization in our pipeline (**Figure 2.2B**), we believe the standard normalization would be enough in this study. Still, the SCTransform does provide a better theoretical foundation and would be beneficial in other studies.

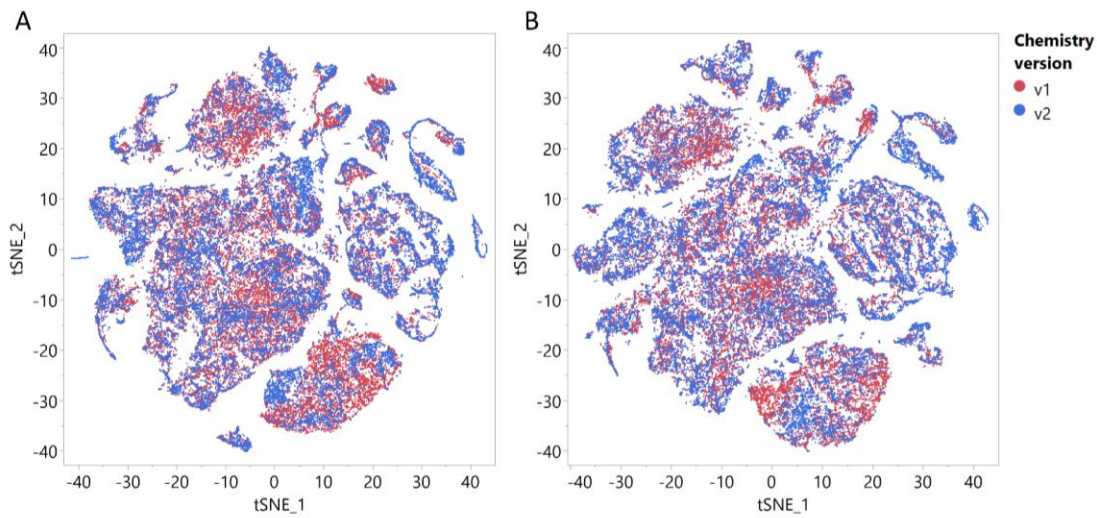


Figure 2.2: CCA significantly improve sample alignment by removing batch effects

Both standard and SCT normalized datasets from two biological replicates show significant improvement in sample alignment.

(A) CCA with standard normalization shows great alignment between two E11.5 datasets.
 (B) CCA with SCT transform normalization shows marginal improvement compared to the standard pipeline.

Another design we made during the library preparation was the dissection of the embryos. We have dissected embryos into three parts: head, heart, and trunk. With an additional undissected whole embryo control, we had four samples from each stage. This design would not only further help us validate the performance of data processing steps, as well as annotate the cellular identities in downstream analyses, but also confirm the robustness of the sample alignment. As shown in **Figure 2.3**, the head, heart, and trunk from two sets of E11.5 embryos showed perfect alignment across replicates, but there was little overlapping between different samples, as expected. The whole embryo control

had a nice coverage of most regions, confirming that the CCA alignment on SCTransformed data successfully minimized the batch effect and kept the biological differences among samples.

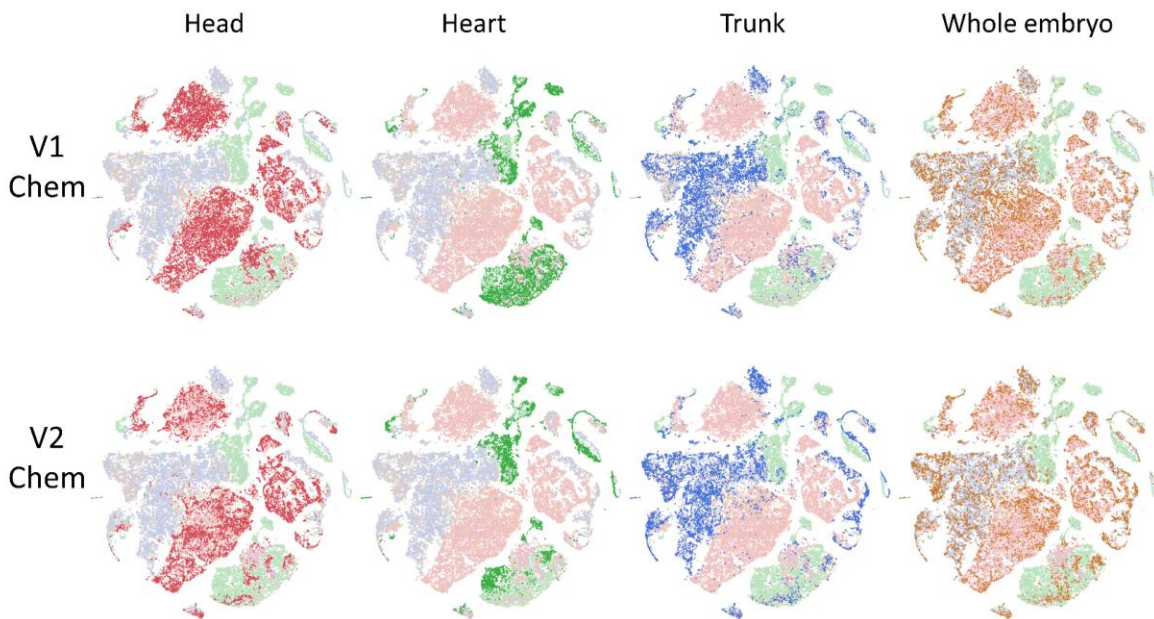


Figure 2.3: Different sample parts of two E11.5 datasets also show good concordance

The three dissected parts (head, heart, and trunk) and whole embryo control from two E11.5 datasets all show good spatial alignment in t-SNE space by CCA (compare top row and bottom row), the whole embryo sample shows a high percentage of overall coverage (last column), but the three different samples show almost no overlap (first three columns).

The datasets from multiple stages also showed better alignment using the CCA approach (**Figure 2.4A**). In a run with six developmental stages ranging from E9.0 to E11.5, we could see most of the clusters shared by all stages. It is worth noting that there was one

cluster from E10.5 that did not align with any other stage (**Figure 2.4A**, circled orange cluster). By analyzing the E10.5 sample independently, we figured out that this misaligned cluster showed an unusual gaussian-like structure in its own t-SNE space. Also, the DGE analysis could not identify meaningful genes, which indicated bad libraries (**Figure 2.4B**). This was confirmed by the unique low-end distribution of this cluster in the number of UMIs per cell (**Figure 2.4C**), indicating that these cells showed a deficient number of transcripts and should be dropped before the analyses. Other than that, we have plotted the nGene-nUMI plot for all libraries as we did in **Figure 2.1C**, and we found that the bad libraries form a clear separate cluster that could be easily differentiated from the good ones, which could be used as a potential approach for better quality control rather than using a direct UMI cutoff (data not shown). In summary, the cluster of bad libraries in the integrated projection was not aligned with any other cells, while most of the other regions showed nice coverage across stages, further confirming the robustness of CCA in integrating multiple datasets. To minimize the damage from bad libraries, all cells from this stage (E10.5) were dropped from downstream analyses.

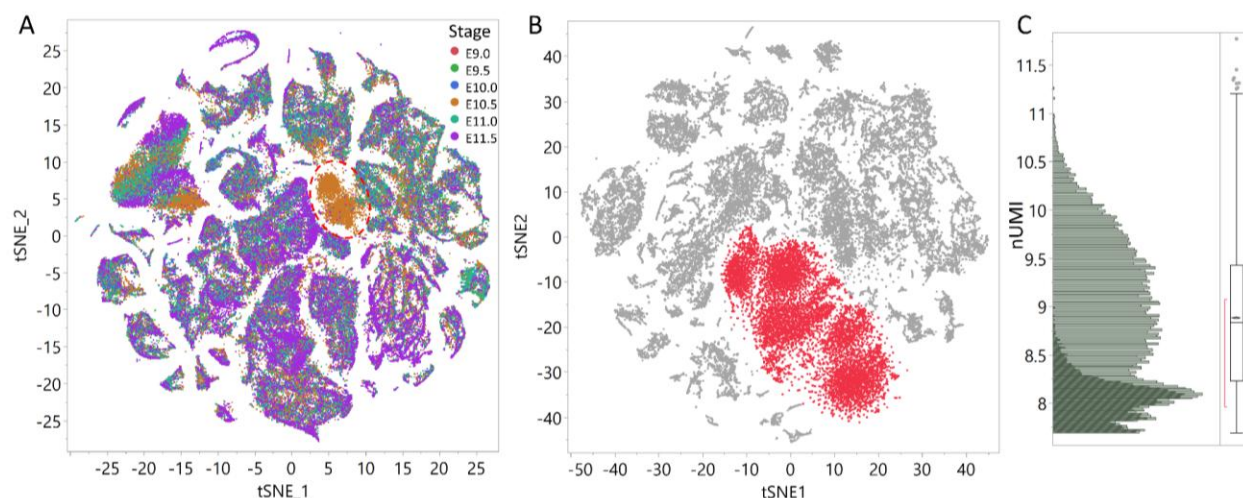


Figure 2.4: Misaligned population from CCA identifies low-quality libraries

One cluster failed to align with others in CCA based time-series dataset was validated as low-quality libraries, confirming the robustness of CCA alignment.

(A) t-SNE projection of samples from E9.0-E11.5 integrated. One population from TS17 shows misalignment. **(B)** t-SNE projection of E10.5 samples only. The misaligned population is highlighted in red, showing a gaussian-like distribution. **(C)** Histogram of nUMIs per cell, the misaligned population highlighted at the low end.

So far, we have presented all the clusters in the t-SNE space. However, there is a potentially better option for visualization known as the UMAP[42]. Compared to the t-SNE, it emphasizes more on global structures, which suits the overall purpose of investigating lineage progression over time. As shown in **Figure 2.5**, the UMAP had a much more compact structure while the t-SNE expanded the local structures more. As a result, the t-SNE plot served as an excellent way to test sample alignment in the preliminary test, while UMAP might benefit us more when moving on with a settled-down pipeline. Note

that both UMAP and t-SNE projections showed very nice stage alignment, as the integration step happens many steps before the secondary dimension reduction.

Like we did in the E11.5 dataset in Chapter 2 and mentioned previously, we dissected embryos into the head, trunk, and heart regions, plus the undissected whole embryo control. The stage and sample distributions in UMAP space were also tested. Other than the same expected stage coverage as in t-SNE, cells from the whole embryo sample had excellent coverage of most regions while other samples occupied different regions with little overlapping, indicating a successful integration of multiple datasets (**Figure S2.1**).

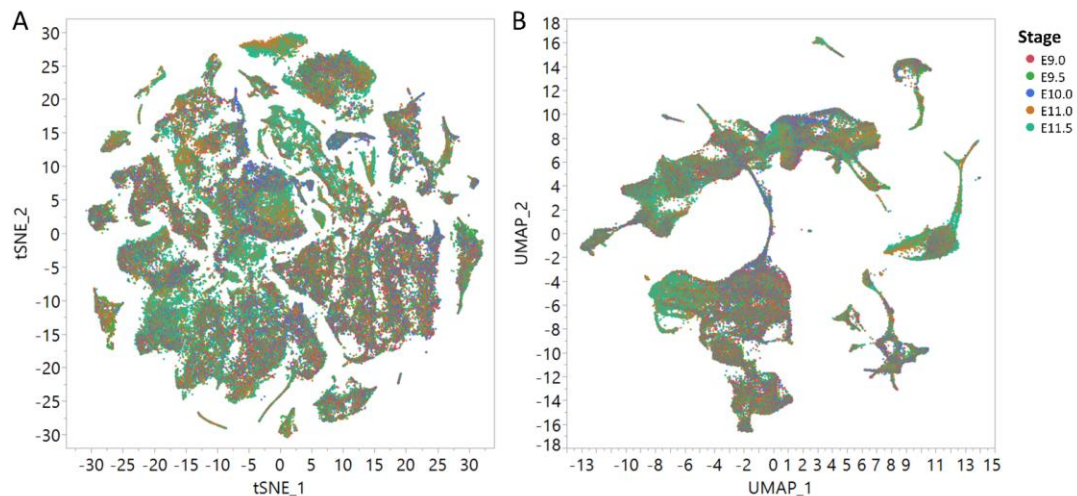


Figure 2.5: Projection of cells from E9.0-E11.5 in low dimensional space, with E10.5 dropped

Projection using UMAP shows a more compacted structure compared to tSNE, which is better suited for viewing global structures in time-series data.

(A) tSNE projection of E9.0-E11.5 datasets integrated by CCA. **(B)** UMAP projection of E9.0-E11.5 datasets integrated by CCA.

2.1.3 Dataset integration: summary and discussion

In this study, we have extended our analyses from one single dataset to a series of developmentally connected datasets, using the established pipeline described in Chapter 1. The major challenge we are facing is the normalization and batch effect correction, without which even the biological replicates couldn't be aligned well (**Figure 2.1**). Efforts made by statisticians and computational biologists helped to make several normalization tools available during the past few years. Using CCA, we were able to successfully remove the batch effects and get integrated low dimensional representations of combined datasets (**Figure 2.2-2.3**). During the process, we have also validated the robustness of such algorithms by accidentally including a group of low-quality libraries (**Figure 2.4**), which turned out to be the left-alone cluster after alignment. This also tells us that the primary quality control from the early pipelines might mistakenly include bad libraries by using a direct UMI cutoff. Instead, we took advantage of the 2D plot using nUMI and nGenes from each cell, from which the bad and good libraries could be clearly separated as they form different clusters (see Material and Methods).

After validating the normalization and batch correction approaches, we integrated datasets from E9.0 to E11.5 into one common low dimensional space using UMAP which better shows the global structural relationship among different cells (**Figure 2.5**). Next, we will demonstrate how such an integrated dataset could be used to identify cell types and understand cell type dynamics.

2.2 A road map for cell type dynamics in mouse embryo from E9.0 to E11.5

2.2.1 The use of integrated approach to performing primary cell identity annotation from E9.0- E11.5 dataset

With cells from different stages integrated into the same space, we were able to identify cellular identities and how they dynamically changed over time.

To obtain more precise annotations, we developed an integrated approach using multiple databases (described in Chapter 1). Briefly speaking, first, we clustered the cells from the integrated dataset in PC space using the graphics-based clustering algorithm from the R package "Seurat". By adjusting the resolution, we ended up with 46 cell clusters. Then we applied differential gene expression (DGE) analysis on each cluster using an algorithm based on the Wilcoxon Rank Sum test from the Seurat package[255]. Next, we referred to multiple databases offered by LifeMap®, including cell type, tissue, phenotype, and Gene Ontology (GO) terms. To be more specific, using the online tool GeneAnalytics (<https://geneanalytics.genecards.org>), we input the differentially expressed genes (DEGs) from a specific cluster, which then can be analyzed using their incorporated databases. We use "Tissue & Cell," "GO Terms," and "MGI Phenotypes" for manual curation and visualization from the output tables. All the results were represented as a matching score based on negative log₂ transformation of the resulting p-value, where higher scores indicate better matches. In parallel, we also compared DGE results with the LifeMap lineage tree (<https://discovery.lifemapsc.com/in-vivo-development/cellular>), a database with over 1000 cell types annotated with known markers and linked by developmental

lineage. We scored each cell type by matching the differentially expressed genes with their markers and visualizing them in a hierarchical tree to help our manual curation. We also referred to previously published single-cell papers[66,67], analyzed all the markers they used to determine cell types, and ranked their expression in our clusters. To visualize the results for better curation, we generated word clouds and phrase clouds from GenAnalytics matching scores (refer to **Figure 1.3E** for example). We also projected the expression levels of the differentially expressed genes into UMAP space to confirm the expression pattern and compared it with specific marker genes to determine cellular identities.

Putting all evidence together, we had a preliminary annotation of all 46 clusters identified in this integrated dataset (**Figure 2.6A**). By looking at the structure of the UMAP projection, we noticed that the majority of the cells were in two big connected clouds, of which the top cloud represents mainly cells from the mesoderm and the bottom from the neuroectoderm. A few isolated islands had unique expression patterns, thus easy to annotate, including blood lineages, epithelial lineages, endothelial lineages, and differentiated neurons (**Figure 2.6B**). To investigate the clusters with similar cell type identity, we pooled similar clusters together and then performed subset analysis on pooled clusters, including ten neuroectoderm clusters (**Figure 2.6B**, cyan), 15 mesoderm clusters (**Figure 2.6B**, orange), four neural crest clusters (**Figure 2.6B**, red), and six epithelial clusters (**Figure 2.6B**, blue). Specifically, we subset the target clusters from the whole dataset to reduce complexity and reran the pipeline with an adjusted number of PC dimensions. In addition to all analyses we had done for the entire dataset, we further

looked at loadings of the first few PCs and combined that with DGE analysis to find genes that differentiate sub-clusters within each category.

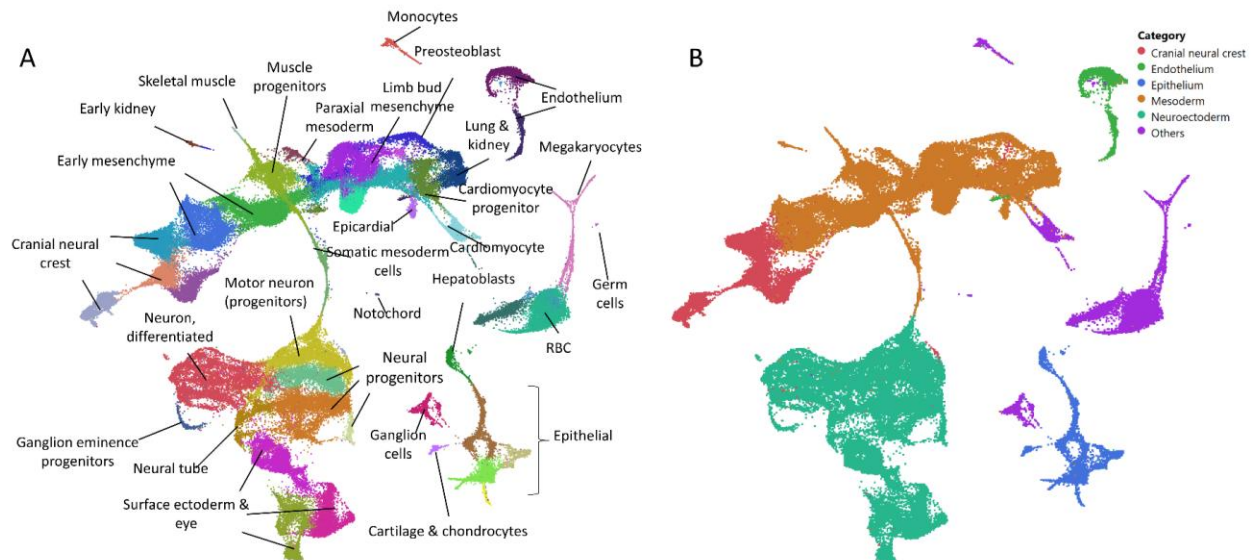


Figure 2.6: Preliminary cellular identities revealed by the integrated annotation approach on the integrated dataset

The first round annotation of the integrated time-series dataset shows five big categories of distinct cell types which are mostly spatially connected.

(A) UMAP projection of the integrated dataset. Labels represent preliminary annotation of 46 clusters identified from the integrated E9-E11.5 dataset. **(B)** Clusters pooled into six major categories for further analyses.

2.2.2 Sub-clustering in neuroectoderm showed the progression of neuronal development

For the ten neuroectoderm clusters captured in our preliminary clustering, we further investigated the differences among these cells with similar identities by analyzing these

clusters independently. Subsetting these clusters revealed more significant details, while the general topology remained unchanged (**Figure 2.7A-B**). Besides all the matrices we used at the global level, we referred to the Abcam neuron marker guide[256] and the Mouse Brain Atlas database[257]. By clustering the subset and applying the annotation, we started with 23 sub-clusters and ended with 11 meaningful sub-clusters after pooling clusters with similar identities, plus two isolated groups of cells with a low number of DEGs and low GSEA p-values which we consider as noise (**Figure 2.7C**).

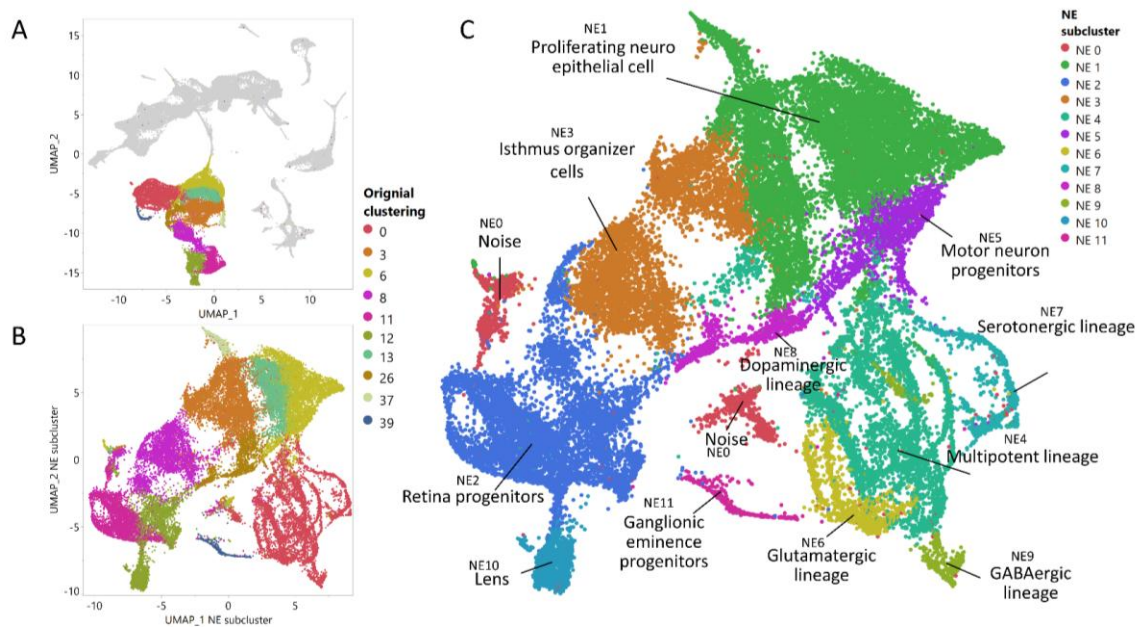


Figure 2.7: Sub-clustering analysis of the neuroectoderm shows the progression of neural development.

Sub-clustering of the neuroectoderm shows a comparable topology, but with higher resolution that helps identify more specific neurons undergoing proliferation and differentiation.

(A) Clusters annotated as neuroectoderm highlighted in global UMAP. **(B)** UMAP projection of neuroectoderm cells after subsetting and reran the pipeline. Colored by

original clustering. **(C)** Subclustering of the neuroectoderm and annotation of cellular identities.

By looking at the distribution of cells from different stages, we observed a bias in which earlier stages occupy the top left of the plot while later stages enrich in the lower right corner (**Figure S2.2A-F**). Neurogenesis is known to start around E10 and peaks at E14[167]; thus, many adult neuroectodermal derived cell types were not expected from our datasets, such as various mature neurons, oligodendrocytes, and precursors which emerge around E12.5[258]. On the other hand, we noticed that although the head samples covered the majority of the population, there's one region (NE1, right half) with almost no cells from the head but the trunk instead (**Figure S2.2G, I**). Interestingly, both the right half of NE1 from trunk cells and the left half from head cells showed strong signals pointing to proliferating neuroepithelial cells, confirmed by some early neural epithelial markers such as *Hes5*[259] (**Figure S2.3**). We have also observed almost no cells from the heart samples and nice coverage of cells from the whole embryo samples (**Figure S2.2H, J**), confirming that the independent processing of the subset did not encounter alignment issues. In addition, NE1 had significantly more cells from earlier stages (**Figure S2.4**), and the GSEA results showed a strong match with cell cycle-related GO:BP terms, as well as the high expression level of multiple early neural epithelial cell markers as *Hes5* and *Notch1*[167,260]. Taken together, this might indicate multiple origins (both head and trunk) of neuroepithelial cells with slightly different expression profiles existed at early organogenesis.

Cluster NE4-9 were cells mostly from later stages, and all showed significantly stronger signals for GO:BP terms related to nerve system differentiation. These cells were potentially postmitotic premature neurons, and we wondered if the finger-like structures (NE7) in these clusters represent different lineages for neuron differentiation. By mapping to known markers for different neuron lineages, we did observe some differentially expressed markers from different lineages, such as *Olig2* at NE5 marks motor neuron progenitor[261], *Fev* at NE7 marks serotonergic lineage[262], *Slc32a1* at NE9 marks GABAergic lineage[263], *Foxa2* at NE8 marks dopaminergic lineage[185,264], *Slc17a6* at NE6 marks glutamatergic lineage[265] (**Figure S2.3**), while the majority part of these clusters shows a mixed expression pattern as we annotated as multipotent lineage (NE4). Using the same approach, we have also identified a cluster of isthmus organizer cells (NE3), a cluster of retina progenitor cells (NE2), a cluster of lens cells (NE10), and a cluster of ganglionic eminence progenitor cells (NE11).

Looking at these cell clusters from a developmental perspective, it is evident that the majority of postmitotic premature neurons come from later stages (NE4, NE6, NE7, NE9) (**Figure S2.3, Figure S2.4**). Note that as an isolated cluster, NE11 had almost no cells from stages before E11.0, and rises sharply afterward. The signal of NE11 is complicated with a mixture of oligodendrocyte progenitor cells and ganglionic eminence cells with high-level expression of *Gad1* and *Gad2*. *Gad1* and *Gad2* are members of the glutamic acid decarboxylase family and are required for the synthesizing of neurotransmitter gamma-aminobutyric acid (GABA), which is utilized by GABAergic neurons originating from the ganglionic eminence regions. The highly enriched expression of *Gad1* and *Gad2*

in the ganglionic eminence region has been published in both canonical RNA *in-situ* and scRNA-seq studies[266,267]. The expression pattern we observed aligned well with published data on ganglionic eminence regions emerging at E11.5, and differentiation of ganglionic eminence basal ganglia neurons were observed at early E12.5[268,269]. Also, it has been addressed that the oligodendrocyte in the brain region was initiated in the medial ganglionic eminence[270]. This indicates NE11 is a newly emerged cell cluster from roughly E11.0, probably in the ganglionic eminence region. Though less obvious, we have also observed a drop in numbers of proliferating neuroepithelial cells at later stages (**Figure S2.4**), which might indicate a transition event in neurogenesis.

In summary, the sub-clustering of the neuroectoderm showed a comparable topology with the cloud from the global plot, but with a much higher resolution of the relationships between different sub-clusters. Similar to what we discovered in Chapter 1, we identified cell types that are undergoing proliferation and differentiation, and expanded that to a time-dependent manner.

2.2.3 Sub-clustering of mesenchyme/mesoderm identified embryonic mesenchymal cell types

Using the same approach applied to neuroectoderm clusters, we sub-clustered 15 clusters showing mesenchymal signatures in our primary clustering (**Figure 2.8A-B**). For cell type identification, given that most of the existing databases have very limited information on developmental stages of cell type markers, we referred to databases with pictures of various expression assays with developmental stage information, of which the

major sources being the EMGA database[271] and the Gene Expression Database (GXD) from Mouse Genome Informatics (MGI)[272]. By manually comparing the top-ranked DE genes with their expression patterns, combined together with published literature, we were able to accurately annotate 24 sub-clusters. After pooling clusters with similar identities, we finally get 13 clusters that cover most of the known mesenchyme cell types, despite a cluster with mixed-signal and scattered distribution considered as noise therefore discarded (ME0) (**Figure 2.8C**). One thing to note is that even though we only show expression plots for specific genes below, the annotation has been done based on a list of genes differentially expressed in each cluster we identified. At early organogenesis, we expect to see all three major divisions of mesoderm: paraxial mesoderm, intermediate mesoderm, and lateral plate mesoderm, which we have identified all of the three in our sub-clustering analysis.

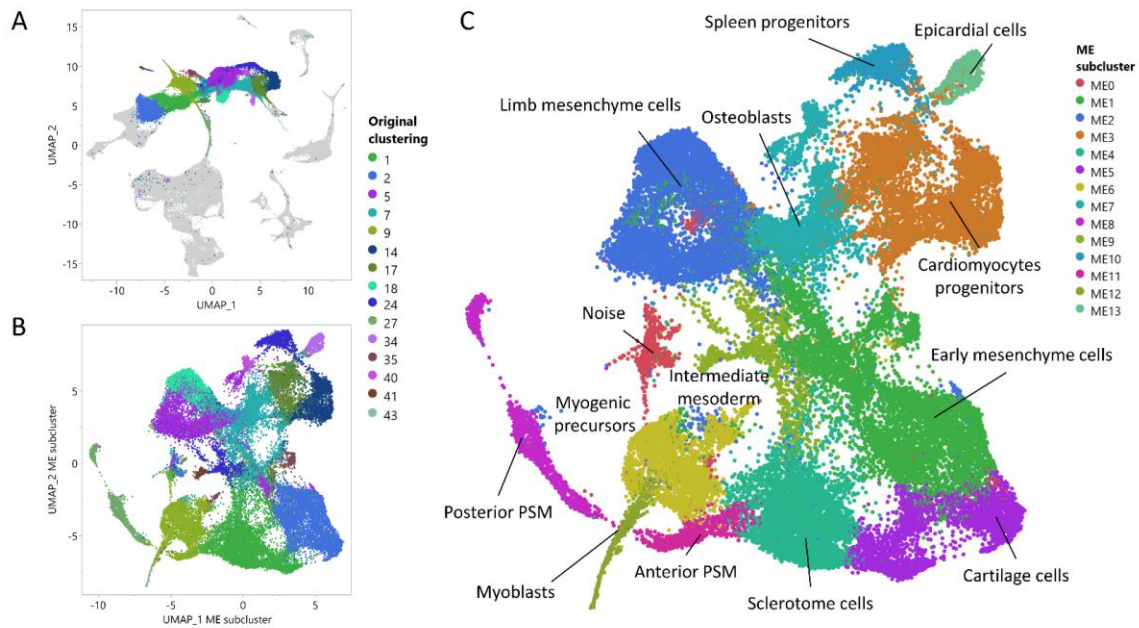


Figure 2.8: Sub-clustering analysis of the mesenchyme shows the progression of mesoderm development.

Sub-clustering of the mesenchyme shows a comparable topology, but with higher resolution that helps identify more specific cell types from multiple regions originated from mesoderm with clear differentiation axis.

(A) Clusters annotated as neuroectoderm highlighted in global UMAP. **(B)** UMAP projection of neuroectoderm cells after subsetting and reran the pipeline. Colored by original clustering. **(C)** Subclustering of the neuroectoderm and annotation of cellular identities.

Intermediate mesoderm are at a very early stage of development at the stages in this study, however, we did see cluster ME9 showing high levels of *Lhx1*, *Pax2*, and *Pax8* compared to other clusters, which are markers for mouse kidney development (**Figure S2.7**)[273–277]. Interestingly, it was the only cluster that was isolated from the

mesenchyme cloud back in the global cell plot before sub-clustering, which might indicate the kidney (to be more specific, nephric duct at these stages) being the most transcriptionally unique organ besides the heart.

The majority of the clusters came from paraxial mesoderm, of which we can see very unique elongated structures in clusters ME8, ME11, and ME12. We believe the elongation of cluster ME11 and ME8 revealed an anterior-posterior axis formation in presomitic mesoderm (PSM) progenitors, given that the anterior PSM markers such as *Add3* and *Fbn2* enriched in ME11, while posterior PSM marker *Chst7* enriched in ME8[121]. Another set of PSM markers *Fgf8*, *Fgf17*, and *Fgf18* also confirmed the same expression trend (**Figure S2.7**)[278]. The other elongated cluster ME12 was annotated as myoblasts with significantly enriched myoblast signals, and markers including *Myf5*, *Myog*, and *Myod1* were enriched (**Figure S2.7**)[125,126,279]. Note the cluster ME6 which connects ME12, shows very high levels of myogenic precursor cell markers such as *Pax3* (**Figure S2.7**)[121]. These two clusters together show a very typical pattern of lineage progression event in development reflected in our cell plot, from a less differentiated precursor to a more differentiated cell type. Besides myotome cells, we also identified sclerotome cells at ME4 with its marker *Pax1* and *Meox1* highly expressed (**Figure S2.7**)[280,281], cartilage cells at ME5 with markers including *Msx2*, *Epha7*, and many collagen genes like *Col11a1* expressed (**Figure S2.7**)[282,283], osteoblasts at ME7 with markers like *Tnc*, *Postn*, and other related collagen genes expressed[284–286].

For lateral plate mesoderm, since the heart is the most developed organ at these stages, we did not include cardiomyocytes in this mesenchyme sub-clustering, knowing that it

was indeed connected to the mesenchyme cloud and showed a very unique expression pattern. However, we identified a cluster with weaker mature cardiomyocytes but stronger cardiac progenitor signals (ME3). We also identified a cluster with specific signals from epicardium (ME13), with both canonical marker *Tbx18* and novel marker *Upk1b/Upk3b* highly expressed[287]. Though it might not be statistically significant, both ME3 and ME13 had a higher coverage from later stages (**Figure S2.6**), which might indicate that a collection of diverse cell types have already been differentiated in the heart (and also rapidly proliferating) while the majority of mesoderm-derived cells are still at their early differentiation stages. We also noticed a high expression of early spleen marker *Nkx2.3* in ME10, aligned with earlier reports that spleen progenitor cells start to condense within the dorsal mesogastrium at approximately E10.5–11[288]. Last but not least, as limb bud emerges around E9, we expect to see limb mesenchyme cells from all stages which we did (ME2). Other than many limb bud markers being expressed, we specifically see a nested expression pattern of some Hox genes (data not shown), which reveals the spatial information on our cell plot. Another lateral plate mesoderm cell lineage is blood, which is much more differentiated, thus, captured during our primary clustering as isolated clusters.

As a conclusion, we were able to identify all expected mesoderm cell types, and the projection of these cell clusters shows a reasonable spatial relationship based on current knowledge of mesoderm development, the differentiation event of several cell types has been clearly indicated.

2.2.4 Sub-clustering of cranial neural crest cells identified cell type dynamics during craniofacial development

Besides the neuroectoderm subset, there was another large region showing enriched head sample derived cells (**Figure S2.1G**, top left), in which we observed strong signals pointing to neural crest (NC) related cell types. The NC is a multipotent cell population that is induced in the dorsal neural tube and then migrates into different parts of the embryo, eventually giving rise to a variety of cell types, including neurons/glia cells, cartilage, bone, smooth muscle, melanocytes, and many more[163,289]. In the cranial region, NC cells contribute to many mesenchymal cell types such as bone, cartilage, and other connective tissues. It has been addressed that the NC cells migrate from the dorsal neural tube to the frontonasal process between E8 and E9.5[290], and by E10.5 NC cells have populated the presumptive cranial base[291]. The migration of NC cells into craniofacial regions maintains spatial segregation, and the anterior-posterior patterning plays a significant role in building the pharyngeal region of the embryo[163,292,293]. Therefore, we hypothesized that we could observe such migration and differentiation events in our study which covered the stages that cranial neural crest (CNC) cells have initiated their migration.

In this study, we identified four clusters with strong GSEA signals of CNC cells as C10, C15, C16, and C19 in our primary clustering (**Figure 2.9A**). These clusters showed similar transcriptional profiles and were closely related to cranial facial features. We subset these clusters and took a further investigation using the similar approach as described above. In this subset, the majority of cells came from the head and whole embryo samples, with

only a small portion from the heart and the trunk samples, as expected (**Figure 2.9B**). Before the subset analysis, cluster C19 showed a unique structure that seems more isolated from the other three clusters, indicating that C19 might be the most unique cell type of the four clusters. In fact, C19 was the only cluster that could be separated after running the dimension reduction independently. In the subset analysis, the original clustering still did a good job of dividing the cells in the new UMAP space, and C19 was an isolated cluster while the other 3 clusters were still connected. As rerunning UMAP and clustering did not provide more useful information, we used the original UMAP coordinate and clustering. Interestingly, the stage distribution did not show significant differences across the clusters except for C10 and C19, which showed a gradient of early-to-late stage shift (**Figure S2.8 D-J**). It's worth noting that half of C19 had almost no head samples (**Figure 2.9B**), which means it was not a CNC cell cluster but another cell type that is closely related, while the other three clusters were mostly from head and trunk samples as described above.

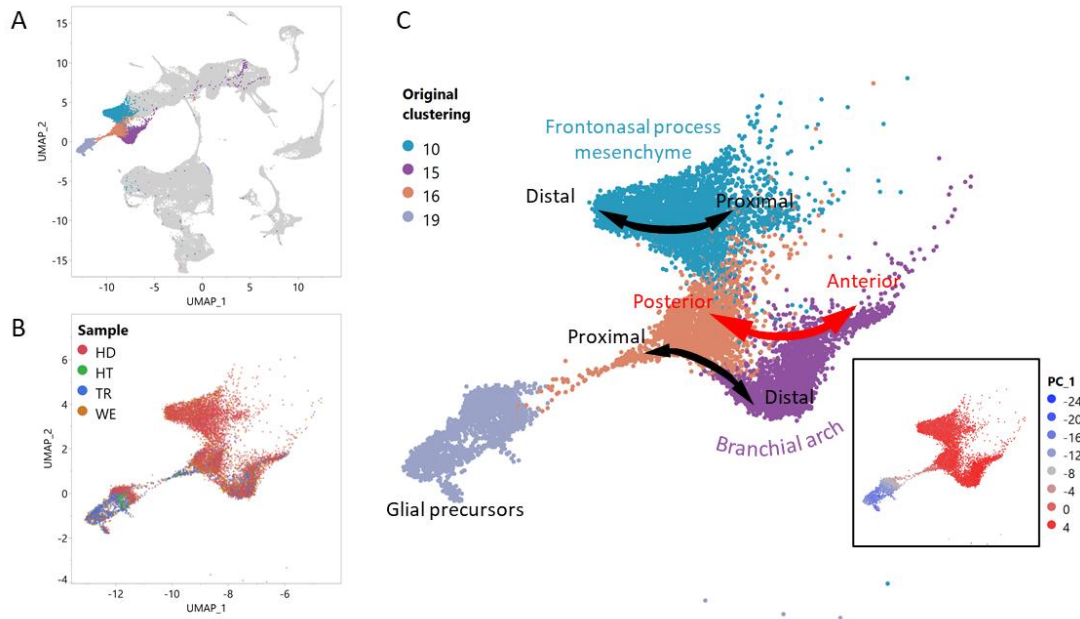


Figure 2.9: Sub-clustering of cranial neural crest (CNC) cells identified cell type dynamics during craniofacial development

Four clusters enriched in head cells show profound craniofacial signatures, subclustering analysis and independent PCA analysis reveal the spatial-related cell type information within this population.

(A) Clusters annotated as CNC highlighted in global UMAP. Colored by original clustering. **(B)** Sample distribution in CNC subset. Colored by sample of origin. **(C)** Annotation of CNC subset. Anterior-posterior axis labeled by a red arrow, proximal-distal axis labeled by black arrows. In the black box: UMAP projection colored by PC1 computed from the subset.

To investigate the nature of this subset, we performed independent PCA analysis and ran GSEA on top loaded genes on the first few PCs. PC1 clearly split C19 from the other three clusters (**Figure 2.9C**, boxed). GSEA results from 100 top-loaded genes of the negative end of PC1 contributing to C19 showed strong matches to neurons in both ‘cell type’ and ‘GO:BP’ terms, while the 100 top-loaded genes of the positive end of PC1 showed more signals for head mesenchyme and bone structures. Based on this we could

hypothesize that C19 might be related to neural crest-derived neurons, while the rest three clusters are responsible for mesenchymal tissues in the head. Additionally, we sub-clustered this subset into eight clusters to find genes responsible for each part at a higher resolution. The strongest signals for cranial neural crest cells come from three of the sub-clusters, which correspond to the distal end of original clusters C15 and C10. That being said, the distal end of C10 and C15 might represent more differentiated cranial neural crest cells while the proximal end of C10, C15, and the cluster C16 in between represents potential precursors that are less differentiated. To further validate the hypothesis above, we investigated the genes differentially expressed in specific regions. C19 has a specific expression pattern of *Sox10*, which is a transcription factor that marks the maturation of glial cells[294,295] (**Figure S2.9**). Gliogenesis is a developmental event that involves many different types of glial cells in both peripheral and central nervous systems (PNS and CNS), of which the Schwann cells from PNS originate from the neural crest cells[296,297]. Interestingly, *Sox10* serves as a marker for both PNS and CNS glial precursors, and we've found other oligodendrocyte precursor markers such as *Plp1* in the same cluster[298], which come from the CNS and do not originate from neural crest cells. That being said, C19 covers glial precursors from both CNS and PNS, though it has been addressed that they have different origins, indicating that at such an early stage of gliogenesis these two populations share similarly transcriptional profiles.

In C15 we identified Dlx family gene members *Dlx1*, *Dlx2*, *Dlx5*, and *Dlx6* to be differentially expressed (**Figure S2.9**). A nested expression pattern has also been observed, with *Dlx5* and *Dlx6* expressed at the more distal end of C15, while *Dlx1*

expressed in more broad regions of C15. Such nested expression pattern of the Dlx family gene has also been reported as 'Dlx code' which revealed the proximal-distal axis formation, showing that *Dlx5/6* are expressed at the distal end of pharyngeal arches while *Dlx1/2* has a broader expression pattern including more proximal regions[163,299]. This shows C15 represents cranial neural crest cells with distal-proximal axis information, with the distal end of C15 being cells at distal pharyngeal arches.

Another set of genes has been reported to mark the anterior-posterior axis in craniofacial development, known as Hox genes[163]. The Hox gene expression was proved to have different expression patterns between NC subpopulations and their rhombomere of origin[163,300–305]. The expression of several Hox genes such as *Hoxa2* and *Hoxb2* in CNC cells have been intensively studied and proved to be involved in the anterior-posterior patterning of pharyngeal arches[306,307]. Similar to Dlx family genes, Hox genes also show a nested expression pattern in pharyngeal arch regions, with paralogue group 2 Hox genes required for pharyngeal arches 2-4, while paralogue group 3 and 4 Hox genes only expressed in more posterior pharyngeal arches (3-4 and 4, respectively)[163]. We projected published Hox genes involved in cranial neural crest Hox code including *Hoxa2/b2/a3/b3/d3/d4* to study their expression pattern in C15 (**Figure S2.9**). The results showed that the anterior Hox genes are broadly expressed in C15, but the posterior Hox genes are much weaker in C15 and relatively strong in the tip of C16. This trend of anterior-posterior Hox gene expression showed an anterior-posterior gradient shifting from C15 to C16. It's also worth noting that the tip of C16 is stage-specific, and lacks the cells from later stages (**Figure 2.9C, S2.8A-E**). Another interesting

observation is the absence of Hox genes at the right tip of C15, which aligned with the observation that pharyngeal arch 1 is devoid of Hox genes[308]. Specifically in pharyngeal arch 1 which is first detectable around E8.5 and continues to group medially[309], the non-Hox homeobox transcription factors such as Dlx family and Hand family genes are differentially expressed[308,310,311]. The projection of *Hand1* and *Hand2* specifically highlighted the right tip of C15, confirming the identity of those cells as the most anterior pharyngeal arch cells (**Figure S2.9**).

The other cluster C10 showed no expression of Hox genes, and very weak expression of Dlx genes. However, we have identified *A/x3* and *A/x4* to be differentially expressed in C10 (**Figure S2.9**). The aristaless-like homeobox gene *A/x3* and *A/x4* play a significant role in CNC differentiation and are expressed specifically in the frontonasal process, typically the distal region[160,309,312]. *A/x4* is reported to be expressed in a more extended proximal region compared to *A/x3* which is stronger in distal[160], though not clearly indicated in our data, might also show a proximal-distal axis in C16 (**Figure 2.9C**).

In conclusion, the expression pattern of multiple transcription factors revealed the anterior-posterior and proximal-distal axis forming along C10, C15, and C16. Here we have confirmed the capability of using scRNA-seq to identify similar cell types even using the entire transcriptome as background. The resolution of the cell type identity is still limited without seeing particular segmentation even using a specific set of genes (data not shown), but the axis formation could be clearly observed. The shift between stages was less obvious as we expected, indicating that the development of CNC cells across these stages might be focusing on cell proliferation rather than differentiation or

emergence of new cell types, which might be a support for the opinion that CNC anterior-posterior positional identity is established before migration[163].

2.2.5 Epithelial cell types and related endoderm differentiation revealed by sub-clustering

In the primary clustering, we observed a highly structured region with six clusters that showed substantially enriched signals for epithelium-related and endoderm-related cell types (**Figure 2.6B**, epithelium at lower right corner). As the name organogenesis suggests, we expect to see various organs emerge from the gut tube such as the lung, the liver, and the pancreas. However, the cell populations from the endoderm are underrepresented from single-cell studies of comparable stages[66], largely due to the low cell volume. Rather than seeing isolated cell clusters from these organs, based on the global primary annotation, we only observed their signals embedded in this highly structured region, together with other epithelium-related cell types. Except for C32 with a clear signal for hepatoblast, it was difficult to determine the identity of the other five clusters in the primary annotation with a mixed-signal of epithelial and endodermal cell types. Though not as intensively studied as the other two germ layers, it is clear that the development of endoderm initiated long before the stages we captured in this study. Formation of the primitive gut beings right after the completion of gastrulation, and at the organogenesis stage (~E9.0) the gut tube is closing by having the foregut pocket, hindgut pocket, and the lateral endodermal walls met at the yolk sac[313,314].

To address how the hepatoblast is related to other endodermal-derived cell types and how endoderm-derived organs differentiate from epithelium, we subset these six clusters and analyzed them independently from other cells of the dataset (**Figure 2.10A**). After running UMAP on the subset, the three clusters 32, 38, and 42 at the tip of the 'fingers' got detached from the main structure (**Figure 2.10B-C**), indicating that they have more unique features compared to the other three clusters. However, the remaining three clusters were still highly structured, thus, we re-clustered the subset into 10 clusters to reveal the meaning of this complicated structure (**Figure 2.10D-E**). The relative topology and spatial relationships of clusters were mostly kept after subclustering (**Figure 2.10D-E**), and the three isolated clusters mentioned above were well reserved. In cluster EPI7 and EPI6, however, more structures formed after subclustering.

In terms of sample origins, the cells from the whole embryo sample showed a nice representation of all clusters. Cells from the trunk also occupied the majority of the clusters except for EPI0 which was mainly cells from the head sample. Interestingly, cells from the heart sample also showed a high percentage representation in several clusters, especially EPI3 which showed strong signals of hepatoblast even before subclustering. Considering the developing liver bud is right next to the cardiac mesoderm when the hepatic specification takes place at ~E8.25[127], it is reasonable to have hepatoblast distributed in both trunk and heart samples. On the other hand, the signals from the cardiac mesoderm are required for liver development[313–315], and the process of cardiogenesis is also suggested to require signals from the endoderm[313,316]. Such communication marks the close relationship between these two cell populations. We have

also observed a heavily biased stage distribution in EPI3 (**Figure S2.11**), with the majority of cells coming from later stages. Such observation aligns with the current understanding of liver development, in which the specification of hepatoblast occurs before E9.0 and remains as a monolayer of endoderm cells, while the expansion and proliferation take place after the liver bud is formed around E10[127]. Multiple hepatoblast markers such as *Afp*, *Alb*, *Ttr*, and *Hhex*[127,317–319] showed strong and specific expression at EPI3 confirmed the identity of this cell population (**Figure S2.12**).

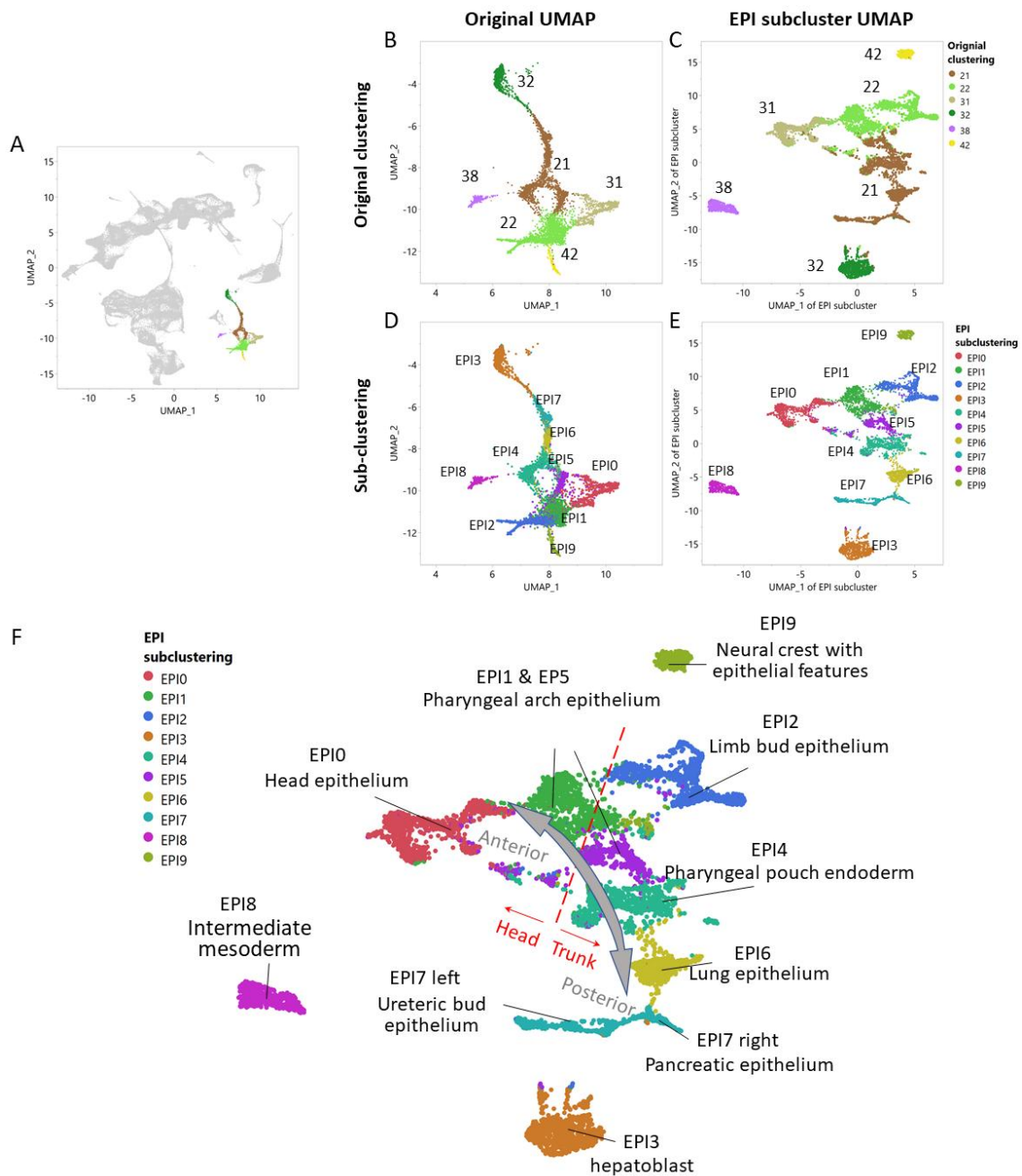


Figure 2.10: Epithelial cell types and related endoderm differentiation revealed by sub-clustering

Subclustering of epithelial cell clusters detaches three more distinct clusters, and gives high resolution views of cell types related to potential endoderm differentiation.

(A) Six clusters identified with epithelial identities, highlighted in the global UMAP plot. **(B-E)** UMAP projection of epithelium subset, before (B, D) and after (C, E) subsetting, colored by original clustering (B, C) and sub-clustering (D, E). **(F)** Annotation of epithelium subset. Anterior-posterior axis labeled by the grey arrow. Head and trunk division marked by the red dotted line.

The other cluster adjacent to the hepatoblast cluster in the original UMAP space also showed enriched endodermal signals (cluster 21 in **Figure 2.10B**). After subclustering, cluster 21 was broken down into four sub-clusters: EPI4, EPI5, EPI6, and EPI7. Note that the subcluster EPI7 showed a branched structure (**Figure 2.10E**), which was not revealed in the original global UMAP coordinate (**Figure 2.10D**). Other than that, EPI6 and EPI4 showed a better separation in the independent subset UMAP coordinate. To address the nature of these clusters, we performed DGE on the subclusters. The enrichment analysis of DEGs from EPI7 suggested that these cells had highly enriched scores for pancreas-related cell types. The formation of the pancreas takes place around the same time as the liver (~E9.0), starting from the dorsal pancreatic epithelium to a proper pancreatic bud[320]. A large number of genes have been described as markers for pancreatic cell types, of which *Pdx1* was first detected in the ventral domain at E8.5, then expands to dorsal and remains specifically expressed through E9.5 to E10.5[320]. Other early endodermal markers like *Nepn* and *Ctps* shared a similar expression pattern with *Pdx1* in the UMAP space (**Figure S2.12**), which was reported as some of the most enriched genes in developing pancreatic bud[321–323]. Note that these pancreatic makers mostly highlight the right half of EPI7, not including the left tip (**Figure S2.12**). By wondering what was the nature of the cells at the left branch, we gathered genes that specifically

highlighted the left branch, among which we found *Duct2*, *Kif26b*, and *Robo2* (**Figure S2.12**). Finding out these genes are all key regulators for kidney development[324–326], it was a bit puzzling at first why the kidney as a mesoderm-derived organ has such a close relationship with the pancreas which is essentially an endoderm-derived organ. But by understanding that the precursors of the pronephric duct actually separate away from the intermediate mesoderm initiated around E8.0, then undergo a mesenchymal-to-epithelial transition to form an epithelial tube of the pronephric duct[277], it makes sense that they share similarities with pancreatic epithelial cells. Notably, there was an understudied gene *2610528A11Rik* that showed specific expression in the left branch of EPI7 (**Figure S2.12**), which was reported as an enriched gene at the caudal endoderm[327]. There was another cluster EPI8 that also showed signals closely related to the kidney, with specific expression patterns of *Gata3*, *Eya1*, and *Pax2* (**Figure S2.12**). Compared to the genes highlighting the EPI7 left branch, these genes function at an earlier point of kidney development. *Pax2* and *Gata3* are key transcription factors that are expressed in nephric duct precursors starting at E8.5[108,273,277]. *Eya1* encodes a transcriptional co-activator that would regulate *Pax2* expression, and is essential for the early differentiation of metanephric mesenchyme[277,328]. The stage distribution of EPI8 also suggested these cells were at earlier stages of differentiation (**Figure S2.11**), lined up with the gene expression pattern. Therefore we concluded that EPI8 represented cells from early intermediate mesoderm, then differentiated into EPI7. It is interesting to see that once the cells from the intermediate mesoderm (EPI8) undergoes mesenchymal-to-epithelial transition, a huge difference occurred represented by the gap between the cells gained epithelial fate (EPI7 left branch) and those before transition (EPI8), while also

made the post-translational cells more alike other epithelial cells from a different origin (EPI7 right branch). Another endodermal-derived organ that has started differentiation during these stages is the lung. The expression of *Nkx2.1* in the anterior foregut endoderm cells at ~E9.0 initiates the specification of the lung; afterward, the formation of two lung buds marks the beginning of the embryonic stage of lung development[135]. Other transcription factors including *Gata6*, *Foxa1*, and *Foxa2* are also involved in endodermal development and lung epithelial cell differentiation[135,329], all these genes together with *Nkx2.1*, showed specific expression in EPI6 (**Figure S2.12**). At a slightly later stage of ~E10.5, the thymus development starts in the posterior part of the third pharyngeal pouch, but is still not fully differentiated by E11.5 which is the latest stage of this study[313,330]. The expression of thymic epithelium key regulators *Pax1*, *Pax9*, and other genes that mark pharyngeal pouch such as *Prrx2* and *Vgll2*[331,332] all showed high expression in EPI4 (**Figure S2.12**). Since the spleen development was just initiated at E11.5[330,333], we did not find a cluster specifically related to it.

So far we have addressed the signals from most endoderm-derived organs, well supported by specific gene expression patterns. The remaining subclusters, however, were more challenging with less featured gene expression and noisier signals. Another outlier cluster, EPI9, was originally connected to EPI2 in the UMAP space (**Figure 2.10D**), but then isolated after subsetting (**Figure 2.10E**), indicating its unique feature. It should be fairly easy to annotate a cluster like this, but then we realized that the DEGs from EPI9 were mostly expressed in multiple clusters in addition to this epithelium subset, and the GSEA signals returned were a mixture of different cell types. Considering the

heterogeneity of the DEGs and the clear epithelium feature of EPI9, it is likely that this cluster could be neural crest-derived cells undergoing epithelial to mesenchymal transition, but still hold their epithelial features. Many of the DEGs were documented as enriched expressors of different neural crest derivatives in later stages, such as *Dpf3*, *Hand1*, and *Lrp2*[334–336]. We also found that many of these genes showed specific expression patterns but were understudied including *Tacstd2* and *Tac2* (**Figure S2.12**). The unique expression of these genes might drive the separation of EPI9 from other cells, but their functions and expression patterns at early embryogenesis remain largely unknown.

The remaining subclusters, EPI0, EPI1, EPI2, and EPI5, showed significant sample diversity. With almost no heart cells in both, EPI0 and EPI1 consisted mainly of head cells, while EPI2 were mostly trunk cells and EPI5 showed an even split between head and trunk (**Figure S2.10**). Some genes reported to be specifically expressed in limb ectoderm were showing specific expression patterns in EPI2, such as *Bhlha9* and *Kremen2*[337,338] (**Figure S2.12**). Chemokine *Cxcl14* highlighted both EPI1 and EPI2 which suggested its role in both trunk cells and head cells, and has been reported to be expressed in both pharyngeal arches and limbs at ectoderm[339]. Considering the sample distribution of these two clusters, it was clear that EPI2 represented limb epithelium while EPI1 represented part of head epithelium, presumably pharyngeal arch epithelium. The adjacent cluster EPI5, however, showed fewer DEGs. As a connection between EPI4 and EPI1, it is more likely that EPI5 shared some similarities with the other two clusters. One gene that showed specific expression patterns at both EPI4 and EPI5,

Vgll2, was reported to be expressed at surface epithelium at pharyngeal arches at ~E9.5[340]. Together, the subcluster EPI1 and EPI5 all showed significant signals related to cells in pharyngeal arches with epithelial features, while EPI4 presented pharyngeal pouch endoderm signatures. The ability to reveal such layered expression patterns by sub-clustering also confirmed the robustness of scRNA-seq in terms of differentiating cell type identities among those with similar expression profiles and anatomical regions. With most cells coming from head samples, EPI0 pointed to cell types in the head region. We found several genes actively expressed in the head region being highly specific in EPI0 such as *Dmrta2*, *Hsex1*, and *Six3*[341,342] (**Figure S2.12**). *Hsex1* and *Six3* were reported to be co-expressed in oral ectoderm for pituitary development around the stages we captured[342]. Other genes such as *Mgarp* marking cornea epithelial cells[343,344], *Aldh1a3* which are highly expressed in nasal placode but not intensively studied[345], and some cranial neural crest genes like *Dlx5/Dlx6*, were also among the DEGs and showed highly specific expression in EPI0 (**Figure S2.12**). Notably, similar to *Cxcl14* which highlighted both EPI1 and EPI2, there was another gene *Sp8* which highlighted both EPI0 and EPI2 (**Figure S2.12**). *Sp8* is known to be a member of zinc finger transcription factors that express in both apical ectodermal ridge and nasal processes[346], which explained why it highlighted both EPI0 and EPI2.

Taken together, from the epithelium subcluster we clearly identified different cell types from various developmental origins even though they shared similar expression profiles. The sub-clustering analysis successfully isolated three clusters that were less similar to the main population (EPI3, EPI8, and EPI9). Related GSEA of these outliers showed that

these three clusters were either at an earlier stage of differentiation (EPI8, intermediate mesoderm), a later stage of differentiation (EPI3, hepatoblast), or undergoing epithelial to mesenchymal transition (EPI9, neural crest with epithelial features). The body of the epithelium subcluster was largely spread out after rerunning UMAP and revealed more features of the sub-population. Specifically, we were able to capture signals from different organs even they were not as differentiated as the hepatoblast at the stages of this study, including kidney (EPI7 left, ureteric bud epithelium), pancreas (EPI7 right, pancreatic epithelium), lung (EPI6 lung epithelium), and the thymus (EPI4, pharyngeal pouch endoderm). These endoderm-derived organs were represented at a low cell population at the stages studied, but successfully captured by our sub-clustering analysis. Besides, more anterior parts of the embryo also had cells with epithelial features, represented by several cell clusters consisting of head originated cells (EPI0, head epithelium; EPI1 & EPI5, pharyngeal arch epithelium). By connecting these clusters together, we were able to draw an anterior-to-posterior arch on the UMAP distribution (**Figure 2.10F**). In addition, we identified a branched structure in EPI7 which was not revealed by global UMAP before the sub-clustering analysis. Two branches of EPI7 pointed to two spatially close organs as the kidney and the pancreas, at the very beginning of their differentiation. The other cluster EPI2 which pointed to another direction and consisted mostly of trunk-derived cells, however, showed signals of the limb. In the mesenchyme subcluster, we have already identified clusters with limb-related marker genes that were annotated as limb mesenchyme cells, but here we found a cluster with similar signals with epithelial features. This is a good example showing how scRNA-seq could differentiate between cells that originated from the same anatomical region, even at very early stages of development. A

similar example could be seen in EPI0 head epithelium, which shared features with the cranial neural crest subset, validated by the projection of various genes that highlighted both regions in the UMAP plot such as the *Dlx* family genes. Viewing this from a different perspective, many gene expression patterns are not unique in scRNA-seq datasets which highlight more than one region, and the investigation of multiple genes could provide valuable insights into the cellular identities. By processing samples from multiple stages, we also observed the transition between the early and late stages of differentiation, represented by the biased distribution of cells from different stages. One obvious example is EPI3 hepatoblasts, with the majority of cells from later stages (**Figure S2.11**), further validating that the formation of liver bud takes place around E10. On the contrary, the other endoderm-derived organs such as lung, kidney, and pancreas, did not show such significant bias towards later stages, suggesting that the differences between these under-differentiated cells were not obvious at the transcriptional level.

2.2.6 Vision on more differentiated cell types: endothelial cells

Next, we moved on to the cell clusters that were not in large 'clouds' with less complicated structures and more specific signals for differentiated cell types.

Like the epithelial cells we just discussed, there was another region (Cluster 20 and 29) showing signals related to endothelial cells (**Figure 2.6B, Figure 2.11A**). It is interesting to see that a small part of cluster 29 was connected to the mesoderm cloud (**Figure 2.11A**, red circle), meaning that part of cluster 29 shared certain characteristics with the mesoderm cells. To further investigate the endothelial cell clusters, we subset these two

clusters from the main population and reran dimension reduction using UMAP. Compared to the connected 'mushroom' structure in the global space, the 'root' (cluster 29) and the 'cap' (cluster 20) were completely separated in the subset space. Surprisingly, the small part of cluster 29 connecting to the mesenchyme cloud was not separated (**Figure 2.11A-B**, red circle), indicating that this small portion was not significantly different from the main part of cluster 20. The subclustering of the subset showed almost identical results compared to original clustering at two cluster resolutions (**Figure 2.11D**), with ENDO1 corresponding to cluster 20 and ENDO2 corresponding to cluster 29. By analyzing the sample of origin for each subcluster, we also observed that the majority of ENDO1 came from the heart sample and the ENDO2 was a mixture of all samples in this study (**Figure 2.11D**). To address the differences between ENDO1 and ENDO2, we performed both independent PCA analysis and DGE analysis between the two subclusters. Apparently, PC1 separated ENDO1 from ENDO2 (**Figure 2.11E**) and revealed significant differences. The top-loaded genes from the positive end of PC1 showed less endothelial-related signals in GO:BP enrichment, while the genes from the negative end of PC1 were more likely to be endothelial and vascular-related (**Figure 2.11F**). The relationship between heart development and endothelial cells has been well explored in previous studies. It has been addressed that the endocardium is unique among all endothelial tissues due to its unique location and specialized functions, reflected at gene expression level[347]. In early development, the endocardium originated from the pericardial area remodels to form the endocardial tube, which is considered the first vasculature structure formed in the embryo[115,348]. Then the vasculogenesis initiates when the heart develops, as the vascular precursor cells expand, the epithelial-to-mesenchymal transformation also takes

place to invade underlying mesoderm regions[115]. Based on this, we hypothesized that the division of ENDO1 and ENDO2 showed the differences between endocardial-derived endothelial cells and more general endothelial cells.

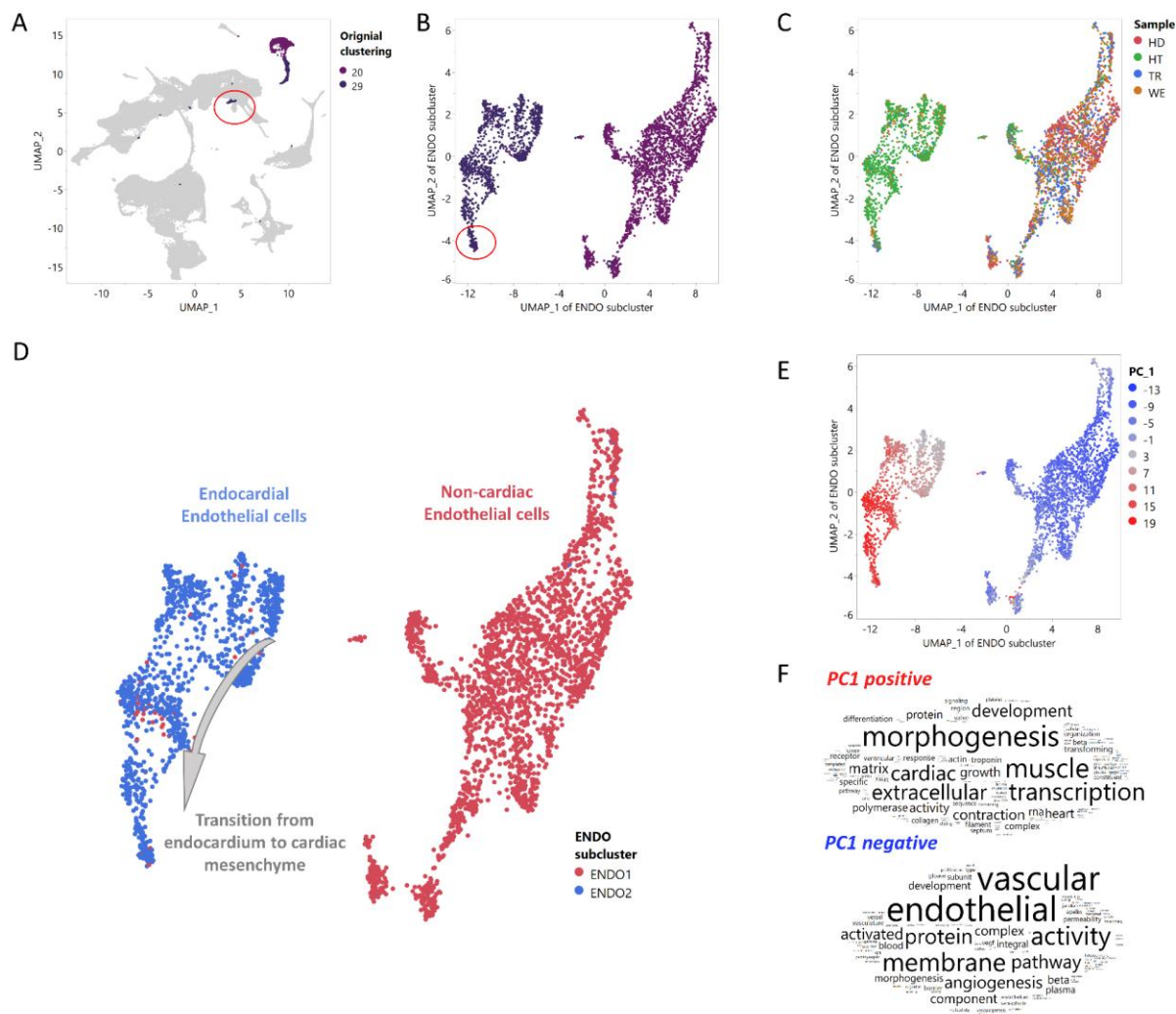


Figure 2.11: Two distinct populations of endothelial cells revealed by sub-clustering analysis

The endothelial cell cluster includes two sub-clusters. Sub-clustering analysis, cell origin, and independent PCA analysis reveal two populations of endothelial cells with different natures.

(A) Two clusters identified with endothelial identities, highlighted in the global UMAP plot. A small population connected to the mesoderm cloud circled in red. **(B)** UMAP projection of endothelial subset. The same population from (A) circled in red. **(C)** Sample of origin colored in UMAP space. **(D)** Sub-clustering of endothelial populations at two cluster resolution, with annotations. **(E)** Independent PCA analysis of endothelial population, colored by PC1. PC1 clearly divided the two sub-clusters. **(F)** Word clouds for enriched GO:BP terms from top genes loaded in the positive (red) and negative (blue) end of PC1.

To test our hypothesis, we looked into specific genes that differentiate ENDO1 from ENDO2. *Hand2* plays a significant role in heart development and was differentially expressed in ENDO2 (**Figure S2.13A**). The expression regions of *Hand2* include cardiomyocytes, epicardial cells, mesenchymal cells, and endothelial cells[349]. *Hand1* is a paralog of *Hand2*, however, showed a slightly different pattern which only highlighted the bottom tip of ENDO1 (**Figure S2.13A**), which is also the small population that was connected to the mesoderm cloud in the global UMAP. *Hand1* was reported to be expressed in the myocardium instead of the endocardium[350], which would explain the nested expression pattern we observed in ENDO2, and suggested that the bottom part of ENDO2 was a cell population bridging between the endothelium and cardiac mesoderm. Another set of paralogs *Gata4* and *Gata6*, which were reported to be expressed in both myocardium and endothelial-derived cardiac cells[113,351], highlighted only ENDO2 (**Figure S2.13A**).

Interestingly, we also found a group of connexin genes, *Gja4* and *Gja5*, with slightly different expression patterns in the subset (**Figure S2.13A**). Connexin genes encode gap junction proteins, which could be detected as early as E8.5 in the cardiovascular system. Though both highly expressed in endothelial cells at early mouse embryonic development, the different expression patterns of *Gja4* and *Gja5* has been illustrated, of which *Gja4* was found to be expressed in both vascular and endocardial endothelial cells, while *Gja5* express only at vascular but not endocardial endothelial cells[83,352]. In our subset, *Gja4* highlighted both ENDO1 and ENDO2, but *Gja5* highlighted only ENDO1 (**Figure S2.13A**). This served as another piece of evidence that ENDO2 were primarily cardiac-derived endothelial cells, while ENDO1 were more broad vascular endothelial cells. This hypothesis was also verified by more general endothelial cell markers such as *Pecam1* and *Chd5*[115] (**Figure S2.13A**), which highlighted both clusters except for the bottom tip of ENDO1. Together with the high percentage of heart-derived cells, it was clear that ENDO2 represented a group of cardiac-derived endothelial cells with the bottom part undergoing a transition between endothelium and mesoderm, and ENDO1 represented more general endothelial cells from the entire embryo. The stage progression of this subset was less informative as only two sub-clusters were addressed, but we still observed a higher percentage of later stages cells in ENDO2 (**Figure S2.13B**), which might be an indication of faster endocardium development at ~E11 compared to other vascular structures of the embryo.

2.2.7 E9.0 to E11.5 cell type dynamics: summary and discussion

Once the integration of time-series datasets have been completed and verified as described in 2.1, we start to make an extended cell type atlas for mouse embryos from E9.0 to E11.5. By applying the integrated annotation method introduced in Chapter 1, we were able to get a primary annotation of different cell types, and pool similar ones into large groups for more detailed analyses (**Figure 2.6**). Using subclustering analysis within each large group, we got a detailed understanding of what cell types are included (**Figure 2.7-2.11**).

Other than the improved chemistry of the scRNA-seq pipeline and increased sample size, in this study, we also incorporated stage information of each cell, which enabled us to find cells that emerge or disappear over the process of embryonic development (**Figure S2.4, S2.6, S2.8, S2.11, S2.13**). Additionally, with all datasets integrated into one common space, we were able to compare the spatial distribution of each stage, and how they change over time by highlighting each stage in the UMAP plot (**Figure S2.1, S2.2, S2.5, S2.8, S2.10**). By combining the stage information and their spatial relationships within the UMAP plot, we have observed the proliferation-differentiation axis, the anterior-posterior axis, and the distal-proximal axis, some of which have been observed in Chapter 1. Most of the findings could be supported by literature, such as the emergence of Glutamatergic lineage cells, ganglion eminence progenitor cells, and hepatoblast. The mentioned three cell types showed low cell counts in E9.0-E10.0 datasets, but increased sharply after E10.0, which aligns well with the studies showing when these cell types first emerged. We have also observed the disappearance of several cell types, one example is the PSM

cells in the mesoderm (**Figure S2.6**). The exact time point when PSM turns into somites is still ambiguous[353–355], and the definition of PSM needs to be specified to address this question. But through our results, we could argue that cells with high expression of PSM markers *Fgf8*, *Fgf17*, and *Fgf18* show two different states before and after E10.0, which might provide an answer to the question.

In terms of the body axis we have observed, we have further confirmed that the structure of the low dimensional plot from scRNA-seq datasets contains spatial information, as we have addressed in Chapter 1. This idea has been well illustrated in the example of the sub-clustering of the cranial neural crest cell population (**Figure 2.9**). The spatial information about the body axis from the cranial regions embedded in the UMAP structure of these cell clusters has been revealed by the expression plot of several sets of marker genes, including *Dlx* family genes, *Alx* family genes, and *Hox* family genes (**Figure S2.9**).

Most of our annotations are also supported by the projection of marker genes (**Figure S2.3, S2.7, S2.9, S2.12, S2.13**), and more information of differentiation trajectories could be inferred by combining the expression pattern and the elongated structures in UMAP space, such as in the epithelial cell clusters sub-clustering (**Figure 2.10**). Analyses that were specifically designed for trajectory inference could be performed to further understand the lineage progression, and our approach is just inspecting this complicated topic superficially.

Note that the stage we used in the previous chapter (E11.5) was also included in this study, but using a different dataset with improved chemistry. Such improvement shows better capturing efficiency and sequencing sensitivity, with significantly higher transcripts captured per cell. We have investigated if such differences would lead to different cell types being identified by comparing the cell clusters between two datasets, and concluded that only minor changes occurred. It's worth noting that with cells from other stages as background, the annotation in this chapter would provide more details, and address some unsolved questions in the previous chapter. For example, in Chapter 1, we did not find cell clusters specifically expressing markers from pancreatic progenitor cells. But in the time-series datasets described in this chapter, we have observed a cluster with pancreatic epithelium signatures (**Figure 2.10F**), indicating that the very early stage of pancreas development has already initiated. With improved chemistry, we were able to get a better detection of pancreatic marker *Pdx1*, which did not show enriched expression in the Chapter 1 dataset, but was identified as DEG in this chapter. In addition, the increased number of cells also provides a better chance to identify cell types of small populations. As a result, the time-series datasets based results presented an increased accuracy of cell types identified.

In summary, we have configured the normalization conditions and successfully integrated time-series datasets from scRNA-seq into one common space for downstream analyses. By applying the annotation approach described in Chapter 1, we showed a detailed catalog of cell types at a specific range of mouse early organogenesis (E9.0 to E11.5). With such information, we were able to find cell types that emerge or disappear during

such a process. However, we have just scratched the surface of the huge amount of information provided by such an integrated dataset, and we hope the annotation of the time-series dataset would serve as a valuable resource for those interested in cell type dynamics of mouse development.

Material and Methods

Animal, sample collection, and scRNA-seq library preparation

Refer to Chapter 1 Material and Methods for details. This study employs the same procedure to make scRNAseq libraries. The only remarkable difference is the version of chemistry from 10x Genomics was updated to version 2, with enhanced library quality and better sequencing sensitivity. In this study, embryos at E9.0, E9.5, E10, E10.5, E11, E11.5 were used for library preparation.

Sequencing, scRNA-seq data analysis, and dataset integration

Once libraries were successfully created, sequencing was performed on an Illumina NextSeq 500. Sequencing data were demultiplexed and processed using CellRanger software (version 1.0, 10x Genomics), using the recommended settings to generate a raw UMI expression matrix. A secondary quality control (QC) step was added as we identified low-quality libraries that passed the CellRanger internal QC. For each cell, the number of genes detected and the number of total UMIs were plotted into 2D space, and the good libraries could be visually identified from low-quality libraries, as they form two separate clusters (except for the erythrocytes, which also form their own cluster due to its unique

feature). We manually split the two populations and kept the good libraries. CellRanger internal QC did a good job for most stages, but not all of them.

To pool datasets from each stage into one combined dataset, we used the R package Seurat (version 3.0, in R environment 3.6)[356]. Specifically, the QC'ed raw UMI of each stage was loaded into Seurat, normalized, and integrated into one pooled dataset using CCA based approach. 3000 variable genes and 30 CCA dimensions were computed by Seurat and used for identifying integration anchors. Once successfully integrated, the pooled dataset was treated as one set and analyzed in a similar way as described in Chapter 1. PCA analysis was applied on the integrated dataset with 100 PCs extracted, and then tSNE/UMAP was used for 2D visualization based on 100 PCs. UMAP was used to represent the global structure of the integrated dataset, and both UMAP and tSNE were used in sub-clustering analysis based on specific scenarios.

Subclustering analysis

Selected clusters for subclustering analysis were subset from the normalized integrated dataset, and ran through the pipeline to generate the projection of cells using UMAP. The PCs used for subclustering analysis were reduced to 30 as the heterogeneity reduces. The clustering was performed using the R package Seurat, different clustering resolutions were tested to get at least five versions of results covering different numbers of resulting clusters. The DGE was also performed using the Seurat package. Cell cluster identification was annotated using the approach described in Chapter 1, and manually compared across different clustering resolutions to determine the best number of clusters.

For cranial neural crest cell clusters, subclustering was performed and evaluated, but not reflected in the main text as little information did they add.

Independent PCA

Independent PCA was applied to the cranial neural crest clusters and the endothelial cell clusters, to identify genes that contribute to the specific part of the structures in reduced dimensional space. The target cell clusters were subset from the normalized integrated dataset, then applied to PCA. The top PCs were examined and top PC loadings from both high and low end were used for enrichment analysis. In both cases, PC1 was able to differentiate the structures we are interested in.

Chapter 3: Re-defining housekeeping genes using scRNA-seq datasets

Introduction

When the concept of a housekeeping gene (HKG) was first mentioned decades ago, it was simply defined as a gene that is always expressed[357]. Later it was modified to refer to genes that are constitutively expressed to maintain basic cellular function, in any cell type under normal conditions. In studies using this definition, a cutoff was usually applied to get rid of a large number of genes that have trace expression in all samples[358,359]. More recently, an extended definition emphasized both constant and stable expression without a cutoff[360], which allows for genes with low expression levels.

The importance of HKGs has become more appreciated as molecular biology and computational biology develops. The most significant application of HKGs is to serve as reference genes in experimental normalization such as qPCR, RT-PCR, Western blotting, and RNA sequencing[361]. With HKGs, or more often known as internal control genes in specific assays, researchers are able to quantitatively measure the expression of genes of interest under different conditions[362]. Usually, this refers to a more strict definition of HKG, which requires a relatively high and stable expression. Genes such as *GAPDH*, *ACTB*, and *B2M* have a long history of being used as HKGs based on their constitutive expression[363]. It has been challenged whether these genes are accurately stable[364], and the attempts of discovering true HKGs never stop[365–368]. HKGs help us better understand cell survival, gene evolution[369], chromatin structure[370,371], and many other fundamental biological questions.

A variety of techniques have been used to address the question ‘Which genes belong to the set of HKGs’ in over a dozen studies, including microarray[370,372–377], RNA-sequencing[358,378], Expressed Sequence Tag (EST)[379], and many others. However, the concordance between studies is generally low. For example, a study that attempted to find consensus HKGs across 15 examined HKG datasets by taking the intersection of over 12,500 genes and ended up with only one gene[380]. The inconsistency across the studies arises from different aspects, such as differences in HKG definition, techniques used to identify gene expression, number of genes covered, and tissue sampling, as summarized in Zhang et. al[380].

Single-cell RNA sequencing (scRNA-seq) is a revolutionary technology that helps to address the variation of the transcriptome at the single-cell level[5]. Compared to microarray and bulk RNA seq, it has much fewer false positives, significantly improves the resolution of expression, and better suits the definition of HKGs in terms of ‘expressed in every single cell’. At the theoretical level, scRNA-seq seems to be the ideal solution for identifying HKGs, but certain challenges make it difficult. The most obvious challenge is the ‘dropout’ event[10,11], which causes a large number of false negatives and makes the expression matrix sparser than it already is. A symptom of this is that a true HKG expressed at a low level would not be detected in many cells. Also, it has been appreciated that genes at the single-cell level show stochastic expression patterns known as transcriptional bursting[10,381], which might challenge the HKG identified in the bulk analysis.

So far the potential of using scRNA-seq datasets to understand HKGs has been largely unexplored[382]. In this study, we show that all scRNA-seq based datasets possess a universal pattern in a graph of expression level by gene detection plot. We explain this pattern's implications on the dropout problem and extract from it a formal definition of HKGs. To simplify the problem, in this study we use a wider definition of HKG, referring to genes that are expressed in all cells regardless of expression level. We validated our definition by using a variety of orthogonal approaches and finally, we used a collection of scRNA-seq datasets to propose genes that are highly likely to be HKGs.

Result and discussion

3.1 From scRNA-seq to housekeeping genes

3.1.1 A unique gene expression pattern observed in scRNA-seq datasets

To study the gene expression of scRNA-seq datasets, we computed the detection frequency (scaled from 0 to 1) and average expression level (per detected cells, log scale). By plotting all genes within a given dataset, we found that the graph contains a curve with a very clear boundary (**Figure 3.1**). This boundary implies that, at a certain expression level, there's a maximum possible detection frequency. However, it has been appreciated that some genes are universally expressed, but at a low expression level[359]. Reflected in the mean-by-frequency plot, there should be genes with a frequency of 1 but a low mean expression level. Rather, we observe a clear edge of the plot, with no genes at a frequency of 1 when log mean expression < 1.5 . We infer that this is a culmination of the dropout event described above, imperfect experimental chemistry, and machine-specific biases. With a significantly large number of zero counts, genes expressed at low levels

are very likely to be missed in some cells during sequencing. Another interesting observation is that the edge of the curve is 'sharp', meaning that there are no outliers on top of the curve. That means these genes on the edge have the highest possible detection frequency at a certain expression level. Therefore, we hypothesize that these genes are ubiquitously expressed in all cells from the sample with a true frequency of 1, but missed by technical limitations at a low expression level. In an ideal experiment with no dropout event, theoretically, the curve should be flat, and true HKGs with a frequency of 1 should be spread along the x-axis at different expression levels. Such a curve with a sharp edge could be observed in all scRNA-seq datasets we tested, from different sequencing platforms, species or cell lines, developmental stages, and tissues/organs (**Figure 3.1**). To efficiently analyze the genes on the edge of the curve, we applied an algorithm that adjusts the observed detection frequency to predict the true frequency using the hypothesis that genes along the boundary are ubiquitously expressed. The algorithm attempts to predict the true frequency of the genes by warping the observed boundary towards the hypothesized ground truth space in the frequency domain. With the understanding that there should be genes with a frequency of 1 at every expression level, we adjust the frequency of each gene based on its vertical distance to the edge of the curve (**Figure 3.2A**, arrows indicate the genes before (red) and after (pink) adjustment). The resultant Adjusted Frequency (AdjFreq) space is also scaled from 0 to 1, the higher AdjFreq means a closer distance to the edge of the curve, along the frequency axis. To analyze the genes on the edge of the curve, we sorted the AdjFreq and picked the top genes with $\text{AdjFreq} > 0.95$ (**Figure 3.2A**).

It's worth noting that in this approach, the genes on the edge of one specific dataset are believed to be ubiquitously expressed only in cells from that dataset. That would explain a certain level of inconsistency, as a gene expressed in all cells from the liver does not necessarily always express in cells from the heart. Therefore, a dataset with higher heterogeneity would be more representative in terms of discovering genes expressed in all cell types. For the same reason, the E11.5 mouse whole embryo dataset was chosen as a reference and used in downstream analyses. For all datasets used in this study, refer to **Table S3.1**.

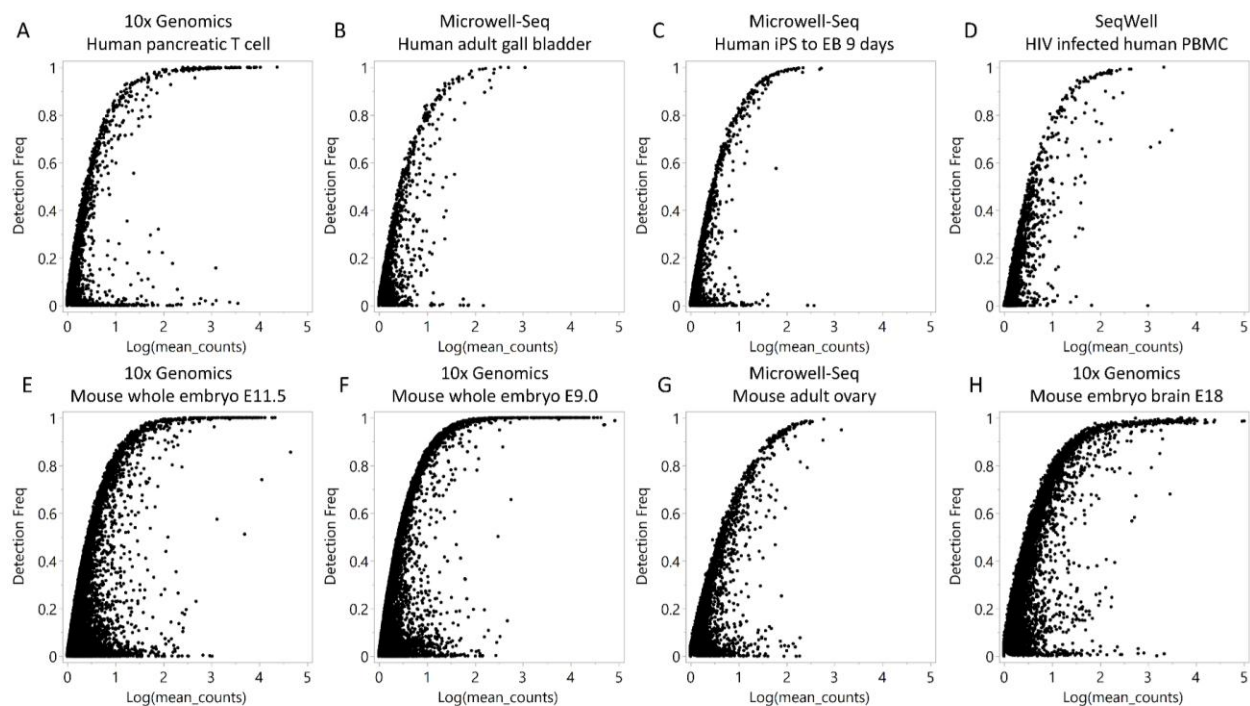


Figure 3.1: A unique pattern with a clear boundary observed across different scRNA-seq datasets

A smooth edge with no outliers on the left can be observed in gene plots generated from all scRNA-seq datasets, regardless of species, tissue type, developmental stage, and platform.

Frequency-by-expression plots from experiments based on different species (**A-D**: human; **E-H**: mouse), sequencing platforms (**A, E, F, and H**: 10x Genomics; **B, C, and G**: Microwell-Seq; **D**: SeqWell), developmental stages (**A, B, D, and G**: adult; **B, E, F and H**: embryo), and tissue/cell types. Refer to **Tab S3.1** for more details.

3.1.2 Genes on the edge show a high level of consistency across different datasets

To test whether the genes with high AdjFreq show consistency across various datasets, we picked a collection of datasets including mouse whole embryo datasets at different developmental stages (**Figure 3.2**, dataset A and D), sample datasets of the mouse brain and heart from 10x Genomics (**Figure 3.2B**, dataset B and E), and datasets of different mouse organs from the mouse cell type atlas *Tabula Muris*[383] (**Figure 3.2C**, dataset C, other samples not shown). We then computed the AdjFreq for each dataset to sort all genes within each dataset. Projections of the top 500 genes onto the E11.5 mouse whole embryo dataset show that they largely fall towards the edge with good concordance (**Figure 3.2B**).

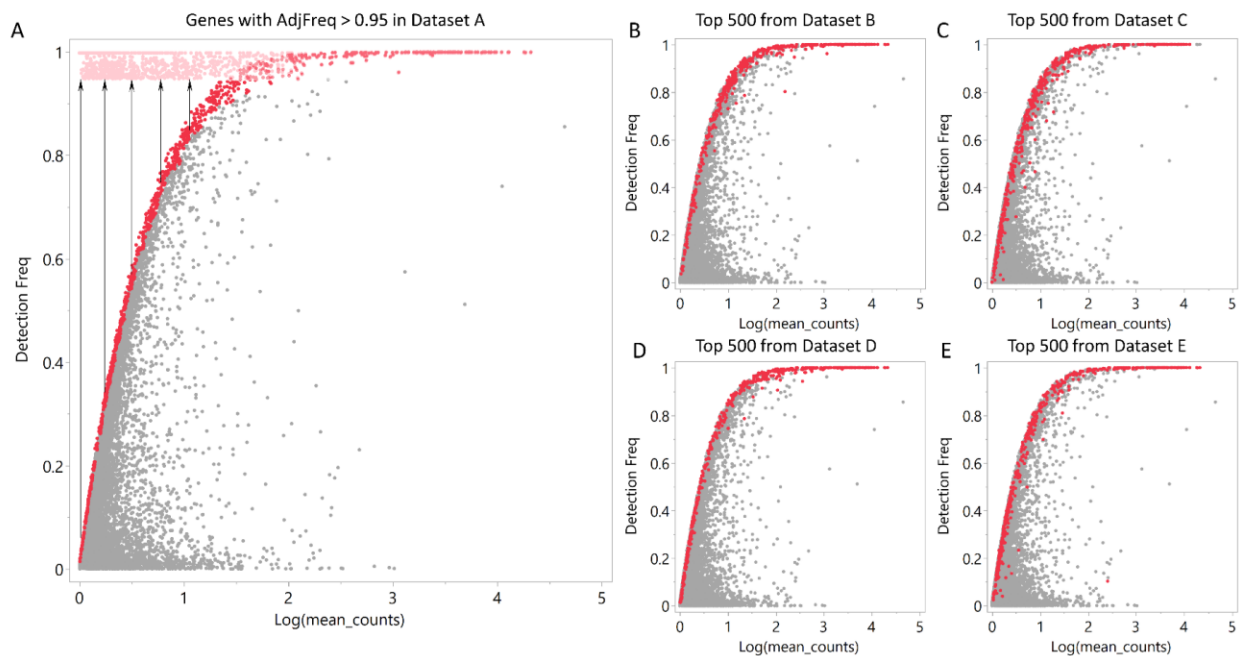


Figure 3.2: Genes on the edge of various datasets show a high level of consistency. Mathematical approach is used to compute AdjFreq which can be used to predict real frequency and extract genes on the edge. Genes with high AdjFreq show concordance across different datasets.

(A) Frequency-by-expression plots from dataset A. Genes with AdjFreq > 0.5 highlighted in red. Black arrows indicate the location of genes before (read) and after (pink) the adjustment of frequency. **(B-E)** Top 500 genes with highest AdjFreq from dataset B-E selected and projected onto the plot of dataset A. Dataset A: mouse whole embryo E11.5, dataset B: mouse embryonic brain, dataset C: mouse adult kidney, dataset D: mouse whole embryo E9.0, dataset E: mouse adult heart. Refer to Tab S1 for more details.

3.1.3 Housekeeping genes from scRNA-seq: summary and discussion

Early in the single-cell transcriptome era, Eiseberg et.al predicted that the new experiments from single-cell resolution might provide significant advantage and ultimately

answer the question ‘which genes are expressed in each and every single cell’[378]. Efforts have been made to identify HKGs under such context, with the appreciation of significant stochasticity and heterogeneity observed in scRNA-seq experiments[382]. Unfortunately, the technical limitations from scRNA-seq makes it difficult to get genes that are truly ubiquitous expressed, especially those at low expression levels. Contrary to the observation from batch experiments that some ubiquitously expressed genes are indeed low expressors[358], these genes are all detected at low frequencies due to the dropout event.

In this study, we propose that by plotting the expression level and detection frequency of each gene within a scRNA-seq experiment, it is possible to find genes that are ubiquitously expressed but missed by technical limitations. In this mean-by-frequency plot, as the mean counts decrease the maximum detection frequency drops below 1 (**Figure 3.1**). Knowing the fact that the expression level of ubiquitously expressed genes spans over a wide range, we predicted that such a plot under a hypothetical perfect condition would be flat, meaning that as mean counts decrease, the HKG remains at detection frequency of 1. The shape of such a curve is unique, with a sharp edge and no outliers above it, indicating that the genes at the edge have the highest detection frequency which should ideally be 1. By testing such curves in multiple scRNA-seq datasets, we concluded that the sharp edge is a universal pattern, regardless of sequencing platforms, species or cell lines, tissues or organs, and developmental stages (**Figure 3.1**). Though all curves have such sharp edges, the shape differs from dataset to dataset. Multiple variables might affect the shape of the curve, such as species, sequencing depth, and RNA level of

different cell types. To avoid such bias, we analyze each dataset individually, without pooling or normalizing different datasets.

Based on such observation, we hypothesized that genes on the edge of the curve are ubiquitously expressed in the cells of the dataset. The distance from a gene to the edge has been quantified using the AdjFreq, which is a representation of predicted true detection frequency. Genes with higher AdjFreq are closer to the edge, while those with low AdjFreq are far apart (**Figure 3.2A**). Comparison of genes on the edge from different datasets showed great concordance (**Figure 3.2B-E**). Cells from specific tissues such as liver and heart show less concordance compared to others, also observed by other researchers possibly due to lower complexity and more specialized functions[358]. Strictly speaking, the genes on the edge are hypothesized to be ubiquitously expressed only in the tested dataset. Therefore, to get genes that are considered HKG we need to analyze a collection of datasets that cover a variety of cell types, or a dataset with great heterogeneity. To minimize the bias from integrating datasets, we used a whole embryo dataset that covers all cell types at mouse embryo E11.5.

In the following part, we are going to provide evidences to support our hypothesis.

3.2 Multiple evidences show genes on the edge are HKGs

3.2.1 Genes on the edge show enriched GO:BPs related to HKG functions

Next, we tested if the genes on the edge are related to housekeeping biological processes by performing enrichment analysis (similar to GSEA described in Chapter 1) on genes with high AdjFreq. We selected the genes with AdjFreq >0.95 and split them into 10 bins

based on their expression level (103 genes per bin) (**Figure 3.3A**). Genes within each bin were subjected to GSEA using ToppCluster (<https://toppcluster.cchmc.org/>) to find related Gene Ontology (GO) terms in the biological process category (GO:BP), then visualized by word cloud of matched GO:BP terms (**Figure 3.3B**). All bins showed at least one significant GO:BP match, manual inspection showed most of the matched GO:BP terms are closely related to HKG functions, including the top hits such as ‘mRNA metabolic process’, ‘electron transport chain’, ‘ATP biosynthetic process’, ‘RNA splicing’ and ‘protein targeting’. Similarly, we performed the same analysis on MGI phenotype database, and many of the bins showed terms related to lethality and embryonic development (**Figure S3.1**), indicating that these genes play housekeeping roles. Note that genes with higher expression levels tend to have better signals, indicating higher confidence with these genes.

To test if the GO:BP terms from different bins are related, we used network analysis to find out shared GO:BP terms across bins. 10 Bins were used as first-level nodes, with GO:BP hits connected to nodes. Network analysis revealed strong relationships among these bins, with multiple housekeeping GO:BPs shared except for bin 7 (**Figure 3.3C**). In summary, the GSEA of the genes on the edge showed high confidence in HKG related functions, and many of these functions are shared across different genes.

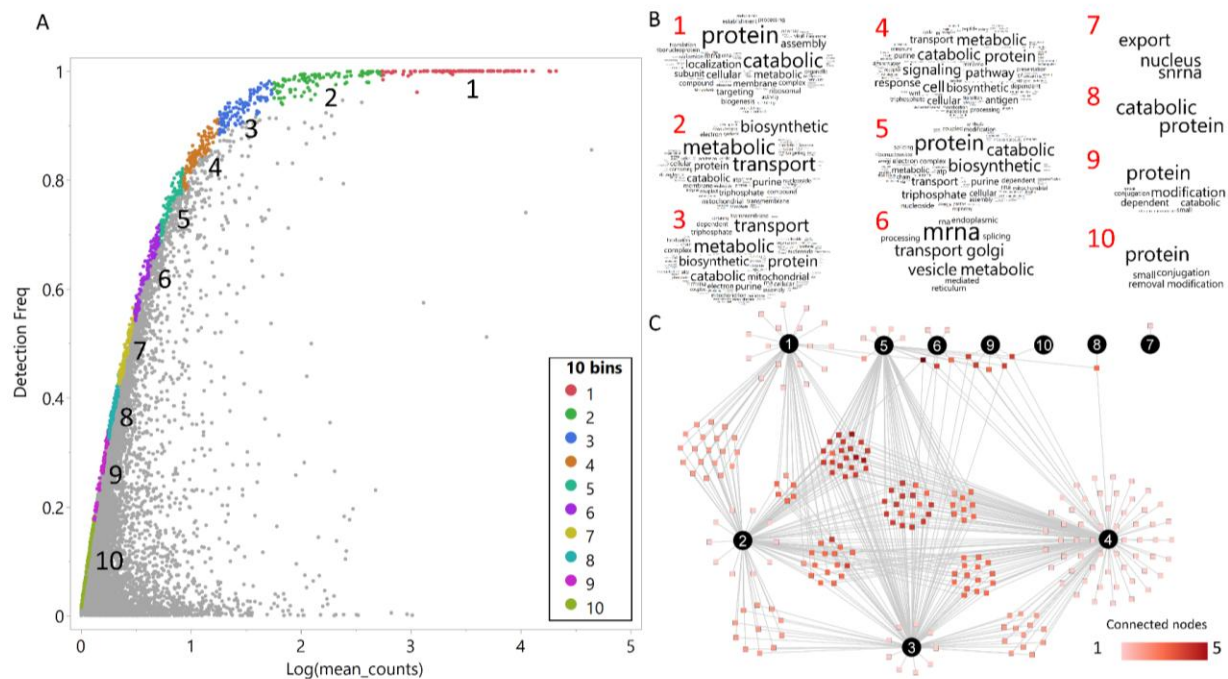


Figure 3.3: Genes on the edge show enriched HKG related GO:BP

Binned genes on the edge of the plot mostly show enriched housekeeping-related GO:BP, and network analysis indicates these bins are heavily correlated.

(A) Frequency-by-expression plots from mouse whole embryo E11.5 dataset. Genes with $\text{AdjFreq} > 0.95$ were divided into 10 bins by expression level and then colored. **(B)** Enriched GO:BP terms from each bin in word clouds. Font size proportional to word frequency. **(C)** Network analysis from 10 bins of genes. GO:BP terms in square nodes colored by the number of connected nodes, gene bins in black circle nodes, edges indicate significant matches between gene bins and GO:BP.

3.2.2 Published HKG lists and cell type specific markers occupy different regions of the curve

We further investigated the relationship between published HKGs and the genes on the edge of our curve. We referred to five published HKG lists for both human and mouse (**Table S3.2**)[358,369,378,382,384]. To compare with multiple lists, we computed a voting score. Each gene got one vote if it is in one of the five HKG lists, therefore all genes were assigned votes from 0 to 5, 0 means not in any of the published HKG lists and 5 means in all the published HKG lists. Since these studies use independent techniques to point at the ground truth, the higher the votes of a specific gene, the higher probability of being a true HKG. The number of votes is highly correlated to the AdjFreq of the gene, indicating the genes recognized by multiple published HKG lists tend to be on the edge (**Figure 3.4A-B**). Across multiple scRNA datasets, ribosomal protein genes showed consistently high detection frequency. These genes encode key protein components of the ribosomal subunits and are essential for protein translation, which should be considered ubiquitously expressed, however, are frequently and intentionally omitted from published HKG lists for reasons that are unclear to us. The genes with high HKG votes and the ribosomal protein genes all tend to fall onto the edge of the curve (**Figure 3.4A**). We also computed cumulative HKG votes from low to high AdjFreq (**Figure 3.4C**). The cumulative counts kept low first and then rose sharply starting from ~ 0.75 , indicating that most of the published HKGs have a high AdjFreq.

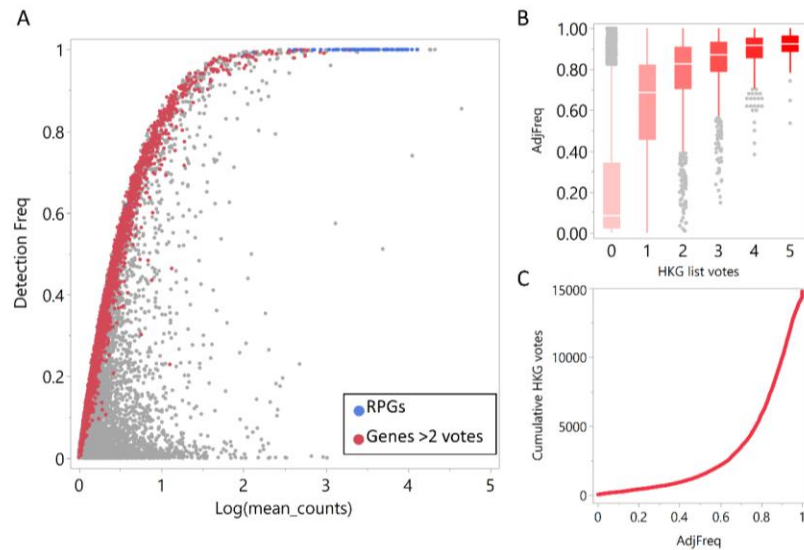


Figure 3.4: Genes from published HKG lists fall onto the edge of the curve

By plotting and comparing the AdjFreq of all genes with published HKG lists, the agreed HKGs from multiple literature tend to fall onto the edge.

(A) Frequency-by-expression plots from mouse whole embryo E11.5 dataset. Genes with >2 HKG votes highlighted in red, ribosomal protein genes (RPGs) highlighted in blue.

(B) Box plot of AdjFreq of genes with different HKG votes. **(C)** Cumulative HKG votes based on AdjFreq.

Viewing the curve from a converse perspective, we believed that cell type specific marker genes are furthest from the edge, reflected by low AdjFreq. To test the hypothesis, we retrieved genes considered as cell type markers from the LifeMap database (<https://discovery.lifemapsc.com/>). They were not only away from the edge, but also presented a significantly lower AdjFreq compared to other genes, showing a completely different pattern on the curve compared to published HKGs (**Figure 3.5A-B**). We then computed cumulative cell type marker counts over AdjFreq, which showed the converse trend as the HKG votes. The marker counts rose sharply at low AdjFreq and then

remained relatively flat at the high end, showing that most cell type markers have a low AdjFreq.

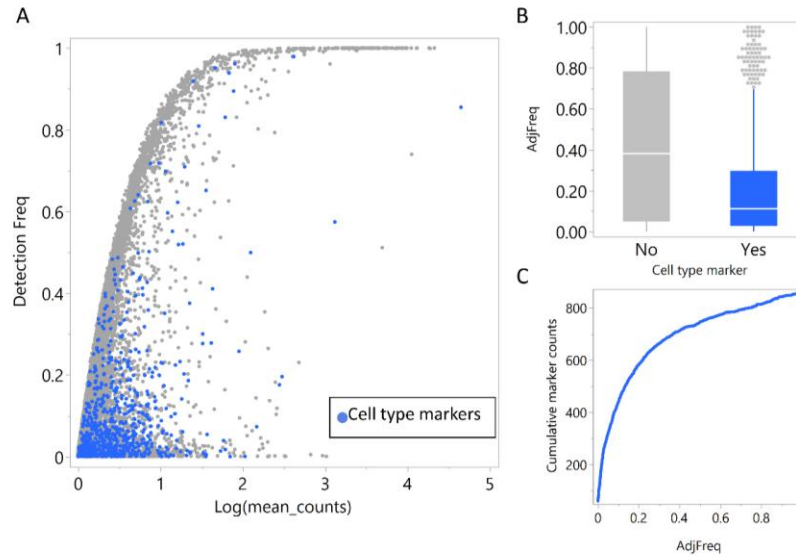


Figure 3.5: Cell type markers do not fall onto the edge of the curve

By plotting and comparing the AdjFreq of all genes with cell type marker gene list, the cell type specific markers tend to fall away from the edge.

(A) Frequency-by-expression plots from mouse whole embryo E11.5 dataset. Genes labeled as cell type markers in blue. **(B)** Box plot of AdjFreq of cell type markers (blue) and other genes (grey). **(C)** Cumulative cell type marker counts based on AdjFreq.

3.2.3 Genes in HKG related GO terms fall onto the edge

Previously we addressed that the genes on the curve show HKG related GO:BP terms, to elaborate this further, we collected genes annotated with various GO terms that are believed to be housekeeping related from the literature[385,386]. It has been addressed that the GO database could be used as a potential resource to discover HKGs[386]. We

hypothesize that genes from such HKG enriched GO terms will also bias towards the edge with high AdjFreq.

By plotting the AdjFreq of genes within these GO terms from all three GO categories, we found most of them are substantially higher than the global AdjFreq, which validated that the genes with housekeeping related behaviors tend to fall onto the edge (**Figure 3.6**).

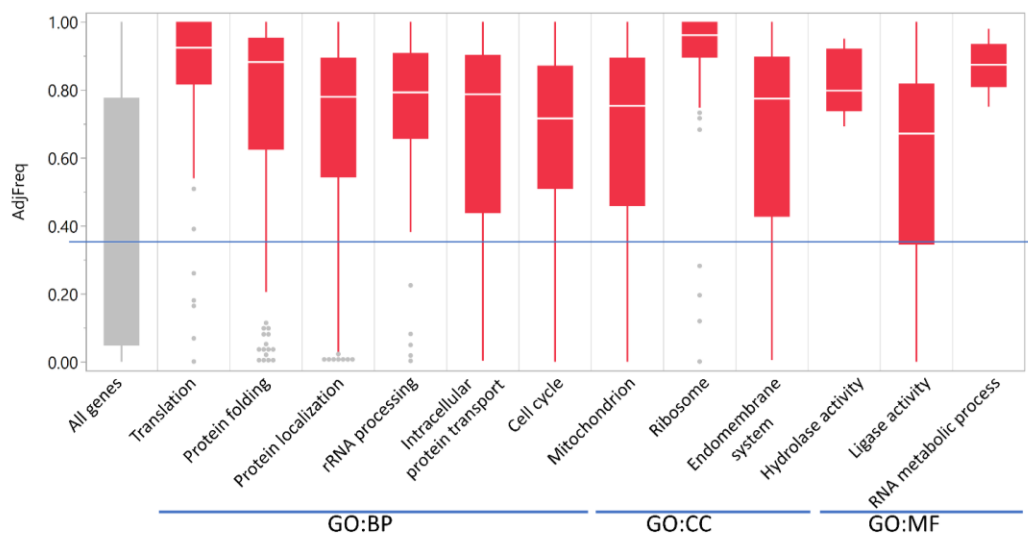


Figure 3.6: Genes in HKG-related GO terms have high AdjFreq

Genes involved in known HKG-related GO terms show higher AdjFreq compared to global level and confirms that HKGs tend to fall onto the edge.

Box plot of AdjFreq of genes in different HKG related GO terms (red). A reference made from all genes at left (grey), the median of which is marked as a blue horizontal line. Specific categories of GO terms labeled at the bottom, including Biological Processes (GO:BP), Cellular Components (GO:CC), Molecular Functions (GO:MF).

3.2.4 Chromatin accessibility also indicates that genes on the edge are more likely to be HKGs

Other than being ubiquitously expressed, HKGs also have other characteristics. It has been published that they have different chromatin organization patterns compared to cell type specific genes, so that the nucleosomes are less stable and DNA more accessible[385,387–389].

To test if the genes on the edge have such chromatin level features, we referred to chromatin accessibility data from a single-cell level study on 13 different tissues[390]. The chromatin accessibility of each gene has been measured as an activity score in over 80,000 tested cells from different tissues. We hypothesized that HKGs would have better chromatin accessibility, meaning a higher activity score.

To validate that HKGs have better chromatin accessibility, we first used the binarized activity scores (either 0 or 1 per cell) to minimize the bias from extreme outliers. We used the sum of binarized activity scores from all cells as the overall activity score of each gene. Ideally, an HKG would be active in all tissues, reflected by a higher overall activity score. By comparing the overall activity scores of all genes with genes from different published HKG lists, we confirmed that genes recognized as HKGs do have higher activity scores (**Figure 3.7A**). The same pattern could also be observed in genes with high AdjFreq from our approach (**Figure 3.7A**, far-right column).

Then we elaborated how the gene activity score would change based on their distance to the edge, reflected by AdjFreq. We split all genes into 10 bins ranked by their overall

activity scores. We showed that genes with high AdjFreq also have higher activity scores, indicating that the genes more likely to be HKGs tend to fall towards the edge. (**Figure 3.7B**)

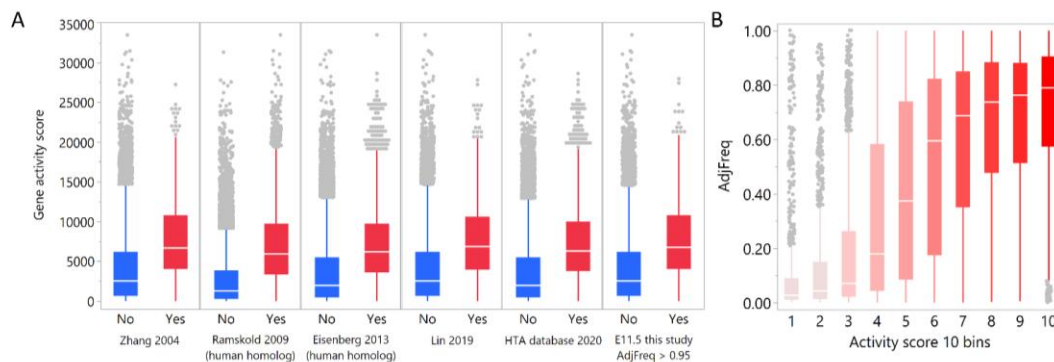


Figure 3.7: Genes with higher chromatin accessibility also have high AdjFreq

Genes within published HKG lists and genes with high AdjFreq both show higher chromatin accessibility. Chromatin accessibility level of all genes also presents a positive correlation with AdjFreq.

(A) Box plot of gene activity scores from different published HKG lists (left five sections) and genes with high AdjFreq in this study (far right section). HKGs in red and non-HKGs in blue. **(B)** Box plot of AdjFreq from genes with different activity scores. Genes were divided into 10 bins based on their activity score (low at left, high at right).

3.2.5 Evidences supporting the HKG hypothesis: summary and discussion

Following the hypothesis we made from the previous section that the genes on the edge of the expression-to-detection plot (or genes with high AdjFreq) are HKGs, here we collected orthogonal evidences to support our hypothesis.

The hypothesis has been tested from different perspectives using four approaches in this study. The GSEA of genes with high AdjFreq confirmed that these genes are enriched in HKG related biological functions, and genes at different expression levels have shared functions (**Figure 3.3**). We also noticed that genes at lower expression levels have much weaker signals returned. The high density of genes and the small initial detection frequency of such genes made the adjustment on the low end significantly noisier than those on the high end, therefore the confidence of a gene on the edge being HKG would drop as its expression level decreases. Other than that, we searched the literature for GO terms that are known to be related to HKGs, which we have proved to be biased towards the edge and have substantially higher AdjFreq than other genes (**Figure 3.6**). Also, we have studied genes that were included in published HKG lists and cell type markers. The concordance among published HKG studies was claimed to be low, so we used HKG list votes to represent the likelihood of a gene being HKG based on previous studies (**Figure 3.4**). Genes revealed by multiple publications as HKG showed high levels of AdjFreq, and the votes turned out to be positively correlated with AdjFreq, suggesting that our approach has a great concordance with the suggested ground truth revealed by previous studies. The cell type markers, on the other hand, have significantly lower AdjFreq than other genes, indicating that the genes off the edge are more likely to be cell type specific (**Figure 3.5**). Finally, we looked into the chromatin structures of the genes. By comparing the gene activity scores from published HKG lists with other genes, we confirmed that HKGs have better chromatin accessibility than other genes reflected by higher gene activity scores (**Figure 3.7A**). We showed that genes with higher activity scores also have

higher AdjFreq (**Figure 3.7B**), such positive correlations would indicate that the genes on the edge are more likely to be HKGs.

In summary, we have provided sufficient evidence showing that our hypothesis is correct, supporting that the genes on the edge of expression-by-detection plot are HKGs. Next, we will present a list of HKGs identified based on this novel approach.

3.3 HKG identified from scRNAseq based approach

3.3.1 AdjFreq from a variety of datasets helps to identify genes likely to be HKGs

Given the hypothesis that the AdjFreq space points at the ground truth, pooling the AdjFreq space across datasets should deviate closer towards that truth, not diverge from it. By averaging the AdjFreq spaces, we observed high agreement (low SD) on either ends of 0 and 1 (HKGs and cell-specific markers) but low agreement (high SD) in the middle (**Figure 3.8A**). By highlighting the published cell type markers which are obviously not HKGs, and ribosomal protein genes (RPGs) which are widely accepted as HKGs (**Figure 3.8A**), we confirmed that the plot of SD versus mean showed two mixed populations, those that are housekeeping genes toward (0, 1), and those that are not toward (0, 0). Gathering numerous datasets should begin to separate these two populations as values become more in accordance with their ground truth. To get a better separation between HKGs and non-HKGs, we applied normal mixture clustering on the plot. We were able to identify one non-HKG population showing low SD and mean AdjFreq, one HKG population showing high SD and mean AdjFreq, and a mixed

population in between (**Figure 3.8B**). The populations identified by normal mixture clustering aligned well with the literature based HKG votes and cell type markers.

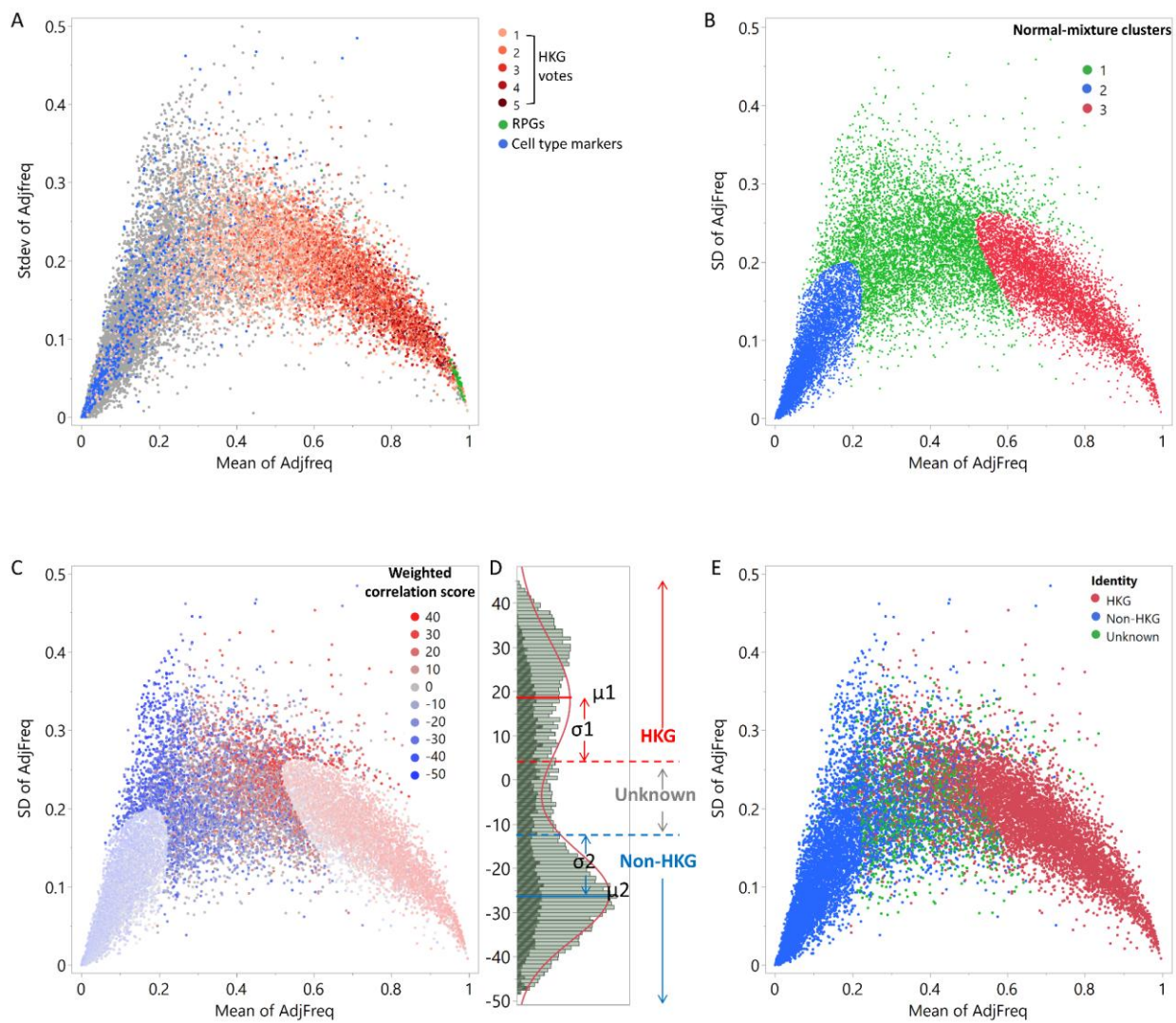


Figure 3.8: AdjFreq from a variety of datasets helps to identify genes likely to be HKGs

By calculating mean and standard deviation of AdjFreq from a collection of datasets, two groups of genes representing HKGs and non-HKGs can be divided using normal mixture clustering in 2D space.

(A) Scatter plot of the standard deviation of AdjFreq by mean of AdjFrq computed from a collection of scRNA-seq datasets. Cell type markers colored in blue, RPGs colored in green, published HKGs with their votes colored in red gradient. **(B)** Normal mixture clustering of the stdev-by-mean plot at the level of four clusters, colored by clusters. **(C)** The SD-by-Mean plot of genes colored by the weighted correlation scores. Genes from normal mixture cluster 1 highlighted and subjected to re-assignment. **(D)** A two-normal-mixture fit to the weighted correlation score. Solid lines indicate the estimated peak of two populations (red: HKG peak, blue: non-HKG peak). Dashed lines indicate one dispersion away from the peak, used as boundaries for HKGs (above the red dashed line), non-HKGs (below the blue dashed line), and unknown genes (between two dashed lines). **(E)** The SD-by-Mean plot of genes colored by their assignment.

3.3.2 Three groups of HKGs with unique functions identified

Next, we further investigated the mixed population. Due to the low sensitivity of scRNA-seq, a true HKG expressed at low level might show fluctuations across the samples reflected by lower mean and higher SD of AdjFreq compared to other highly expressed HKGs. To solve this problem, we looked into the pairwise gene-gene correlations. The basic idea is that if a gene in the mixed population shows high levels of correlations with genes in the HKG population, it is likely to be a HKG. We investigated the top 50 correlated genes of each gene, the membership of these 50 neighbours (HKG, non-HKG or mixed) were integrated and weighted by the correlation and finally yielded a weighted score (**Figure 3.8C**). This score tells us whether the highly correlated neighbours of a gene are likely to be HKGs or not. The distribution of such scores shows a clear bimodal distribution. Based on such distribution, we were able to re-assign the membership of

genes in the mixed population. Eventually we got three groups of genes, with 7424 genes in the HKG group.

Lastly, we asked what biological functions do these putative HKG genes play? To address this question, we clustered the gene-gene correlations of all these HKG genes we identified into eight clusters, and extracted enriched GO:BP terms and related pathways from each cluster (**Figure 3.9**). At the first glance we were able to observe two large groups of genes that are highly correlated, one responsible for nucleic acid related functions (cluster 1 and 2), another responsible for protein related functions (cluster 5-8). Specifically, cluster 1 covered the genes involved in DNA repair, RNA transcription and post-transcriptional modifications with enriched GO:BP 'mRNA splicing', 'mRNA processing', 'DNA repair' and 'mRNA transport'. With the majority of RPGs covered in cluster 2, genes in this cluster are mainly responsible for rRNA processing and translation, reflected by enriched GO:BP 'Translation', 'rRNA processing', 'Ribosome biogenesis' and 'tRNA processing'. In other group of genes (cluster 5-8), cluster 5 had a high correlation with cluster 1 and shows extremely high enrichment score for GO:CC 'Nucleoplasm', 'Nucleus', superpathway 'Chromatin regulation/acetylation', 'Gene expression', 'Translational control', and GO:BP for both translational and transcriptional functions, indicating that cluster 5 included genes encoding proteins located in nucleus and play significant roles in regulating gene expression. Cluster 6-8 were all closely related to protein metabolism. All of these three clusters showed enriched genes in superpathway 'Vesicle-mediated transport', 'Metabolisms of proteins' and GO:BP 'Protein transport', but possibly at different steps. Cluster 6 showed significantly higher scores for GO:CC

'Nucleoplasm', 'Cytoplasm', while cluster 7 focused on 'Golgi apparatus', 'Lysosome', and cluster 8 had much higher scores for 'Endoplasmic reticulum' as well as 'Mitochondrion'. This suggests that these four gene clusters are likely HKGs responsible for protein modification and transport at different cellular compartments. Besides these two large groups, cluster 3 and 4 showed weak correlations with other HKGs, however, also represented unique functions that are widely considered to be housekeeping. Cluster 3 showed enriched functions that are closely related to metabolism, especially degradation, reflected by GO:BP terms such as 'Autophagy', 'Autophagosome maturation', and superpathway 'Lysosome'. Cluster 4 showed enriched GO terms related to mitochondrial functions that are required for cellular respiration, such as GO:BP 'Oxidation-reduction process', 'Mitochondrial electron transport', 'Oxidative phosphorylation', GO:CC 'Mitochondrion', 'Respiratory Chain', and superpathway 'Respiratory electron transport'.

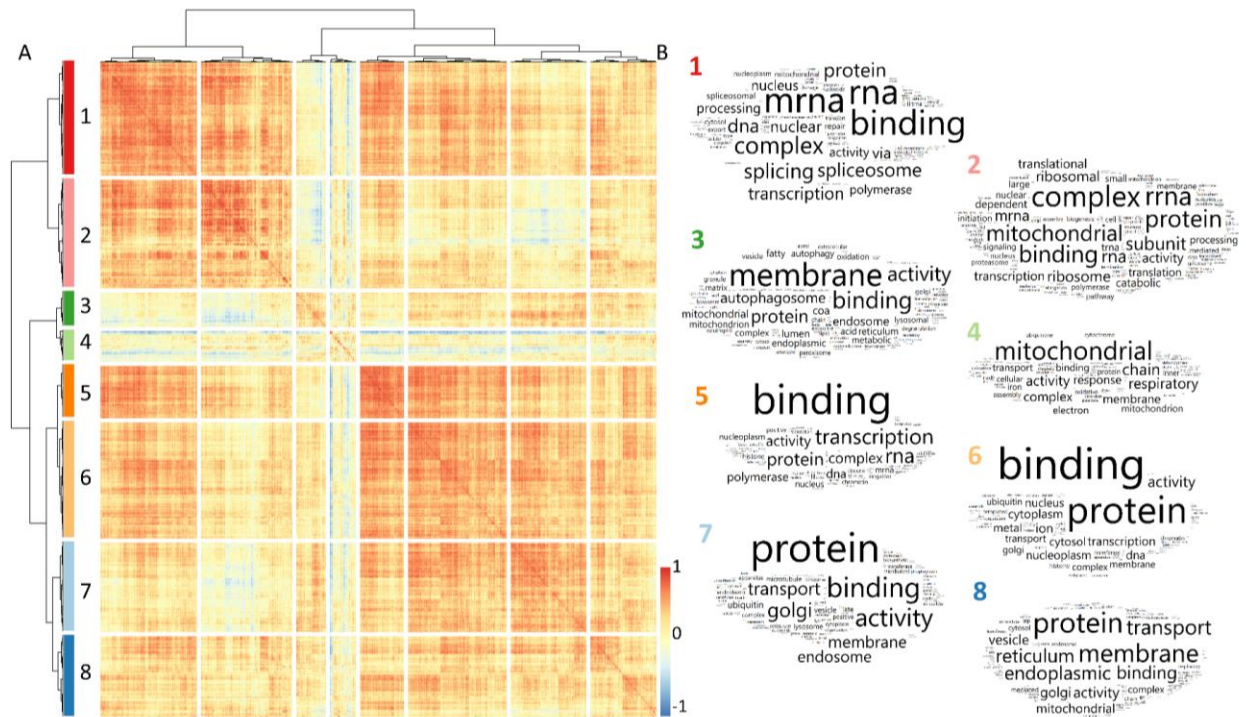


Figure 3.9: AdjFreq-derived HKGs reveal a variety of biological functions.

Clustering of putative HKGs based on gene-gene correlation shows clusters of HKGs with different functions, two large primary groups include nucleic acid related functions and protein related functions.

(A) Hierarchical clustering of gene-gene correlation from the 7424 putative HKGs in this study. Rows colored by 8 clusters (at left). **(B)** Enriched GO:BP terms from each cluster represented by word clouds.

3.3.3 HKG list identified from scRNA-seq datasets: summary and discussion

In the last part of this study, we tried to use our approach to get a list of highly confident HKGs. In our simplified model, we believe that a true HKG has the following features: high average AdjFreq across multiple datasets, and relatively low SD of AdjFreq. Based

on this, we collected a set of datasets that covered various organs and developmental stages, and computed the mean and SD of AdjFreq for every gene. By highlighting the ground truth HKGs and non-HKGs (**Figure 3.8A**) we have confirmed that they are two distinct populations on two ends of the plot. By using clustering and correlation-based modeling, we finally got a list of 7424 genes that we believed to behave like HKGs. Compared to other publications this is still a relatively long list, and we can easily shrink the list by applying arbitrary thresholds or using rankings to get top genes, but we decided to report the list and let the readers decide the resolution. We have also tried to answer the question ‘what are the biological functions of HKGs’ by analyzing the HKG list we got and clustering them into highly correlated groups (**Figure 3.9**). As expected, the clustering of HKGs showed fewer structures compared to those non-HKGs, but we were still able to visually capture three large groups. These three groups showed significant signals to nucleic acid related functions (cluster 1 and 2), cellular respiration functions (cluster 3 and 4), and protein functions (cluster 5-8), respectively. Also there was a positive correlation between cluster 1 and cluster 5-6, which might indicate the connection between transcription and translation.

Though we have tested a collection of datasets from different sources, we were unable to perform an exhaustive aggregation of all available datasets. Based on the results shown, we do believe that we are pointing at the ground truth. To move this forward, more and more datasets could be averaged using AdjFreq and an HKG list should continue to emerge. In addition, our approach might also be valuable in scRNA-seq data analysis pipeline development. A variety of imputation tools have been developed to overcome the

dropout event, for better data normalization and integration. One obvious challenge is the difficulty to find a universal solution for all datasets. Being able to determine ubiquitously expressed genes, especially such genes within a specific dataset, would potentially provide dataset-specific normalization parameters. By comparing such genes across different cell type clusters, it's also possible to find genes ubiquitously expressed in specific cell types which could be potential cell markers.

There are also remaining challenges to be addressed. First, we used the simplest definition of HKG in this study, which refers to the genes expressed in all cells regardless of their expression levels. Other branches of HKG studies might have additional requirements such as relatively stable expression in different cell types, which requires a robust normalization approach in scRNA-seq datasets. Second, we did not address the alternative splicing or any protein-level modifications, both largely extend the diversity of gene products[380]. The fast-developing multi-omics single cell techniques might provide a more comprehensive understanding in the future. Third, it is still difficult to provide a definitive list of HKGs. Using our approach we were able to identify two major populations from all genes (**Figure 3.8**), which are still quite continuous and lack a clear division. Using some arbitrary cutoffs could be an easy and straightforward solution, but not necessarily precise.

In summary, we have observed a unique gene expression pattern in scRNAseq datasets represented by the detection-by-frequency plot, and we have confirmed that genes on the edge of such curves (with high AdjFreq) present HKG-like behaviors by various approaches. Using a collection of various datasets, we have reported a list of HKGs

revealed by extracting genes with high AdjFreq, as well as the biological functions of these genes. We believe that the use of the scRNAseq dataset could shed light on the long-standing debate ‘which genes are HKGs?’.

Material and Methods

scRNA-seq datasets processing

All scRNA-seq datasets used in this study were listed in **Table S3.1**. The raw UMI count matrix was used as input, the detection frequency of each gene was determined by the number of detected cells over the total number of cells. The expression level of each gene was determined by the total number of UMIs over the number of detected cells, then log transformed.

The AdjFreq was determined as following:

Let A be an array such that $A \in R^{m \times 2}$ where m is the number of genes under consideration. The columns of this array are the average UMI counts and frequency of cells with counts above zero, denoted A_{avgUMI} and A_{freq} , where $A_{avgUMI}, A_{freq} \in R$. Re-index the sequence A in monotone increasing order with respect to A_{avgUMI} . From A , select an array of points B with monotonically increasing frequency representing the left boundary of the distribution where $B = (x_n \mid x_{n_{freq}} > \max_{freq}((x_0, \dots, x_{n-1})))$. Using the points in B as bounds, bin all the points in A such that they fall between any given pair of points $(x_{n_{avgUMI}}, x_{n+1_{avgUMI}})$. For each point, measure the vertical distance between it and the line formed by the two boundary points. The fractional distance of this point to the

boundary becomes the new adjusted frequency. m denotes the slope between the two points and b denotes the intercept.

$$p_{AdjFreq} = \frac{p_{freq}}{p_{avgUMI} * m_{x_n, x_{n+1}} + b_{x_n, x_{n+1}}}$$

GSEA on gene lists and network analysis

The GSEA was performed using ToppCluster (<https://toppcluster.cchmc.org/>). The GO:BP was set as the target query database, with Bonferroni as the correction method and p-value cutoff at 0.05. To make the word cloud, all significant hits for one cluster were collected and split into words; common words ‘process’ and ‘regulation’ were removed for clear representation. Word clouds were made in JMP® (version 14.0), with font size proportional to word counts. To perform network analysis, the network generator output was selected from ToppCluster to generate an XGMML file and analyzed in Cytoscape[391]. The gene sets (bins) and returned GO:BP were used as nodes, significant hits were edges between GO:BP and gene sets, the GO:BP nodes were colored by the number of connected nodes. To better understand the unique functions of HKGs, the GSEA on the HKGs identified in this study shown in **Figure 3.9** was performed using GeneAnalytics (<https://ga.genecards.org/>)

Reference HKG lists

A total of 5 published HKG lists used in this study were listed in **Table S3.2**. Zhang et.al provided a transcript-based list, which was converted to genes using the author provided conversion table. Ramskold et.al and Eisenberg et.al provided human gene lists, which

were converted to mouse genes using vertebrate homology database from MGI (<http://www.informatics.jax.org/homology.shtml>). All gene IDs/symbols were converted to current gene symbols using MGI batch query tool (<http://www.informatics.jax.org/batch>). Genes appear more than once after conversion were counted only once, only protein coding genes were used for downstream analyses.

Chromatin accessibility data processing

Data of chromatin accessibility collected from Cusanovich et.al[390]. Binarized activity score matrix was used as input (20,782 genes by 81,172 cells), the activity score of each gene was computed by the sum across all cells. The gene symbols were converted as described above. All genes were sorted by their activity scores and divided into 10 bins with equal size to compare AdjFreq (**Figure 3.7B**).

HKG identify assignment based on a collection of dataset

All datasets shown in Table S1 were used to compute AdjFreq. Genes detected in more than one dataset were used in downstream analysis. The mean of AdjFreq and the standard deviation of AdjFreq across all datasets were plotted in **Figure 3.8A**, then clustered using normal mixtures clustering. All genes were clustered into four populations, two of which were overlapped and covered the lower left corner therefore pooled into one as the non-HKGs (cluster 2 in blue), another cluster as HKG (cluster 3 in red), and the cluster in between as mixed (cluster 1 in green) (**Figure 3.8B**). The gene-gene correlations were computed from whole embryo datasets E9-E11.5 to avoid possible batch effects. Specifically, the UMI matrices of these sets were inner joined first then

normalized by SCT transformation using R package ‘Seurat’[392],. The joined normalized dataset was then transposed to treat genes as samples, and PCA used to reduce the dimensionality of cells into 100 PCs. Pairwise correlations then computed from the gene-by-PC table. The top 50 highly correlated genes were chosen for each gene as its neighbours. A score Sn of each cluster shown in **Figure 3.8A** was computed for each gene based on its top 50 neighbours as the following:

$$Sn = \sum_{i=0}^k Prob_i Corr_i^2$$

$$S = S3 - (S1 + S2 + S4)$$

Here Sn represents the score for cluster n , k is the number of neighbours in cluster n , $Prob_i$ is the probability of neighbour i in cluster n based on normal mixture model, $Corr_i$ is the gene-gene correlation between neighbour i and target gene. S is the overall weighted score for the target gene. The distribution of weighted scores were then fitted to a two-normal mixture model. Genes within one dispersion from the top population were considered to be HKG, genes within one dispersion from the bottom population were considered non-HKG, and genes in between were considered unknown (**Figure 3.8C**). We used such criteria to re-assign the identity of genes from the mixed population (C1 of **Figure 3.8B**), which gave 7424 HKGs, 10344 non-HKGs, and 1629 unknowns.

Chapter 4: Metagene: a scRNA-seq-based new approach to systematically understand gene functions

Introduction

A key objective of system biology research is to systematically understand gene functions and how different genes interact. It has been a long history that classical genetics works from the phenotype to determine the function of specific genes. Or reversely, starting from specific genes and looking for related phenotypes so that the functions could be inferred[393]. Through decades of hard work, scientists have gained knowledge about thousands of genes through genetic approaches, such as in vitro mutagenesis and gene knockdown/knockout. However, our understanding of the genome is still limited. Of all the protein-coding genes in mice, a large number of genes have no experimental-based annotations[394]. Even for the annotated genes, many of which have only a few Gene Ontology (GO) terms related, indicating a lack of insights into gene functions. Having a systematic approach that helps predict gene function is crucial to understanding our genome.

Nowadays, with the development of genome technology and high throughput sequencing, we have access to massive genomic data which enables us to analyze the secret of gene expression at a higher level. Furthermore, we are able to study the interactions between different genes. One of the attempts scientists have made is investigating patterns of gene expression. Back in 1998, scientists confirmed that by clustering gene expression data from microarrays, genes with similar functions could be clustered together[395]. After that, various approaches based on gene co-expression have been developed to study conserved genetic modules[396], find biological relevant genes to predict gene

function[397–400], characterize cellular state[401], and study gene evolution[402]. A human RNA-seq-based gene co-expression database has also been established[403]. Attempts have been made to use co-expression from microarray-based data to predict gene functions, with great success[404]. Such approaches to understanding gene functions from sequencing data have been nicely summarized[405].

However, these methods have certain limitations in their ability to capture full structures of the data, and the bulk RNA seq based expression level has been averaged across all sample cells thus considered as low resolution. Recently, with scRNA-seq technology, it is possible to overcome these hurdles by obtaining sequencing data with a larger scale and higher resolution. From the scRNA-seq, we are able to get transcriptional profiles at single-cell resolution, thus able to cluster similar cells into groups. Knowing that the input data frame for the scRNA-seq analysis pipeline is a cell-by-gene table, we hypothesize that using the same rationale, we are able to cluster genes playing similar roles into groups. Certain challenges remain to be overcome: at the single-cell level, fewer genes are expressed compared to ~10k genes per cell obtained from batch RNA-seq[358] due to a variety of reasons such as stochastic expression[10], transcriptional bursting[381,406], and technical limitations like dropout event[26].

Inspired by cell-oriented studies described in previous chapters and related work in gene co-expression networks, we find that by transposing the expression matrix and considering genes as samples while cells as variables, we are able to project genes onto a reduced dimensional space just as we did for cells, therefore cluster genes with similar roles into groups called ‘metagene’. Based on the known genes within each metagene

together with other evidence, we could annotate the biological functions of metagenes just as we did in cell type annotation in most scRNA-seq studies. This will provide a systematic approach that helps to understand gene functions.

Here we introduce a complete pipeline to extract co-expressed gene clusters (metagene) from the E11.5 mouse embryo scRNA-seq dataset, with the biological insights of these metagenes. Also, we provide examples of using metagenes to understand gene functions related to cell cycle regulation and find novel cell type markers.

Result

4.1 Setup metagene extraction workflow based on canonical scRNA-seq pipeline

Inspired by scRNA-seq, we noticed that the output of most pipelines was a scatter plot of cells in a reduced dimensional space (such as tSNE or UMAP), with transcriptionally similar cells grouped into clusters. Considering that the input data frame for such output was a 2D expression matrix with genes and cells as columns and rows, we hypothesized that by swapping the ‘variable’ and ‘sample’, we are able to create a similar scatter plot with genes as data points rather than cells. Unlike the classical scRNA-seq in which we have to perform differential gene expression to extract genes differentially expressed to understand the identity of each cell cluster, in this gene plot we can directly apply bioinformatics tools to our gene clusters. With some well-studied genes in the clusters, we are able to infer the biological functions that this gene cluster may play, and eventually provide opportunities to understand functions on unknown genes. Here we name the gene clusters identified by this approach ‘metagene’, referring to a group of genes that play related biological roles.

The frame of metagene extraction is similar to cell cluster extraction from the canonical scRNA-seq pipeline, generally includes the following parts:

- I. Data normalization
- II. Dimension reduction
- III. Clustering
- IV. Biological interpretations

Based on this framework, we start with a transposed expression matrix with genes as 'samples' and cells as 'variables'. A few modifications were made to achieve better quality metagenes.

To test the quality of metagene extraction with different modifications, we first identified three marker gene groups that are considered highly correlated:

- a). 76 ribosomal protein genes
- b). 42 G1/S phase marker genes
- c). 54 G2/M phase marker genes

The ribosomal protein gene set is ubiquitously expressed in all cells at a high level, gene-gene correlation within this set could be used to assess library quality. The other two sets of genes are also essential for proper cellular functions. If the metagene is properly extracted, the gene-gene correlation within each set should also be high. In our preliminary test to evaluate different variables in steps I and II, metagenes were extracted by Ward Hierarchical clustering of resulting tSNE coordinates, gene set enrichment

analysis (GSEA) was then applied to all extracted metagenes, and number of enriched Gene Ontology (GO) terms were compared across different variables we tested. We focused on the gene-gene correlation within the ribosomal protein gene and cell cycle marker group, as well as the number of GO:BP terms returned from GSEA results, to determine if certain modifications give better results.

The major goal of this study was to understand gene co-expression in early embryonic organogenesis. We decided to use the E11.5 mouse whole embryo scRNA-seq dataset as the starting point. To ensure the best quality of the input dataset, we selected the set with a newer version of chemistry mentioned in Chapter 2, which in theory is a biological replicate of the dataset used in Chapter 1, but with higher sensitivity and better library qualities. The details of the selection of the input dataset will be further discussed in section 4.1.4.

The key steps of the workflow are described below:

4.1.1 Normalization data for MG extraction

Since sequencing depth and sensitivity vary between libraries, normalization is one important step in the scRNA-seq pipeline to remove such bias[407]. The significance of such normalization has also been emphasized in past gene co-expression network studies based on microarray data. It has been addressed that different normalization methods show comparable distribution of expression and the network performance shows minor differences among methods[408], while another study pointed out that bias based on expression level has been observed in co-expression analysis[409].

To test if normalization could improve the quality of metagene, we compared results from the dataset processed in three different ways: non-normalized, median log-normalized[38], SCT-normalized[255]. In the non-normalized trial, the raw gene-by-cell UMI matrix was directly transposed and subjected to downstream analyses. Log-normalization is one of the most straightforward but powerful approaches in scRNA-seq. In this method, the UMI matrix was first scaled towards a scaling factor (the median library size in this study) to exclude bias from different library sizes, then log-transformed and transposed as input for downstream analyses. SCT-normalization is a model-based method that is specifically designed for scRNA-seq datasets using regularized negative binomial regression, which has been addressed to successfully remove the influence of technical bias but preserve biological heterogeneity[255]. In this method, we used the SCTransform function from the R package 'Seurat' to perform transformation and scaling[27], then send the transposed output to downstream analyses. To test the effect of expression level on our metagenes, we projected the expression level of each gene in our reduced dimensional space to check if clustering has been affected by such variables. Since the effectiveness of normalization could only be measured by results from the downstream analyses, we performed dimension reduction introduced in the next section. tSNE projections of all three sets were made following PCA, then colored by Ward hierarchical clustering of SCT-normalized dataset (**Figure 4.1A-C**). Based on colored tSNE projection, the structure of some isolated clusters shows good concordance across all three sets, while the body part of the non-normalized dataset shows a significantly different structure compared to the other two. For gene-gene correlation, the ribosomal

protein genes showed high correlations in all three sets, but cell cycle genes correlation from non-normalized data showed a much longer tail compared to normalized datasets (**Figure 4.1D**). Then metagenes were extracted from each dataset from the corresponding tSNE and evaluated by GSEA (**Figure 4.1E**). Again, the non-normalized dataset showed both low mean and median number of GO:BP terms returned. In conclusion, we believe that normalization is necessary to remove bias between different libraries for metagene extraction. Using SCT compared to canonical log normalization only presents marginal differences based on our tests, but since it provides a better theoretical foundation and tighter clusters, we chose SCT normalization in our workflow.

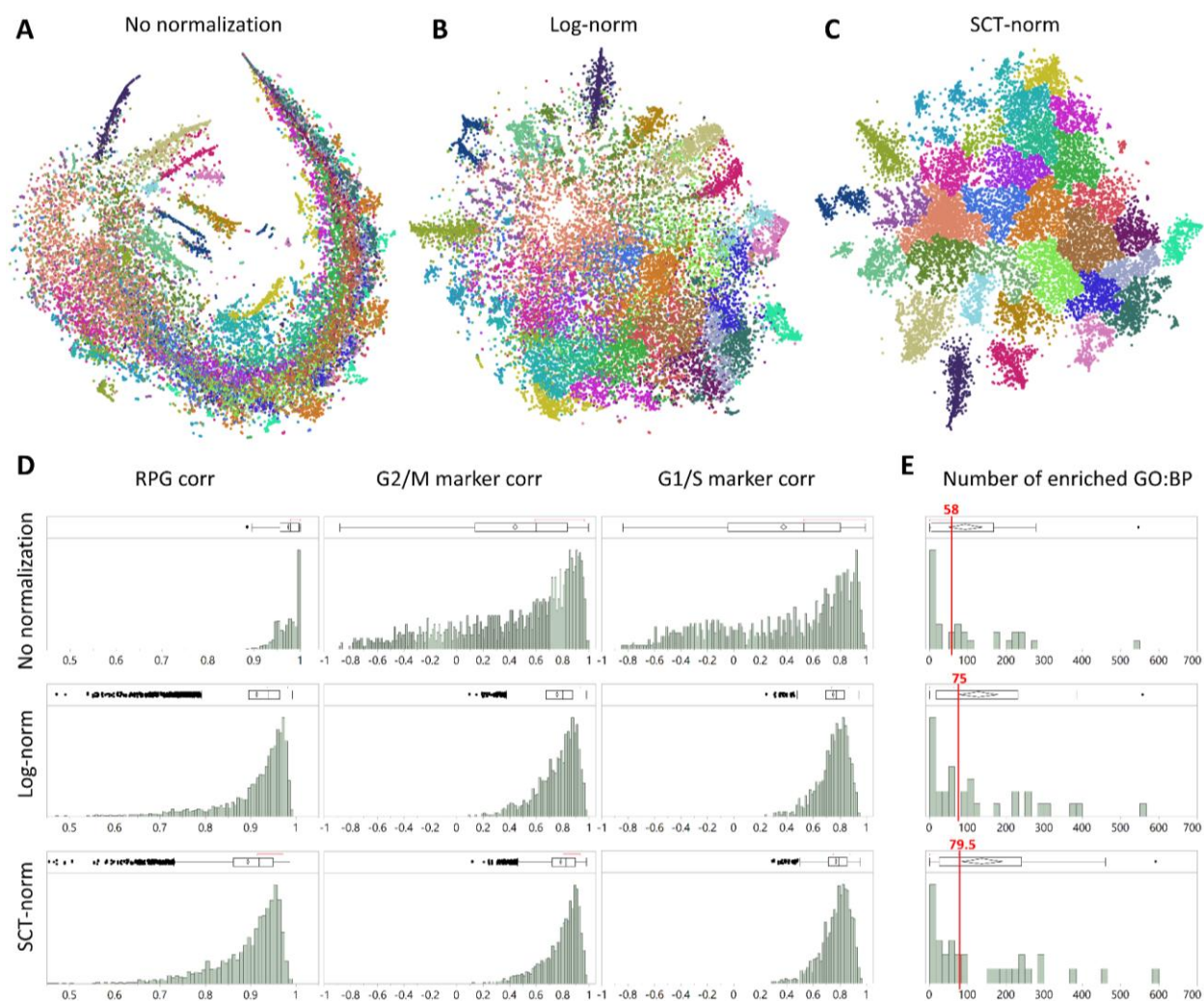


Figure 4.1: Comparison of different normalization approaches used in metagene extraction

SCT-norm outperforms non-norm and log-norm in extracting metagenes from E11.5 mouse whole embryo dataset, by analyzing benchmark gene correlation and enrichment analysis results.

(A-C) tSNE projection of genes with no normalization, log normalization, and SCT normalization. Genes are clustered in tSNE space of SCT normalization into 32 clusters by Ward Hierarchical clustering, then projected onto all three plots. **(D)** Histogram of gene-gene correlation of ribosomal protein genes (left panel) and cell cycle marker genes (middle and right panel), extracted from no normalization (top), log normalization (middle) and SCT normalization (bottom). **(E)** Histogram of number of significantly enriched

GO:BP terms for each cluster, extracted from no normalization (top), log normalization (middle) and SCT normalization (bottom). Clusters were determined by 32 Ward Hierarchical clustering from each tSNE space, and enriched GO terms evaluated by Topcluster. GO terms with adjusted p-value < 0.05 presented, median number highlighted by red line.

4.1.2 Dimension reduction and visualization

In the canonical scRNA-seq pipeline, various approaches have been used to reduce the dimensionality of the input UMI matrix, major variants involve Principal Component Analysis (PCA)[33], diffusion map[34], Independent Component Analysis (ICA)[410], and Non-negative matrix factorization (NMF)[411]. The general purpose of this step is to reduce noise and ease the computational workload. In a classical scRNA-seq based study, thousands of cells will be sequenced and expression levels of over 20,000 genes will be presented in the original expression matrix. With each gene as a dimension, a huge computational burden is expected. However, not all these dimensions are valuable, some of the genes might be playing similar roles, while some others are detected at extremely low levels and should be considered noise. The dimension reduction step aims at reducing the high dimensionality of all genes into a manageable number of lower dimensions, while keeping most of the biological heterogeneities. In our study, the UMI matrix was transposed after normalization, with cells as variables instead of genes. Since the goal is to study genes, we reduced the dimensionality of cells so that similar cells within one cell type would be collapsed. It's worth mentioning that the number of cells in our dataset is roughly the same as genes. From a variety of options, we chose PCA as

our primary dimension reduction tool, which has been widely used in other scRNA-seq studies.

A secondary dimension reduction step is often applied for visualization purposes. Usually tens of PCs might be kept after PCA, though this might be a manageable number for the computer, it is hard to visualize and get an intuitive understanding of the dataset for humans. In this step, a non-linear method such as t-distributed stochastic neighbor embedding (tSNE)[40] and Uniform manifold approximation and projection (UMAP)[42]. These methods will further reduce the dimensionality from tens of PCs to 2 or 3 dimensions to be visualized in a scatter plot. Knowing that the UMAP emphasizes more on global structure and tSNE spreads the local structure more (see Chapter 2 for more details), we selected tSNE as our secondary dimension reduction tool to generate a 2D scatter plot of genes.

Before performing tSNE, it is important to determine the number of PCs used for downstream processing, of which the most widely used heuristic approach is ranking the percent variance explained by each PC. Usually, the number of PCs is dataset-dependent and should have explained a large percentage of variance. To optimize the number of PCs used in our metagene extraction workflow, we compared tSNE structures computed from different numbers of PCs. By projecting Ward Hierarchical clustering assignment from 100 PC derived tSNE to other tSNE space shows no significant difference and vice versa (**Figure 4.2A-C**). However, when we dive into each individual cluster, we find examples that are only identified when more PCs are used (**Figure 4.2D-F**). Further GSEA on these clusters showed the genes have certain GO:BP terms enriched,

indicating that these genes were not random noise. For example, one cluster enriched in spermatogenesis GO:BP (enrichment score >48) can only be identified when 100 PCs were used (**Figure 4.2D-F**, blue cluster). Another cluster enriched in cell cycle related GO:BP (enrichment score >166) also did not isolate if only 25 PCs were used (**Figure 4.2D-F**, red cluster). However, the gene-gene correlation based on the number of PCs used did not present a significant difference (**Figure 4.2G**), as well as the number of enriched GO:BP terms returned (**Figure 4.2H**).

Although determining the dimensionality of a dataset is challenging and subjective, we concluded that compared to the canonical scRNA-seq pipeline, more PCs are needed to precisely capture functionally related genes (100PCs used in this study).

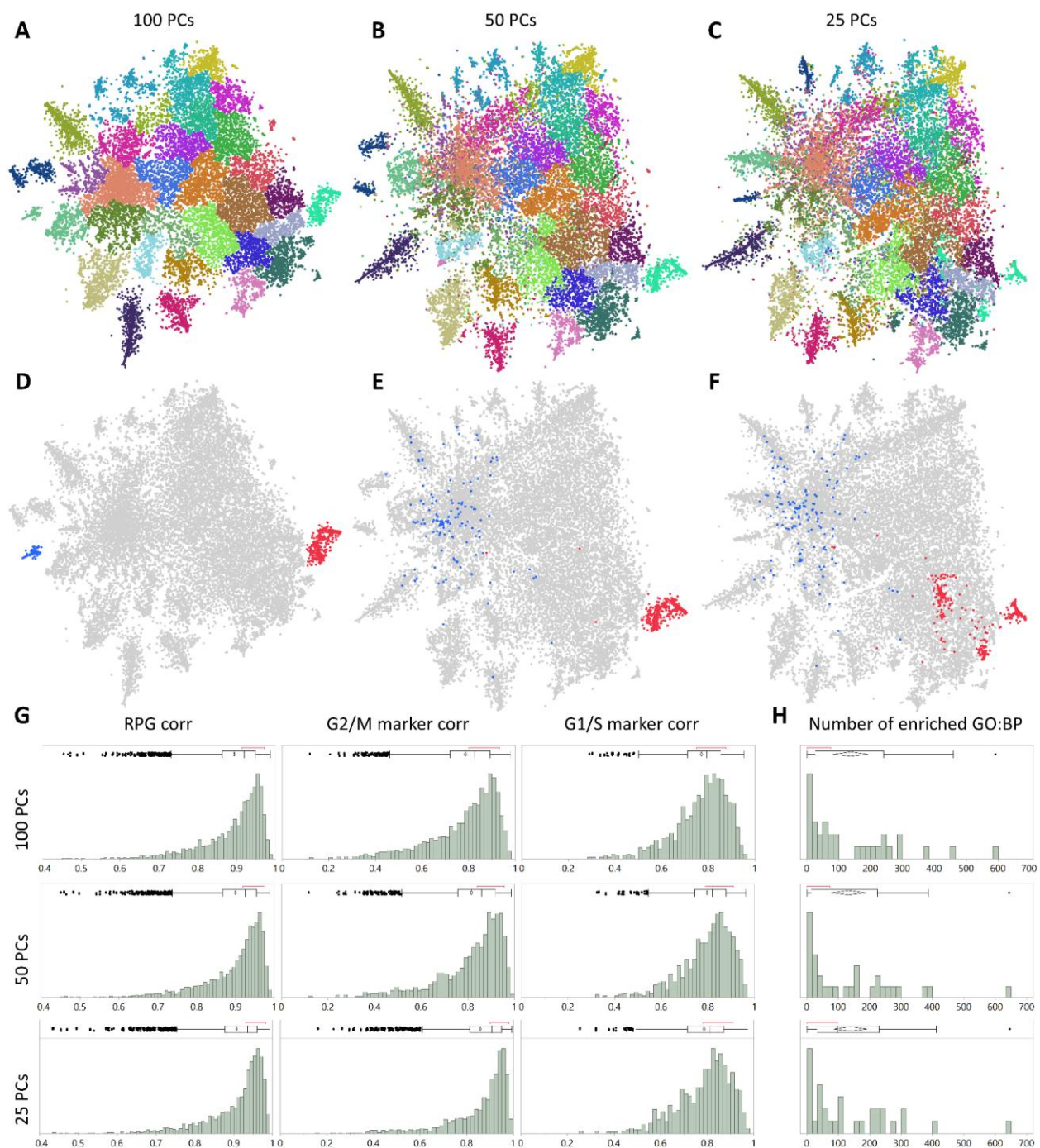


Figure 4.2: Optimization of the number of PCs used

Decreasing the number of PCs used in metagenome extraction causes loss of informative metagenes, with non-significant changes of gene-gene correlation and enrichment analysis results.

(A-C) tSNE projection of genes generated from 100 PCs (A), 50 PCs (B) and 25 PCs (C). Genes were colored by the 32 Ward Hierarchical clusters from 100 PCs, then projected onto each plot. **(D-F)** Two representative clusters are lost when reducing the number of PCs from 100 (D) to 50 (E) and 25 (F), colored in blue and red. **(G)** Histogram of gene-gene correlation of ribosomal protein genes (left panel) and cell cycle marker genes (middle and right panel), extracted from no normalization (top), log normalization (middle), and SCT normalization (bottom). **(H)** Histogram of the number of significantly enriched GO:BP terms for each cluster, extracted from no normalization (top), log normalization (middle), and SCT normalization (bottom). Clusters were determined by 32 Ward Hierarchical clustering from each tSNE space, and enriched GO terms were evaluated by Topcluster. GO terms with adjusted p-value < 0.05 presented.

4.1.3 Clustering of genes is different from clustering of cells

Once the dimensionality of the input dataset is properly reduced and visualized in tSNE space, the next challenge is to put genes into meaningful groups by clustering. Clustering approaches for cells have been introduced in Chapter 1, but a few modifications are needed for clustering genes.

First, the reduced dimensional space of genes is heavily biased by gene expression and detection level (**Figure 4.3A-B**), genes with low expression/detection form a cloud in the left-center while genes with high expression detection form another continuous cloud on the other side. As functionally related genes don't necessarily have the same expression/detection level, clustering in PC space is also biased (**Figure 4.3C**). After several attempts, we still could not find an effective approach to correct such bias in upstream steps.

Second, the structure of genes in tSNE space is much more complicated than that of cells, with a large number of continuous data points presented. Given that functional gene

groups could be varied in size, canonical Ward Hierarchical clustering in tSNE space does not work well (**Figure 4.3D**).

Instead of clustering in the reduced dimensions (PC or tSNE), we choose to compute the pairwise correlations of genes from PCs and cluster on such correlations, after comparing the results from all these approaches. To evaluate the quality of clustering based on different inputs, we measured the number of GO:BP terms returned by each trail; more enriched GO:BP terms were captured in correlation-based clustering compared to clustering in PC or tSNE space (**Figure 4.3E**). Other than that, we used a modified Ward Hierarchical clustering approach named Iterative Binary Splitting (IBS) to perform clustering (**Figure 4.3F**). In this approach, the dataset is split by Ward Hierarchical clustering into two clusters, each resulting cluster is then split again by Ward Hierarchical clustering, iteratively. By doing so we will get 2^N clusters, N represents the number of iterations. There are several advantages of IBS compared to canonical Ward:

1. No pre-assumption of cluster numbers is needed.
2. Each split is an independent event and will not be affected by other clusters.
3. Flexible cluster size and stopping point are possible. Splitting of one branch could be stopped while other branches keep splitting.

Here we applied IBS to our dataset with up to 10 iterations (1024 clusters). The clustering result was then trimmed by manually assigning stop points to each split, based on the following general rules:

1. If two daughter clusters show similar GO:BP enrichment results, they should be stopped at the mother cluster.
2. If both daughter clusters show significantly lower GO:BP enrichment p-values, they should be stopped at the mother cluster.
3. For hard-to-determine clusters, genes from the enrichment analysis were manually inspected and determined if further split is necessary.
4. After determining the preliminary terminal nodes, we made biological interpretations of each node and further modified the merging and splitting (Refer to 4.1.5 about how we determined the identities of all metagenes).

We finalized with 63 gene clusters, which showed improved GO:BP enrichment results compared to Ward (**Figure 4.3G**).

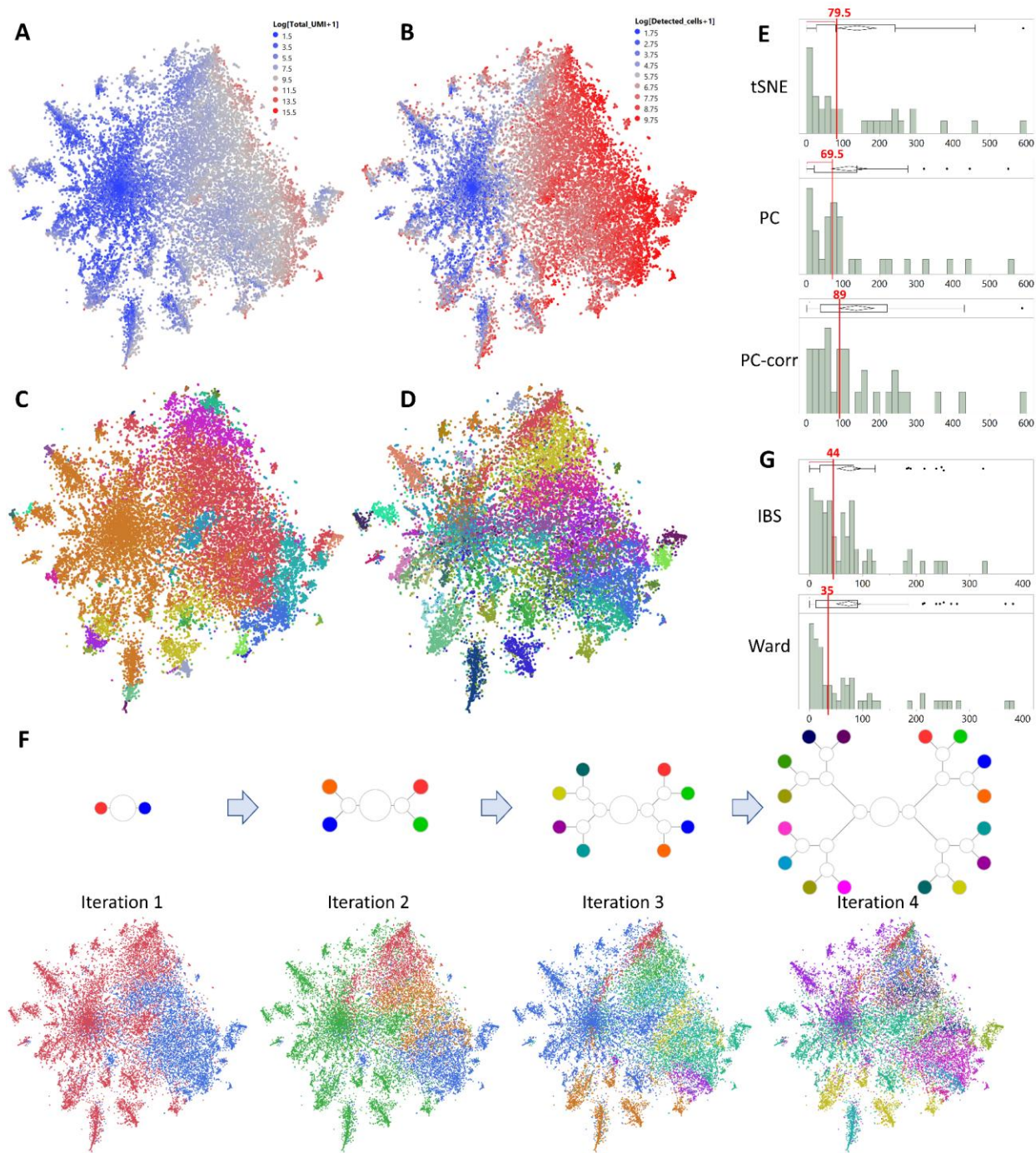


Figure 4.3: Comparison of different clustering approaches in metagene extraction
tSNE projection of metagene is affected by gene expression and detection level. To best cluster such a dataset, we use IBS, a modified hierarchical clustering based on PC correlations, which outperforms other clustering approaches.

(A) tSNE projection of genes colored by $\log(\text{UMI} + 1)$. **(B)** tSNE projection of genes colored by $\log(\text{Detected cells} + 1)$. **(C)** tSNE projection of genes, colored by clustering in PC space using Ward Hierarchical clustering. **(D)** tSNE projection of genes, colored by clustering in tSNE space using Ward Hierarchical clustering. **(E)** Histogram of the number of significantly enriched GO:BP terms for each cluster, extracted from clustering in tSNE space (top), PC space (middle) and PC-correlation space (bottom). Clusters were determined by 32 Ward Hierarchical clustering from each tSNE space, and enriched GO terms were evaluated by Toppcluster. GO terms with adjusted p-value < 0.05 presented, median number highlighted by red line. **(F)** Schematic view of IBS clustering, with four iterations. **(G)** Histogram of the number of significantly enriched GO:BP terms for each cluster, extracted from clustering in PC-correlation space using IBS 63 clusters (top) and 64 Ward hierarchical clusters (bottom). Enriched GO terms were evaluated by Toppcluster. GO terms with adjusted p-value < 0.05 presented, median number highlighted by the red line.

4.1.4 Highly heterogeneous dataset is needed for metagene extraction

It is known that certain genes are only expressed in specific cell types, therefore cannot be identified if there's no such cell type in the dataset. In canonical scRNA-seq analysis, we have the entire transcriptome sequenced so that all cells could be annotated by their transcriptional profiles. In our study, the UMI matrix was transposed as our input, if the same rationale could be applied here, then in theory all cell types should be sequenced to annotate the entire transcriptome. Considering this, the sequencing dataset we use should cover as many cell types as possible.

To verify this, we performed metagene extraction on a dataset with only red blood cells (RBC), compared to our E11.5 whole embryo dataset. Considering reduced variance in the less heterogeneous dataset, we tested both 25 PCs and 100PCs. Compared to the whole embryo dataset, the tSNE projection from the RBC-only dataset showed fewer structures. Specifically, the cluster-like structures seen in the RBC projection were not

real meaningful functional-related genes based on enrichment results, which came from tSNE artifacts when inappropriate dimensions were selected (**Figure 4.4A-B**). Lots of genes were dropped because of low expression/detection in RBC (**Figure 4.4C**), and further analysis of enrichment results from the RBC dataset did not identify gene clusters with meaningful functions other than only a few housekeeping gene clusters (**Figure 4.4A-B**, arrow of RPG cluster). As shown in **Figure 4.4D-E**, only RPG correlations were kept at a high level, while the cell cycle markers correlations dropped significantly using only RBC as the source data, and the enrichment analysis results from RBC only dataset were much worse. We have also tested a whole embryo dataset at an earlier stage of development. As we expected, with less differentiated cell types in this dataset, we were not able to extract adequate meaningful metagenes (data not shown).

In conclusion, in order to classify genes into functional groups, a highly heterogeneous dataset is required. To prevent possible batch effects when combining datasets from different sources, the whole embryo sequencing would be ideal.

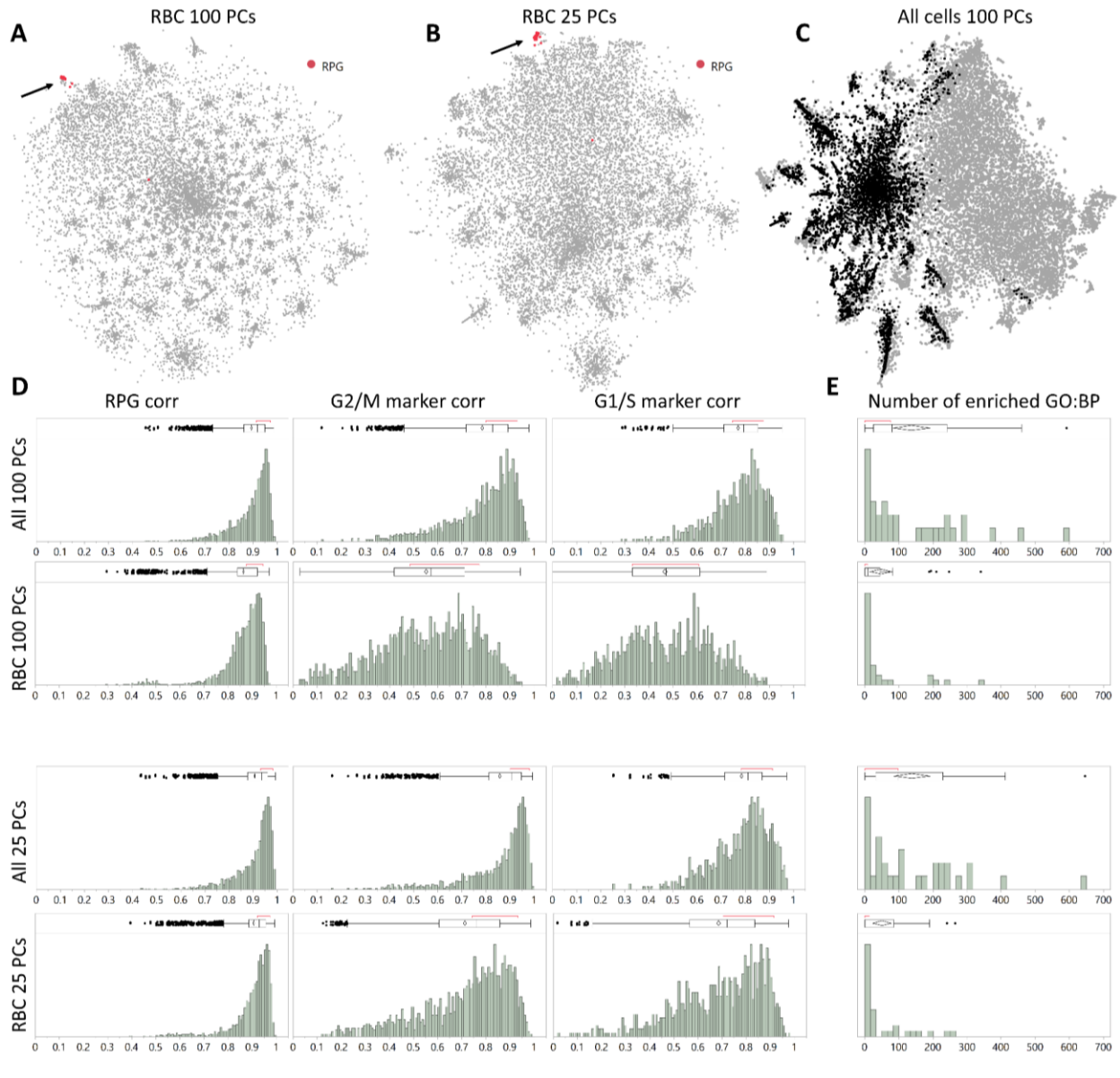


Figure 4.4: Highly heterogeneous dataset is needed for metagene extraction

Dataset with low heterogeneity (RBC) results in uninformative metagenes, and inappropriate selection of PCs causes fake clusters. To successfully extract metagene, highly heterogeneous datasets are preferred.

(A-B) tSNE projection of genes from only RBCs, using 100 PCs (A) and 25 PCs (B). (C) tSNE projection of genes from all cells, highlighted genes were not detected in RBC, therefore, dropped in A and B. (D) Histogram of gene-gene correlation of ribosomal protein genes (left panel) and cell cycle marker genes (middle and right panel), extracted

from all cells with 100 PCs(top), and only RBC with 100 PCs (bottom). **(E)** Histogram of the number of significantly enriched GO:BP terms for each cluster, extracted from all cells with 100 PCs(top), and only RBC with 100 PCs (bottom). Clusters were determined by 32 Ward Hierarchical clustering from each tSNE space, and enriched GO terms were evaluated by Topppcluster. GO terms with adjusted p-value < 0.05 presented, median number highlighted by the red line.

4.1.5 Biological interpretation

Once the clustering has been determined from the dataset, the cluster needs to be annotated with biological meanings. Unlike the cell cluster annotation in most scRNA-seq pipelines, in this study, we were trying to understand the functions of the gene clusters rather than cell type identities of the cell clusters. Since the clustering has been applied directly to genes, there's no need to perform any analysis to find significant genes such as differential gene expression (DGE), which is widely used in most scRNA-seq studies. Instead, we could directly query the genes within each cluster to find related biological signals through enrichment analysis. There are multiple databases we can use to query our metagenes, such as cell type marker gene databases which tell us what cell type might be the metagene expressed, gene-gene interaction databases which tell us how the genes within each metagene interact with each other, and gene ontology databases which contain gene annotations from three different categories (see Chapter 1 for more details). In the first round of biological interpretation, we queried GeneAnalytics (see Chapter 1 for more details) to get enrichment analysis results on both cell type and GO:BP. These results were used in the previous step to determine the stopping point for the IBS hierarchical tree, by using the p-values as references. To get biological insights of the

metagenes, we made word clouds/phrase clouds from each enrichment analysis for visualization purposes, and then manually inspected the enriched terms. The font sizes of word clouds were proportional to the log-transformed p-values, indicating the level of association of the term with queried genes. A preliminary conclusion was made based on the top hits returned by the enrichment analysis. By analyzing all the enrichment analysis results, all the metagene could roughly be divided into two categories: housekeeping genes and cell type specific genes.

Three approaches were used to further understand the metagenes. First, we looked into the gene lists of each metagene, and manually checked the well-studied genes such as cell type marker genes, key regulators of pathways, and famous genes for specific biological processes. Based on such genes (and genes related to such genes) from each metagene, we then searched the literature for more details and summarized the general biological functions of the metagene. For example, metagene 32 (MG32) with 330 genes highlights part of one specific arm of the tSNE projection (**Figure 4.5A**), and the returned enrichment analysis showed strong signatures pointing to liver functions, indicated by liver associated cell types in cell type query (hepatocytes and hepatoblasts), as well as liver metabolic associated GO:BPs such as 'Lipoprotein metabolic process', 'Lipid metabolic process', 'Cholesterol homeostasis' and so on (**Figure 4.5B**). By further looking into the members of the metagene, we found multiple hepatocytes markers including *Afp*, *Alb*, and *Ttr*[318,319], as well as key regulators in the liver metabolic process, such as apolipoprotein *Apoc2* and *Apoa1* in cholesterol metabolism[412], and acyl-coA thioesterase *Acot1* and *Acot4* in lipid metabolism[413].

Second, we also obtained markers for specific cell types or key regulators for a specific biological process from the literature and projected them onto the gene tSNE plot to compare the overlapping with our metagenes. In this case, we've collected 18 hepatocytes markers, with 16 of them located on the same region of our MG32 (**Figure 4.5C**).

Third, we took advantage of the cell plot generated from the same source dataset and made interpretations of the metagenes (described in the following section).

In summary, we developed this semi-automated approach to gather biological signatures for each metagene and manually inspected these signatures to find related biological functions. Since the interpretation is a subjective process, three researchers performed the interpretation independently, and then discussed the unmatched metagenes to reach a consensus.

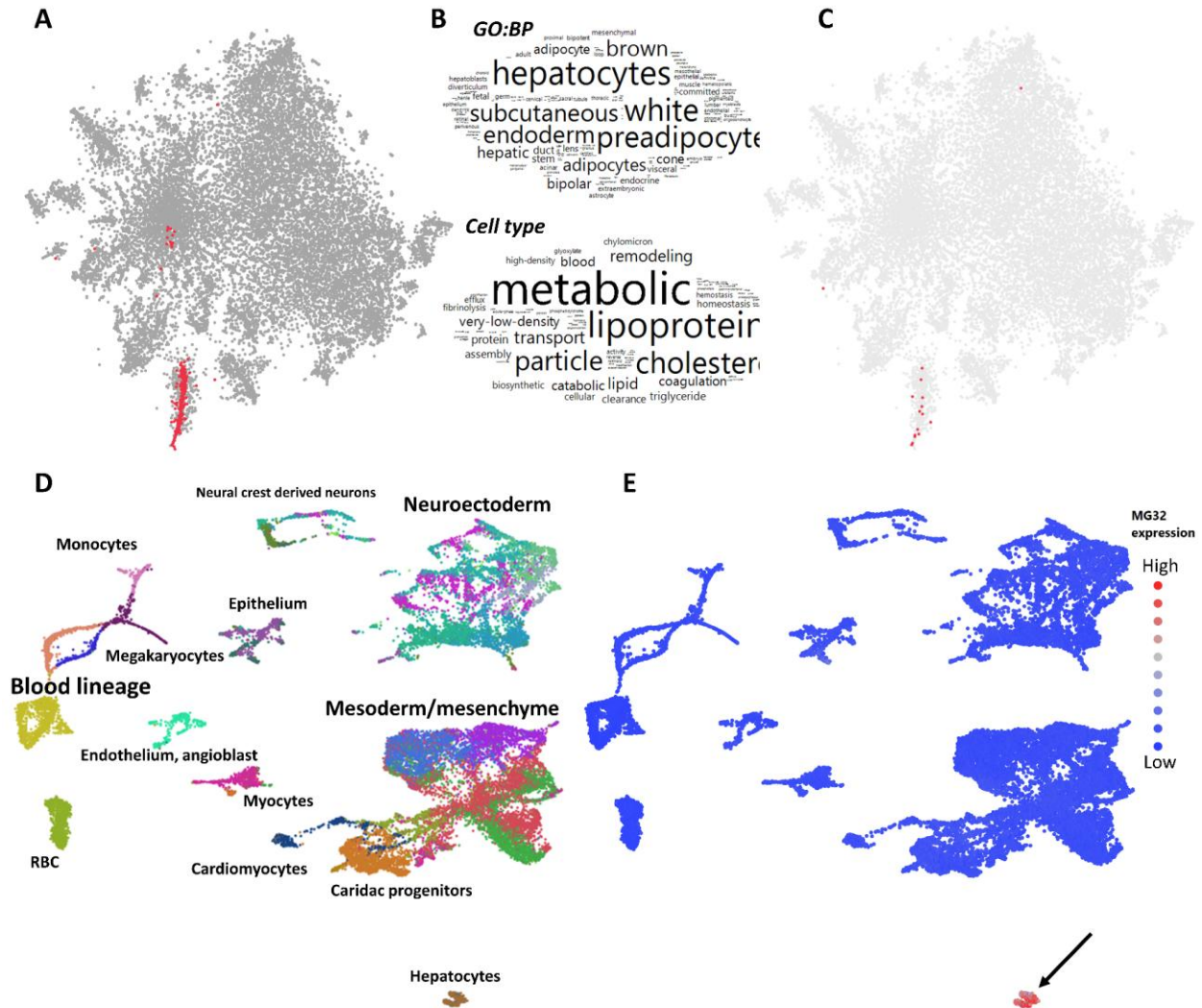


Figure 4.5: Biological interpretation of metagene functions

Bioinformatics database query, marker gene lists, and annotation of cells from the same source dataset help the interpretation of metagene.

(A) tSNE projection of metagene extracted from E11.5 mouse whole embryo, metagene 33 highlighted in red. (B) The word cloud is based on GO:BP (top) and cell type (bottom) from genes within metagene 33. (C) The cell plot from the source dataset was used for the metagene extraction. Colors indicate different cell types, with brief text labels. (D) The expression level of genes from metagene 33 projected onto the cell plot.

4.1.6 Cell cluster annotation helps to identify cell type specific genes

In addition to the bioinformatics approaches that target directly at the metagenes, we've also taken advantage of the widely used canonical scRNA-seq pipeline which targets cell type identification. By looking into the expression levels of each metagene in cell clusters extracted from the same source data, we were able to relate the metagene to specific cell types, especially helpful to interpret the metagenes that play cell type specific functions.

To be more specific, first, we applied the pipeline described in Chapter 1 to process the input dataset for this study, to get a cell plot with annotations. As mentioned before, here we used the E11.5 dataset with a newer version of chemistry compared to the one used in Chapter 1, though in theory they are biological replicates and should result in comparable cell type clusters. The cell tSNE plot generated from the newer version E11.5 dataset turned out to be topologically similar to what we got from the older version (**Figure 4.5D**). The dataset contained two large continuous clouds which represent the developing mesoderm and neuroectoderm, as well as more differentiated cell types represented as isolated islands. A one-to-one comparison was conducted between this dataset and the one we presented in Chapter 1, with comparable results (data not shown). Second, to explore the relationship between the cell clusters and our metagenes, we computed the expression level of each metagene by summing up all genes within the metagene and projected the expression level onto the cell plot. By examining the expression across different cell clusters, we were able to determine if a metagene is related to a specific cell type. Following the example of MG32 described in the previous section, we have already concluded that this metagene showed strong signatures related to liver metabolic

processes and liver-associated cell types, by looking at the expression level of these genes in different cell types we could obtain another orthogonal evidence of the biological functions. By projecting this metagene onto the cell space, we observed explicit expression in one specific cell cluster, which we annotated as hepatocytes (**Figure 4.5D-E**). This further confirmed that MG32 includes genes that are responsible for liver metabolism, hepatocytes function, and are highly expressed in hepatocytes. Using this approach for all metagenes extracted, we were able to get a more precise understanding of the cell type specific metagenes, and on the other side, confirmed that the housekeeping metagenes were not differentially expressed in any specific clusters.

4.1.7 Metagene extraction: summary and discussion

In this section, we have investigated the scRNA-seq dataset from a completely different perspective, focusing on genes rather than cells. Inspired by the canonical scRNA-seq pipelines which group cells with similar identities together, we asked the question if we can do the same thing with genes. By transposing the input UMI matrix, we were able to treat the genes as samples and cells as variables. A series of manipulation is needed to obtain informative results, including the high heterogeneity of the input data matrix (**Figure 4.4**), proper normalization (**Figure 4.1**), parameterization of dimension reduction (**Figure 4.2**), and clustering approach (**Figure 4.3**). By comparing different algorithms at different steps and tuning the parameters, we ended up dividing the entire transcriptome from the E11.5 mouse whole embryo dataset into 63 groups of genes, named 'metagenes'. Within each metagene, the members are highly correlated, and believed to play similar biological roles.

In summary, our metagene extraction workflow can be summarized as below:

1. Highly heterogeneous dataset (E11.5 mouse whole embryo) obtained from scRNA-seq as input.
2. Data normalization by SCT transformation and transposition.
3. Dimension reduction by PCA, then tSNE from top 100 PCs.
4. Clustering from the gene-gene correlation of 100 PCs using IBS.
5. Get the biological signatures of all IBS nodes (1024 preliminary metagenes) through enrichment analysis.
6. Trim the IBS tree by comparing enrichment analysis results of daughter nodes, then stop dividing at specific mother nodes. Terminal nodes were considered as preliminary metagenes.
7. Analysis of preliminary metagenes by manual inspection of returned enrichment analysis, members of gene lists, and literature.
8. Matching the cell type annotation of the same input dataset and relate to preliminary metagenes.
9. Finalized metagene list with 63 members.

The clustering of the genes has been one of the biggest challenges. Not only the clustering algorithm, the question ‘what should we cluster on’ is more of a problem. Unlike the canonical scRNA-seq pipelines that focus on cells, clustering directly on PC space is heavily biased by low expressors, and clustering on reduced dimensional space such as tSNE is also not feasible due to the complexity of the plot. We’ve also tried to drop the low expressors, which did not work well and failed to fulfill the goal of annotating the entire

transcriptome as large numbers of genes from scRNA-seq are expressed at low levels. Finally, after clustering the dataset on a variety of spaces and analyzing the output, we found that clustering on the correlations of PCs would give the most accurate results. The gene coexpression network has a much more complicated relationship compared to that in the cell space from canonical scRNA-seq pipelines, therefore we've modified the widely used Ward Hierarchical clustering to get more precise results. The IBS approach we used would minimize the effects from neighboring clusters and provide the possibility to flexibly adjust the number of final clusters (**Figure 4.3F**).

We have also noticed that the source dataset used for metagene extraction needs to be highly heterogeneous, otherwise noisy metagenes with meaningless biological information will be extracted (**Figure 4.4**). We have tested to extract metagenes from less differentiated embryos, or samples from specific tissues or organs, which both gave us unsatisfactory results, as many genes don't express at all in low heterogeneity samples.

After we have completed the clustering of metagenes from the gene plot, we then attempted to annotate the relevant biological functions of each metagene, similar to how we annotate the cell cluster identity in Chapter 1. We first took advantage of the enrichment analysis results by using the genes within each metagene as input to gain biological signatures from various databases, therefore we got a preliminary understanding of possible biological functions. Then we investigate the marker genes included, and make projections onto the cell plot generated from the same source dataset. We also compare to the annotations made from the cell plot to understand what cells might express certain metagene. One example for a metagene involved in hepatocytes

functions is shown in **Figure 4.5**. Such approach would allow precise annotations for metagenes that are cell type specific, as the information from cell plot provide strong support, while less informative for those that are housekeeping-related.

In summary, we have presented a complete pipeline for extraction and identification of metagene based on a highly heterogeneous scRNA-seq dataset. Next, we will present a complete list of metagenes extracted from a E11.5 mouse embryo scRNA-seq dataset.

4.2 Metagenes from E11.5 mouse embryo scRNA-seq

4.2.1 Cell type specific genes identified from E11.5 mouse embryo metagenes

Using the workflow described above, we extracted 63 metagenes from the E11.5 dataset. Except for 9 clusters with weak or mixed signals assigned as ‘undefined’, enrichment results for all metagenes showed there were two big categories of metagenes: 26 cell type specific metagene and 28 ubiquitously expressed metagene (or housekeeping metagenes) (**Figure 4.6, Figure S4.1A**). Comparing enrichment analysis results with gene detection/expression plot showed a good match: housekeeping metagenes were expressed in various cell clusters while cell type specific metagenes showed cell cluster specific expression patterns. In tSNE space, the cell type specific metagenes were mostly represented by well-isolated clusters, while housekeeping metagenes were connected except for two specific clusters annotated as cell cycle metagene and ribosomal protein metagene. Cell type specific genes showed unique patterns in PC space as their expression levels in certain cell types differed significantly depending on the cellular identities, therefore easier to be isolated. Most housekeeping genes can be detected in

almost all cells with one exception as cell cycle genes, which could also be isolated from housekeeping metagene clouds (**Figure S4.1A**, red dashed line). Another exception was the metagene of ribosomal genes, as their expression levels were so high that they reached almost 100% detection rate, they were tightly correlated and tightly clustered (**Figure S4.1A**, purple dashed line).

To better visualize the data structure, we created an IBS tree, showing 63 terminal nodes after trimming (**Figure 4.6**, inner part). To better understand each metagene, we made a cell tSNE plot using the canonical scRNA-seq pipeline for the same source, and projected the gene expression level of each metagene onto this cell tSNE plot, as described in the previous section. As expected, most cell type specific metagenes highlighted certain cell clusters in cell tSNE space, while housekeeping metagenes highlighted most of the cell clusters.

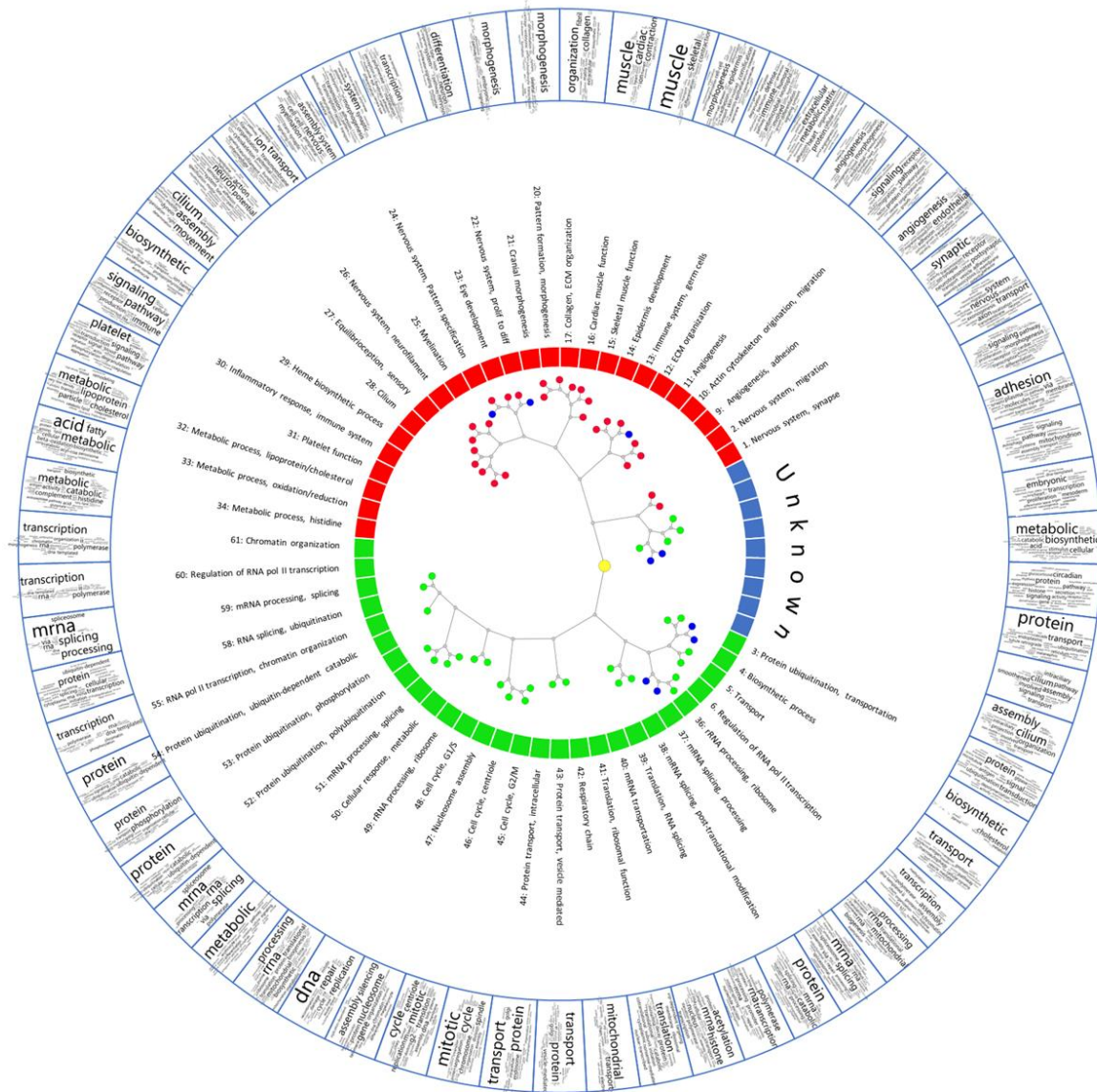


Figure 4.6: Metagenes extracted from E11.5 mouse whole embryo scRNA-seq

Inside: the trimmed hierarchical tree of all genes, the yellow nodes indicate the starting point with all genes, each edge indicates a split into two daughter clusters. Final clusters colored by green (housekeeping), red (cell type specific) or blue (unknown) nodes, Middle ring: annotation of all extracted metagenes, divided into the same categories used in the hierarchical tree. Outer ring: word cloud made from the enriched GO:BP terms of each metagene.

For 26 cell type specific metagenes we extracted, we identified metagene 1, 2, 22, 23, 24, 25, 26, 27, and 28 to be associated with nervous system development (**Figure 4.7A**). Enrichment analysis of these metagenes showed a strong match to nervous system related GO:BP. By projecting the expression level of these metagenes to the tSNE space of cells, all of the metagenes highlight the neuroectoderm cloud (**Figure 4.7A**, right panel). The split of these metagenes could be explained as they focus on different parts of nervous system development. Metagene 1, 2, and 24 highlighted the top half of the neuroectoderm cloud (**Figure 4.7B, E-G**), indicating these genes are expressed in more differentiated cells. Note that all these three metagenes were next to each other in the gene plot (**Figure 4.7B**), meaning that the spatial information on the gene plot also provided information about whether two groups of genes are related. Interestingly, though all the three metagenes pointed to the differentiating neuroectoderm related functions, we clearly observed differences across the three. MG24 had specifically enriched GO:BPs involved in pattern formation such as ‘Anterior/posterior pattern specification’. We found that MG24 might cover genes for different functions, including a lot of Hox family genes, which contribute to morphogenesis during mouse development[414], as well as key genes that are involved in brain and neuron development such as glutamate decarboxylase *Gad1* and *Gad2*[415], several solute carrier family members *Slc18a2*, *Slc6a2* and *Slc6a5*[416–418]. MG1, on the other hand, had the highest enrichment score for synapse-related functions, indicated by the GO terms ‘Chemical synaptic transmission’, ‘Synapse assembly’ and ‘Synaptic vesicle exocytosis. Multiple genes related to different neurotransmitters can be found in this metagene, including GABA receptors *Gabbr1*, *Gabra5*, *Gabrb3*, and *Gabrg2*[419,420], glutamate receptors *Gria1*,

Gria2, *Gria3*, *Grik3*, and *Grin2b*[421–423], as well as calcium channel and potassium channel genes *Cacna1b*, *Cacnb1*, *Cacng2*, *Kcnd2*, *Kcnq2* and *Kctd8*[424–428]. And in MG2, which also had signals related to synaptic transmission with several neurotransmitters genes found, showed other genes that are involved in axon and dendrite development functions, such as genes responsible for tubulin, microtubules, and kinesin functions including *Tubb3*, *Map1b*, and *Kif1b*[429–431]. Clearly, these three metagenes are involved in neurotransmitter functions, but focus on more specific subcategories respectively. MG22 highlighted the bottom half of the neuroectoderm cloud which we annotated as cells transitioning from proliferation to differentiation (**Figure 4.7C, H**), also showed high enrichment scores of GO:BP terms such as ‘Neuron fate commitment’ and ‘Neuron differentiation’. Key neuron development regulators have been identified in this metagene such as *Notch1* and *Dll1* in the Notch Delta pathway[432], *Wnt7a*, and *Wnt7b* in the Wnt pathway[433], as well as other transcription factors like *Neurog1*, *Pax3*, and *Pax7*[434–436]. Combining the cell expression plots with the enrichment results, we concluded that MG22 is involved in the neuronal cell differentiation process. Note that many of these transcription factors play multiple roles during mouse development, such as *Pax3* and *Pax7* are also known to be regulators of myogenesis[437]. Our results suggest that these two genes at the specific developmental time point tested are more involved in one role than another, which could also be valuable to understand multi-functional genes at a finer resolution. MG23 overlaps the region of low expressors and turns out to be noisy (**Figure 4.7C, I**), but it covered most of the marker genes involved in eye development, such as *Tyr*, *Tyrp1*, *Six6*, *Vsx2*, *Cryaa*, *Rax*, *Pax6*, *Rorb* and *Crybb1*[198,438–442]. The expression of metagene 25, 26, 27, and 28

are more focused on specific cell clusters rather than large regions in the cloud (**Figure 4.7D, J-M**), and their enriched GO:BP terms also pointed to more specific cell type related functions. Specifically, MG25 showed a restricted expression pattern in the cluster annotated as oligodendrocytes, with various oligodendrocytes markers and myelination regulators such as *Cnp*, *Plp1*, *Zfp488*, *Mpz*, and *Myrf1*[70–72,443]. MG26 and MG27 highlighted the same cell cluster in tSNE space annotated as sensory neurons. MG26 showed more enriched signals for axon and neurofilament functions, and all three neurofilament protein-coding genes were included in this metagene: *Nefl*, *Nefm*, and *Nefh*[444]. MG27, however, showed more specific signatures related to sensory neurons, with genes involved in photoreceptors and sensory nervous systems such as *Nrl*, *Pou4f1*, and *Isl1*[445–447]. MG28 is unique as it highlighted a relatively small part of the neuroectoderm cloud, which was subset and analyzed to be neuroectoderm cells with more specialized epithelial signatures. The enrichment results from MG28 showed strong signatures of cilia functions such as ‘Cilium movement’ and ‘Cilium assembly’, with key genes including *Wdpcp*, *Pifo*, and *Fam161a*[448–450]. It has been addressed that most neurons possess primary cilia, and ciliary GPCR signaling is involved in neural development[451], which could connect the results with the cell expression plot. This is one example of how different pieces of evidence could be put together to explain the relationship between a metagene and a specific cell cluster (in this case, neuroectodermal cells, epithelial signatures, and cilia functions). With more information provided, such observations would strengthen our understanding of how biological processes are involved in development.

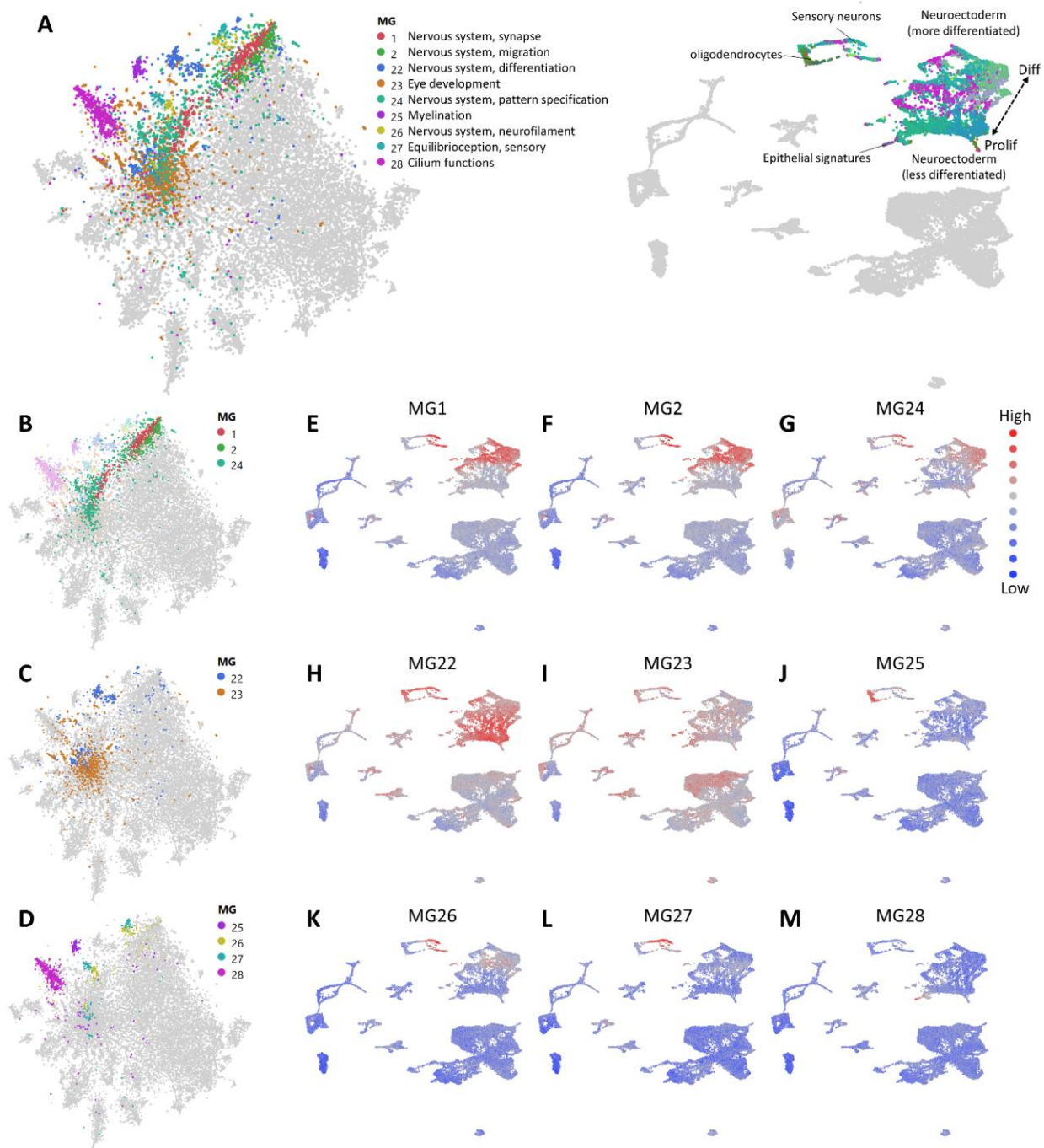


Figure 4.7: Neuroectoderm-related cell type specific metagenes extracted from E11.5 mouse whole embryo

Nine metagenes are annotated to be involved in neuroectoderm-related cell types, with their expression plots highlighting the neuroectoderm cell cloud from the same source dataset.

(A) tSNE projection of genes from E11.5 mouse whole embryo (left) with nervous system development related metagenes highlighted and colored, UMAP projection of cells from E11.5 mouse whole embryo (right) with nervous system related cell clusters highlighted and colored. **(B)** Metagene 1, 2, and 4 are highlighted and colored in tSNE projection, representing genes involved in differentiating neuronal cells. **(C)** Metagene 22 and 23 are highlighted and colored in tSNE projection, representing genes involved in proliferating neuronal cells. **(D)** Metagene 25, 26, 27, and 28, highlighted and colored in tSNE projection, representing genes involved in differentiated cell types. **(E-M)** tSNE projection of cells from E11.5 mouse whole embryo, colored by the expression level of each metagene.

We also identified several metagenes related to mesoderm development, including metagene 9, 10, 11, 12, 14, 15, 16, 17, 20, and 21 (**Figure 4.8A-B**). As mentioned above, MG24 contributes to nervous system development. However, it has a subgroup that covers 25 Hox family genes. MG20 covered the rest of Hox family genes, and other limb morphogenesis regulators including *Tbx4*, *Fgf10*, *Bmp4*, *Sox5*, and *Sox6*[452–456]. The genes within MG20 highlighted the entire mesoderm cloud, indicating their broad expression in all mesoderm-related cell types (**Figure 4.8 B, G**). By looking into these Hox genes, we found that MG24 covered mostly anterior Hox genes (*Hox1-Hox8*), while MG20 covered posterior Hox genes (*Hox9-Hox13*). Hox genes are well studied to show an anterior-posterior expression pattern in developing embryos[457,458], it is interesting to see that our approach is able to divide Hox genes by their anterior-posterior expression pattern, and precisely group them with other neuroectoderm genes (MG24) and mesoderm genes (MG20).

MG12 and MG17 both showed enriched GO:BP related to extracellular matrix (ECM) organization, with their gene expressions covering a broad range of cell types (**Figure**

4.8B, E-F). Of the two metagenes, MG12 showed a clear connection with early heart development, including some proepicardium markers like *Wt1* and *Tcf21*[459,460], as well as cardiomyocytes progenitors like *Gata6* and *Gata4*[461,462], which may indicate a relationship between active ECM formation and heart development. Details of MG12 will be discussed in a later section. MG17 showed no signatures related to cardiac cells and highlighted more of the cranial region in the cell plot (**Figure 4.8F**), but included a large number of collagen genes and other markers such as *Col12a1*, *Col5a2*, *Col6a1*, *Col6a2*, *Fmod*, and *Lum*[463–465]. MG21 showed a specific expression pattern in head mesenchyme (**Figure 4.8H**), the enrichment results of these genes showed GO:BP related to morphogenesis and craniofacial structure formation, which are heavily related to MG17 both biologically and spatially in tSNE space (**Figure 4.8B**), notable markers include *Alx1*, *Eya1* and *Msx1*[466–468].

Metagene 9, 10, and 11 showed specific expression in cell clusters which we identified as endothelium and angioblasts (**Figure 4.8C, I-K**), enrichment analysis of these metagenes showed a strong match to angiogenesis and vasculogenesis. MG9 had the strongest signal for angiogenesis, and angioblast marker genes *Cd34*, *Abcb1a*, *Pecam1*, *Flt1*, and *Nrp2* are members of this metagene[115,469]. MG10 showed a good match to the actin cytoskeleton and cell migration functions but much weaker angiogenesis signatures, with a number of key genes regulating cell morphology such as *Shroom2*, *Amotl1*, and *Fmn13*[470–472]. MG11 presented an even more restricted expression pattern (**Figure 4.8K**), and enrichment analysis detected signals from endocardial cells with marker genes *Etv2m*, *Nfactc1* and *Tek*[347]. We've also observed endothelial cells

with signatures from different origins (refer to Chapter 2 for more details), and the same division could be observed at metagenes here. In summary, these three metagenes covered genes involved in angiogenesis related functions from different origins and different sub-categories. MG14 showed strong signatures of ‘epidermis development’, and cell projection shows specific expression in epithelial cell clusters (**Figure 4.8L**). 13 keratin genes and 10 claudin genes are found in this metagene, confirming it is related to epithelial cell functions.

MG 15 and MG16 are involved in muscle functions of different sub-categories. We have already identified muscle-related cell types as one of the most differentiated cell types in the E11.5 mouse embryo (refer to Chapter 1), and the related metagenes also showed the most isolated pattern in tSNE space (**Figure 4.8D**), indicating the genes involved in muscle functions have unique expression patterns. Specifically, MG15 showed signatures related to skeletal muscle, with key markers and regulators such as *Myf5*, *Myod1*, and *Myog*[124,473], while MG16 showed cardiac muscle signatures including markers *Tnni3*, *Tnnt2*, *Myl4*, *Actc1*, and *Des*[93,474], confirmed by their distinct expression patterns in the cell tSNE space (**Figure 4.8M-N**). Note that even though we observed a clear split between the two metagenes, they were still spatially connected, which further confirmed that our approach could precisely reveal the relationships among different groups of genes.

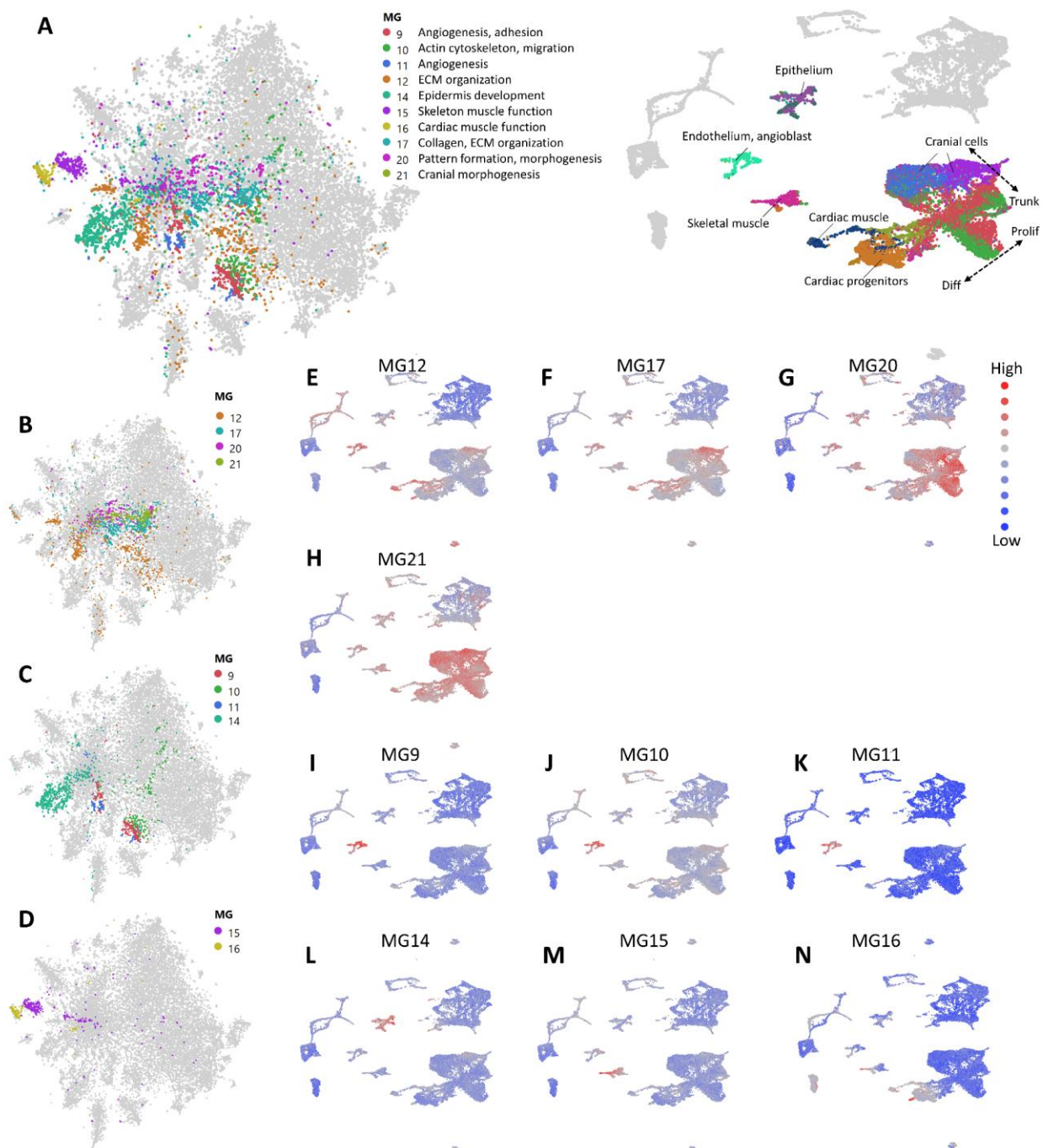


Figure 4.8: Mesoderm-related cell type specific metagenes extracted from E11.5 mouse whole embryo

Ten metagenes are annotated to be involved in mesoderm-related cell types, with their expression plots highlighting the mesoderm cell cloud and clusters from the same source dataset.

(A) tSNE projection of genes from E11.5 mouse whole embryo (left) with mesoderm cell types related metagenes highlighted and colored, UMAP projection of cells from E11.5 mouse whole embryo (right) with mesoderm related cell cloud/clusters highlighted and colored. **(B)** Metagene 12, 17, 20, and 21 are highlighted and colored in tSNE projection, representing genes involved in ECM functions, pattern formation, and morphogenesis. **(C)** Metagene 9, 10, 11, and 14 are highlighted and colored in tSNE projection, representing genes involved in endothelium and epithelium. **(D)** Metagene 15 and 16, highlighted and colored in tSNE projection, represent genes involved in cardiac and skeletal muscle cells. **(E-N)** tSNE projection of cells from E11.5 mouse whole embryo, colored by the expression level of each metagene.

Other than neuroectoderm and mesoderm cell type related metagenes, we've discovered metagenes involved in hematopoietic cell functions and hepatocytes metabolism (**Figure 4.9A-B**). Similar to what we presented in Chapter 1, the blood lineage cells showed very unique patterns in cell tSNE space, showing monocytes, megakaryocytes, less differentiated precursors (HSCs), and more differentiated erythrocytes (RBCs). With increased sequencing sensitivity, each of the HSC and RBC groups was divided into two subgroups, representing cells from different dissected parts and RBCs at different stages of maturation (**Figure 4.9B**). MG13 and MG30 expression plots highlighted one branch of the hematopoietic cell fork which we annotated as monocytes and other immune cells (**Figure 4.9D, F**). In the gene tSNE space, MG13 was clearly divided into two parts (**Figure 4.9B**), one of which is spatially connected to MG30 and contributed to the immune-related signatures. We analyzed the other isolated part of MG13 independently, and found it is highly enriched in germ cell signatures, with key regulators such as *Dnd1*, *Pou5f1*, and *Dazl*[475]. It has been well studied that the formation of the precursors of the primordial germ cells has initiated long before the stage we've tested[475,476], with a very small cell population. As illustrated in **Figure 4.2D-F**, this small sub-cluster only

remained when a large number of PCs were kept for the metagene pipeline, probably because of the small cell population. On the other hand, MG30 showed clear signatures for immune functions, marker genes of multiple immune cell types were included such as *Itgam*, *Cd72*, *Cd28*, *Ptprc*, and *Cx3cr1*[477–480]. In summary, MG13 and MG30 covered genes responsible for the immune system, with a small subgroup responsible for germ cell functions.

MG29 showed a strong signal of heme biosynthetic processes, and expression of this metagene highlighted both mature and immature erythrocytes (**Figure 4.9E**). All 10 hemoglobin genes were included in this metagene, as well as other marker genes required for heme biosynthetic pathways such as *Alad*, *Alas2*, *Abcb10*, and *Fech*[481–483]. Notably, MG29 also included two divided subgroups, with one subgroup adjacent to the ‘low expression’ core showing noisy signals without any enriched signatures. MG31 showed a strong signal of platelet functions, with high expression in another branch of the hematopoietic cell fork (**Figure 4.9G**). Selective markers for megakaryocytes and earlier precursors like *Itga2b*, *Itgb3*, *Mpl*, and *Runx2* were included in this metagene[484–486], indicating this metagene plays a role in platelet maturation. Metagene 32, 33, and 34 came from the same finger of gene tSNE projection, and their expression projections all highlighted one cell cluster in cell tSNE space annotated as hepatocytes (**Figure 4.9C, H-J**). We did not pool them into one metagene as the enrichment analysis of each metagene showed that these metagenes are all involved in liver metabolic processes, but focused on different chemical compounds. MG32 covers many hepatocytes markers including *Afp*, *Alb*, and *Hnf4a*[318], as well as key genes needed for lipid and cholesterol

metabolism such as multiple members from apolipoprotein family *Apoa1*, *Apoa2*, and *Apoa5*[487], enzymes like *Lcat*, *Soat2*, and *Hmgcs2*[488–490]. MG33 showed a more specific match to oxidation-reduction and fatty acid processes, including multiple members from the acyl-CoA dehydrogenase family, aldehyde dehydrogenase family, and flavin-containing monooxygenase family[491–493]. MG34 showed more specific matches to histidine metabolism, including *Uroc1*, *Hal*, and *Ftcd*[494–496].

To wrap up all the cell type specific metagenes, for all cell type clusters identified by cell clustering from a canonical pipeline, we were able to extract at least one corresponding metagene. Furthermore, genes related to specific sub-categories can be revealed for one cell cluster (MG32-34), biologically relevant functions also can be well separated using our approach (MG15-16). In summary, the successful mapping between our cell type specific metagenes with the cell clusters confirms that metagene is both sensitive and reliable in dividing genes into different groups based on their biological functions.

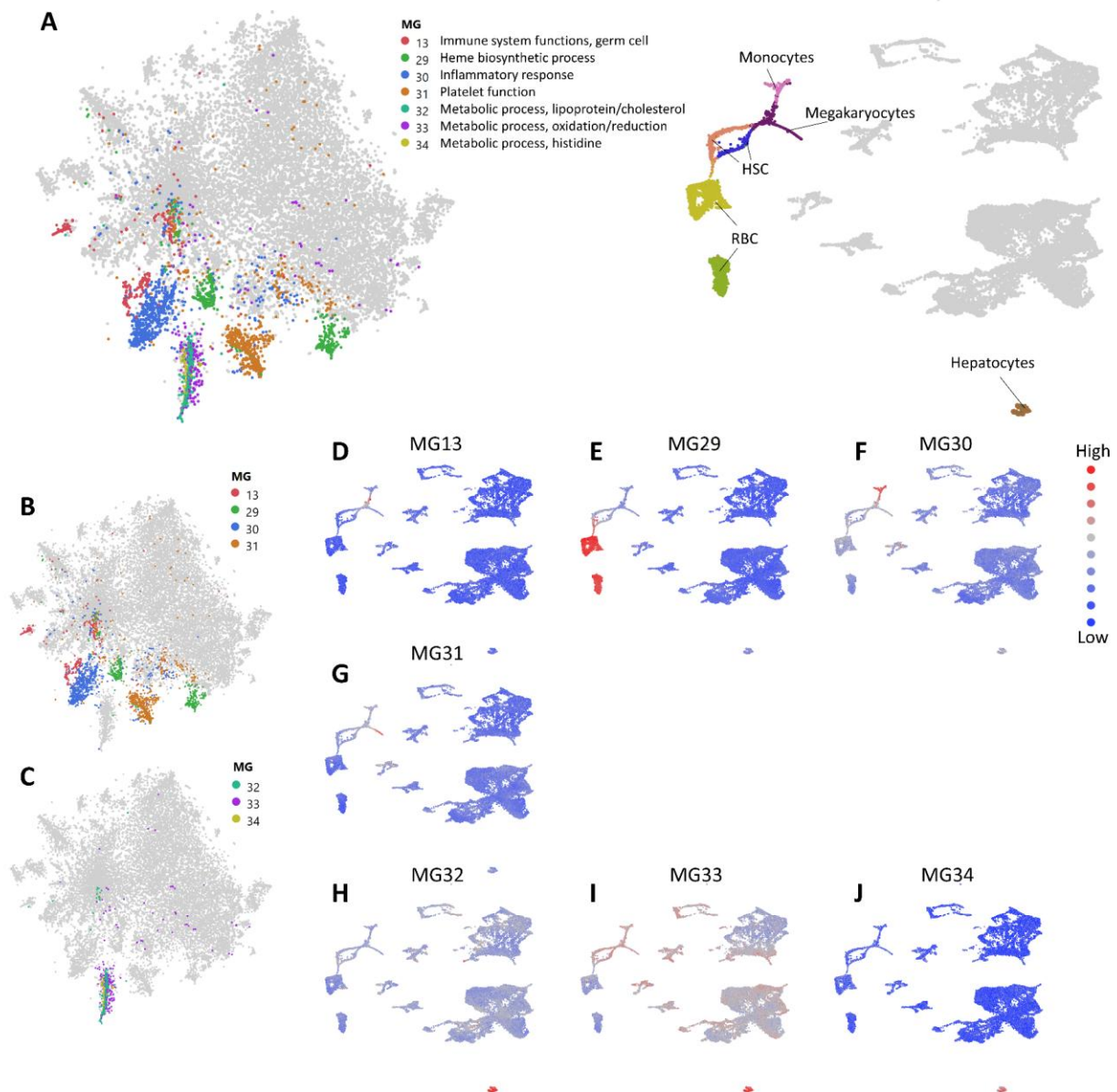


Figure 4.9: Blood and liver related cell type specific metagenes extracted from E11.5 mouse whole embryo

Seven metagenes are annotated to be involved in blood and liver related cell types, with their expression plots highlighting the corresponding cell clusters from the same source dataset.

(A) tSNE projection of genes from E11.5 mouse whole embryo (left) with metagenes involved in blood and liver related cell types highlighted and colored, UMAP projection of cells from E11.5 mouse whole embryo (right) with corresponding cell clusters highlighted

and colored. **(B)** Metagene 13, 29, 30, and 31 are highlighted and colored in tSNE projection, representing genes involved in blood-related cell types. **(C)** Metagene 32, 33, and 34, highlighted and colored in tSNE projection, representing genes involved in liver-related cell types. **(D-I)** tSNE projection of cells from E11.5 mouse whole embryo, colored by the expression level of each metagene.

4.2.2 Housekeeping genes identified from E11.5 mouse embryo metagenes

For 28 housekeeping metagenes we extracted, we were able to recover most of the processes ubiquitously needed for almost all cell types. Different from the cell type specific metagenes, these housekeeping metagenes formed a continuous cloud in gene tSNE space (**Figure 4.10A, Figure S4.1A**). The expression plot of these metagenes highlighted multiple cell clusters rather than specific ones, as expected (**Figure 4.10B, Figure S4.2**). In this case, the cell plot is less helpful for us to understand the functions of these metagenes. To gain more insights from the preliminary enrichment analysis and to classify all these housekeeping metagenes, we computed the correlation of GO:BP enrichment scores among all metagenes, then performed hierarchical clustering on the correlation between each metagene (**Figure 4.10A**). To test if this approach can successfully group metagenes with similar functions, we first looked at the cell type specific metagenes that we annotated in the previous section. It is clear that most of the relevant cell type specific metagenes showed high correlation and were grouped together, such as the nervous system related metagenes, liver function metagenes, immune function metagenes, muscle function metagenes, angiogenesis metagenes, and morphogenesis metagenes (**Figure 4.10A, red labels**). This indicates that such a clustering approach can effectively group metagenes with similar functions together.

Moving to the housekeeping genes, we found that the majority of them could be grouped into five broad sub-categories: transcription, post-transcriptional modification, translation, post-translational modification, and cell cycle (**Figure 4.10A, green labels**). By highlighting these five sub-categories in the gene tSNE plot, they were all spatially connected, except for the 'post-transcriptional modification' (**Figure S4.2A**) which was split into two separated parts. Note that in the hierarchical tree view from our IBS clustering, metagenes from the same sub-categories were not always in the same neighborhood (**Figure 4.10C**), indicating that within each sub-category there are still different groups of genes playing different roles. This is also a reason why we chose only to pool sister nodes when trimming the hierarchical tree, rather than pooling nodes from different mothers.

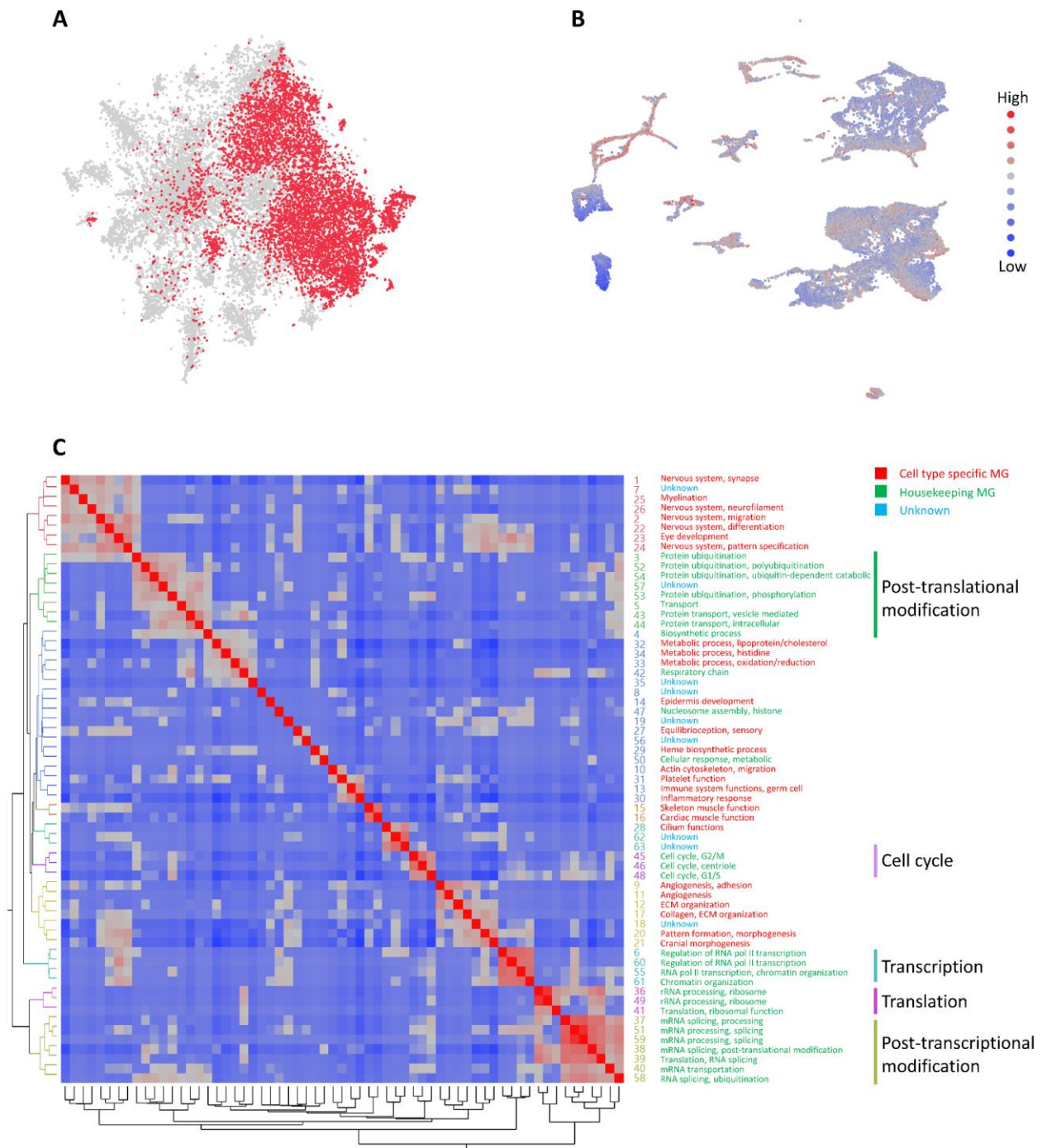


Figure 4.10: Housekeeping metagenes extracted from E11.5 mouse whole embryo 28 metagenes are annotated to be involved in housekeeping functions, with their expression plots highlighting multiple cell clusters from the same source dataset. Clustering analysis shows five big categories of housekeeping metagenes with related functions.

(A) tSNE projection of genes from E11.5 mouse whole embryo, with all 28 housekeeping metagenes highlighted and presents a continuous structure. (B) The average expression level of all housekeeping metagenes, highlights multiple clusters with the RBC clusters at a low level. (C) Clustering dendrogram based on the GO:BP enrichment scores from all metagenes, with all metagenes labeled. Red: cell type specific metagenes; Green: housekeeping metagenes; Blue: unknown. Five broad categories of metagenes are marked at the far right, showing nice clustering structures.

Zoom into each sub-category, the post-translational modification is the biggest one, and the only sub-category that showed a divided structure in tSNE space (**Figure S4.2A**). The bottom half showed clear signatures with protein ubiquitination, phosphorylation, and transportation, including metagene 43, 44, 52, 53, and 54. MG43 and 44 both showed strong signals of protein transport, but target different organelles. MG43 was highly enriched in the endoplasmic reticulum (ER) and Golgi related protein transport, including multiple COPI coat complex subunit protein-coding genes *Cope*, *Copb1*, *Copb2*, and *Copg1*[497], various genes coding for transmembrane protein located at ER and Golgi as cargo receptors such as *Tmed1*, *Tmed2*, *Tmed7*, *Tmed9*, and *Tmed10*[498,499]. MG44 included genes from more downstream organelles transport like endosome and lysosome, such as *Ap1s1*, *Slc17a5*, and *Tbc1d5*[500–502]. MG52, 53, and 54 all showed protein ubiquitination and phosphorylation signatures, but at different locations. MG52 and MG54 focused on protein ubiquitination in the cytoplasm, while MG53 showed protein phosphorylation in the nucleoplasm. The other half including metagene 3, 4, and 5, however, showed mixed signatures of both protein ubiquitination, protein transport, and metabolic process. In MG3 we found genes involved in dynein functions like *Dync1h1*, *Dync1i2*, *Dync1li2*, *Dynlt1b*, and *Dynlt1c*[503], also kinesin-related genes *Kif3c* and

Klc2[504], microtubule-associated protein genes *Map1a* and *Mapre2*[505], all of which contribute to intracellular transportation. Other than that, multiple kinases were also found, like *Mapkbp1*, *Mapk8ip3*, *Mapk8*, and *Map4k4*[506] in the MAPK pathway. MG4 showed enriched signatures of metabolic processes for various compounds, with key enzymes such as *Acta2*, *Nsdhl*, *Msmo1*, and *Dhcr7*[507,508], as well as peroxisome genes *Pex5*, *Pex19*, and *Zfand6*[509,510]. Different from the liver-specific MG32-34 introduced before, MG4 showed almost no cell type specific marker, showing that such metabolic genes are probably expressed universally in all cell types. MG5 focused more on transportation, with vesicular transport protein genes *Trappc2l* and *Trappc4*[511,512], multiple members from trafficking protein particle complex genes, and dynein genes[513]. Unlike the cell type specific metagenes, the annotation of these metagenes was largely dependent on the bioinformatics from enrichment analysis, but clearly, we have observed different sub-categories within the post-translational modification section, and the details remained to be explored.

From our gene tSNE projection, we are able to get two well-isolated clusters, annotated as cell cycle regulation metagene and ribosomal protein metagene. Further splitting the cell cycle metagene gives us metagene 45, 46, and 48, which correspond to different phases of the cell cycle regulation (**Figure S4.2B**). By enrichment analysis and comparing to cell cycle marker list, we annotated MG45 as G2/M cell cycle metagene and MG48 as G1/S cell cycle metagene, which will be described in detail afterward. MG46 is also related to the G2/M phase, but more specifically on centriole functions.

Another metagene showed an isolated structure is MG41, annotated as ribosomal protein metagene, as all 76 ribosomal protein genes are within this metagene (**Figure S4.2C**). Other members of this cluster involve some pseudogenes for ribosomal proteins, as well as other genes involved in ribosomal functions such as translation initiation/elongation factors. Two neighboring metagenes, MG36 and MG49, also showed signatures related to ribosomal functions (**Figure S4.2C**). MG36 and MG49 both presented signals of translation initiation, and various initiation factors could be found in both metagenes, including *Eif2d*, *Eif3a*, *Eif3b*, *Eif5*, and *Eif5b* in MG36, as well as *Eif1*, *Eif2b1*, *Eif2b4* and *Eif5a* in MG49[514,515]. Also, most of the mitochondrial ribosomal protein genes could be found in these two metagenes. One difference is that MG36 covered genes involved in tRNA functions, such as *Trmt1*, *Trmt10a*, *Trmt10c*, and *Trmt11*[516,517].

Within the continuous cloud of other housekeeping genes, another sub-category we identified was 'transcription'. Metagene 6, 55, 60, and 61 were annotated as transcription regulation metagenes, which are spatially connected in the gene tSNE projection, and their expressions level reflected on the clusters in the cell tSNE are globally similar with exceptions of extremely high levels in some hematopoietic cell clusters (**Figure S4.2D**). MG55 and MG61 were more related to chromatin structure remodeling, including genes such as chromatin helicase, methyltransferase, and acetyltransferase. MG6 included a variety of DNA binding protein genes such as the bromodomain gene family and zinc finger protein gene family, while MG61 included various transcription factors that regulate gene expression.

Metagenes 37, 38, 39, 40, 51, 58, and 59 were identified as post-transcriptional modification metagenes, which were also spatially connected in tSNE space (**Figure S4.2E**). MG37 and MG38 consist of most of the Snrp family genes which are needed for spliceosome assembly. MG51 has most of the Srsf family genes, which are splicing factors needed for mRNA splicing and transport[518]. MG51 and MG38 also consist of most of the Hnrnp family genes, known as components of ribonucleosomes needed for pre-mRNA processing[519]. MG38 also covered most of the Psm family genes which encode genes for proteasome complexes[520]. The proteasome is known as a key player in mRNA turnover and surveillance after transcription[521]. MG59 has other genes related to mRNA processing but not in gene families mentioned above. MG40 has multiple nucleoporins and their related genes, which are important for mRNA transportation. Metagene 39 has a relatively weaker and mixed-signal from mRNA processing, translation, and transcription.

By analyzing all metagene GSEA results and projections, we confirmed that our pipeline can precisely extract metagene from scRNA-seq data. One extreme evidence is a cluster of histone genes MG47 with only 35 genes. Both detection and expression levels of most histone genes were very low due to the natural limitation of scRNA-seq technology, on average only 1% of the cells detect these genes, and only 1 transcript per detected cell. But through our pipeline, we identified MG47 with 60% histone genes (21 out of 35), which shows high sensitivity and explains why MG47 did not cluster with other housekeeping metagenes.

4.2.3 Metagene from E11.5 mouse embryo: summary and discussion

Using the pipeline described in 4.1, we have identified 63 metagenes from a scRNA-seq dataset based on E11.5 mouse embryo. The resultant metagenes could be roughly divided into two categories, housekeeping metagenes and cell type specific metagenes (**Figure 4.6**). It is interesting to see that the division of these two groups happens at iteration 1 in our IBS clustering (**Figure 4.6**, inner part), indicating that scRNA-seq is able to easily differentiate between the states of these two types of genes. Taking advantage of the annotation described in Chapter 1, we were able to directly treat the metagene as an input gene list for enrichment analysis. By comparing the documented marker genes with our metagenes, as well as annotated cell plots from the same source dataset, we were able to annotate the cell type specific metagenes without too much trouble (**Figure 4.5, 4.7-4.9**). Notably, these genes form tight clusters that are isolated from the main cloud, just like what we observed as more differentiated cell clusters in Chapter 1.

When analyzing the housekeeping metagenes, we could not use the cell plot as a reference as their expressions cover the majority of the cells, with only one exception as the erythrocytes (**Figure 4.10B**). As a special cell type, the enucleation of erythrocytes makes them unable to proliferate[552]. The process takes place early in embryonic development, presumably at E12.5 in mice with other steps taking place earlier[209]. Our data would suggest that the transcription of housekeeping related mRNAs has been largely decreased at least one day before E12.5.

Without the help from the cell plots, we took advantage of the bioinformatics database from GO, and clustered all our metagenes based on the enrichment scores returned by GO, which also gave us results with clear structures (**Figure 4.10C**). Most cell type specific metagenes form tight clusters using the GO scores, while the housekeeping metagenes could be divided into five sub-groups: 'Transcription', 'Post-transcriptional modification', 'Translation', 'Post-translational modification', and 'Cell cycle', all of which can be visually identified from the dendrogram.

For each metagene, we have investigated the genes involved, and revealed the differences among metagenes under the same category. This gave us a higher resolution of understanding what biological functions are involved in each category, both from housekeeping and cell type specific metagenes. For example, the three metagenes related to liver cell functions (MG32-34, **Figure 4.9**) all showed explicit expression in hepatoblast, but are responsible for metabolic functions of different compounds. In housekeeping metagenes, more than one metagenes showed signatures of protein transport, but by investigating the details of metagene lists we found that the organelles involved in each metagene are different (**Figure S4.2**). Once we finished annotating the metagenes, we found that related metagenes also show nice spatial connections in tSNE space (**Figure S4.2**), indicating that the spatial information embedded in low dimensional space generated by our approach provides valuable insights of gene functions.

With this complete list of metagenes, we have confirmed the robustness and precision of our approach, and we believe that such approach would allow for further understanding

of unknown genes that get assigned to these annotated metagenes, which will be discussed in the following sections.

4.3 Identifying novel cell cycle regulators from metagene

4.3.1 Cell cycle regulation

In eukaryotes, the cell cycle is directly related to several important biological processes, involving DNA replication, chromosome segregation, and cell growth. Typically, eukaryotic cell cycles can be divided into four phases: G1 phase, S phase, G2 phase, and M phase. In a living organism, most cells are at the G1 phase, during which cells increase their size and prepare for DNA synthesis[522]. At the S phase, DNA replication occurs. G2 phase is the gap between S (synthesis) phase and M (mitosis) phase, where cells continue to grow and prepare for mitosis. At the M phase, mitosis takes place and cells get divided into two daughter cells. Through the past few decades, scientists revealed several dependencies that control cell cycle[523], such as cell size[524–526], DNA damage[527,528], completion of DNA replication[529–532], microtubule formation[533,534] and chromosome segregation[535].

The fact that cells multiply through cell division has been observed over a century ago, but not until the last few decades we were able to understand the molecular mechanism that regulates the cell cycle which enables cell division. In 2001, the Nobel Prize in Physiology or Medicine was awarded jointly to Leland H. Hartwell, Tim Hunt, and Sir Paul M. Nurse for their discoveries of key regulators of the cell cycle[536]. Back in the early 1970s, Dr. Hartwell carried out a series of experiments in yeast and succeeded to identify

over 100 genes specifically involved in cell cycle control, by applying time-lapse photomicroscopy on mutated yeast cells[537]. Later, Paul Nurse discovered a specific gene *cdc2* using genetics approaches in fission yeast, whose homolog in human Cdk1 was identified later in 1987, as a member of family cyclin-dependent kinase (CDK) which control the transition between cell cycle phases[538,539]. Another set of key cell cycle regulators which bind to the CDKs, cyclin, was discovered in the early 1980s by Tim Hunt[540]. Together, these discoveries in the early days based on traditional genetics and biochemistry laid the groundwork for discovering more genes responsible for cell cycle regulation. Later, a general biochemistry approach to isolate cell cycle-regulated genes was developed[541], with the prediction that ~250 cell cycle-regulated genes might exist. With the development of modern biology, more advanced approaches at the genome level such as microarray, GRO-seq, and RNA-seq were also used to identify cell cycle regulators[542,543], but there remains a long way to go until we fully understand the mechanisms of cell cycle regulation.

4.3.2 Metagenes identified with enriched cell cycle-related functions

Of all the metagenes we identified, three of which showed strong signals of cell cycle related GO:BP terms, and they are spatially adjacent in tSNE space (**Figure 4.11A**). More specifically, GSEA from MG48 returns enriched G1S phase GO:BP terms, MG45 returns enriched G2M phase GO:BP terms and MG46 returns weaker but mixed phases GO:BP terms.

To further verify the identity of these metagenes, we compared these metagenes with 96 known cell marker genes obtained from literature, of which 42 markers are expressed in the G1/S phase and 54 markers in the G2/M phase. Projections of these marker genes in the tSNE space show high spatial concordance with our cell cycle metagenes (**Figure 4.11B**). MG45 largely overlaps with G2/M markers (53 out of 54), MG48 largely overlaps with G1/S markers (39 out of 42), while MG46 has almost none of either markers. To get a precise understanding of all the 3 cell cycle metagenes and their relationship with cell cycle markers, we subset each metagene together with 96 markers and reran our pipeline independently (**Figure S4.3**). Within each trial, only the selected metagene and cell cycle markers were used to generate tSNE plots based on their expression in all cells in the dataset (**Figure S4.3**, left column). As expected, we observe a separation between G1/S and G2/M markers in all 3 trials. Aligned with the observations before, in the tSNE plot of the MG45 trial, we observe that all G2M markers are embedded in the MG45 genes while G1S markers are outliers (**Figure S4.3A**, left panel). In MG48, most G1S markers are embedded in the MG48 genes and G2M makers are outliers (**Figure S4.3C**, left panel). In the MG46 trial, we observe 3 groups with little overlapping (**Figure S4.3B**, left panel). We also computed the average pairwise gene-gene correlation between each gene and two sets of markers(**Figure S4.3**, middle column), which again confirmed that MG45 is a metagene highly related to G2/M phase functions and MG48 is highly related to G1/S phase functions (**Figure S4.3A, C**, middle and right column). Though MG46 showed enriched cell cycle GO:BP terms, it seems that these genes are not highly correlated to any specific cell cycle phase (**Figure S4.3B**, middle and right panel), and we decided to process MG45 and 48 only for further investigation.

We then investigate all genes within the metagene but not annotated as marker genes. By ranking the average pairwise correlation with two marker lists, we can tell which genes are potentially highly co-expressed with known cell cycle markers. We manually analyzed the top 150 genes correlated to G1/S phase markers by GO terms and literature mining, 125 genes show functions directly related to the G1/S phase, 11 genes show some clues but no direct evidence related to the G1/S phase, 14 genes show almost no available data or unrelated functions. For the top 150 genes correlated to G2/M phase markers, 124 genes show functions directly related to the G2/M phase, 9 genes show some clues but no direct evidence related to the G1/S phase, 12 genes show almost no available data or unrelated functions, and 5 genes turned out to be non protein-coding genes.

4.3.3 Novel cell cycle regulators could be identified from metagenes and confirmed by FACS and RNAi knockdown

Based on the observations described above, we hypothesize that MG45 includes genes responsible for G1/S cell cycle regulation, and MG48 includes genes responsible for G2/M cell cycle regulation. To test our hypothesis, we selected several genes with unknown functions or functions unrelated to cell cycle regulation, and tested if the cell cycle regulation is affected when these candidates were knocked down. By filtering out low expressors, ranking genes by their correlations with G2/M markers, and manual literature search, we selected *Ccdc77*, *C330027C09Rik* (marked as C33Rik in figures), *Arhgap11a*, *Arhgef39*, *Rnf26*, *Lrr1*, and *Dnajc9* as our candidate genes. Some of the candidate genes have no experimental data, and their functions are barely known, such as *Ccdc77* and *Dnajc9*. Some of the candidate genes have limited experimental data, but

are unrelated to the cell cycle, including *C330027C09Rik*, *Arhgef39*, and *Lrr1*. The rest showed a relationship with cancer, but were not extensively studied in terms of the cell cycle, such as *Arhgap11a* and *Rnf26*.

In addition, we selected three well-studied genes without known cell cycle related functions as negative controls, including *Col1a1*, *Sod1*, and *Spa17*. We've also selected five cell cycle markers from our metagenes as positive controls, including *Ccna2*, *Ccnb2*, *Cdca8*, *Cdc6*, and *Cdc45*. The positive control genes are shown to regulate the cell cycle, and knocking them down would exhibit significant changes in cell cycle progression[544–546].

First, we introduced synthetic siRNA with lipofectamine into mouse fibroblast cells to knockdown target genes. By qRT-PCR, we confirmed successful knockdown with 70%-90% reduction of mRNA expression level (**Figure 4.11C**). Then we analyzed the distribution of cells at each cell cycle stage by DAPI staining and Fluorescence-activated cell sorting (FACS) analysis. A histogram of DNA content was generated for each group, then the percentage of cells falling into each phase was calculated. As expected, the negative control group showed no significant changes compared to the mock (**Figure 4.11D**), three of the positive controls showed significant changes in all three phases, and the other showed significant change in the M phase (**Figure 4.11E**). In our experimental group, all genes showed significant change in at least one cell cycle phase (**Figure 4.11F**). Other than the number and statistics, the shape of the DNA content histogram also presented different structures in many of the candidates. It's worth noting that we have explored the proper statistical analysis to use for our cell cycle phase distribution. By going

through publications with a similar design, we found most studies used one-way ANOVA and t-test[547,548], or even no statistical analysis[549]. We have reported our results following the conventions using the t-test, but we have explored the possibility of using the chi-square test, which we believe is the more appropriate test to be used for analyzing distributions of cells into different phases. The major problem with such a test is the large sample size, as the number of cells tested in FACS is far over the recommended sample size for the chi-square test, the p-values for the tests all turned out to be significant (even the replicates), known as large sample effect[550]. To support the results from the t-test, we have calculated the Cramer's V for each test to deal with the extremely large sample size [551], which leads to the same conclusion (data not shown). We have also compared the cell proliferation and apoptosis rate by counting the number of cells and percentage of live cells at 48 hours post-transfection. Though many groups showed non-significant changes, the knockdowns from both positive controls and experimental presented a trend of decreased cell proliferation rate with a smaller number of cells compared to the mock **(Figure S4.4)**.

In conclusion, the knockdown analysis from MG45 and MG48 proved that metagenes could be used as a powerful approach to identify the functions of unknown genes in a systematic way.

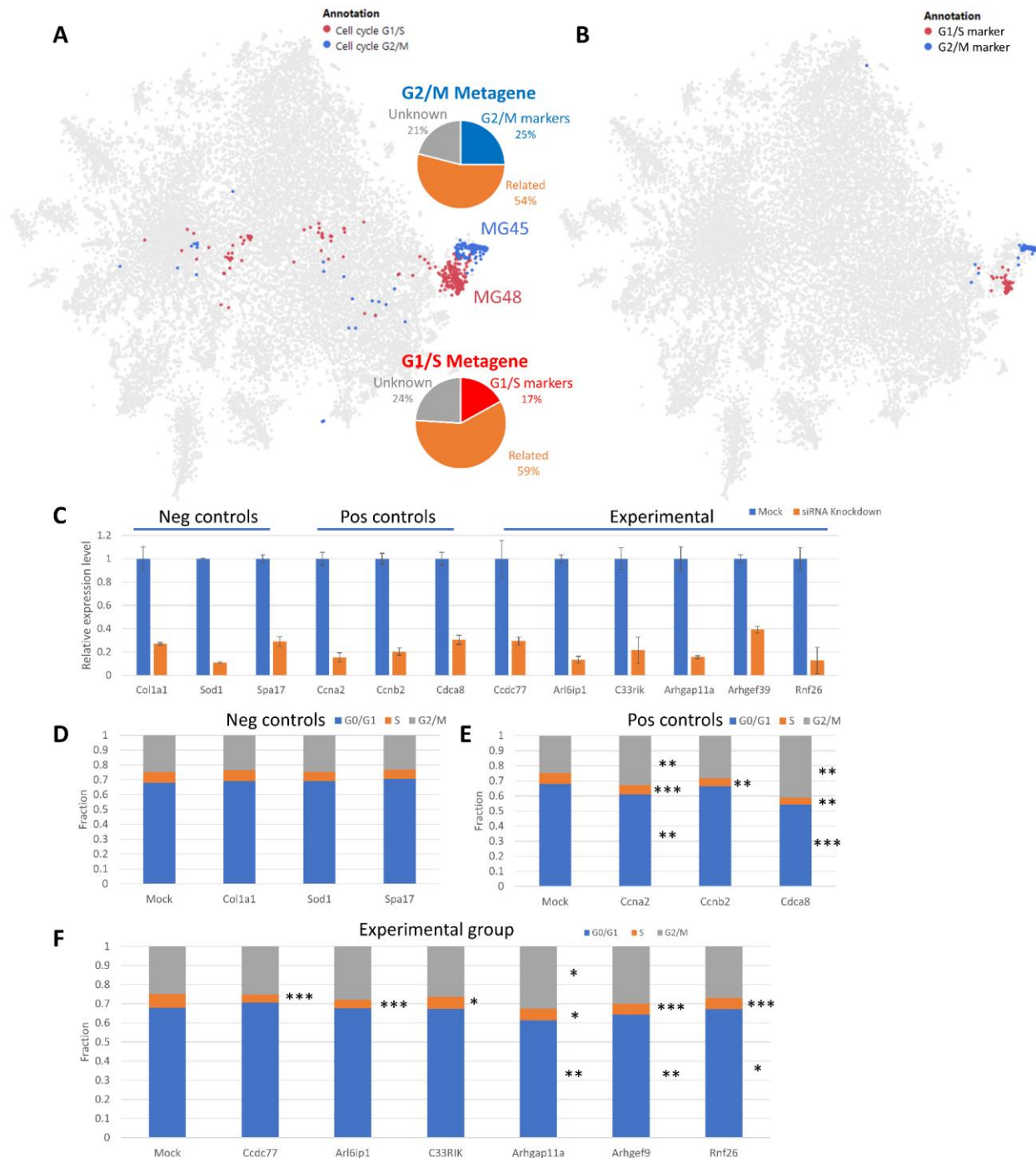


Figure 4.11: Cell cycle metagene identified from scRNA-seq used to identify potential cell cycle regulators

Two metagenes are annotated as genes involved in functions of different cell cycle phases with good alignment to known cell cycle markers, knock-down of unknown genes from these metagenes identifies potential genes involved in cell cycle regulation.

(A) tSNE projection of genes from E11.5 mouse whole embryo, with MG45 (blue) and MG48 (red) highlighted. The pie chart shows the percentage of marker genes, related-genes, and unknown genes in each metagene. **(B)** tSNE projection of genes from E11.5 mouse whole embryo, with G1/S markers (red) and G2/M markers (blue) highlighted. **(C)** siRNA knock-down of negative controls, positive controls, and experimental genes. **(D)** Percentage of cells in each cell cycle phase, from each of the negative control knockdown. **(E)** Percentage of cells in each cell cycle phase, from each of the positive control knockdown. **(F)** Percentage of cells in each cell cycle phase, from each of the experimental knockdowns. P-values generated from t-test, marked by $p < 0.1$ (*), $p < 0.05$ (**), and $p < 0.01$ (***). Two biological replicates were included in all experiments.

4.4 Identifying novel cell type markers from metagene

4.4.1 A metagene with a unique gene expression pattern

Of all the cell type specific metagenes identified, most of them showed a tight cluster in tSNE space, with a few exceptions (**Figure 4.8-4.10**). We further investigated these metagenes, and found MG12 is particularly interesting which highlighted the cardiac region with strong ECM signatures. The genes within MG12 are more scattered compared to other cell type specific ones, and by sub-clustering, we found a subset of MG12 showing explicit expression in a small population of cells, while the enrichment analysis results were ambiguous and provided little insight into the cell type identification (**Figure 4.12A-B**). By carefully investigating the genes involved in this subset, we found that several epicardial marker genes are included in the subset such as *Tbx18* and *Wt1*[107,459], as well as a few other genes involved in heart development, which highlighted the same cell population in cell plot (**Figure 4.12C**). Working from the other way around, we also identified the differentially expressed genes from the cell plot, using

the pipeline described previously (refer to 4.1.6). As expected, the cell cluster shows epicardial signatures and was annotated as epicardial cells.

Knowing that the GO enrichment of this gene subset shows a poor match to specific cell types, probably due to diluted signatures, we believe that other genes within this set might be potential undiscovered epicardial specific markers, as genes within each metagene are highly correlated and play similar biological roles.

4.4.2 Novel epicardial-specific genes could be identified from metagenes and confirmed by in-situ hybridization

To test our hypothesis that the subset of MG12 is a group of genes involved in epicardial development and includes potential epicardial specific markers, we performed in-situ hybridization on genes selected from this subset. To prioritize the candidates, we compared the MG12 subset with the differentially expressed genes obtained from the corresponding cell cluster, and found the genes within both as our candidates. Then we performed the literature search to get rid of genes known to be expressed in epicardial cells. Lastly, we inspected the detection level of the remaining genes and visualized the expression pattern in the cell plot (**Figure 4.12E**), ranking all candidates by their specificity of expression to generate a ranked list of candidates.

To perform the in-situ hybridization, we harvested mouse embryos from E11.5. First, *Tbx18* was selected as positive control which highlighted the epicardium as expected (**Figure 4.12D**). By testing top candidates from our list, we identified a few genes show explicit expression in the epicardium such as *Tmem151a* and *Bcam* (**Figure 4.12F**),

together with a few others showing expression in the epicardium but not specific such as *Gm4208* and *Lama5* (data not shown).

In conclusion, cell type specific metagenes could be used as a potential approach for identifying genes that are specifically expressed in certain cell types, providing spatial information that would help us better understand gene functions.

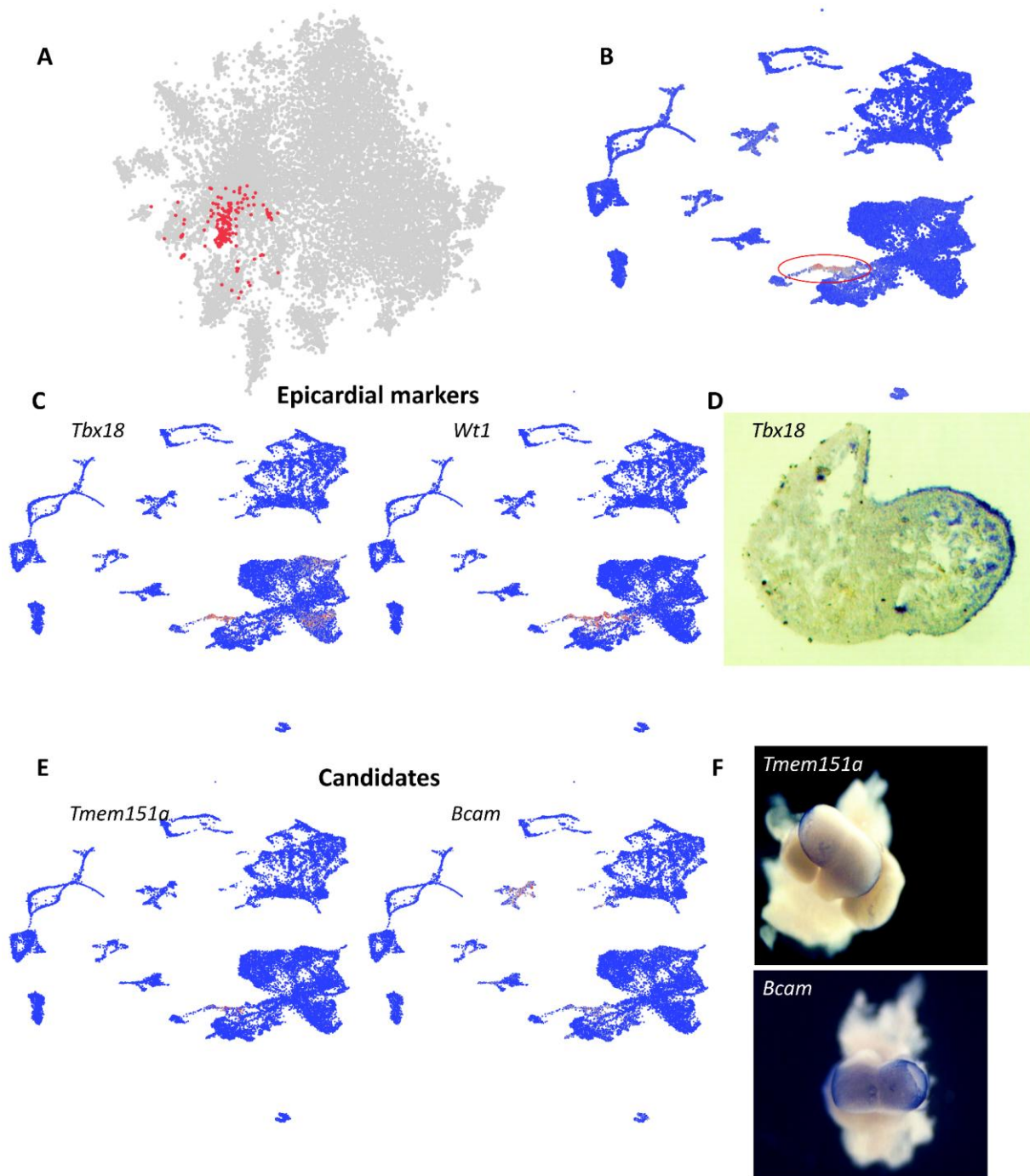


Figure 4.12: Epicardial metagene identified from scRNA-seq used to identify new cell type marker

A subset of MG12 shows explicit expression in the epicardial cell population with a few epicardial markers included. In-situ hybridization shows other genes in the metagenes might be new epicardial markers.

(A) tSNE projection of genes from E11.5 mouse whole embryo, with a subset of MG12 highlighted. (B) UMAP projection of cells from E11.5 mouse whole embryo, colored by the expression level of genes from a subset of MG12 indicated in A. (C) UMAP projection same as B, colored by epicardial markers *Tbx18* and *Wt1*. (D) In-situ hybridization of *Tbx18* in E11.5 mouse embryo heart. (E) UMAP projection same as B, colored by candidate genes *Tmem151a* and *Bcam*. (F) In-situ hybridization of *Tmem151a* and *Bcam* in E11.5 mouse embryo heart.

4.5 Identifying functions of novel genes: summary and discussion

To illustrate how metagene helps us to understand genes with unknown functions, we focused on two metagenes that showed a strong match to cell cycle signatures. By comparing the metagene with the well-studied cell cycle markers, we found the mapping of G1/S and G2/M markers to each metagene is near-perfect (**Figure 4.11A-B**). By literature search of genes highly correlated with the markers within each metagene, we identified candidates that have no data on their functions or functions that are unrelated to cell cycle regulation. By knocking down these candidates and comparing the distribution of cells from each cell cycle phase, we confirmed that many of the candidates showed significant changes in cell cycle distribution (**Figure 4.11C-F**). This confirms the possibility of using metagene as a new approach to systematically understand gene functions. On the other hand, the cell type specific metagenes have also been confirmed using in-situ hybridization (**Figure 4.12**), with multiple positive results showing the robustness of our metagene approach.

In conclusion, in this chapter, we have developed a novel approach that focuses on the genes from a highly heterogeneous scRNA-seq dataset, and grouped genes with similar biological functions into 63 metagenes. We have provided thorough annotations on all the

metagene extracted, using the knowledge from the cells of the same source data and various bioinformatics tools. Finally, we've shown that the metagene could be used as a powerful tool to infer functions of unknown genes, both housekeeping and cell type specific. This would provide a new approach that helps us understand the gene functions from a novel and systematic perspective.

Material and Methods

Animal, sample collection, and scRNA-seq library preparation

Refer to Chapter 1 Material and Methods for details. This study employs the same procedure to make scRNA-seq libraries. The only remarkable difference is the version of chemistry from 10x Genomics was updated to version 2, with enhanced library quality and better sequencing sensitivity.

Preliminary metagene extraction

Once libraries were successfully created, sequencing was performed on an Illumina NextSeq 500. Sequencing data were demultiplexed and processed using CellRanger software (version 1.0, 10x Genomics), using the recommended settings to generate a raw UMI expression matrix.

The raw UMI matrix was subjected to the secondary quality control (QC) step, refer to Chapter 2 for more details. In this study, we picked the E11.5 mouse whole embryo dataset performed in 10x Genomics version 2 chemistry.

The QC'ed raw UMI matrix was normalized and scaled by SCT-normalization, using R package Seurat (version 3.0, in R environment 3.6)[356]. From this point, the dataset was

transposed, so that the genes are now observations and cells are variables. PCA using 100 PCs and tSNE were then applied to generate a 2D visualization of the dataset, with genes being plotted.

The pairwise gene-gene correlation was computed from the PC table (all genes by 100 PCs) using the Pearson method. The clustering was performed on this pairwise gene correlation table using the Iterative Binary Splitting (IBS) approach. To be more specific, the input matrix was divided into two clusters using Ward Hierarchical Clustering, then each of the daughter clusters was then divided into another two clusters using the same approach, iteratively. By making N iterations, we ended up with 2^N clusters. In this study, we used up to 10 iterations with 1024 clusters to ensure each branch was fully separated into unique biological functions. Within each iteration, the mother-daughter relationship can be clearly maintained to make the resultant hierarchical tree named 'IBS tree', which was subjected to biological interpretation and further trimming. Each cluster (or node, in a hierarchical tree perspective) in this case is considered a preliminary metagene.

Biological annotation, IBS tree trimming, and finalized metagenes

Starting from a hierarchical tree with 1024 metagenes (nodes), further trimming is required, as many of the splits were unnecessary and the resultant daughter nodes need to be pooled. The annotation and pooling were performed manually by looking into the enrichment analysis results of each cluster. To be more specific, genes within each cluster were analyzed using GeneAnalytics (<http://geneanalytics.genecards.org>). For each cluster, scores of related matches for LM:Cell Types, GO:BP, and LM:Phenotypes

were extracted. The background noise of the enrichment analysis was cleaned using random gene lists query (see Chapter 1 Material and Methods for more details). On the other hand, we processed the dataset using the approach described in Chapter 1 to generate a precise annotation of cell types from the input E11.5 mouse embryo dataset. With the upgraded 10x Genomics pipeline, we got similar results presented in Chapter 1, and we were able to use such information for gene cluster annotation. Basically, we computed the expression level of each gene cluster, and projected it to the cell plot. This helped us to separate genes into cell type specific genes and housekeeping genes, by looking at their expression pattern in cell plot. Genes with restricted expression at specific cell clusters were considered as 'cell type specific genes', and those without such patterns were 'housekeeping genes'. The biological annotation of the cell cluster also gave information about the gene clusters that show specific expression patterns, which added to the enrichment analysis results.

To pool gene clusters with similar identities, the enrichment score and biological annotation of each cluster were first manually compared between two daughter clusters, if determined to be similar, then the splitting was stopped at the mother cluster. If the daughter cluster shows unclear biological signatures or very low enrichment scores, the cluster is then inspected in more detail, and in most cases stopped at the mother cluster. The vast majority of the nodes were over split, which should be stopped at interaction 4 or 5. After manual inspection, we ended up with 63 clusters which we believe represent unique biological functions.

Metagene based cell cycle analysis, qPCR, and FACS

To select potential cell cycle regulators from the cell cycle related metagenes, we first selected 96 known cell cycle markers for both G1/S and G2/M phases. For other genes within the cluster, we computed their correlation with markers and then manually checked literature about related information on cell cycle regulation. We focused on the genes without any documentation about cell cycle functions but showed a high correlation with cell cycle markers and a relatively high number of transcripts detected, marking them as candidates.

To test whether candidate genes regulate cell cycle, we performed RNAi knockdown of the candidates in MEFs cell, with two biological replicates. To be more specific, MEFs cells were seeded at 60% - 80% confluency in a 10cm cell culture dish prior to RNAi, using DMEM with 10% FBS and 1% non-essential amino acid. At time zero, 125 pmol siRNA oligo and 37.5 μ l RNAiMax Lipofectamine mixture were added to cells. Media were changed at time 24hr, and cells were harvested at 48hr. Half of the harvested cells were subjected to mRNA extraction, and the expression levels of candidate genes were inspected by qRT-PCR using Taqman probes. Cell counts and percentage of live cells were generated using Countess II automated cell counter (Thermo Fisher) for each sample, with two technical replicates. To perform FACS analysis on cell cycle, the other half of the harvested cells were rinsed in PBS, then fixed in 70% ethanol overnight at -20°C. Fixed cells were washed in PBS twice for 15 minutes, then 1 ml DAPI working solution (1 μ g/ml in 0.1% Triton X100 PBS solution) was added to cells and incubated for 15 - 30 minutes at room temperature. Cells were filtered through a 40 μ m mesh filter

before loading onto FACS. BD LSRFortessa X20 (BD biosciences) was used to perform Fluorescence cell sorter (FACS) analysis, signal intensity histogram of DAPI was used to estimate the distribution of cells at different cell cycle stages. 30,000 events with two technical replicates were collected for each sample, and results were analyzed by BD FACSDiva software (BD biosciences). The cell cycle stage distribution of all controls and experimental groups were generated, and compared to the mock using t-test.

In-situ hybridization

In-situ hybridization protocol is adapted from GEISHA with a few modifications (<http://geisha.arizona.edu/geisha/protocols.jsp>). Mouse embryos were harvested and staged according to stage definition from EmouseAtlas[553]. Probes were generated by PCR from cDNA made from mRNA extracted from mouse embryos followed by in-vitro transcription. Embryos were washed with PBS and fixed in 4% paraformaldehyde overnight. Fixative was then removed with two PBS washes, then applied to cryosectioning or continued as whole mount. Embryos were then stored in 100% MeOH overnight, and rehydrated into PBT through a graded MeOH. After two 10 min wash in PBT, embryos were digested with 10 µg/ml Proteinase K in PBS for 30 min, then washed 10 min in PBT twice and fixed in 4% paraformaldehyde for 30 min. Embryos were then washed in PBT twice, and incubated in the prehybridization buffer (50% deionized formamide, 5X SSC, 2% blocking powder, 0.1% Triton X-100, 0.1% CHAPS, 50 µg/ml yeast RNA, 5mM EDTA, 50 µg/ml heparin) at 65°C for 2 hours. Antisense probes were added to the embryo at 100-1000 ng/ml concentration and incubated at 65°C for 48 hours. Embryos were washed with 2X SSC with 0.1% CHAPS for 20 min three times at 65°C,

then repeated in 0.2X SSC with 0.1% CHAPS, rinsed twice with KTBP (50 mM Tris-HCl, 150 mM NaCl, 10 mM KCl, 1% Triton X-100) at room temperature for 10 minutes. Embryos were then blocked in KTBT containing 20% heat-inactivated sheep serum overnight at 4°C. Anti-DIG-AP Fab fragment was diluted 1:500 in 20% sheep serum and then incubated for 2 hours at room temperature with rocking. The preabsorbed anti-DIG was then diluted to a final concentration of 1:2000 in KTBT with 20% sheep serum. Embryos were incubated in the antibody overnight at 4°C, washed five times with KTBT for 1 hour on a shaker, then overnight at 4°C. Embryos were then rinsed in fresh NTMT (100 mM NaCl, 100 mM Tris pH 9.5, 50 mM MgCl₂, 0.1% Triton X-100) twice for 15 min, applied to the color reagent in dark. The hybridization signals were monitored, and the reaction was stopped by washing embryos in KTBT followed by BPS. The embryos after color reaction were then applied to a microscope with images captured.

Chapter 5: Conclusions

In this thesis work, we have mainly focused on the technique of single-cell RNA sequencing, which provides transcriptional profiles at single-cell resolution, which could be used as a powerful tool to address various questions. The major challenges of applying scRNA-seq fall into two parts: the sequencing pipeline and the data analysis pipeline. As a fast-developing technology, the commercialization of scRNA-seq showed great success over the past few years, mature scRNA-seq platforms emerged such as 10x Genomics Chromium, Fluidigm C1, Smart-seq2, and so on. The chemistry and engineering behind these pipelines also develop to enable higher capture efficiency, sequencing sensitivity/depth, and throughput. For example, in our study, we have performed experiments on both version 1 and version 2 chemistry from 10x, with significant improvement observed in the latter. On the other hand, the data analysis pipeline, however, is the part believed to be more important in most biological studies. With help from statisticians, mathematicians, bioinformaticians, and computer scientists, various data analysis pipelines are now available, such as ‘Seurat’, ‘Limma’, ‘CellRanger’, and even some web-based tools. Researchers nowadays only need a few clicks to go from a raw expression matrix to a well-clustered low-dimensional representation. However, customization and parameterization are still necessary in order to get accurate results, which is well-illustrated as part of this thesis work.

The general aim of our work is to use scRNA-seq to understand the development and gene regulation of mouse embryos. To accomplish this aim, we started from a scRNA-seq dataset obtained from both dissected and whole mouse embryos, and ended with

four closely related projects represented as four chapters of this thesis work. In Chapter 1, we introduced our pipeline establishment and an integrated approach for precise cell type annotation, and eventually presented a detailed cell type catalog from the E11.5 mouse embryo (**Figure 5.1**, first column). In Chapter 2, we expanded the pipeline to time-series datasets collected from different time points of mouse early organogenesis, which were integrated into one dataset by solving the batch correction problem, and analyzed the dynamics of cell type change over time in detail (**Figure 5.1**, second column). We consider these two projects as ‘cell-oriented’ projects, and the main challenges we faced were parameterization for large-scale high-heterogeneity datasets, and manual inspection of the semi-automated annotations of each cluster. In Chapter 3, we presented a unique pattern that we have observed in all scRNA-seq datasets, and use that as a novel approach to identify housekeeping genes (**Figure 5.1**, third column). Lastly, in Chapter 4 we viewed the scRNA-seq datasets from a different perspective by transposing the input matrix and treating genes as samples, eventually got all genes clustered into groups with similar biological functions named ‘metagene’, and use metagene as a novel approach to infer functions of unknown genes (**Figure 5.1**, fourth column). We consider these two chapters as ‘gene-oriented’ projects, as they mainly focus on how to interpret gene functions and coexpressions using scRNA-seq data.

As we completed all these four chapters and looked back, we have realized that most of the published studies using scRNA-seq have only interpreted a very small part of the information revealed by such a sophisticated technique. The amount, precision, and sensitivity of the information embedded in scRNA-seq data have been significantly

undervalued. When we look into one dataset from one angle, a large number of interesting biological observations could be made, and the explanation could be revealed by looking at the data from a different perspective. In the following section, we will walk through the conclusions we made from each chapter, and the connections between different observations when viewing the data from different angles.

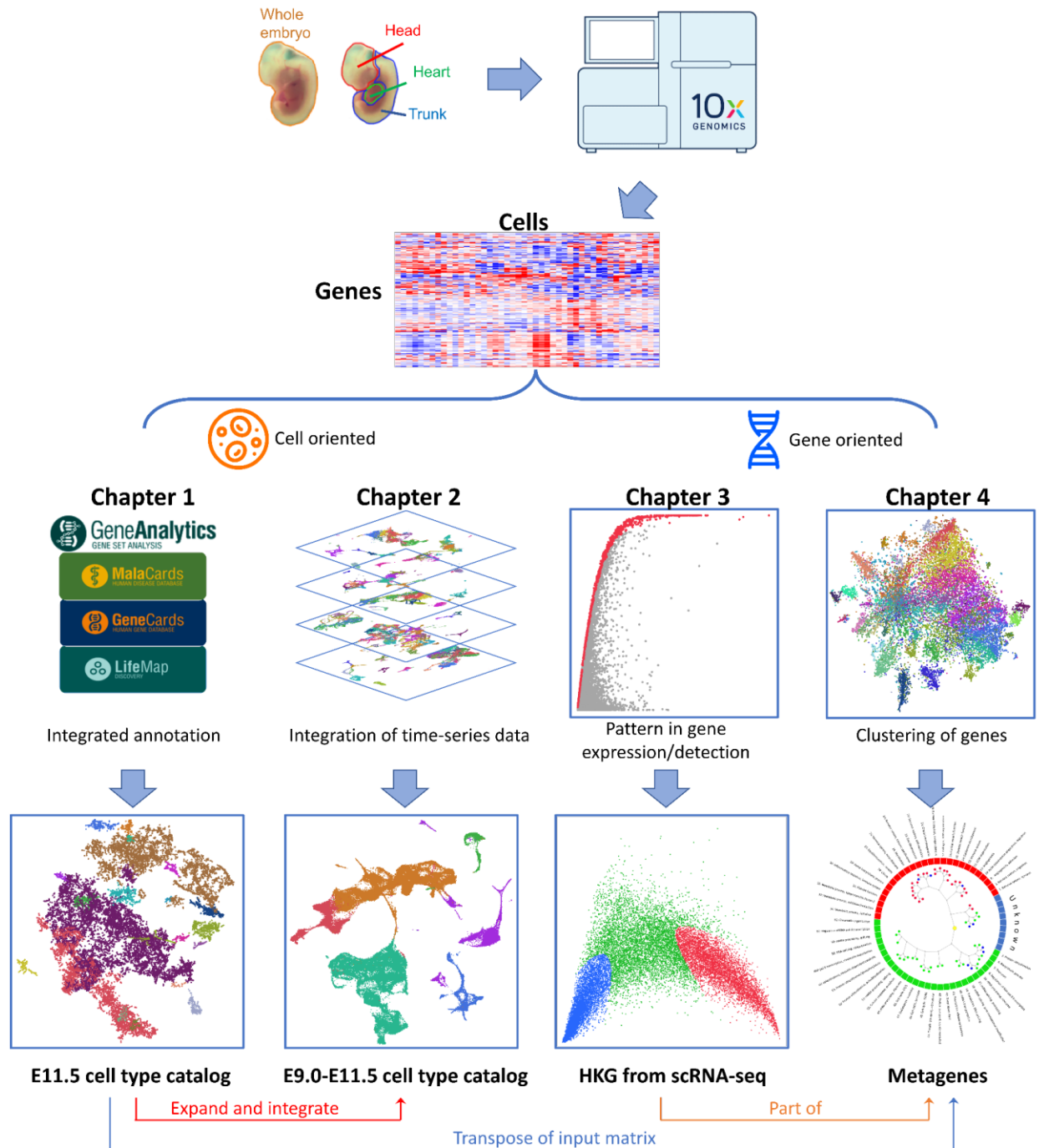


Figure 5.1: Overview of all four chapters

This thesis work primarily focuses on using scRNA-seq to understand mouse development and gene regulation. Starting from dissected embryos and scRNA-seq, we have established an integrated data analysis and annotation pipeline for large-scale scRNA-seq datasets, and presented a cell type catalog for E11.5 mouse embryos (first

column). We then expanded the pipeline to integrate multiple datasets from a time-series experiment (second column). We also identified a unique pattern in scRNA-seq profile and used it to identify housekeeping genes (third column). Finally, we transposed the input matrix and viewed the data from a different perspective, extracted metagene, and used that to understand gene functions (last column).

In Chapter 1, we have successfully established the data analysis pipeline for analyzing large-scale scRNA-seq datasets, by optimizing the algorithms and parameters selected at each step including quality control, normalization, dimension reduction, differential gene expression, and clustering (**Figure 1.1**). We have explored different approaches that could help to improve the precision of cell identity annotation, and provided an integrated approach that combines dissection information, GSEA query, marker genes projection, and cell lineage map scoring (**Figure 1.2-1.5**). Such an approach has been extended to the sub-clustering of less differentiated parts of the cell plot, revealing more details of cell types within undifferentiated neuroectoderm and mesoderm (**Figure 1.6-1.8**). By processing and analyzing the data from the E11.5 mouse embryo, we concluded that there are 20 unique cell types that are more differentiated, together with two large groups of cells from neural ectoderm and mesoderm that are less differentiated at this stage of mouse development (**Figure 1.6**). A few interesting observations were made from this dataset, we concluded that the tSNE plot from scRNA-seq contains spatial information by showing the nested expression pattern of Hox family genes (**Figure 1.10**). We've also observed multiple elongated structures in tSNE space, which we concluded such structures indicate cell type differentiation or cell types differentiated from the same

ancestor (**Figure 1.9, 1.11**), validated by sub-clustering analysis with marker gene projection and independent PCA.

In Chapter 2, we have illustrated the batch effects in combining multiple scRNA-seq datasets (**Figure 2.1**), and we concluded that CCA successfully removes the batch effects from our time-series datasets (**Figure 2.2-2.5**). Based on the approach from Chapter 1, we analyzed the integrated dataset from multiple mouse embryonic stages and presented a detailed roadmap of cell types with temporal information (**Figure 2.6-2.11**). With such information, we conclude that certain cell types emerged or disappeared during mouse development, and confirmed that scRNA-seq could be used as a powerful tool to understand cell differentiation.

In Chapter 3, we started from a unique pattern observed in all scRNA-seq datasets by plotting the gene expression level by their detection frequency (**Figure 3.1-3.2**). By analyzing Gene Ontology data, literature of housekeeping genes/cell type markers, and data from chromatin accessibility studies, we concluded that the genes from the edge of the curve are housekeeping genes (**Figure 3.3-3.7**). We then gathered information from a collection of scRNA-seq datasets and presented a list of housekeeping genes using this novel approach (**Figure 3.8-3.9**).

In Chapter 4, we viewed the scRNA-seq dataset from a new perspective by transposing the input expression matrix and treating the genes as observations and cells as variables. We concluded that such an approach can be used to cluster genes into groups with similar biological functions named 'metagene'. We established a pipeline for extracting

metagenes and optimized the parameters by using sets of benchmark genes and analyzing enrichment analysis results (**Figure 4.1-4.4**), and presented a list of metagenes extracted from mouse E11.5 embryo (**Figure 4.5**). We concluded that genes could be divided into housekeeping genes and cell type specific genes, and provided detailed annotation of all metagenes extracted (**Figure 4.6-4.10**). We also concluded that metagene could be used as a novel approach to infer functions of unknown genes, by knocking down unknown genes in cell cycle metagene following phenotypic analysis and in-situ hybridization of unknown genes in epicardial metagene (**Figure 4.11-4.12**).

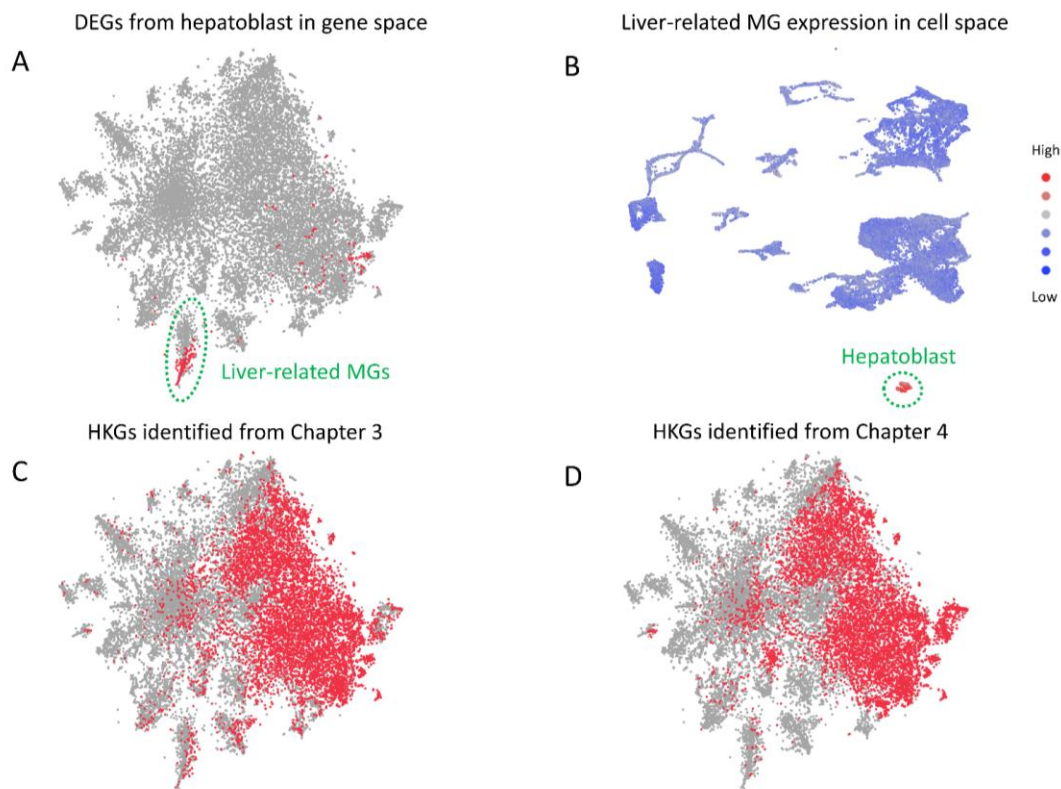


Figure 5.2 Connection between chapters: how metagenes relate to cell type identity and housekeeping genes

Metagenes extracted from Chapter 4 show a strong connection with cell types identified from Chapter 1, with DEGs identified from cell clusters aligning well with liver-related metagenes. HKGs identified from Chapter 3 and Chapter 4 show high concordance.

(A) tSNE projection of genes from E11.5 mouse whole embryo from Chapter 4. Differentially expressed genes (DEG) from the hepatoblast cell cluster in Chapter 1 are highlighted in red. Liver-related metagenes identified in Chapter 4 are circled by a green dashed line. **(B)** UMAP projection of cells generated by approaches described in Chapter 1. Cells colored by the expression level of genes in liver-related metagene identified in Chapter 4. A cluster of hepatoblast is circled in green dashed line. **(C)** tSNE projection of genes from E11.5 mouse whole embryo from Chapter 4, with genes identified as HKGs in Chapter 3 highlighted. **(D)** tSNE projection of genes from E11.5 mouse whole embryo from Chapter 4, with genes annotated as HKGs in Chapter 4 highlighted.

By looking at the four chapters together, we find that these different projects are indeed tightly related. For example, we have introduced how a cell plot could be used to verify the identity of a cell type specific metagene (**Figure 4.5**). Taking this one step further, we also verified that the differentially expressed genes (DEGs) from the hepatoblast cell cluster also presented nice concordance with the liver-related metagenes (**Figure 5.2A-B**). Knowing that the DEGs and metagenes were extracted from two completely different approaches, such a relationship would further validate the robustness of both. We have also compared the housekeeping genes identified from both Chapter 3 and Chapter 4 (**Figure 5.2C-D**), again the nice concordance supported the conclusion from both. As we mentioned in Chapter 4, many of the metagenes showed similar signatures. After careful inspection, we decided not to pool them as there are still subtle differences between them. When we compare these metagenes with even larger scale datasets, such as the integrated one from Chapter 2, we can see how our metagenes help to explain the identity of similar cell types. One example is MG9 and MG11, both showed strong signatures of

angiogenesis but one of them is clearly cardiac-related (**Figure 4.8A, C, Figure 5.3A**). When we project the gene expression level of these two metagenes into the integrated cell space from Chapter 2, they highlighted two neighboring cell clusters (**Figure 5.3B-C**), which we have already identified as endothelial cells and cardiac-derived endothelial cells (**Figure 5.3D**). By further looking at the stage distribution of cells within each cluster, we are able to relate our observations with the differentiation of endocardium cells along the stages (**Figure 5.3E**). Another pair of closely related metagenes are MG1 and MG22, which are believed to be related to functions of more neurons that are more differentiated and those undergoing a transition from proliferation to differentiation (**Figure 4.7A-C, Figure 5.3F**). Projecting to our integrated UMAP space, they highlighted different regions of the neuroectoderm cloud (**Figure 5.3G-H**), and the cell clusters from these regions were highlighted by different metagenes (**Figure 5.3I**). NE6 and NE9 were annotated as more differentiated cells for specific lineage, and they had higher expression of MG1 (more differentiated neurons) (**Figure 5.3G**). On the other hand, the multipotent lineage NE4 had a higher expression of MG22 (neurons transitioning from proliferation to differentiation) (**Figure 5.3H**). All three clusters showed significantly more cells from later stages (**Figure 5.3J**), indicating the initiation of neurogenesis taking place[167]. Taken together, when looking at all four chapters as a whole, the conclusions we draw can often be verified by analyzing the data from a different perspective, and as they come into one piece, they get supported by multiple orthogonal pieces of evidence.

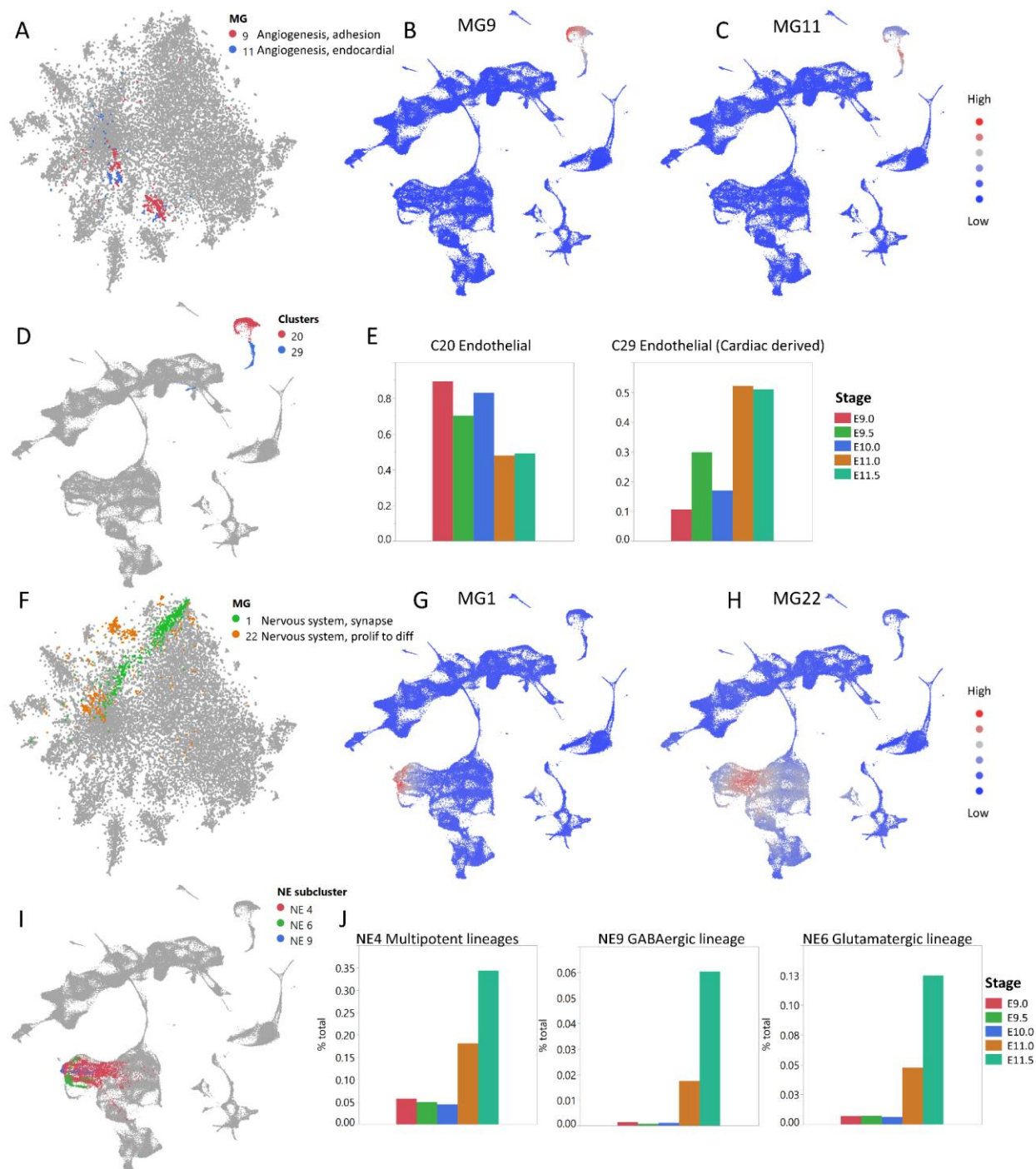


Figure 5.3 Connection between chapters: how metagenes relate to cell type dynamics and differentiation

Some of the metagenes show similar identity and highlight closely-related cell types in integrated space, which might contribute to the understanding of cell type dynamics

(A) tSNE projection of genes from E11.5 mouse whole embryo from Chapter 4, with MG9 and MG11 highlighted. (B-C) UMAP projection of cells from the integrated dataset in Chapter 2. Colored by the expression level of genes in MG9 and MG11. (D) The same UMAP space as B-C, with cell clusters 20 and 29 highlighted. (E) Distribution of cells from each stage in clusters 20 and 29. (F) tSNE projection of genes from E11.5 mouse whole embryo from Chapter 4, with MG1 and MG22 highlighted. (G-H) UMAP projection of cells from the integrated dataset in Chapter 2. Colored by the expression level of genes in MG1 and MG22. (I) The same UMAP space as G-H, with cell clusters NE4, NE6, and NE9 highlighted. (J) Distribution of cells from each stage in clusters NE4, NE6, and NE9.

One thing that's interesting to think about, and still remains a challenge for us, is the genes with multiple functions. It has been well illustrated that genes could be involved in multiple biological processes. For example, *Pax3* is known to be a key regulator in both skeletal muscle development and neurogenesis[435,554]. However, in our metagene approach, we assigned each gene to only one metagene. As shown in **Figure 5.4A**, *Pax3* was a member of metagene 22 responsible for neurons transitioning from proliferation to differentiation, but not in metagene 15 for skeleton muscle function. It is interesting to see the expression of *Pax3* in cell space, which indeed highlighted the neuroectoderm much more than the skeleton muscle (**Figure 5.4B**). It is possible that *Pax3* at this specific stage of development is more involved in nervous system development, and it may shift to a different metagene at a later stage. Similar patterns could be identified in housekeeping genes as well. Dynein genes are known to be involved in protein transport especially in developing neurons, and also responsible for cell cycle regulation[555–557], but multiple members of dynein genes such as *Dync1h1* turned out to be a part of MG3 (protein ubiquitination, transportation), rather than the two cell cycle metagenes (**Figure 5.4C**). The expression plot of *Dync1h1* showed higher expression in some neuronal populations,

rather than other cell cycle regulators which highlighted multiple clusters (**Figure 5.4D**). It seems that our approach is able to assign a gene to metagene based on its primary function, but it would be ideal if we are able to assign multi-functional genes to different metagenes. We have tested a couple of approaches such as biclustering and different matrix decomposition methods[558], but none of those gave us satisfactory results.

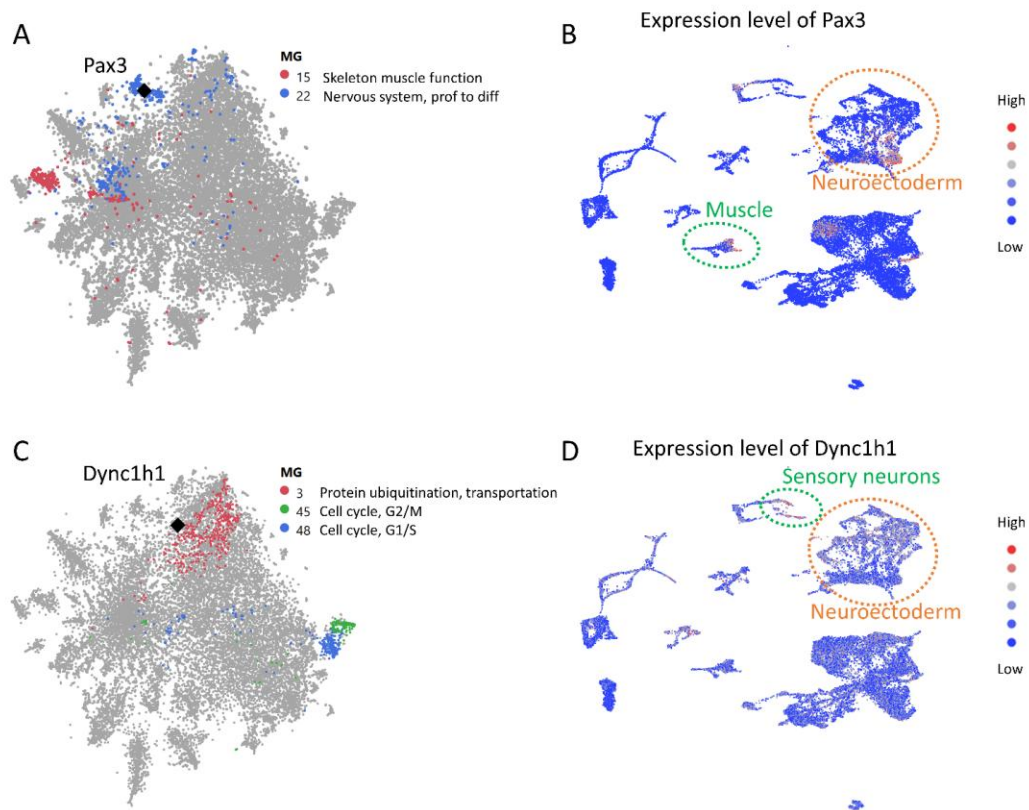


Figure 5.4 Metagene include genes with multiple functions

Genes with more than one function get assigned to one metagene, which might indicate their primary role in the sample tested, also remain a challenge for metagene.

(A) tSNE projection of genes from E11.5 mouse whole embryo from Chapter 4, with MG15, MG22, and gene *Pax3* highlighted. (B) UMAP projection of cells from the same source dataset, colored by the expression level of *Pax3*. (C) tSNE projection of genes from E11.5 mouse whole embryo from Chapter 4, with MG3, MG45, MG48, and gene *Dync1h1*

highlighted. **(D)** UMAP projection of cells from the same source dataset, colored by the expression level of *Dync1h1*.

In summary, we have presented a complete data analysis pipeline for analyzing large-scale scRNA-seq datasets, not only from the perspectives of cells but also shed light on gene co-expression and regulation. We have generated valuable results for the scientific community with a complete catalog of cell types at early mouse organogenesis, as well as an entire transcriptome classified into annotated functions. We have also provided and verified novel approaches to identify housekeeping genes and functions of unknown genes, using scRNA-seq datasets. We believe that the work described in this thesis would inspire other researchers about things to learn from the transcriptomic studies based on scRNA-seq.

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Appendix II: Supplemental Figures and Tables

Figure S1.1: Two populations identified in E11.5 scRNA-seq, quality control and PCA elbow plot

(A) nGene_nUMI plot of E11.5 mouse whole embryo scRNA-seq, showing an outlier group with a low number of genes detected. **(B)** An example of 3-normal distribution during quality control. **(C)** Elbow plot for PCA analysis.

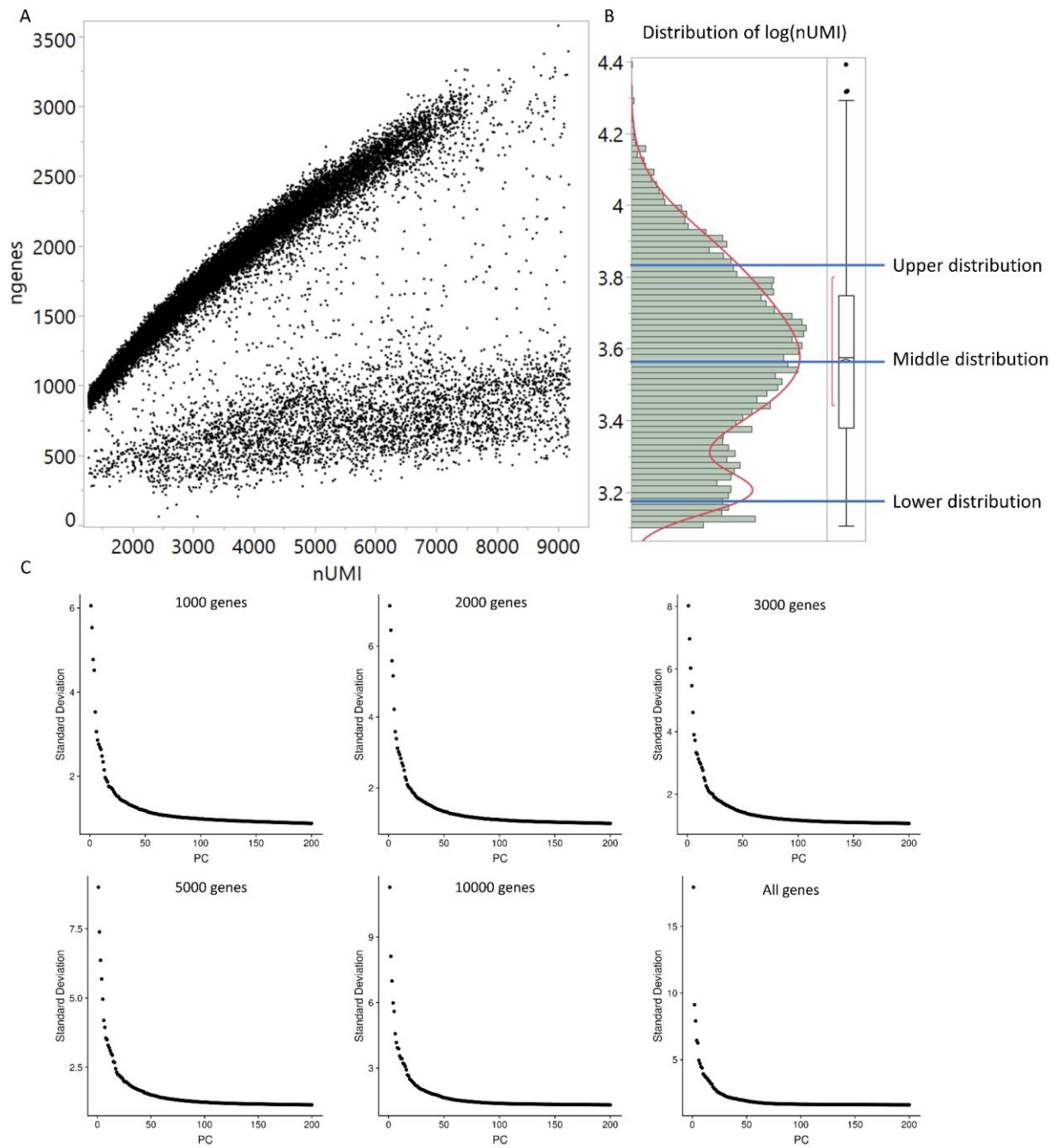


Figure S1.2: Examining changes in the t-SNE structure depending on the number of genes and PCs used.

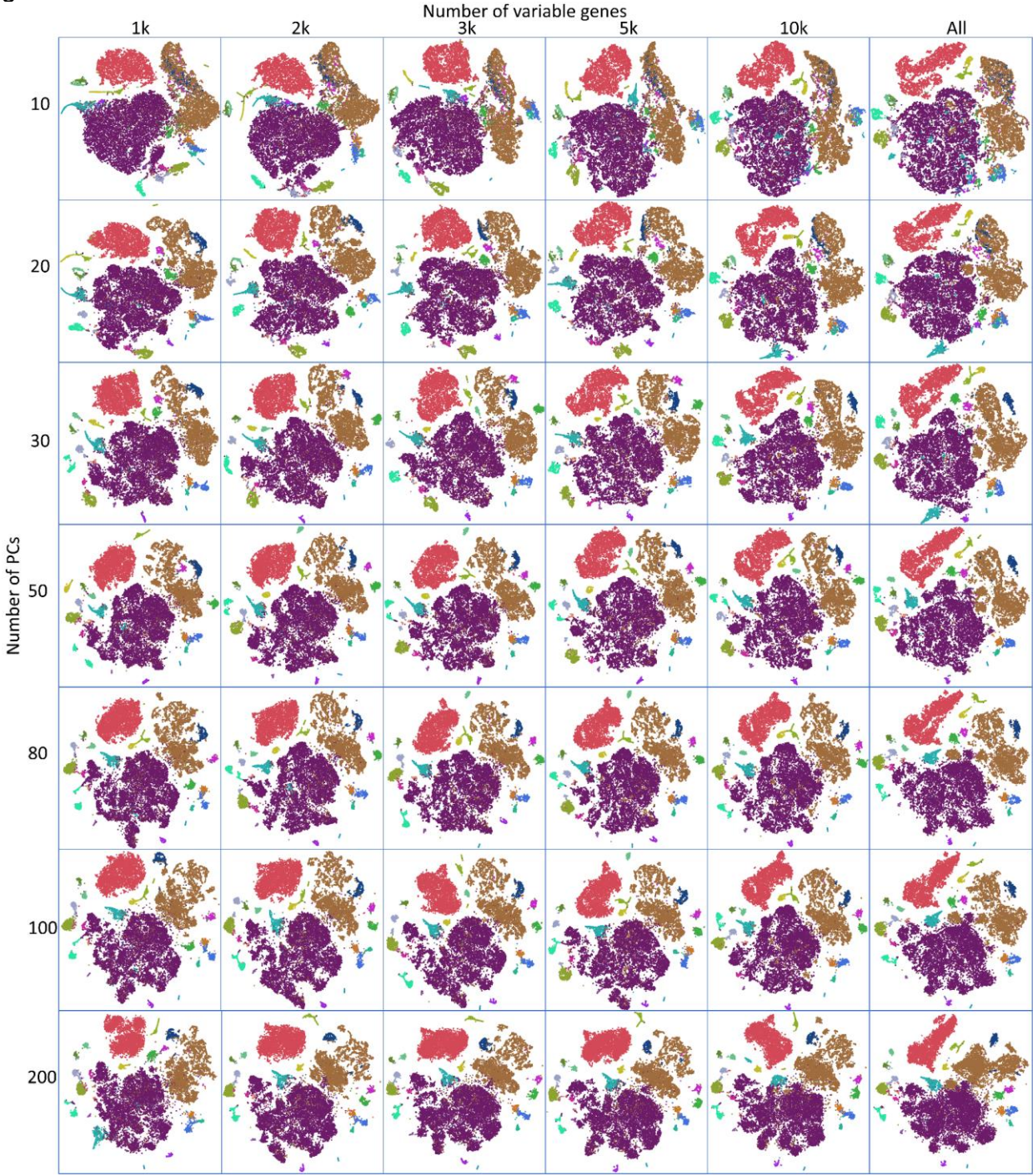


Figure S1.3: Examining changes in the UMAP structure depending on the number of genes and PCs used.

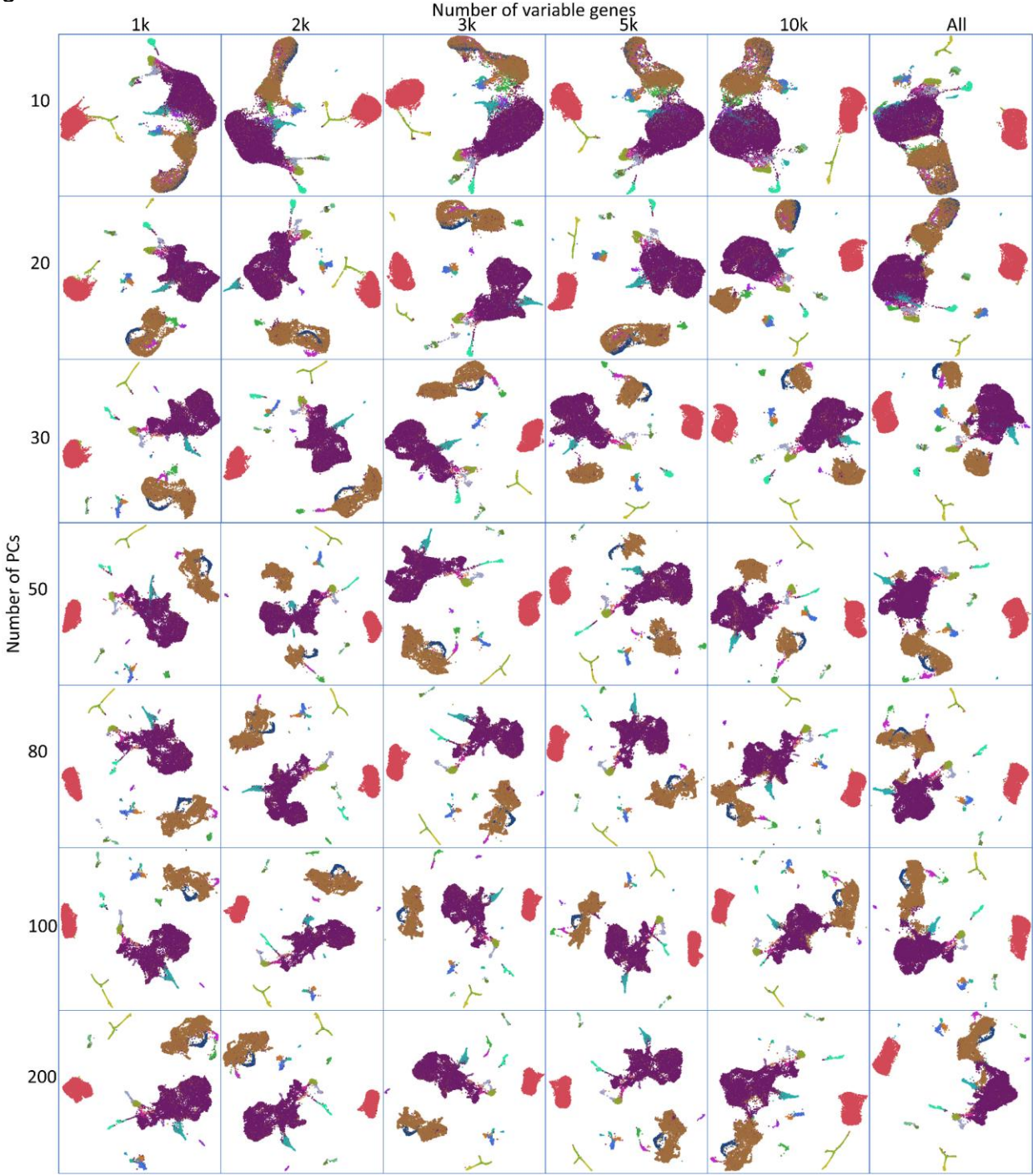


Figure S1.4: Examining changes in the t-SNE structure with different levels of perplexity

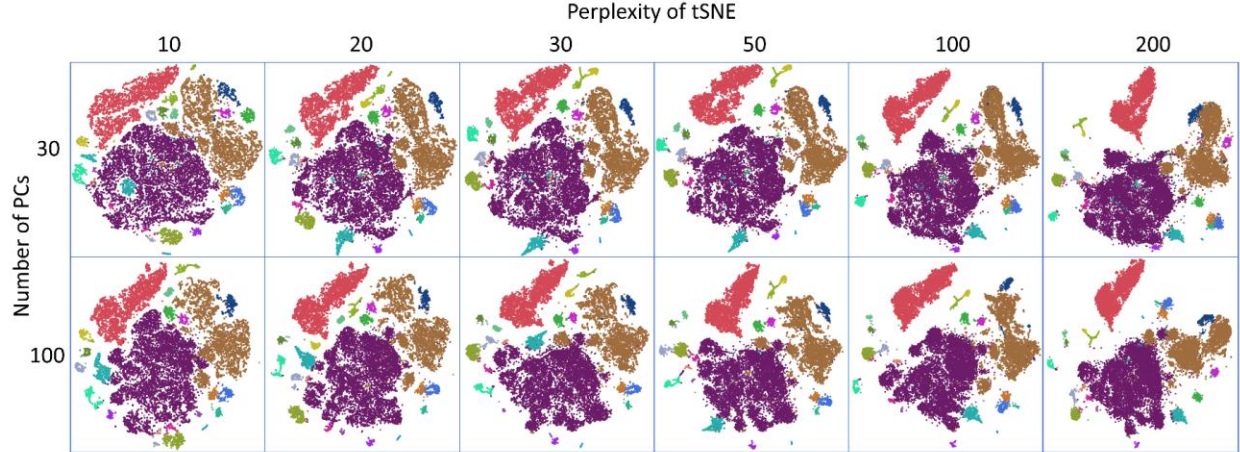


Figure S1.5: Comparison of 2D-t-SNE and 3D-t-SNE
(A) 2D-t-SNE from top 100 PCs. **(B)** 3D-t-SNE from top 100 PCs.

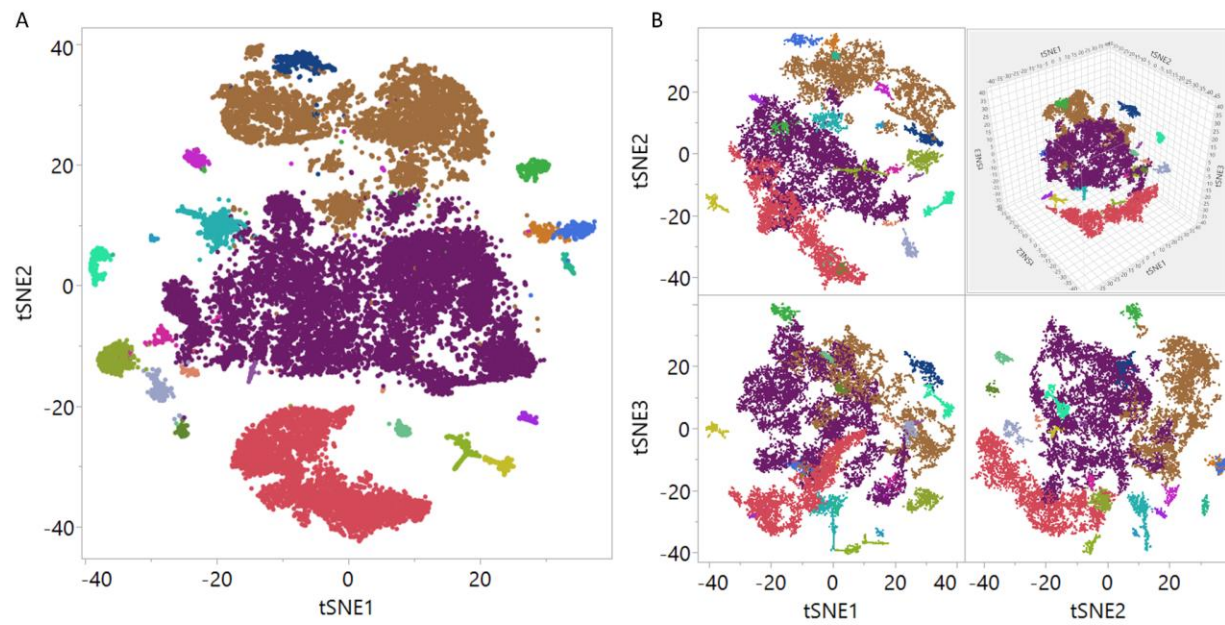


Figure S1.6: Examining different clustering algorithms in both PC and t-SNE space
(A) Clustering in PC space and colored in t-SNE space. **(B)** Clustering in t-SNE space and colored in t-SNE space.

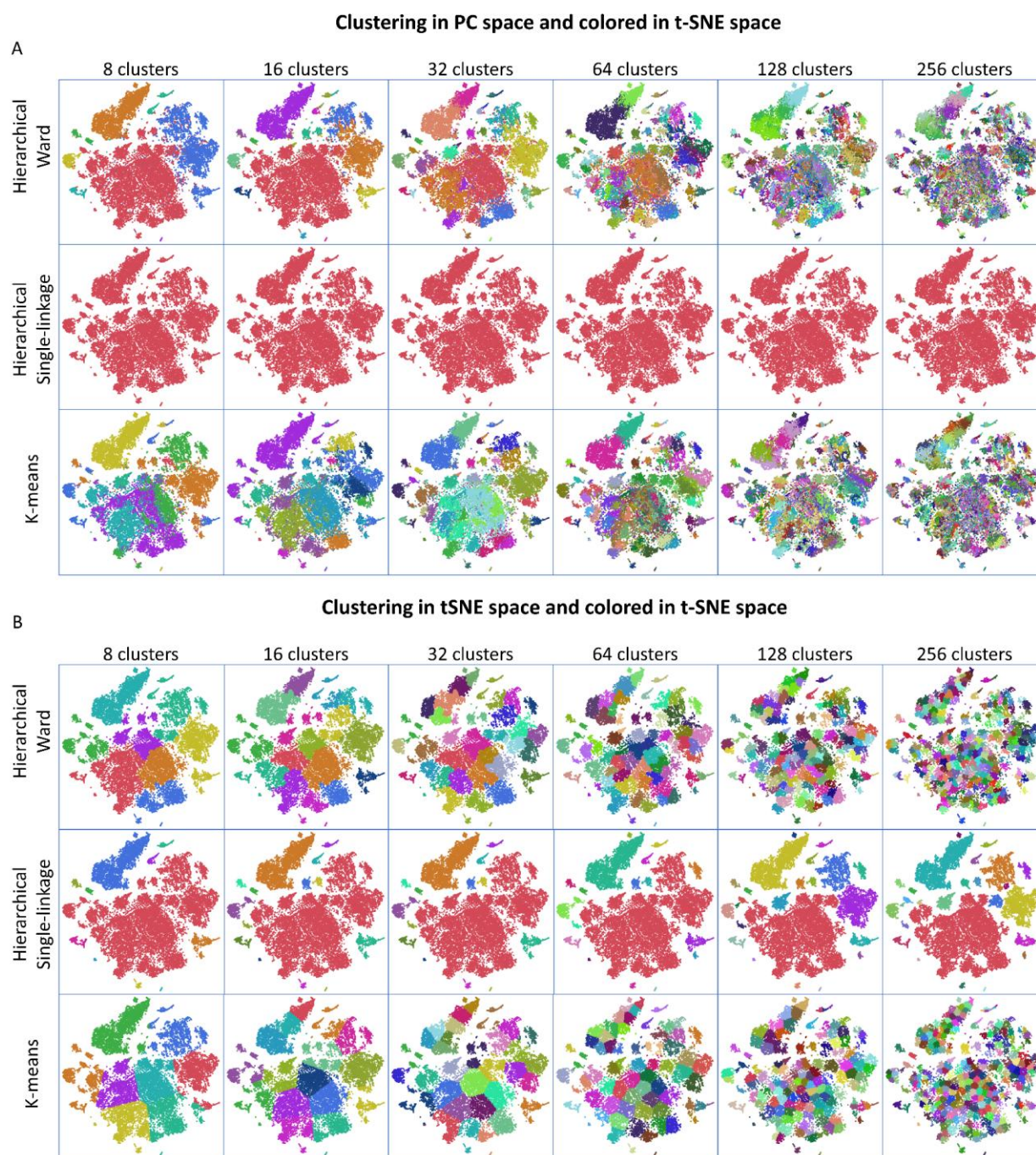


Figure S1.7: Selected marker gene expression plots for unique clusters



Figure S1.7: Continued



Figure S1.7: Continued

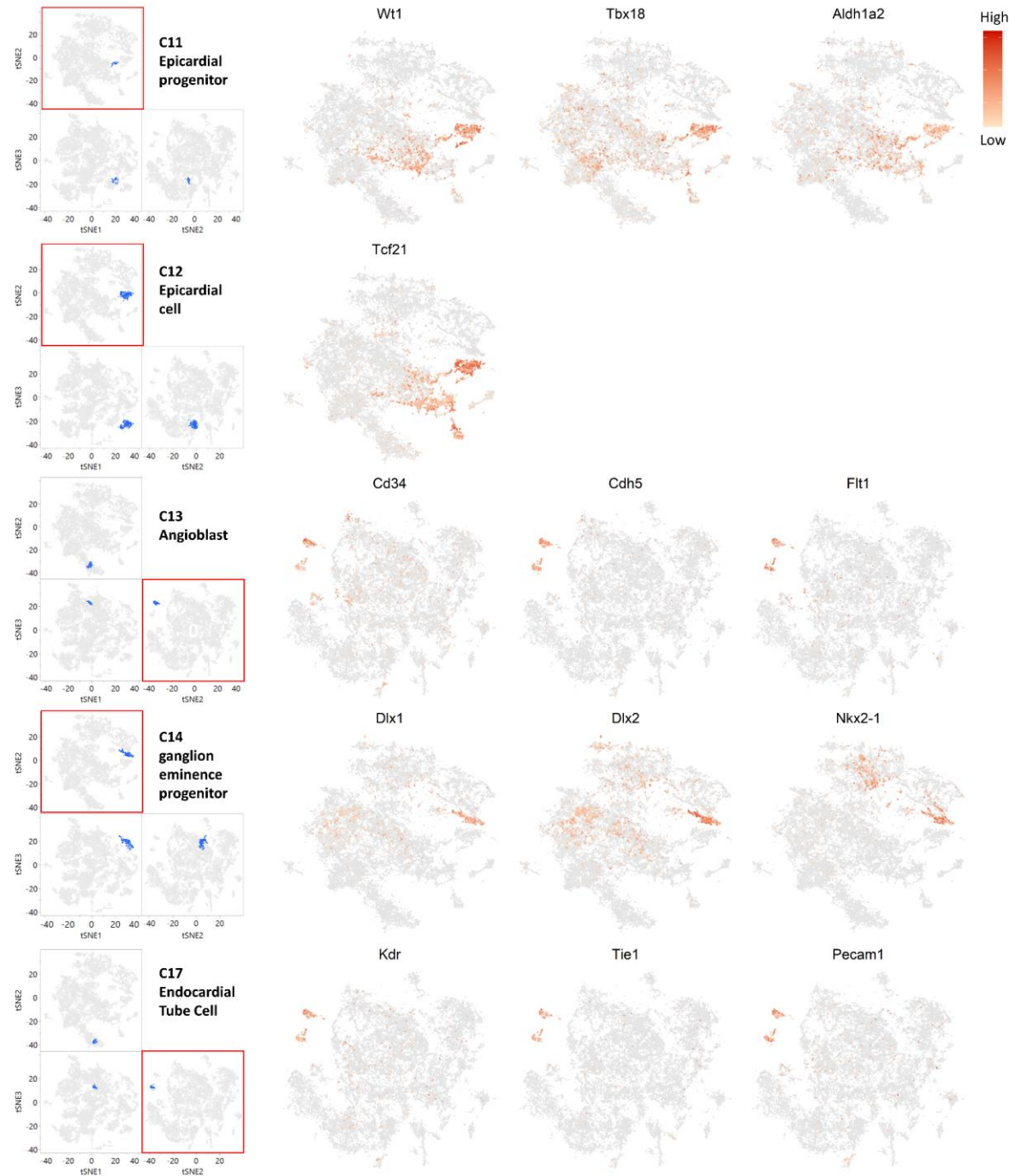


Figure S1.7: Continued

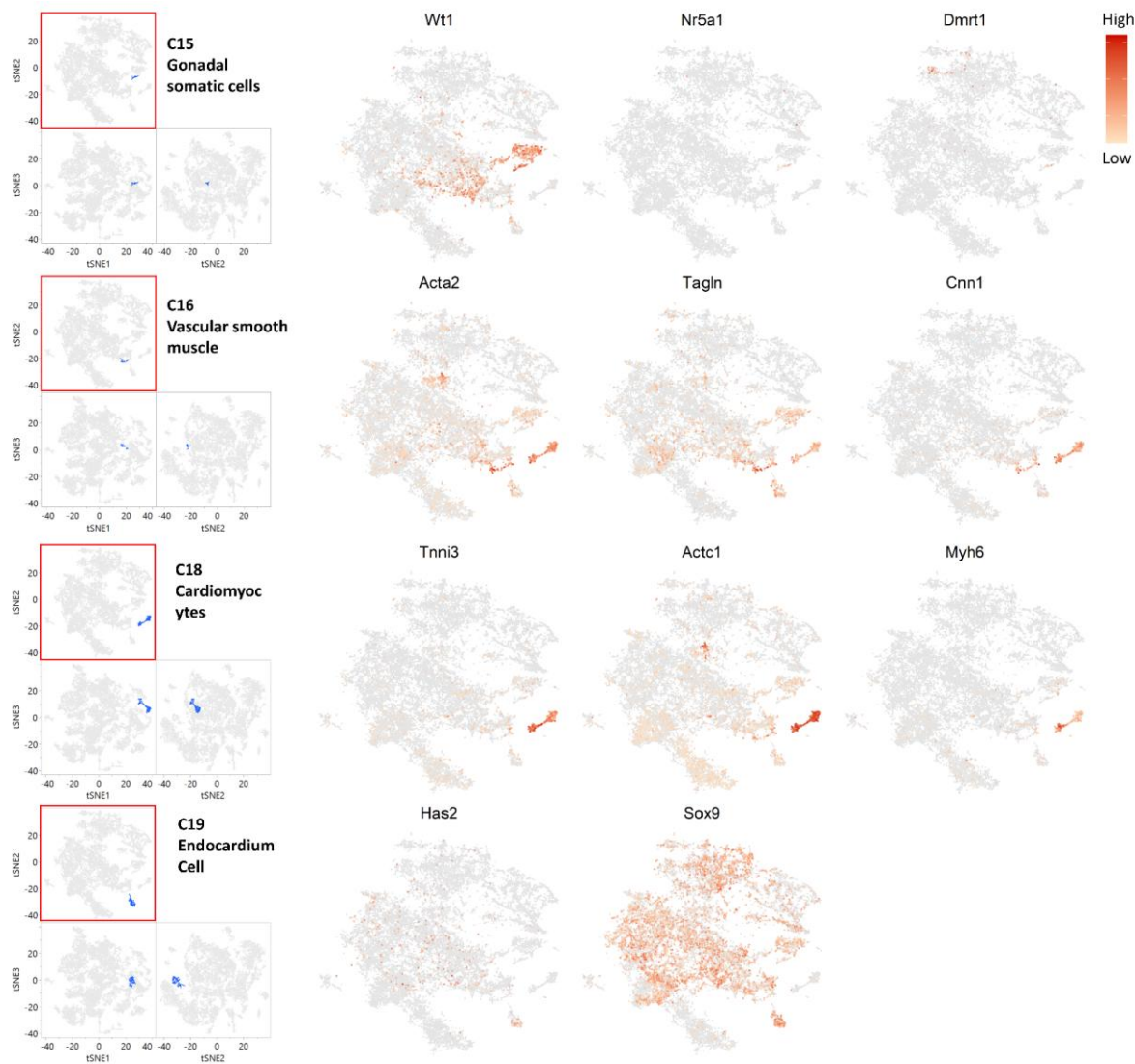


Figure S1.8: Marker gene expression plots for the mesenchyme cloud sub-cluster

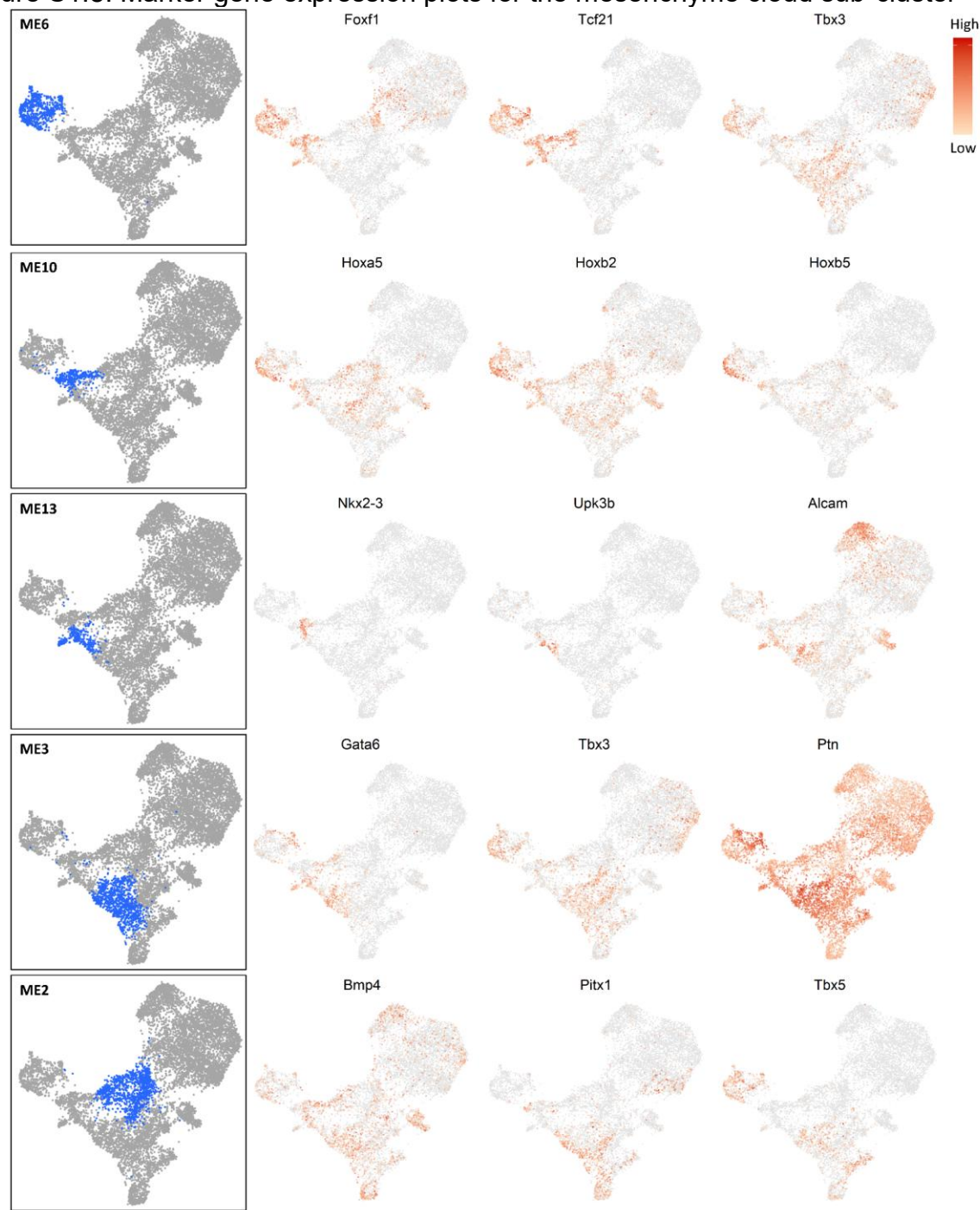


Figure S1.8: Continued

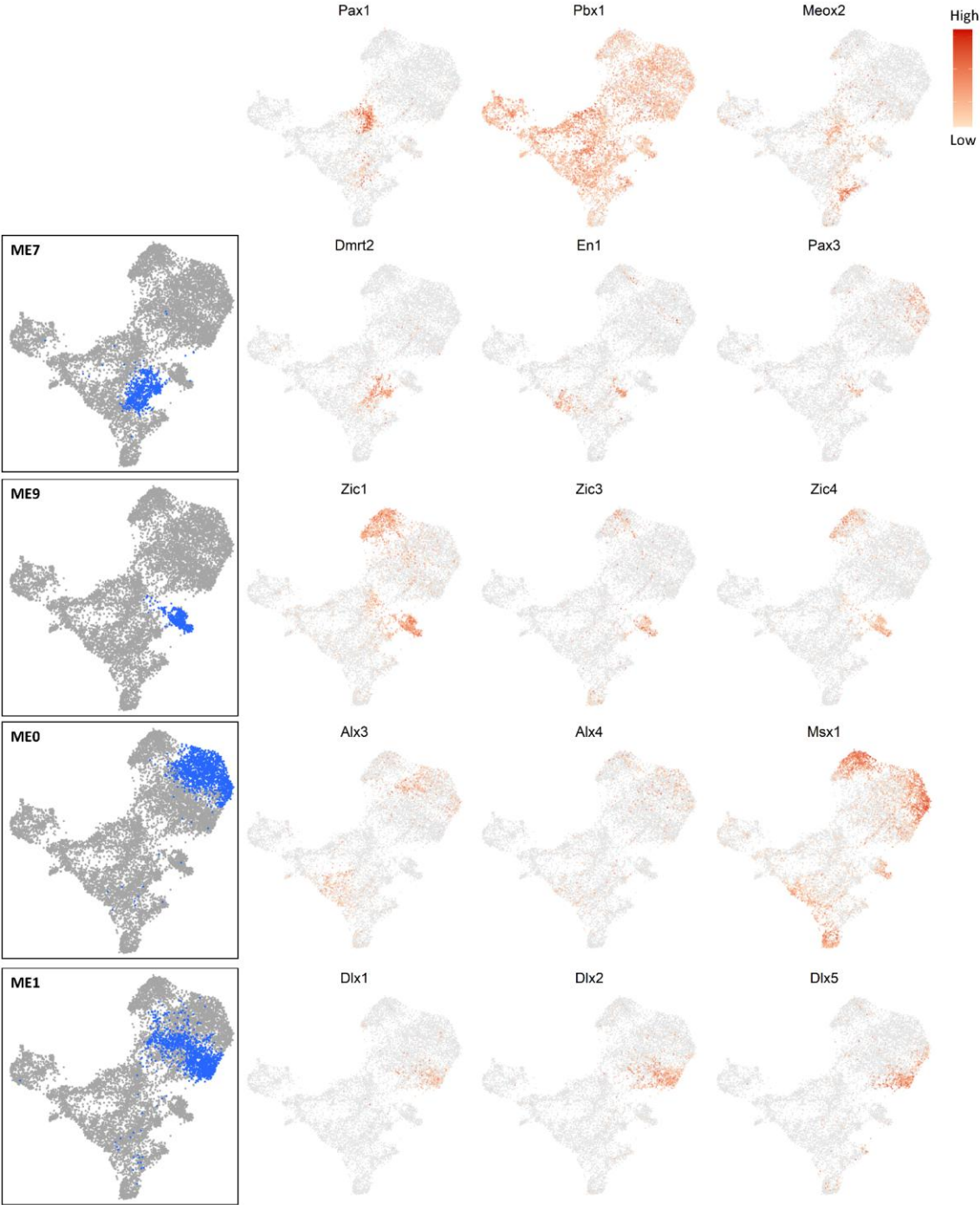


Figure S1.8: Continued

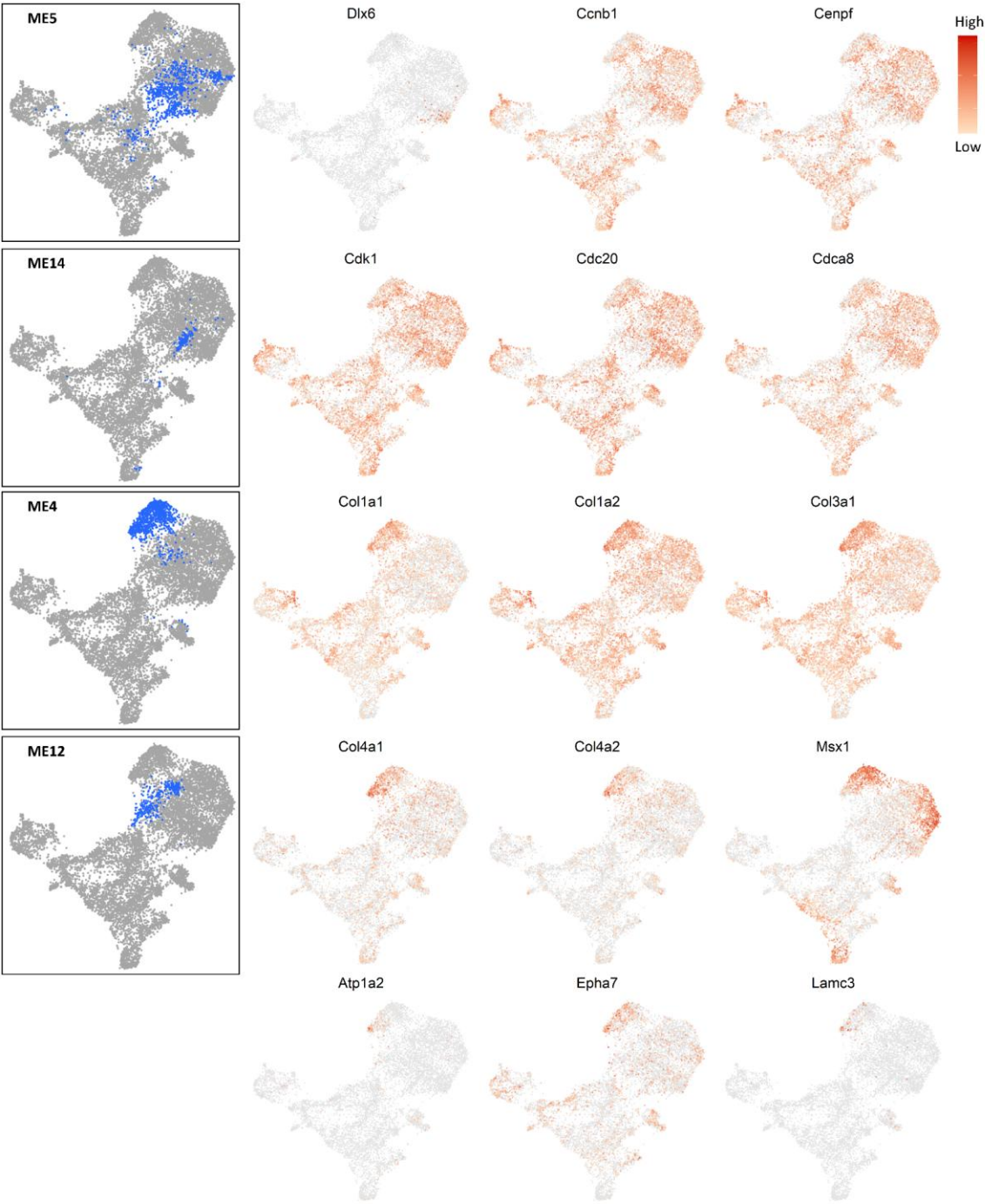


Figure S1.9: Hox gene expression plot for mesenchyme cloud sub-cluster

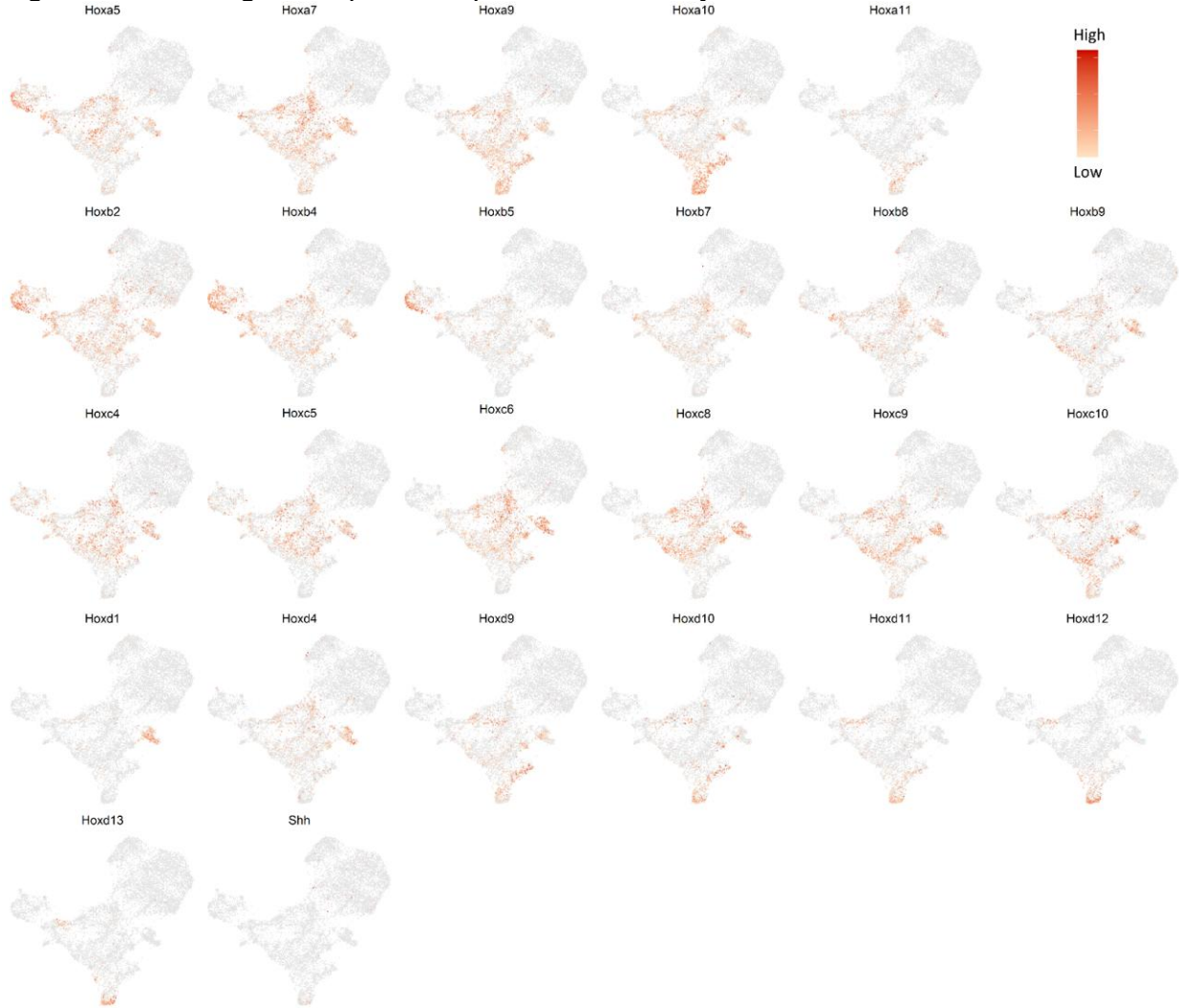


Figure S1.10: Selected marker gene expression plot for neuroectoderm cloud sub-cluster

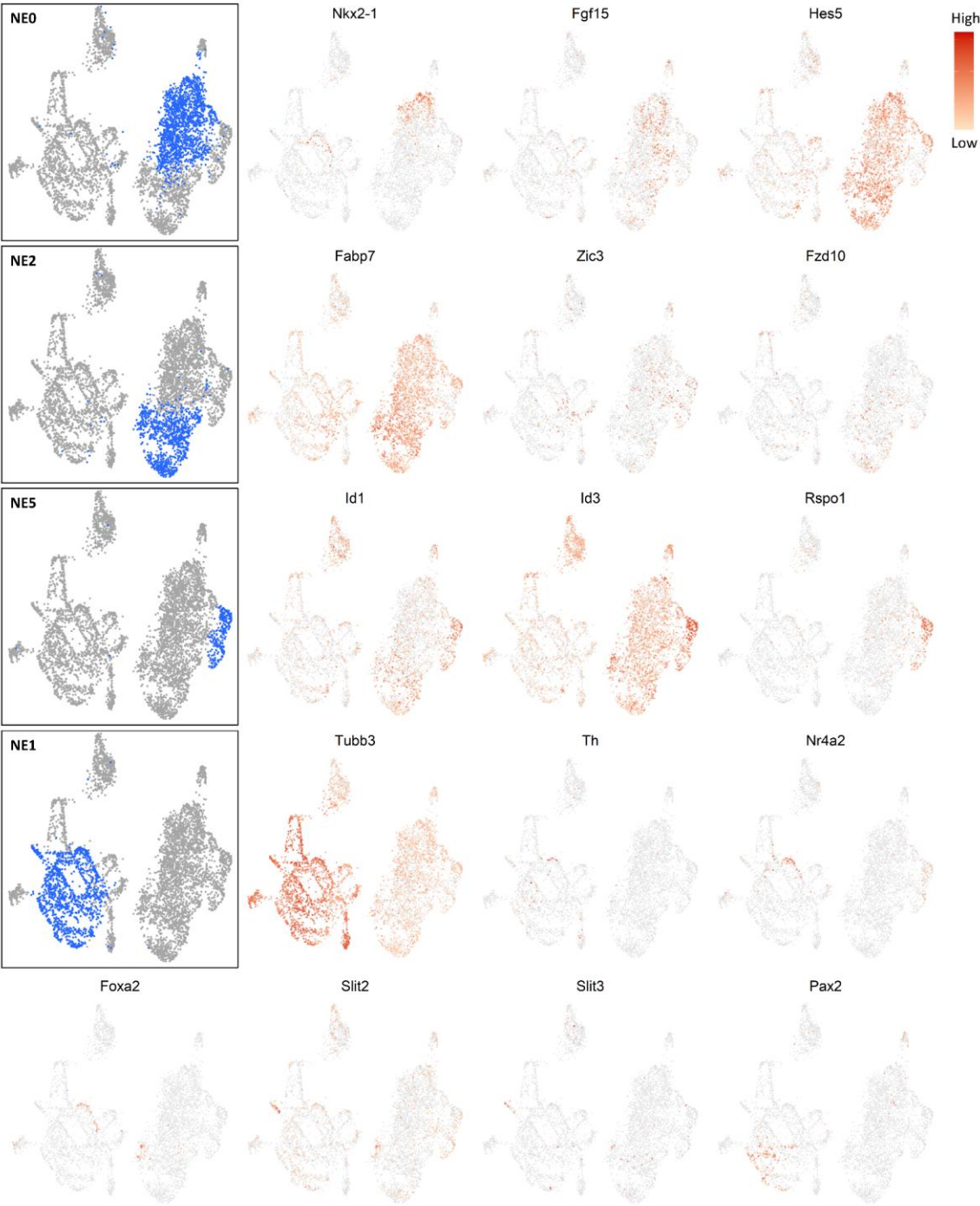


Figure S10: Continued

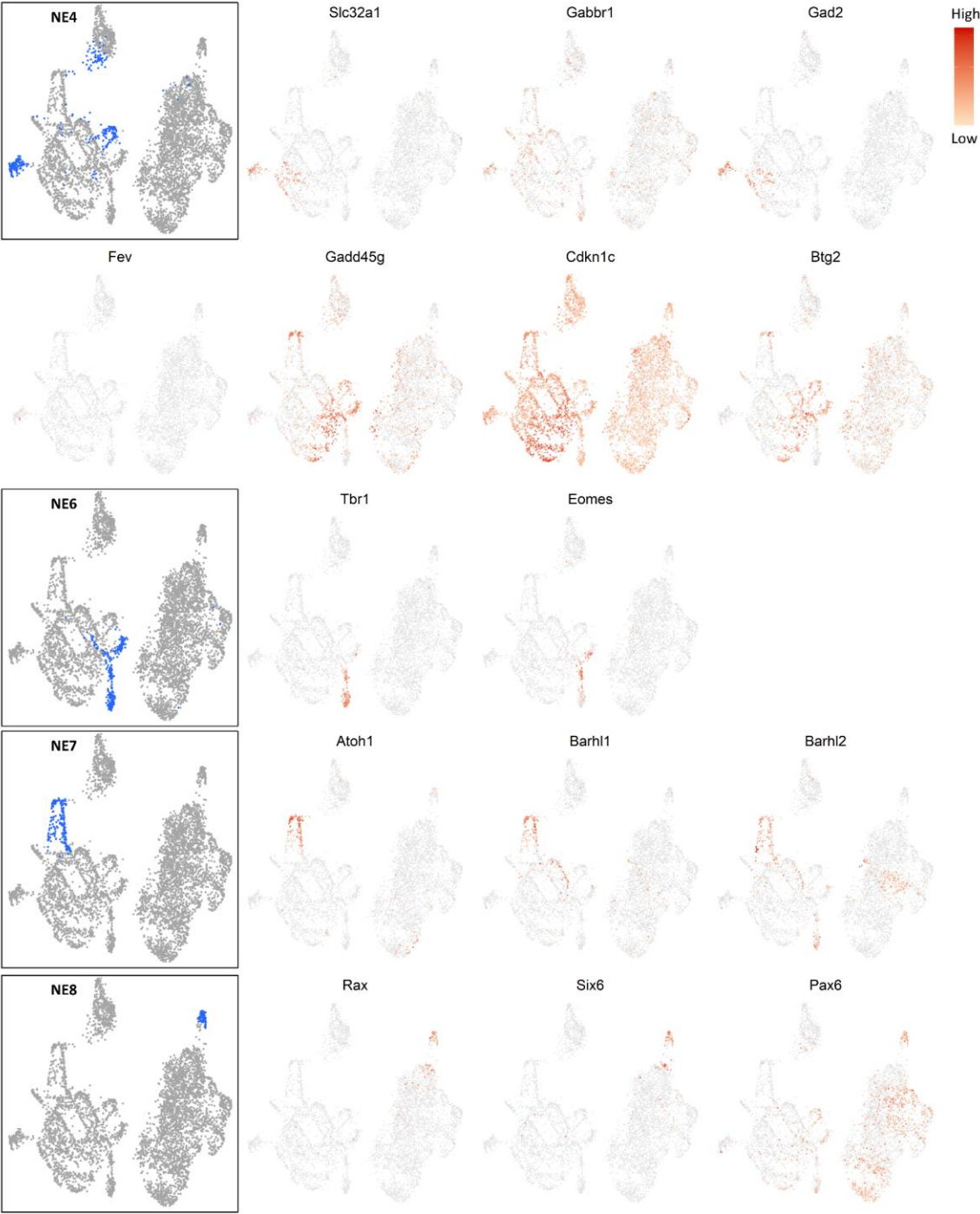


Figure S2.1: Coverage of cells from different stages and samples in UMAP space
(A-F) UMAP projection of E9.0-E11.5 integrated dataset, cells colored by developmental stages, most regions cover all stages. **(G-K)** UMAP projection of E9.0-E11.5 integrated dataset, cells colored by sample of origin.

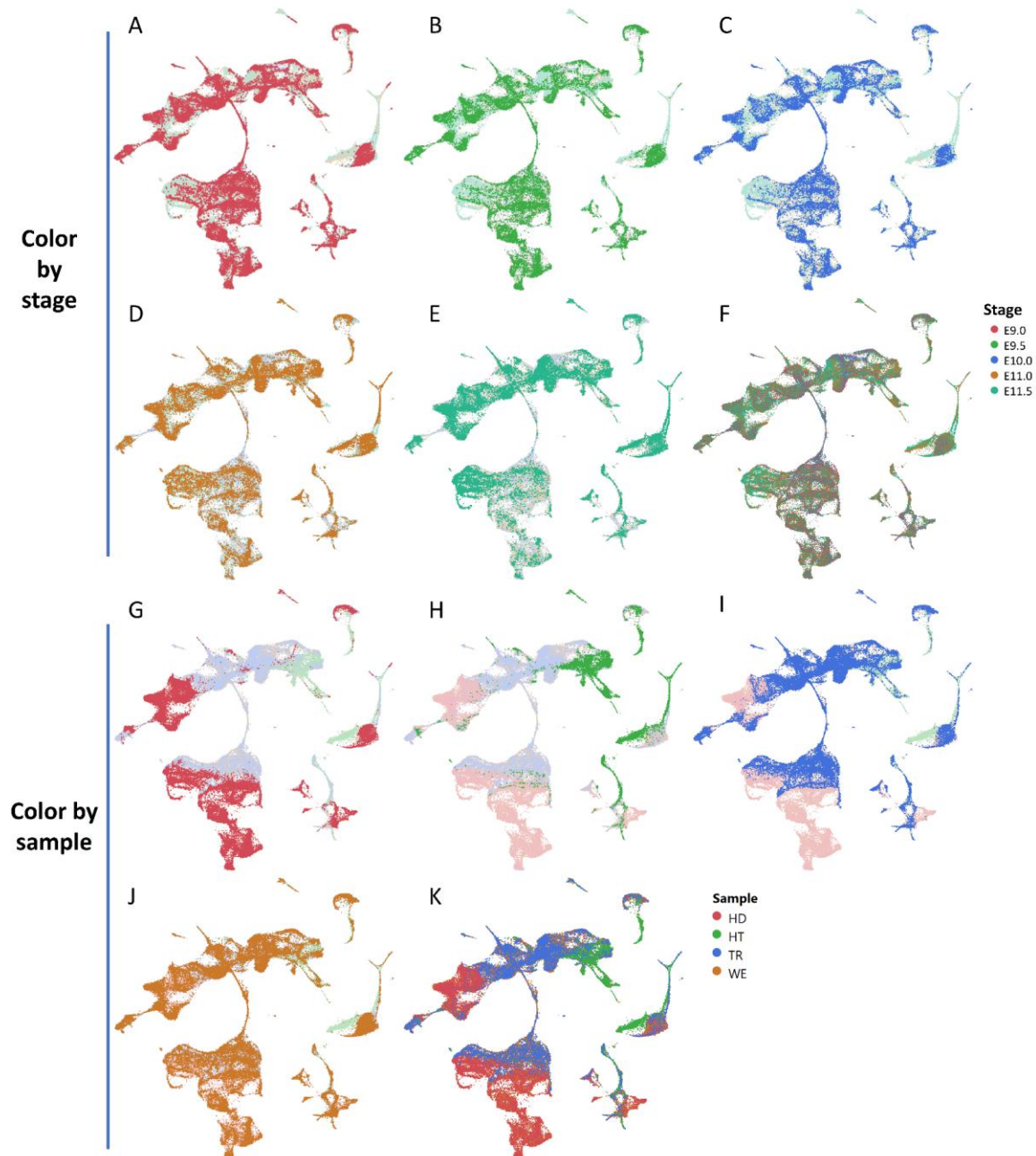


Figure S2.2: A trend of early-to-late stages has been observed in neuroectoderm subset
(A-F) UMAP projection of E9.0-E11.5 integrated dataset neuroectoderm subset, cells colored by developmental stages **(G-K)** UMAP projection of E9.0-E11.5 integrated dataset neuroectoderm subset, cells colored by sample of origin.

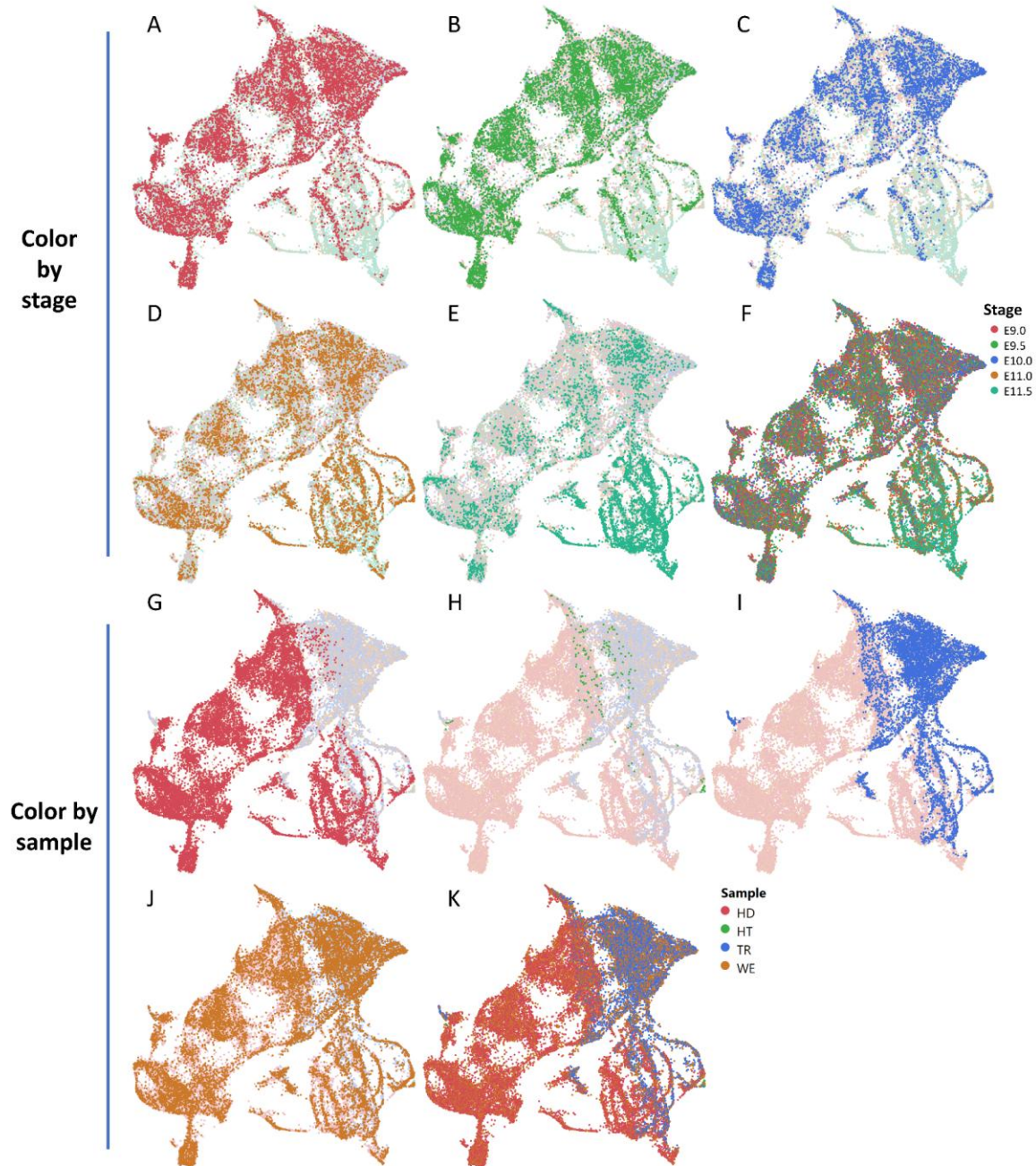


Figure S2.3: Various neural makers identify groups of neuroectoderm cells with different fates

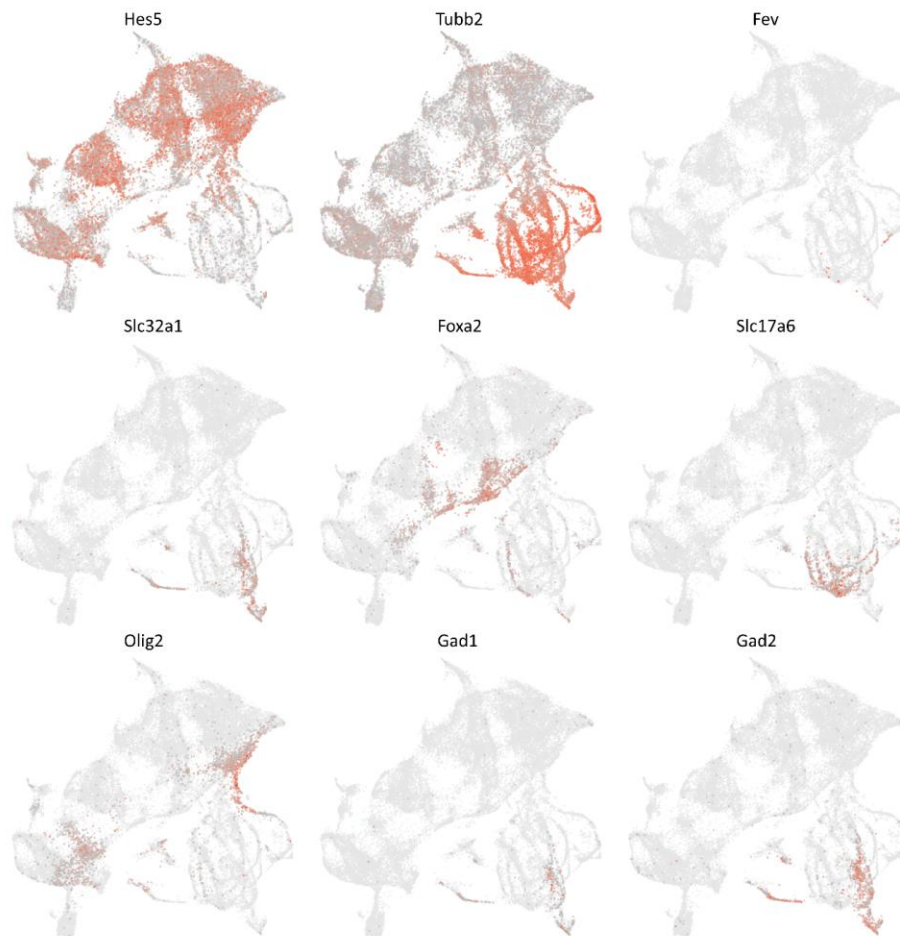


Figure S2.4: Stage distribution of neuroectoderm sub-clusters

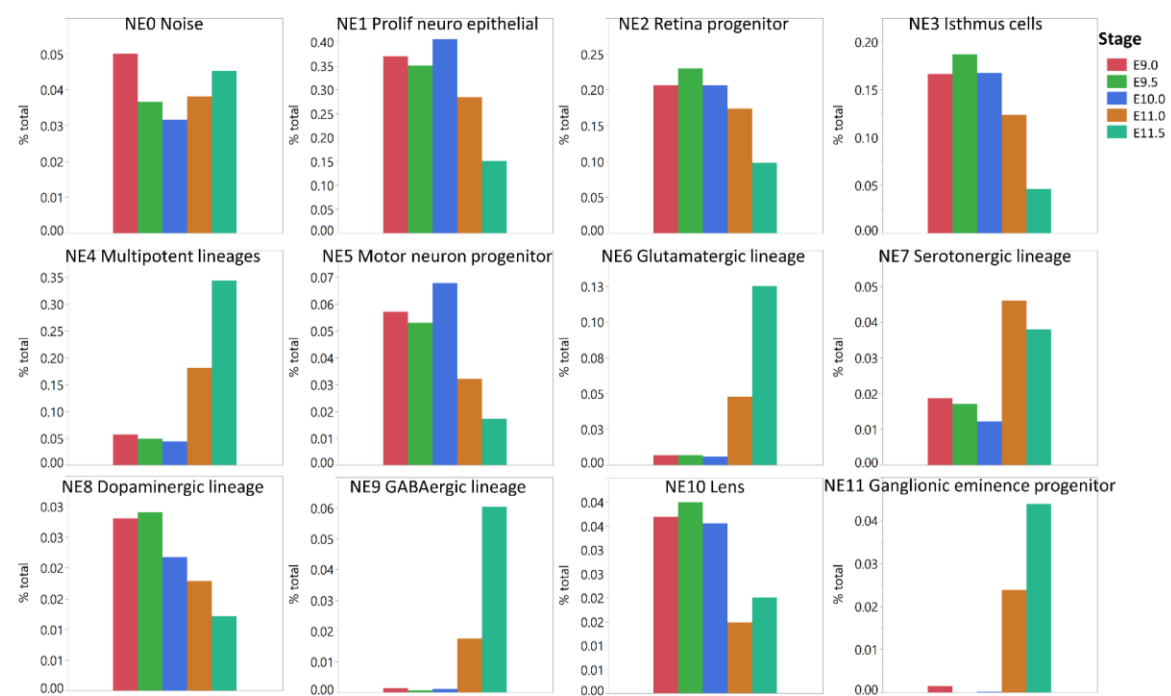


Figure S2.5: Coverage of cells from different stages and samples in UMAP space of neuroectoderm subset

(A-F) UMAP projection of E9.0-E11.5 integrated dataset mesoderm subset, cells colored by developmental stages **(G-K)** UMAP projection of E9.0-E11.5 integrated dataset mesoderm subset, cells colored by sample of origin

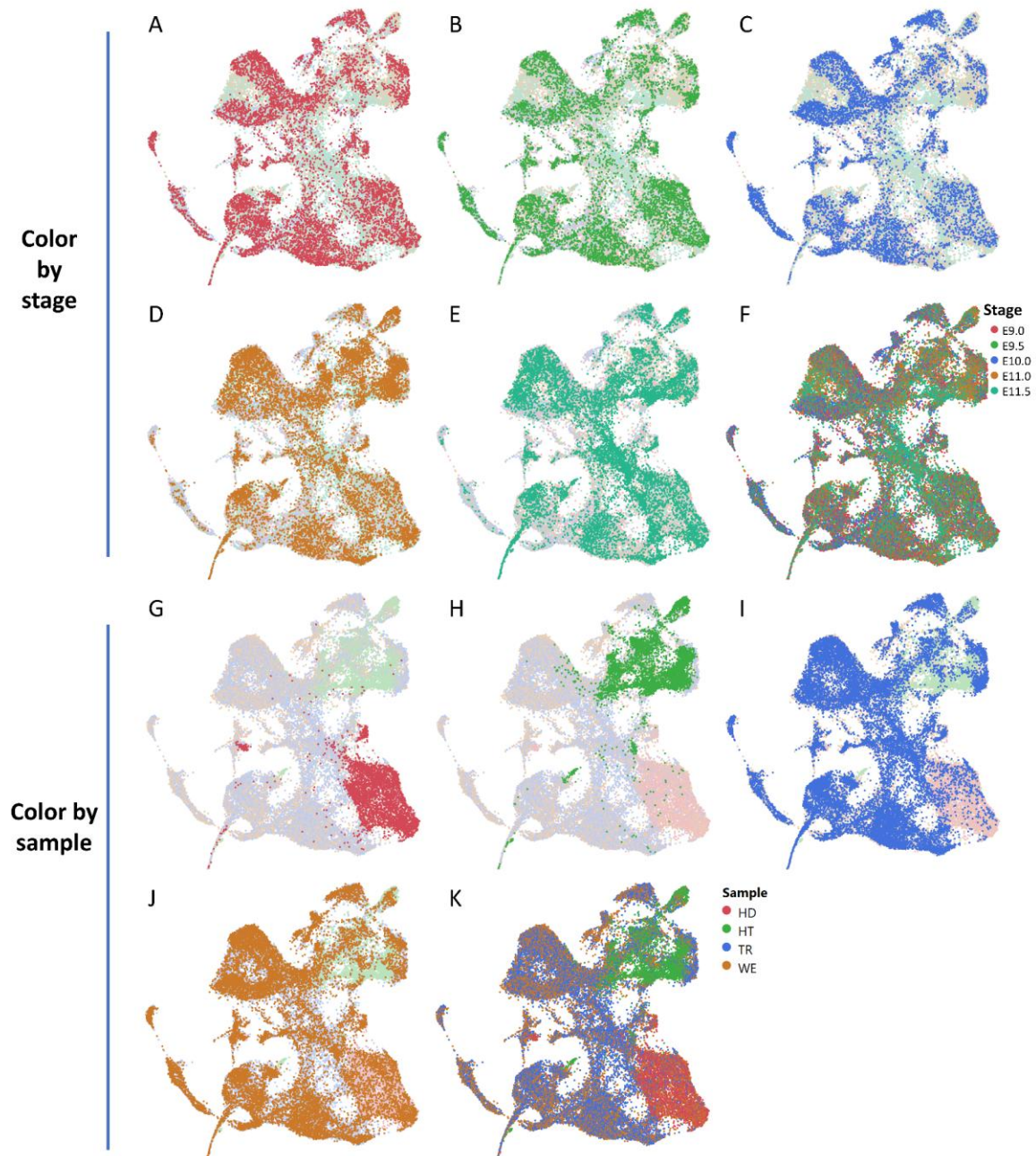


Figure S2.6: Stage distribution of mesoderm sub-clusters

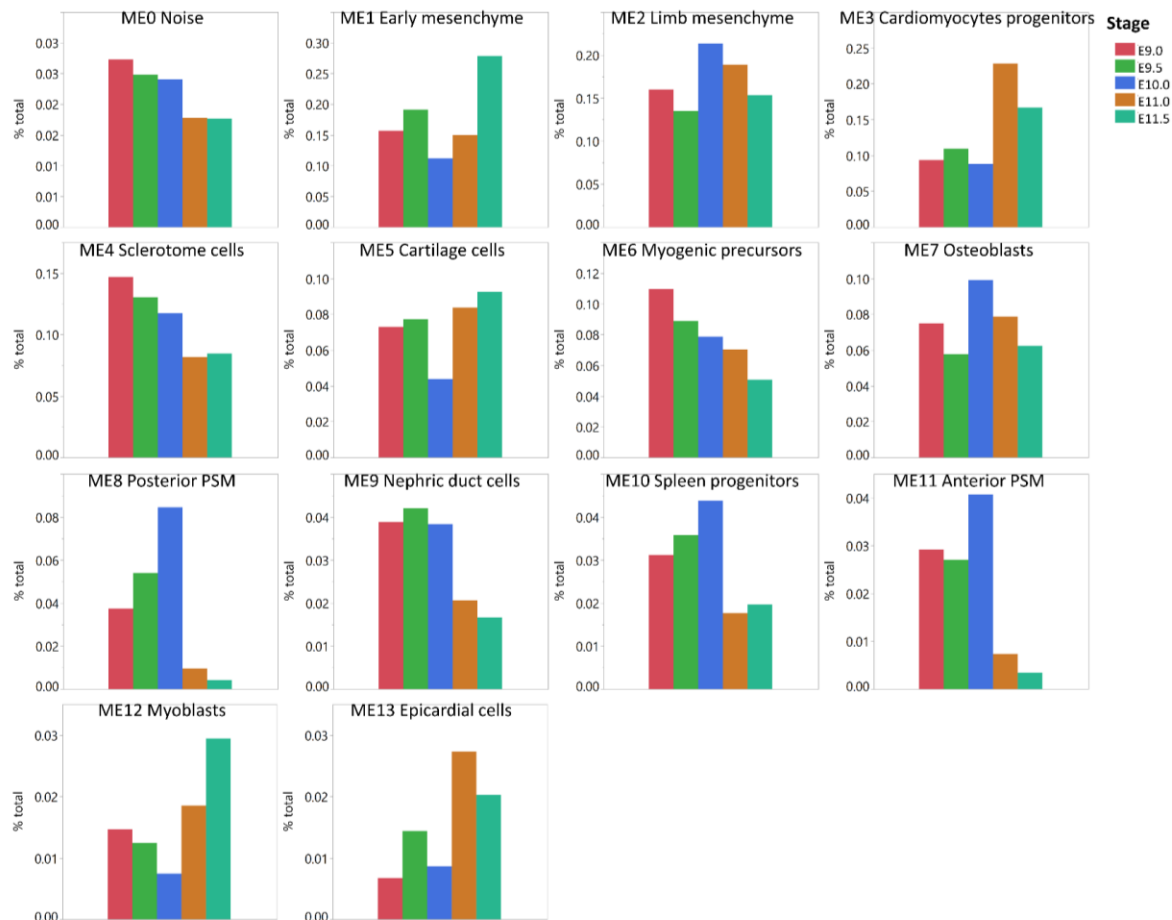


Figure S2.7: Various neural makers identify groups of neuroectoderm cells with different fates

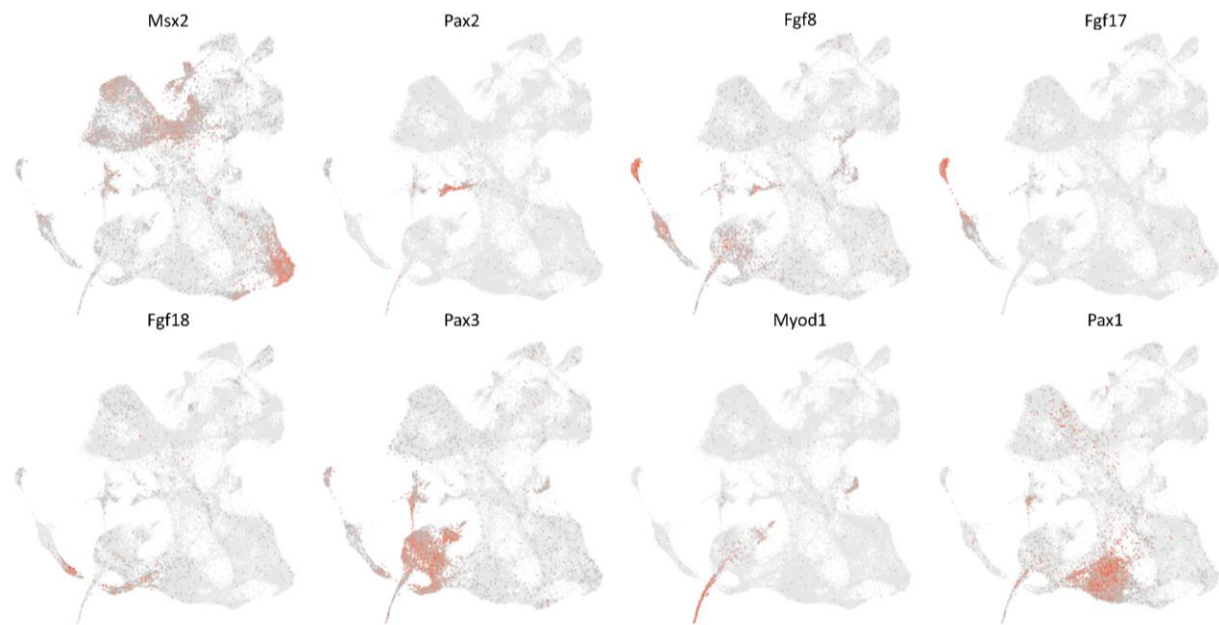


Figure S2.8: Coverage of cells from different stages in UMAP space and stage distribution of cranial neural crest (CNC) subset

(A-F) UMAP projection of E9.0-E11.5 integrated dataset CNC subset, cells colored by developmental stages **(G)** Stage distribution of CNC clusters

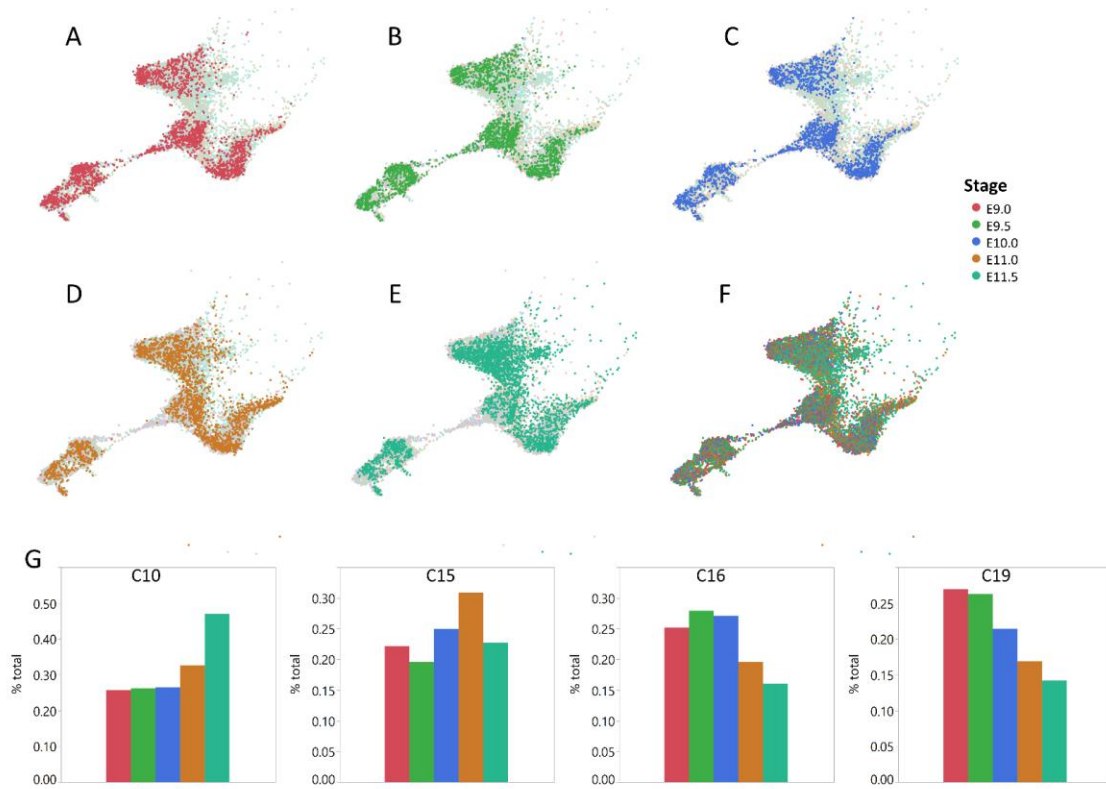


Figure S2.9: Various neural crest makers identify groups of cranial neural crest cells with different fates

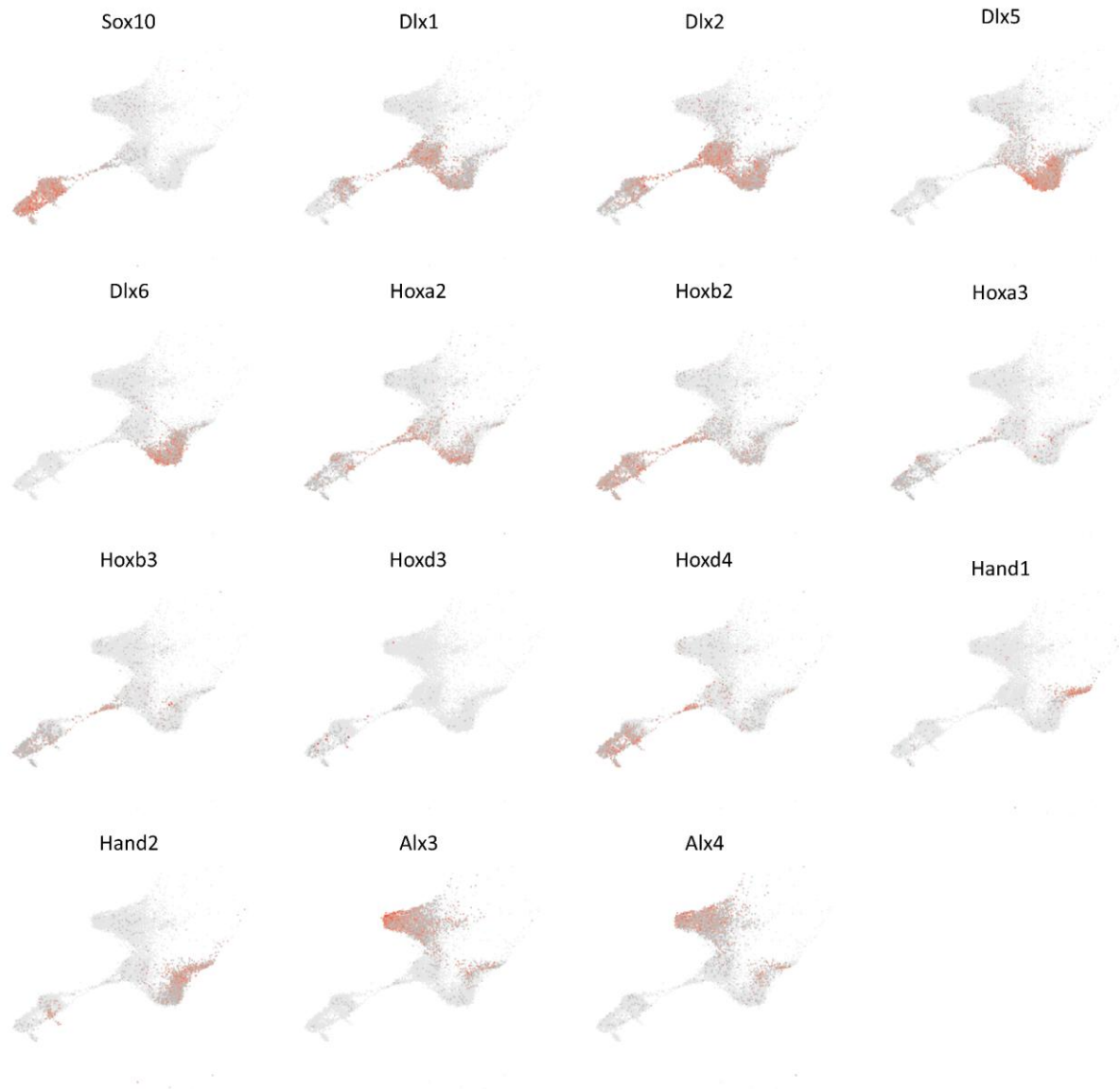


Figure S2.10: Coverage of cells from different stages and samples in UMAP space of epithelium subset

(A-F) UMAP projection of E9.0-E11.5 integrated dataset epithelium subset, cells colored by developmental stages **(G-K)** UMAP projection of E9.0-E11.5 integrated dataset epithelium subset, cells colored by sample of origin

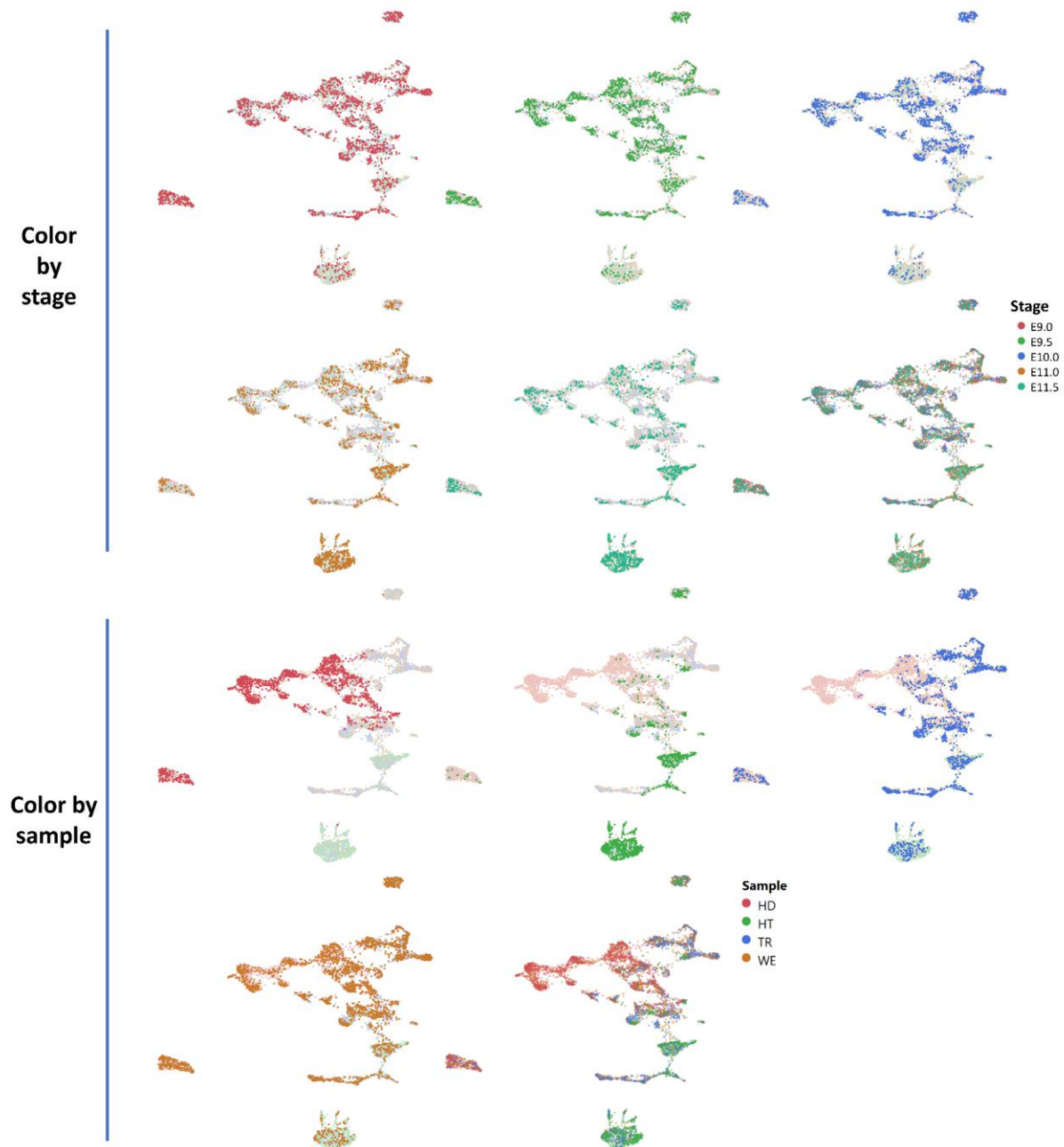


Figure S2.11: Stage distribution of epithelium sub-clusters

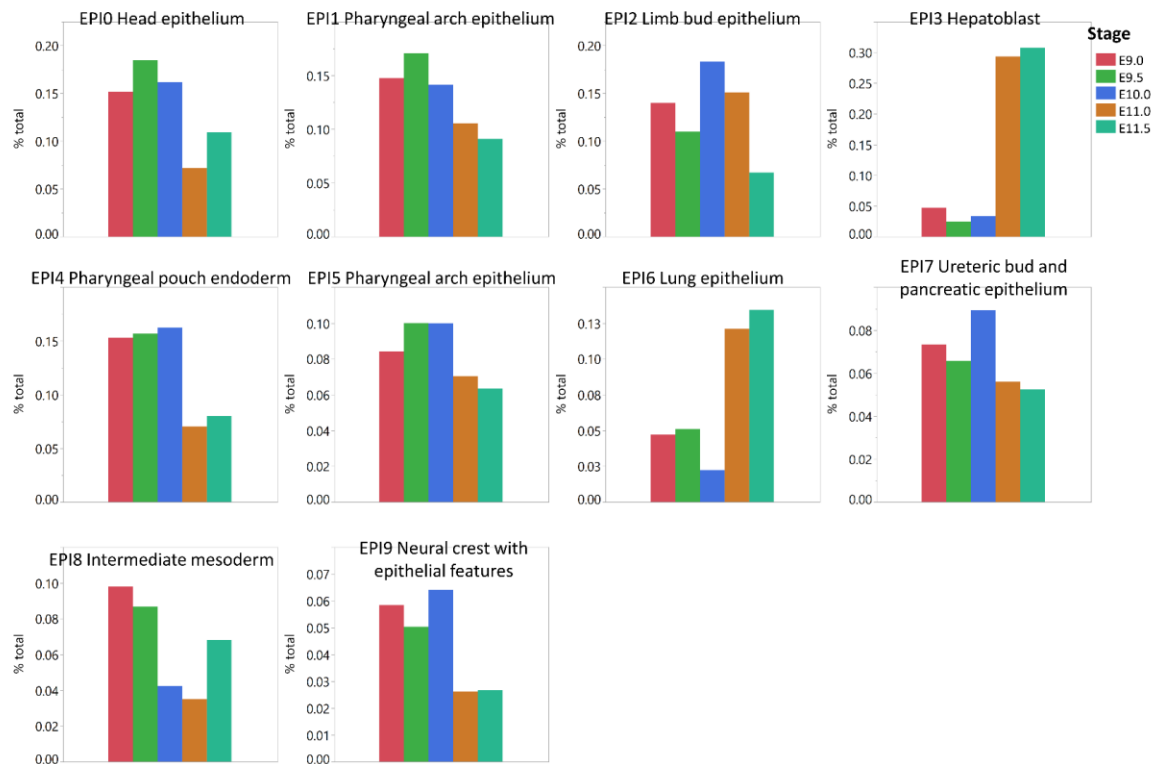


Figure S2.12: Various makers identify groups of epithelial and endodermal cell types with different fates



Figure S2.12: Continued



Figure S2.13: Various makers and stage progression identify differences between endothelial subcluster

(A) Marker gene projection in endothelial subcluster. **(B)** Stage distribution of epithelium sub-clusters

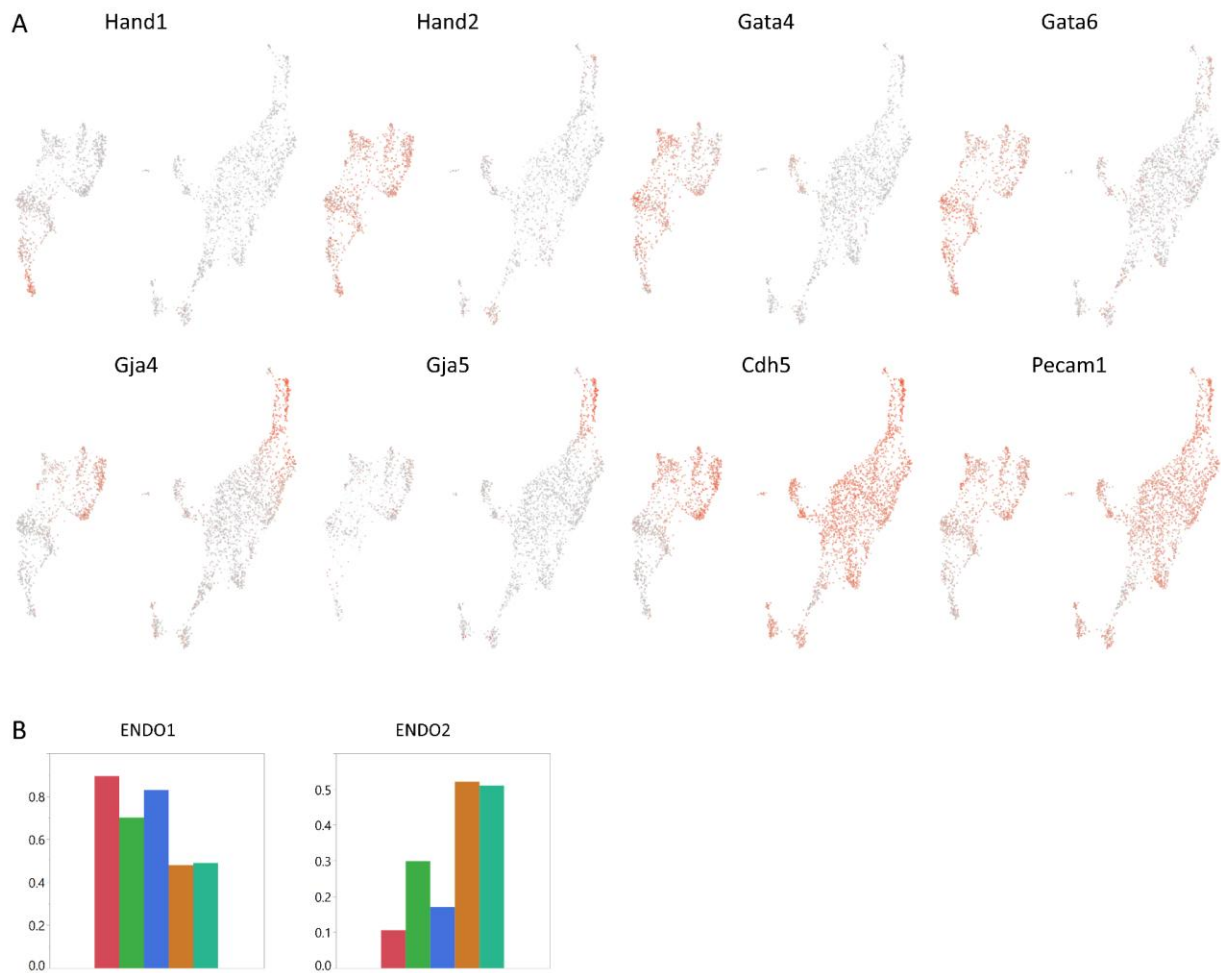


Figure S3.1: Enriched MGI Phenotype terms of each bin in word clouds.

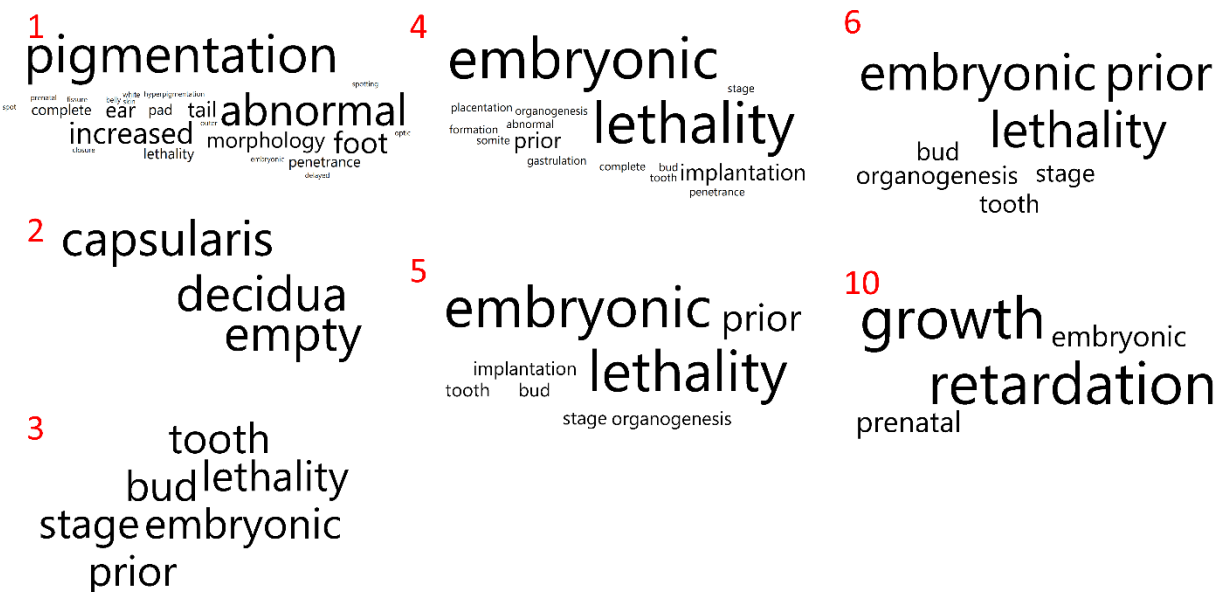


Figure S4.1: Both housekeeping and cell type specific metagenes identified from scRNA-seq dataset

(A) tSNE projection of genes from E11.5 mouse whole embryo, colored by cell type specific metagenes (CTSG), housekeeping metagenes (HKG) and unknown. **(B)** The same tSNE projection as A, colored by 63 metagenes, with annotation of each metagene.

Fig S1

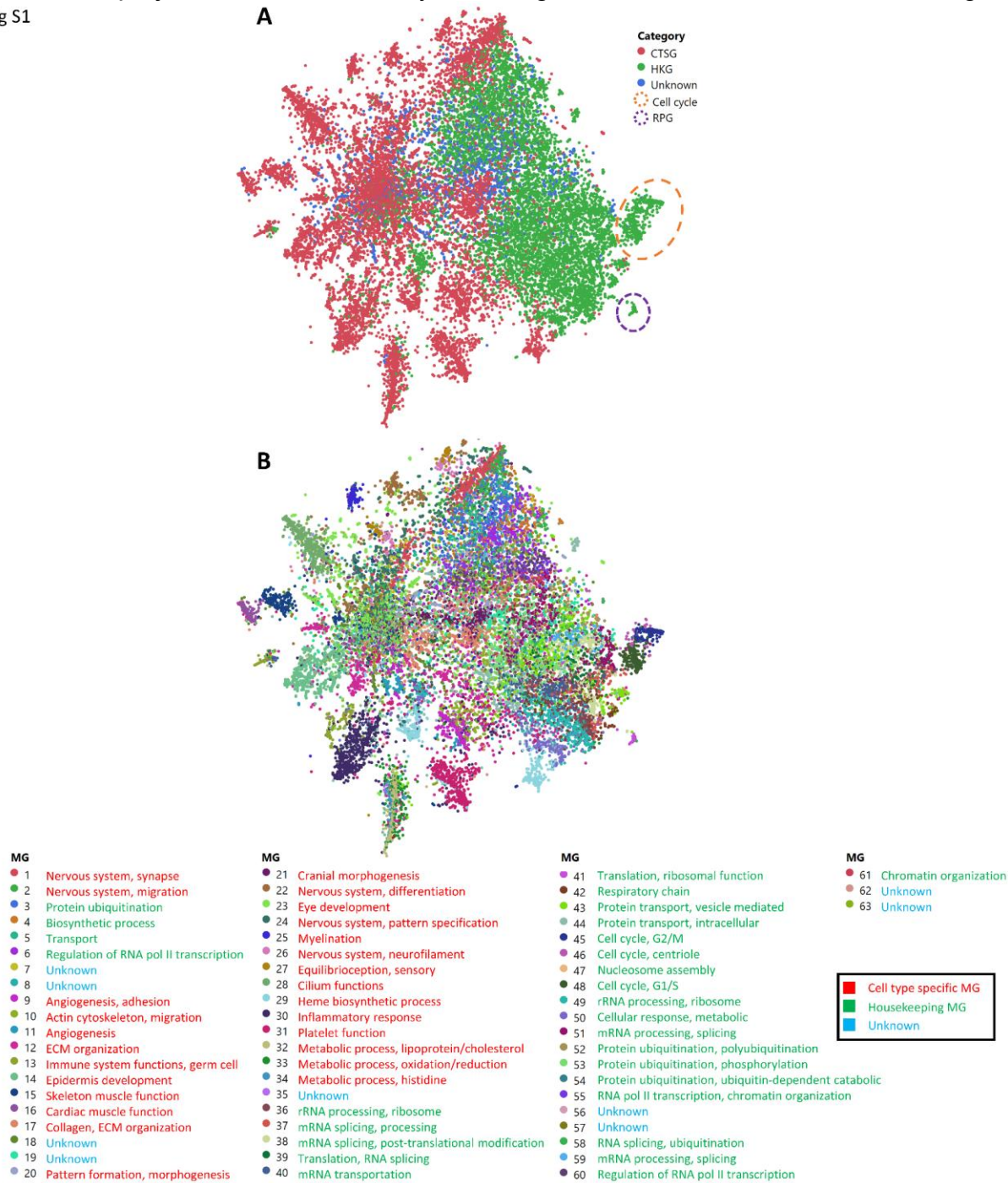


Figure S4.2: Housekeeping metagenes extracted from E11.5 mouse whole embryo
28 metagenes are annotated to be involved housekeeping functions, with their expression plots highlighting various cell clusters from the same source dataset.

(A-E) Housekeeping metagenes in five different broad categories. A: post-translational modification; B: cell cycle; C: transcription; D: translation; E: post-transcriptional modification. Left column: tSNE projection of genes from E11.5 mouse whole embryo, with each of the five broad categories highlighted and colored; Second column: annotation and color legend of the first column; Third column: Metagene tree, with nodes from each of the five broad categories highlighted in green; Right column: UMAP projection of the cells from the same source dataset, colored by average expression level from each of the five broad categories.

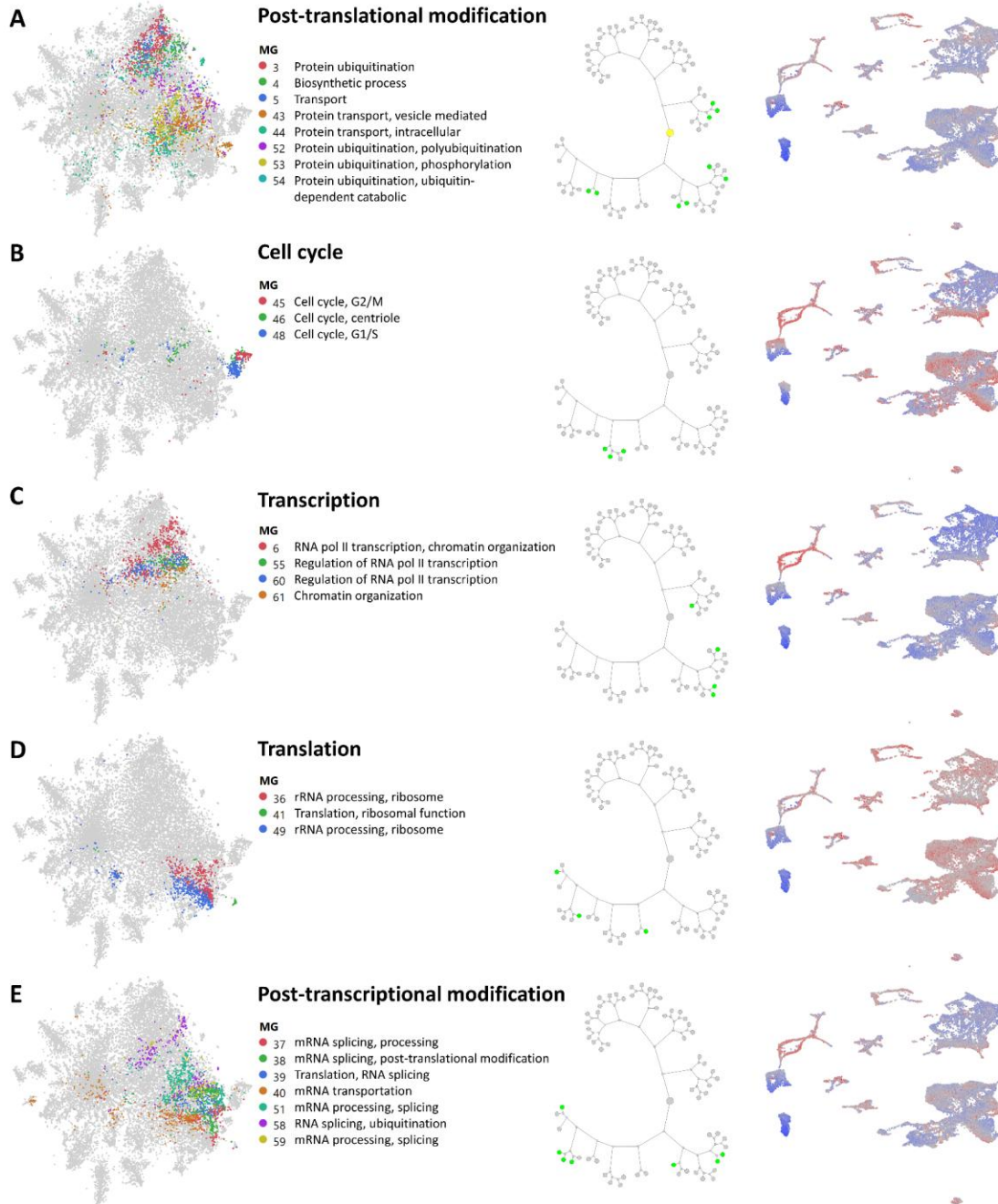


Figure S4.3: Three metagenes mapped to two sets of cell cycle markers

By analyzing three metagenes independently with two sets of cell cycle markers, MG45 shows good alignment with G2/M markers, MG48 shows good alignment with G1/S markers, and MG46 shows none.

(A) tSNE projection made from only MG45 and two sets of cell cycle markers, based on E11.5 mouse whole embryo scRNA-seq dataset, colored by gene sets (left panel), correlation with G1/S markers (middle panel), and correlation with G2/M markers (right panel). **(B)** Same setup as A but tSNE made from MG46 and two sets of cell cycle markers. **(C)** Same setup as A but tSNE made from MG48 and two sets of cell cycle markers.

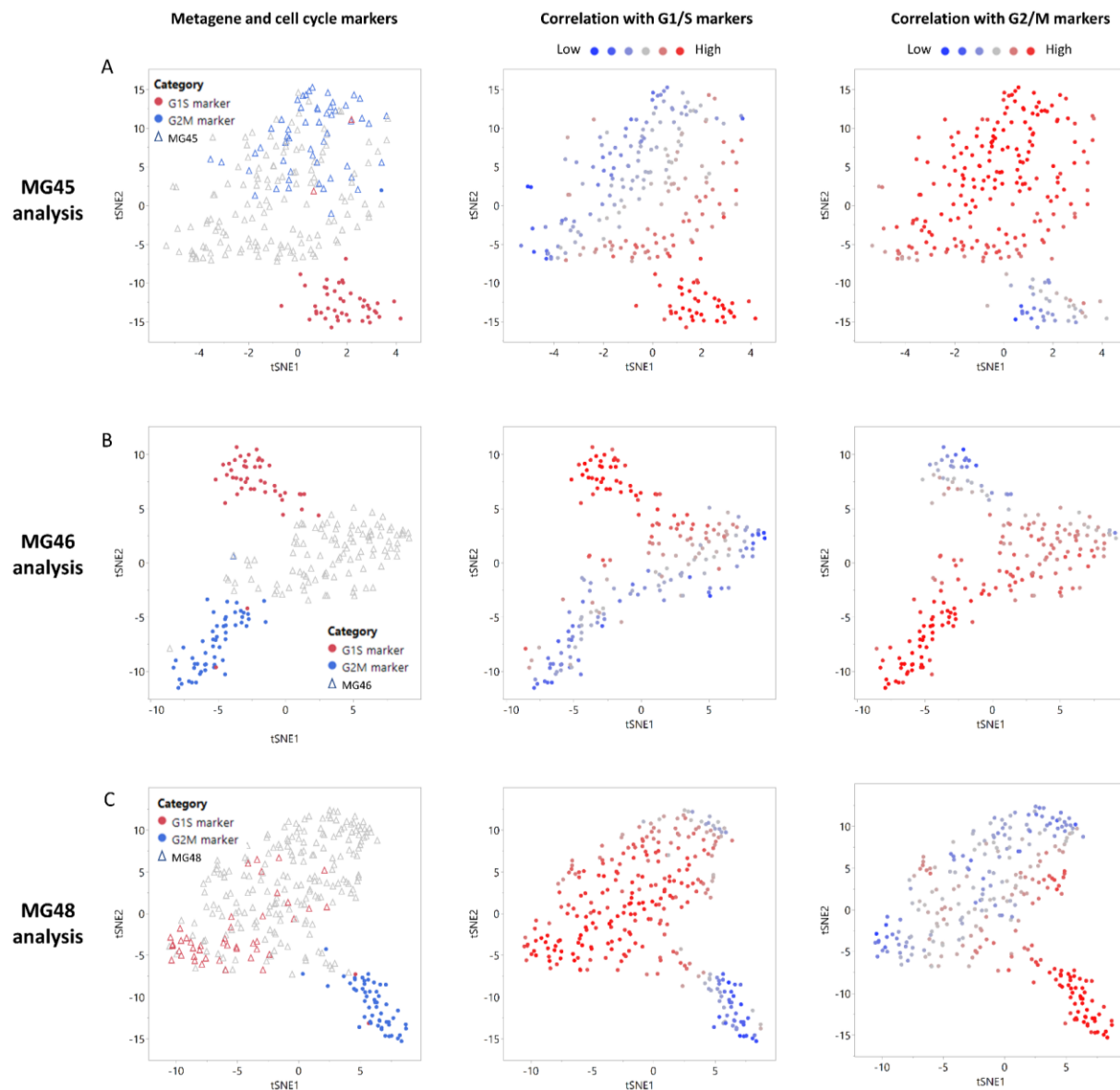


Figure S4.4 Cell proliferation of all KD cells
 The number of total cells (blue bar) and percentage of live cells (orange bar) for all knockdown cells. Numbers have been normalized to the Mock, two biological and two technical replicates were used.

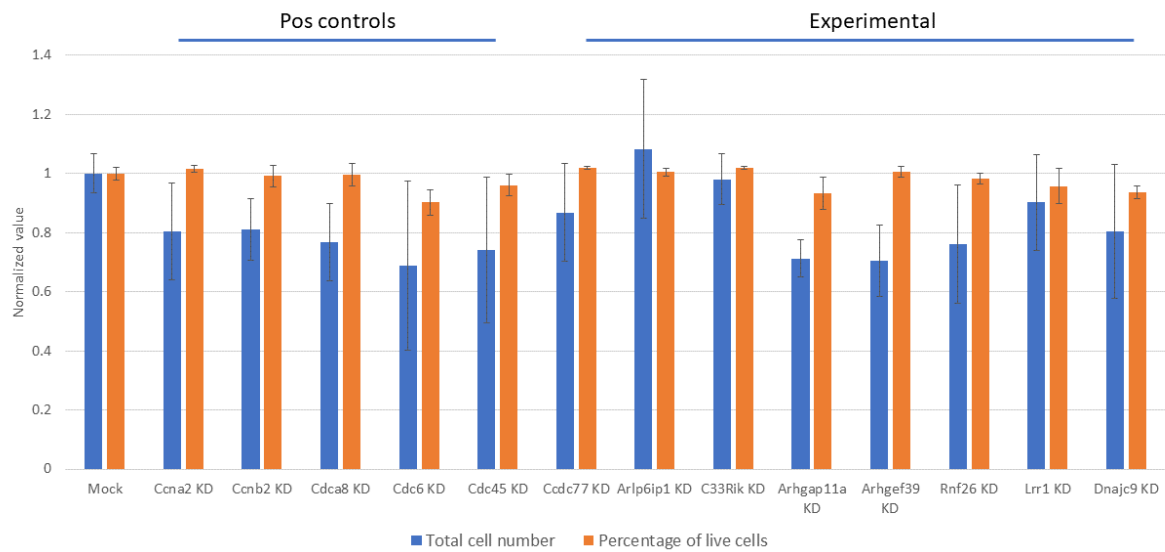


Table S3.1: Details of all datasets used in this study

Description	Number of cells	Availability
Mouse Cell Atlas: Mouse adult ovary	2459	GSM2906456
HIV infected human PBMC	59286	https://singlecell.broadinstitute.org/single_cell/study/SCP256/integrated-single-cell-analysis-of-multicellular-immune-dynamics-during-hyper-acute-hiv-1-infection
Human Cell Landscape: Human iPS-to-EB_9Day	25000	GSM4008719
Human Cell Landscape: Adult-Gall-Bladder1	10000	GSM3980137
Chromium demonstration dataset: 4k Pan T cells from a healthy human donor	4538	https://support.10xgenomics.com/single-cell-gene-expression/datasets/2.1.0/t_4k?
Chromium demonstration dataset: 1k Brain cells from an E18 mouse	931	https://support.10xgenomics.com/single-cell-gene-expression/datasets/2.1.0/neurons_900
Chromium demonstration dataset: 1k Heart cells from an E18 mouse	712	https://support.10xgenomics.com/single-cell-gene-expression/datasets/3.0.0/heart_1k_v2
Chromium demonstration dataset: 10k Heart cells from an E18 mouse	7713	https://support.10xgenomics.com/single-cell-gene-expression/datasets/3.0.0/heart_10k_v3
Tabula Muris: Bladder_P4_3	149	GSE109774
Tabula Muris: Bladder_P4_4	1783	GSE109774
Tabula Muris: Bladder_P7_7	568	GSE109774
Tabula Muris: Heart_and_Aorta_P7_4	654	GSE109774
Tabula Muris: Kidney_P4_5	610	GSE109774
Tabula Muris: Kidney_P4_6	908	GSE109774
Tabula Muris: Kidney_P7_5	1264	GSE109774
Tabula Muris: Limb_Muscle_P7_14	2243	GSE109774

Tabula Limb_Muscle_P7_15	Muris:	2301	GSE109774
Tabula Muris: Liver_P4_2		1006	GSE109774
Tabula Muris: Liver_P7_0		596	GSE109774
Tabula Muris: Liver_P7_1		322	GSE109774
Tabula Mammary_Gland_P7_12	Muris:	1938	GSE109774
Tabula Mammary_Gland_P7_13	Muris:	2543	GSE109774
Tabula Muris: Tongue_P4_0		3101	GSE109774
Tabula Muris: Tongue_P4_1		2488	GSE109774
Tabula Muris: Tongue_P7_10		1949	GSE109774
Mouse whole embryo E9		26440	
Mouse whole embryo E9.5		24678	
Mouse whole embryo E10		19242	
Mouse whole embryo E11		26126	
Mouse whole embryo E11.5		25978	

Table S3.2: Published HKG lists used in this study

Publication	Organism	Originally published HKG No.	Converted mouse gene No. used in this study	Year
Zhang 2004	Mouse	Transcript based	856	2004
Ramskold 2009	Human	7897	6420	2009
Eisenberg 2013	Human	3803	3381	2013
Lin 2019	Mouse	913	874	2019
Houkpe 2020	Human and Mouse	3277	3268	2020